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(54) **NOVEL CRISPR-CAS12I SYSTEMS AND
USES THEREOF**

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C12N 15/90 (2006.01)

(52) **U.S. Cl.**

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15/88 (2013.01); *C12N 15/907* (2013.01);
A61K 48/00 (2013.01); *C07K 2319/09*
(2013.01); *C12N 2310/20* (2017.05); *C12N*
2800/80 (2013.01)

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§ 371 (c)(1),

(2) Date: **Oct. 24, 2024**

(57)

ABSTRACT

The disclosure provides Cas12i polypeptides, fusion proteins comprising such Cas12i polypeptides, CRISPR-Cas12i systems comprising such Cas12i polypeptides or fusion proteins, and methods of using the same.

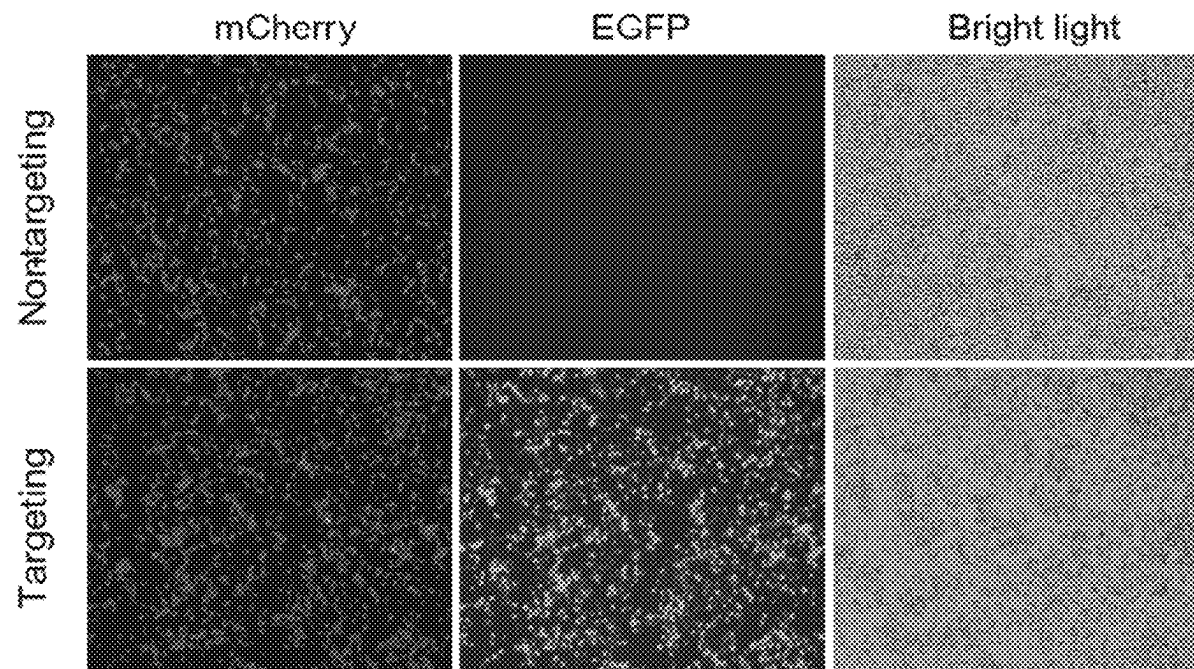
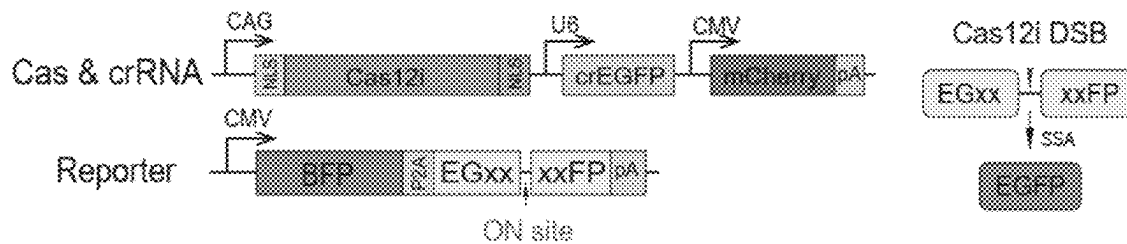
Specification includes a Sequence Listing.

(30) **Foreign Application Priority Data**

Apr. 25, 2022 (WO) PCT/CN2022/089074

Nov. 2, 2022 (WO) PCT/CN2022/129376

Jan. 20, 2023 (WO) PCT/CN2023/073420



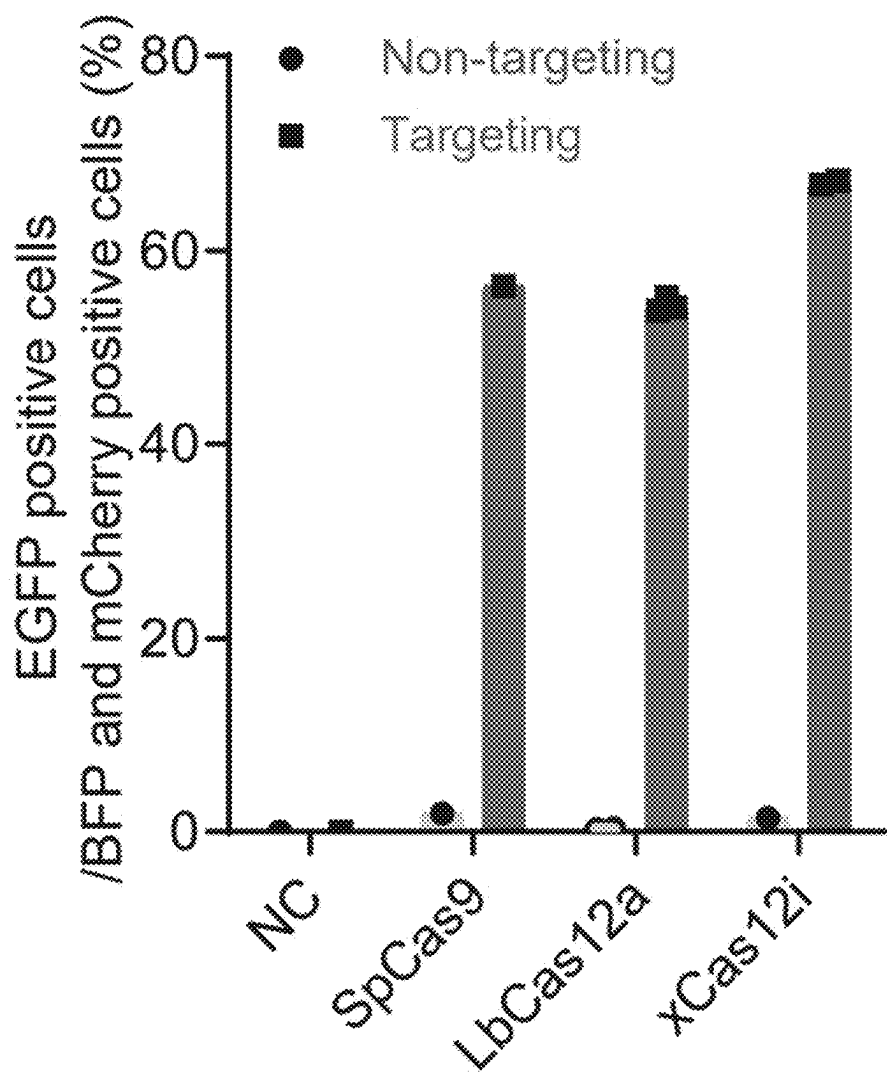


FIG. 1A

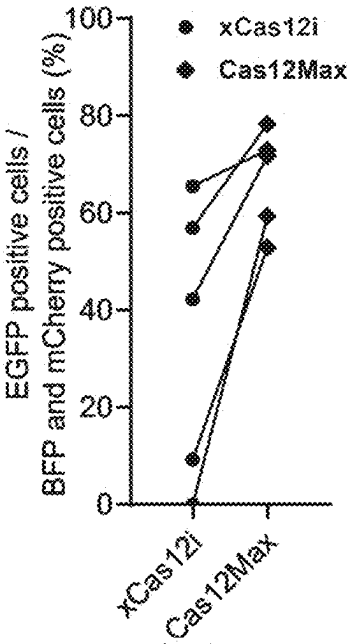


FIG. 1C

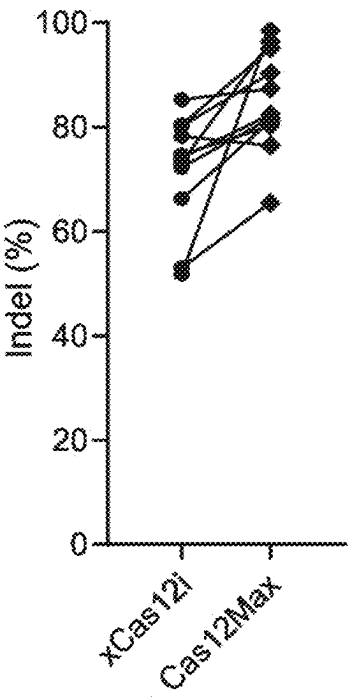


FIG. 1D

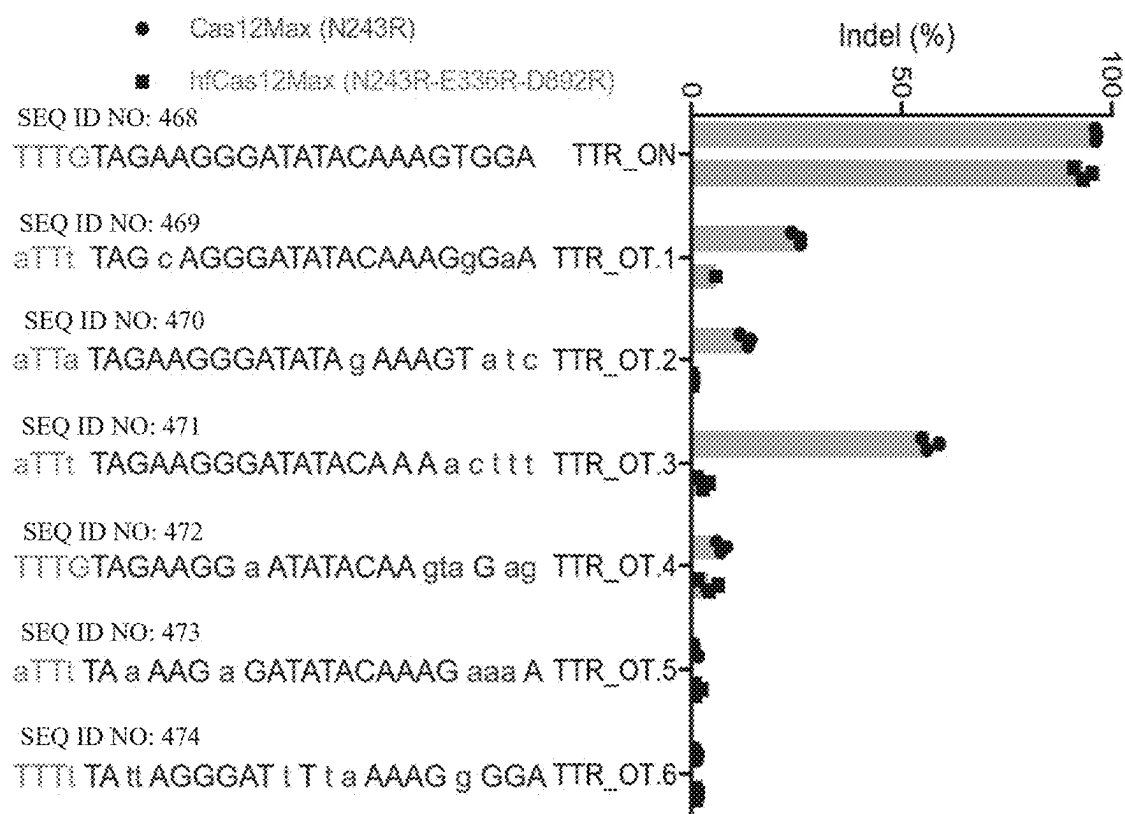
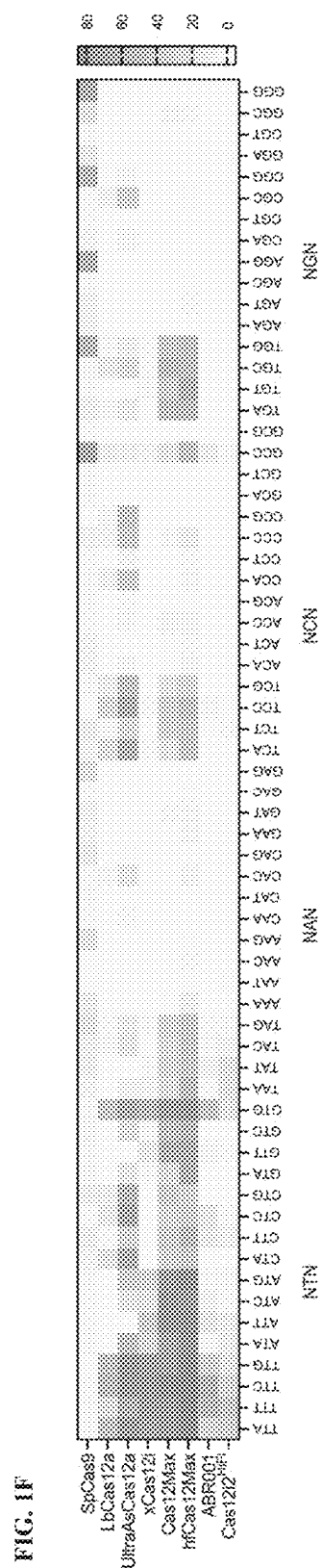


FIG. 1E



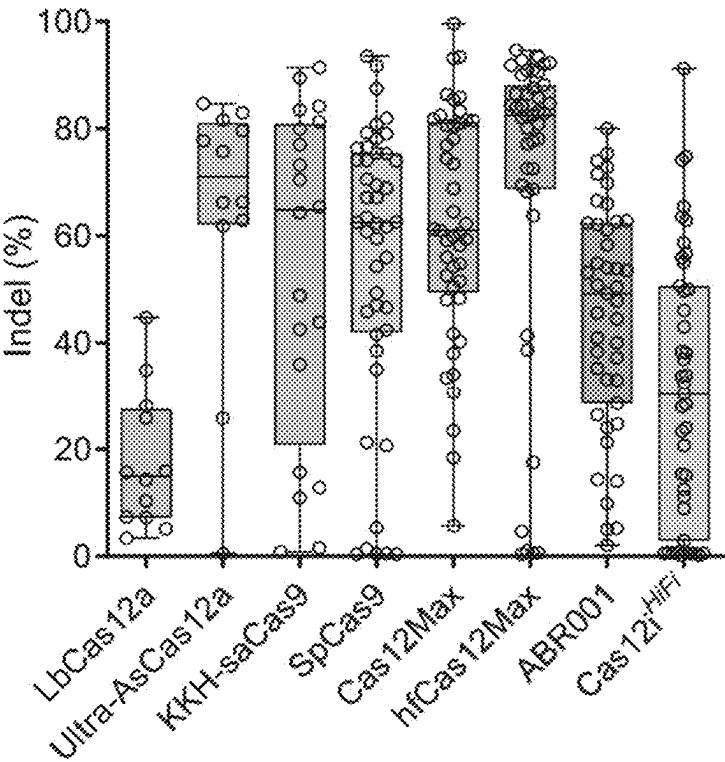


FIG. 1G



FIG. 1H

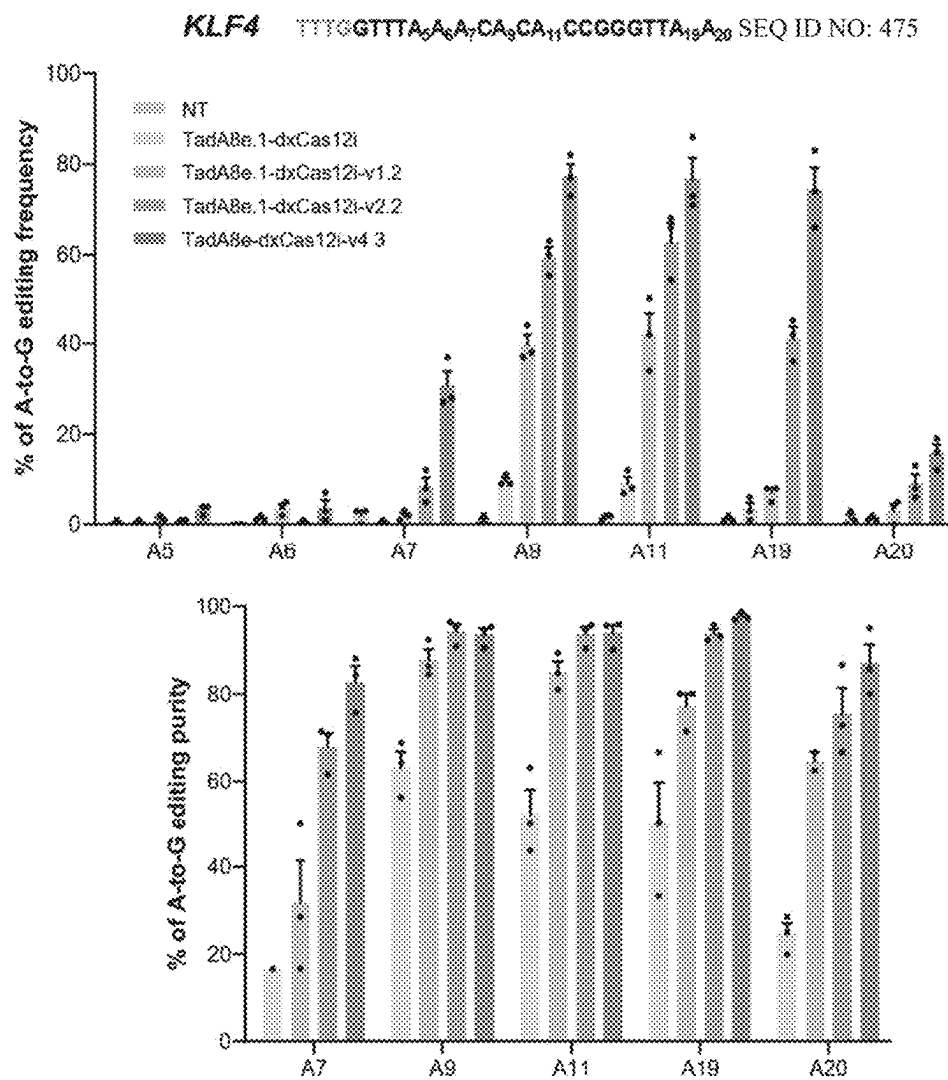


FIG. 11

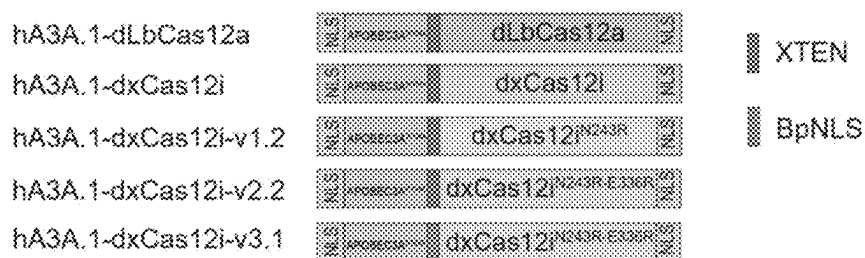


FIG. 1J

RUNX1 TTTCTTC₃TC₅C₆C₇C₈TC₁₀TGC₁₃TGGATAC₂₀ SEQ ID NO: 476

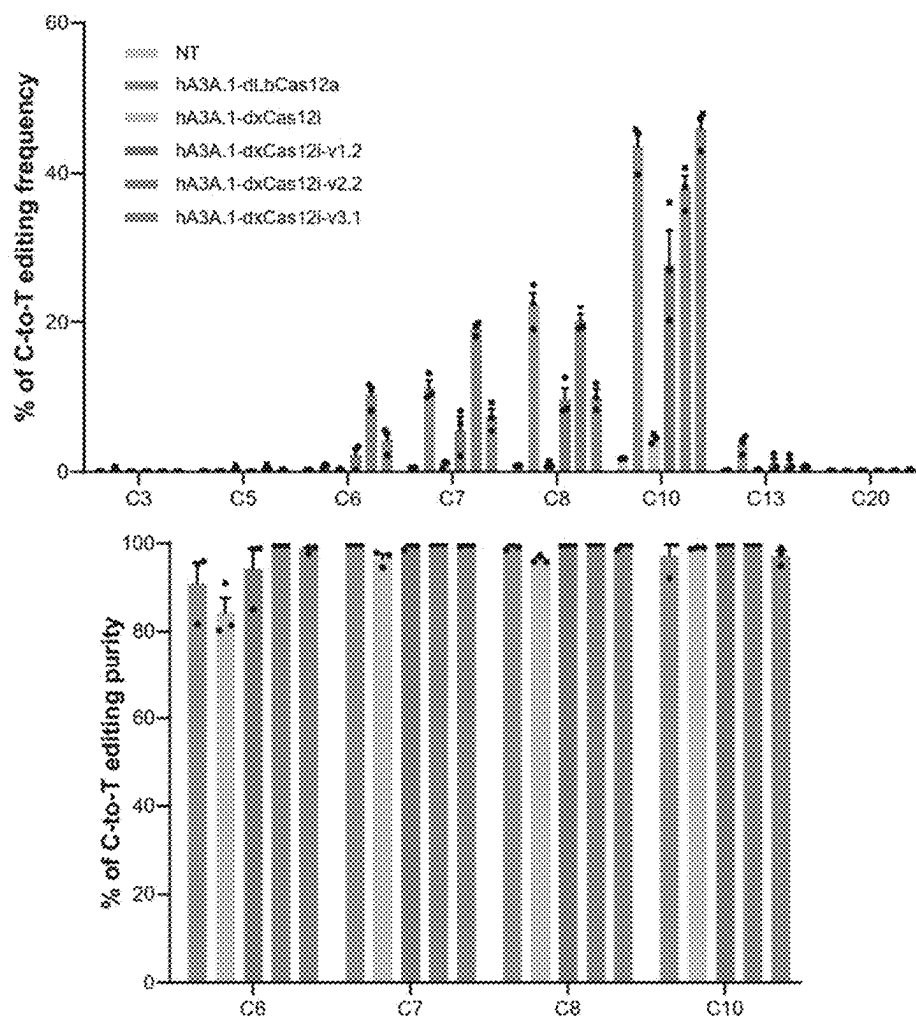


FIG. 1K

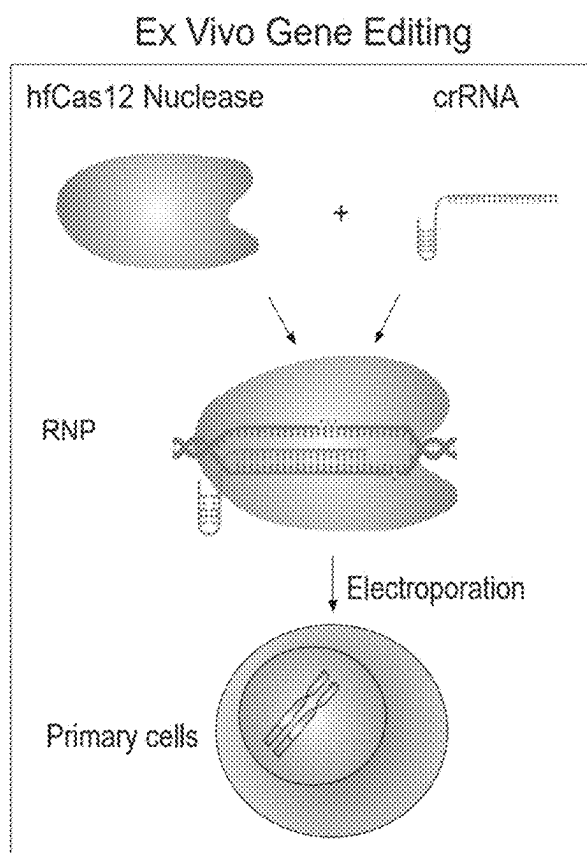


FIG. 2A

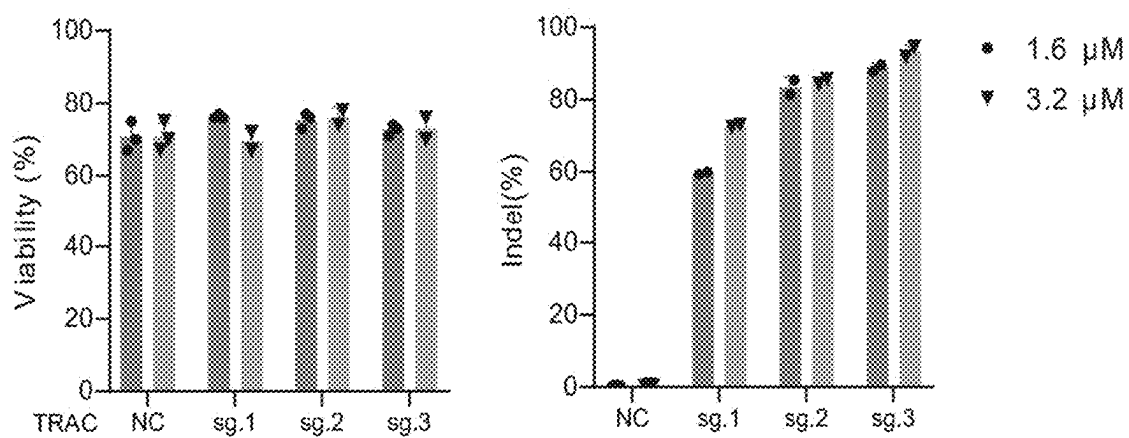
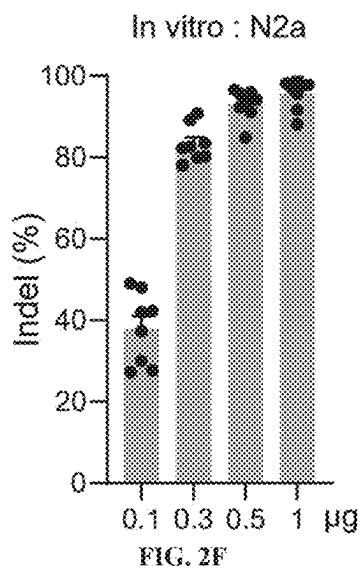
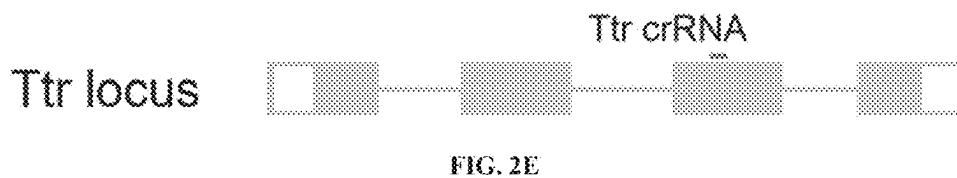
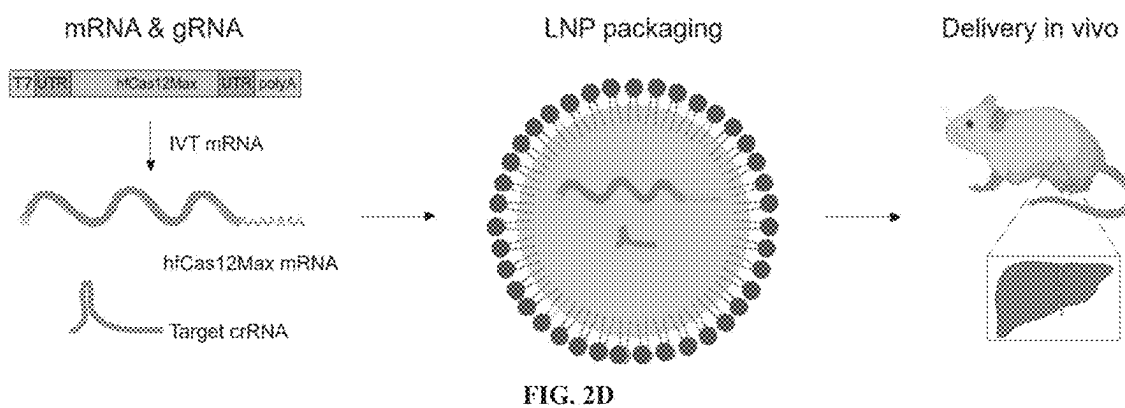
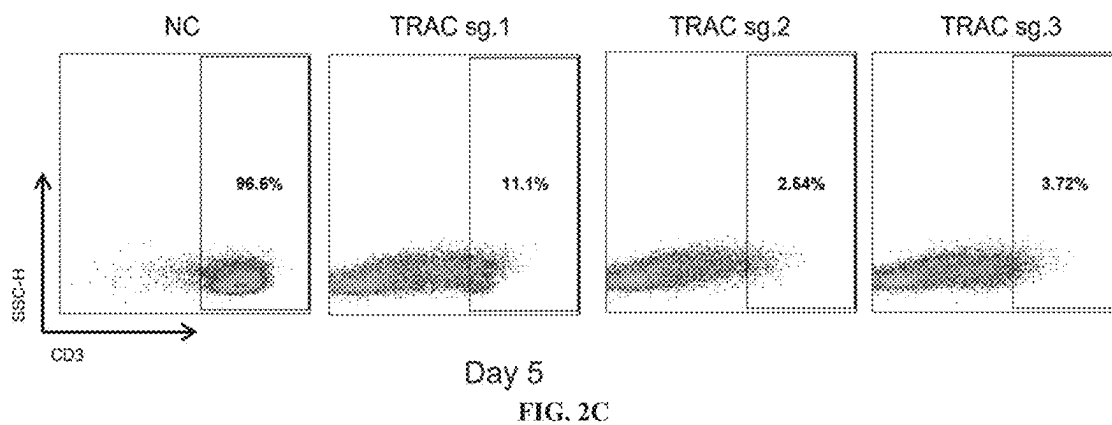


FIG. 2B



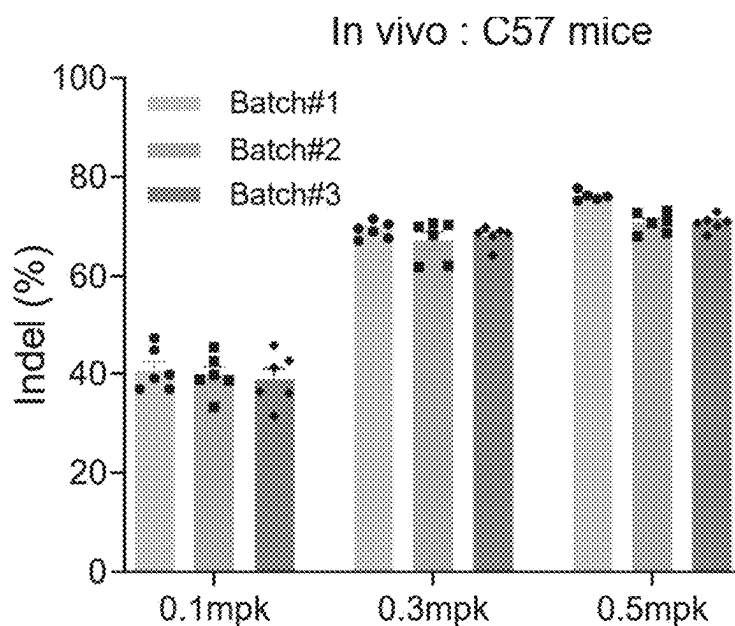


FIG. 2G

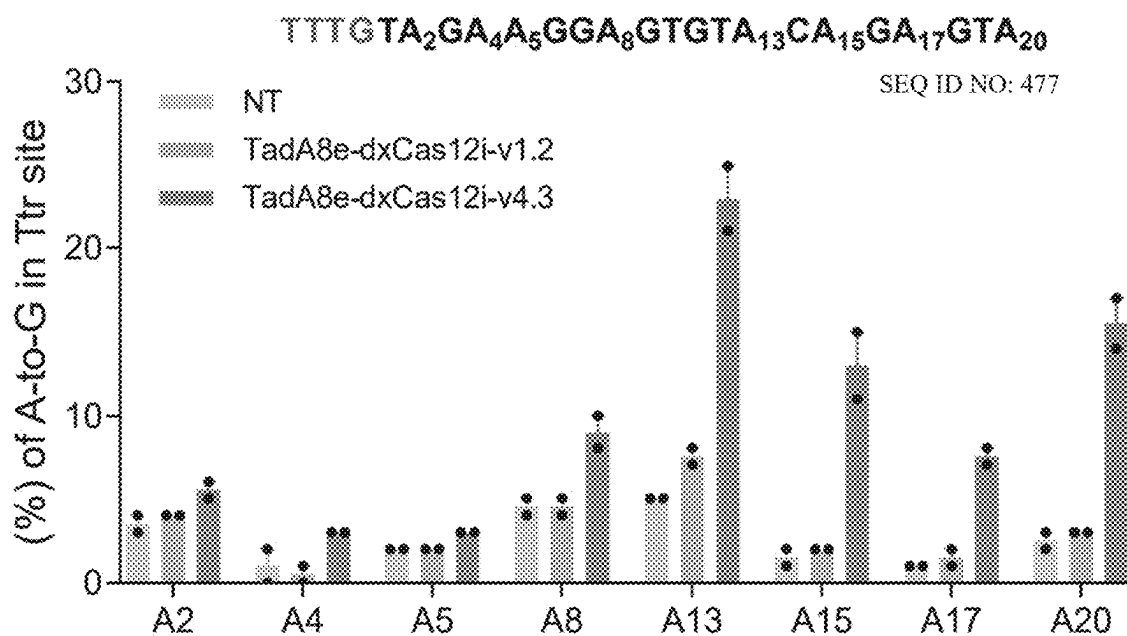


FIG. 2H

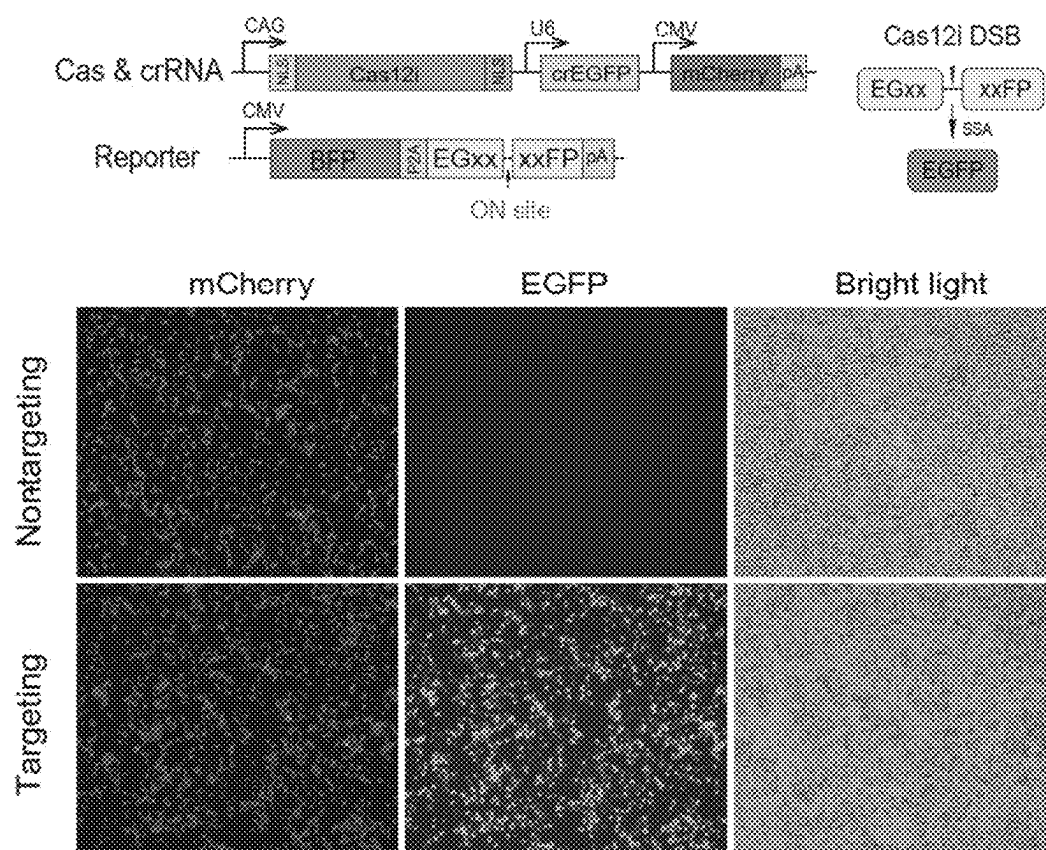


FIG. 3A

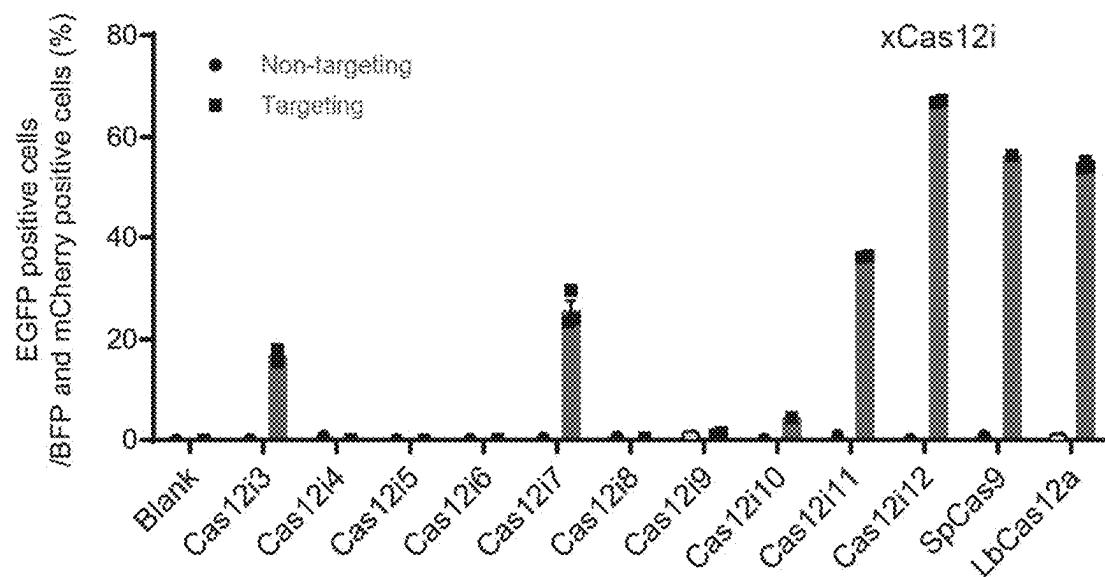


FIG. 3B

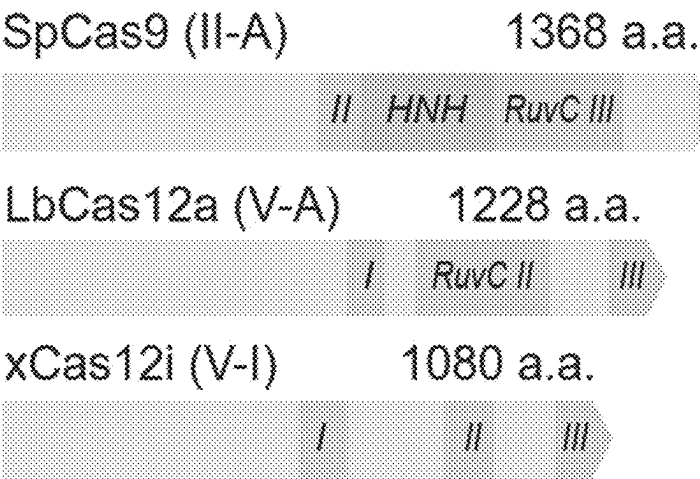


FIG. 4A

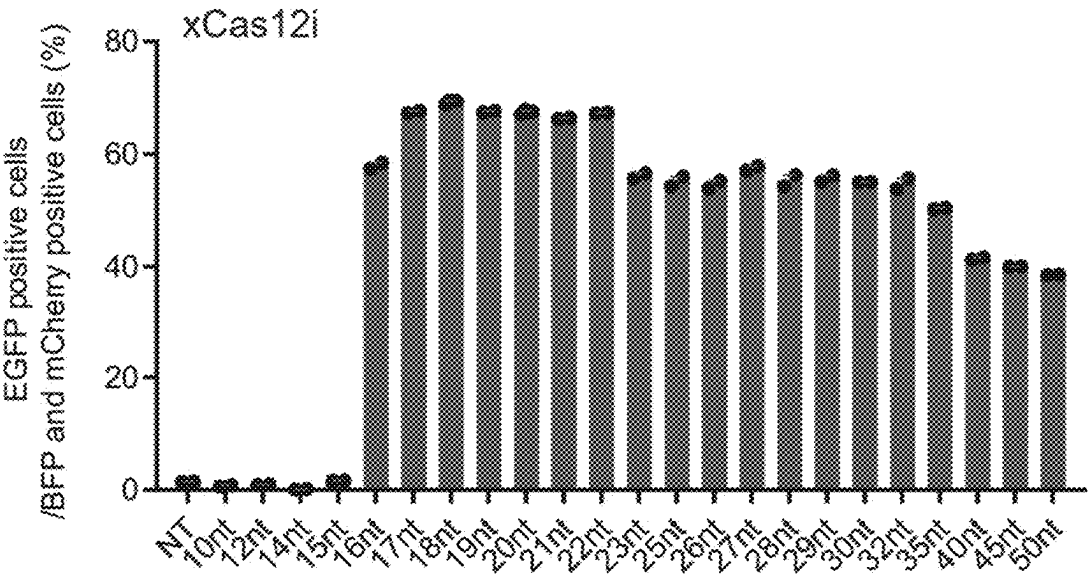


FIG. 4B

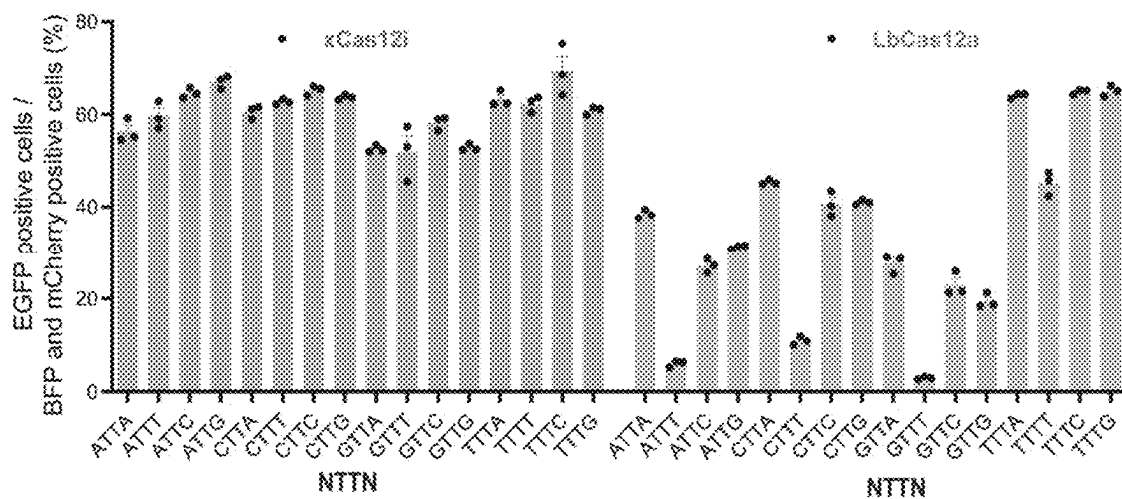


FIG. 4C

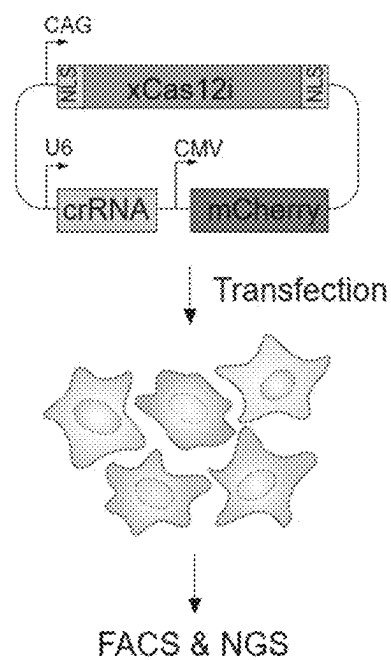


FIG. 4D

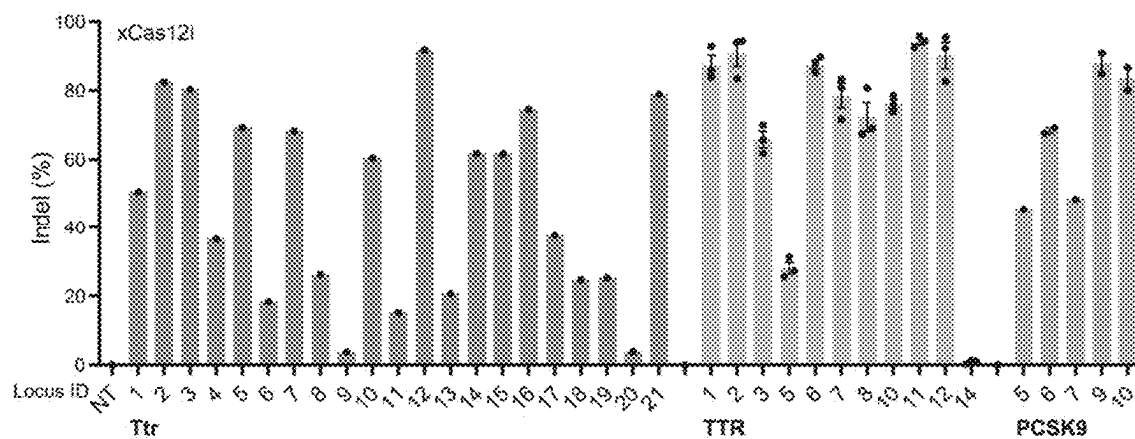


FIG. 4E

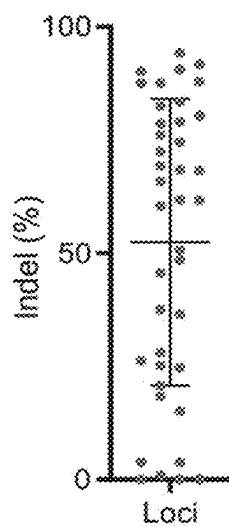


FIG. 4F

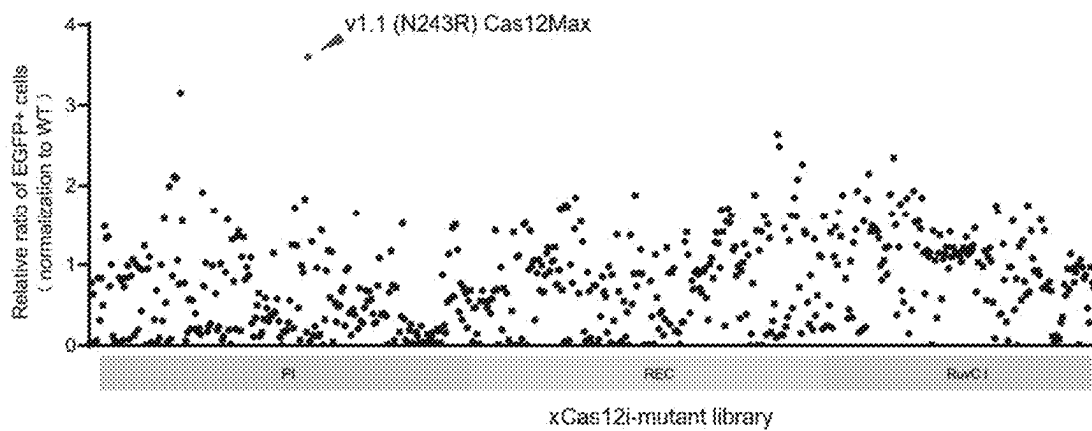


FIG. 5

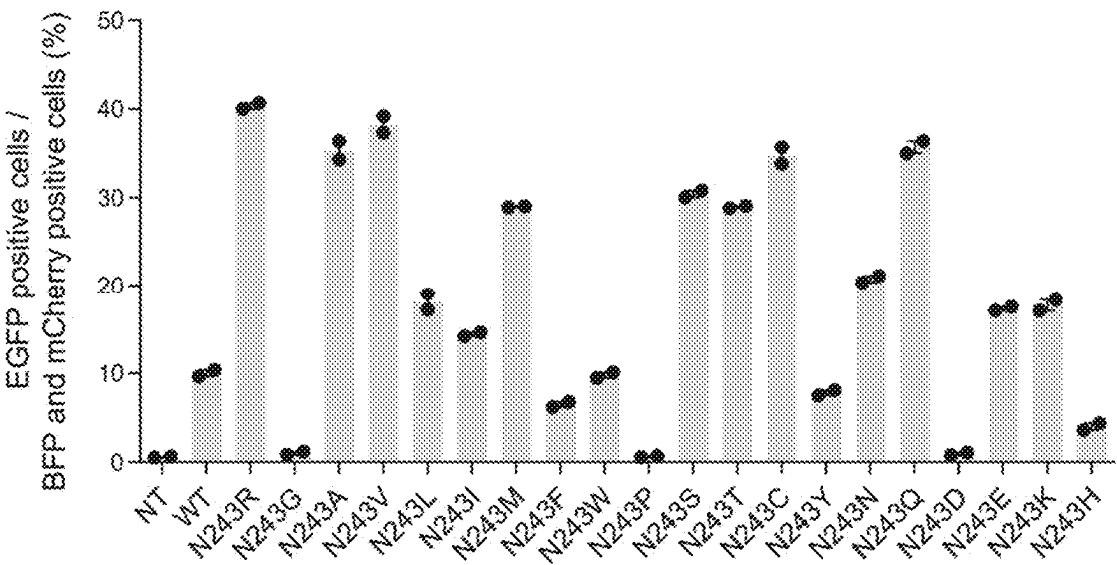


FIG. 6A

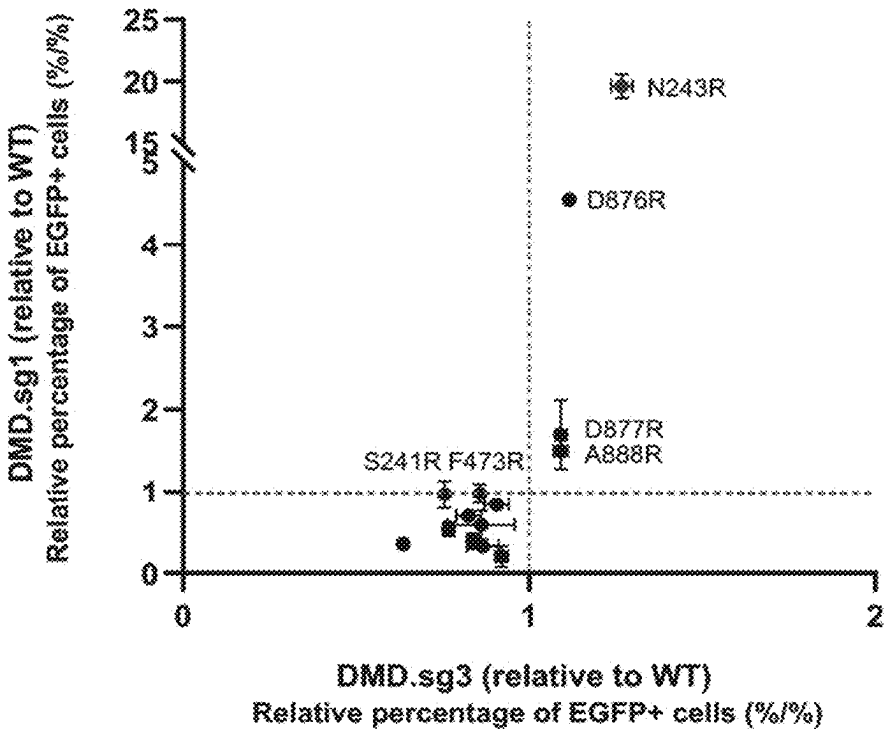


FIG. 6B

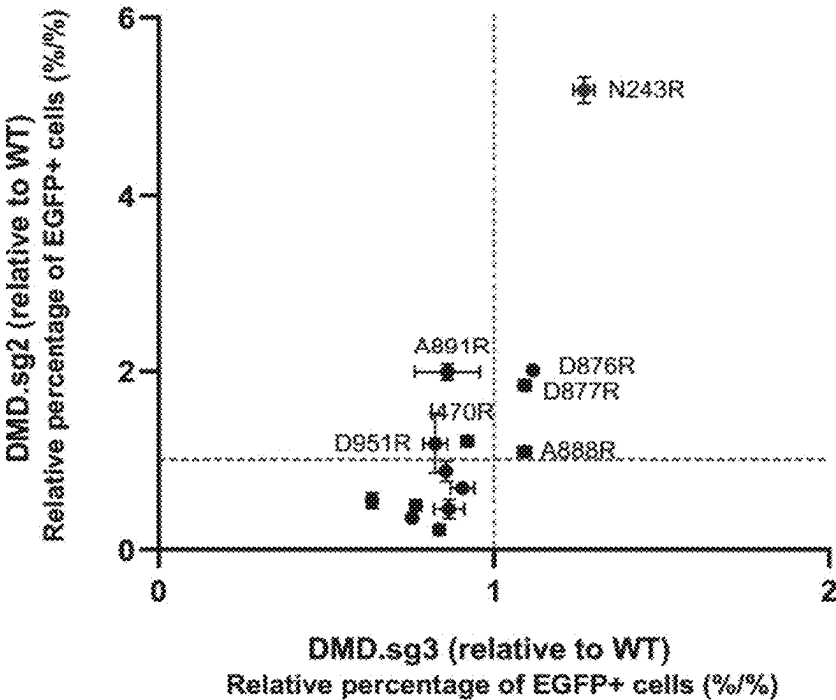


FIG. 6C

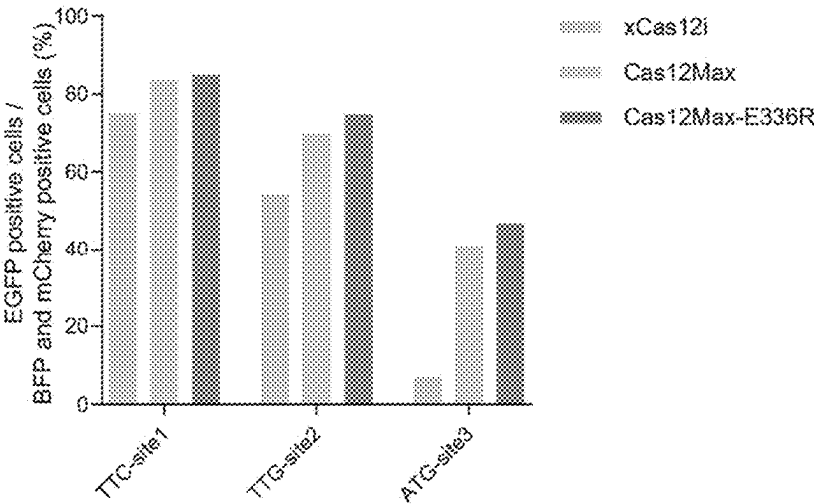


FIG. 6D

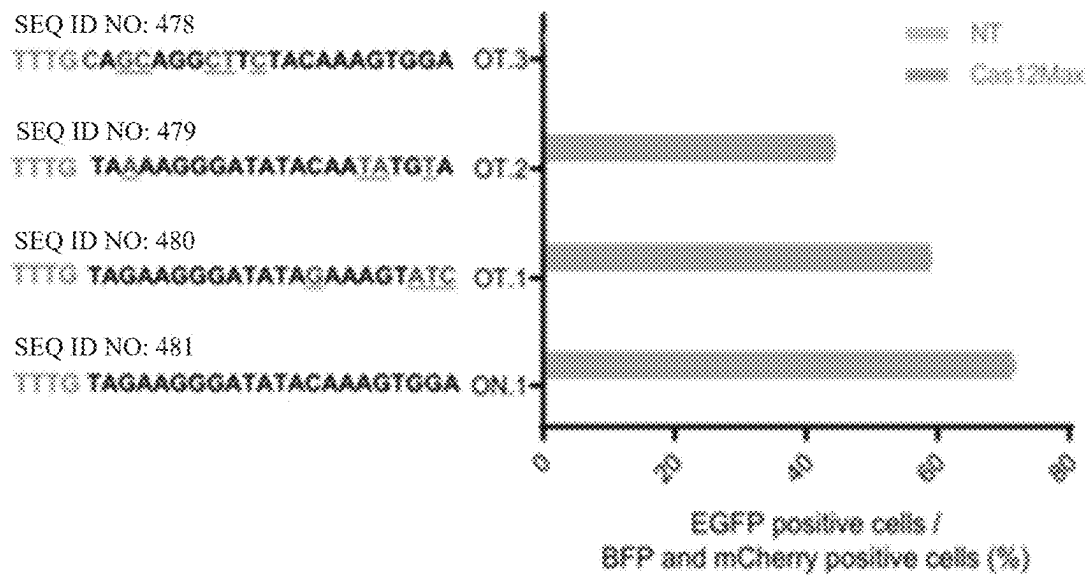


FIG. 7A

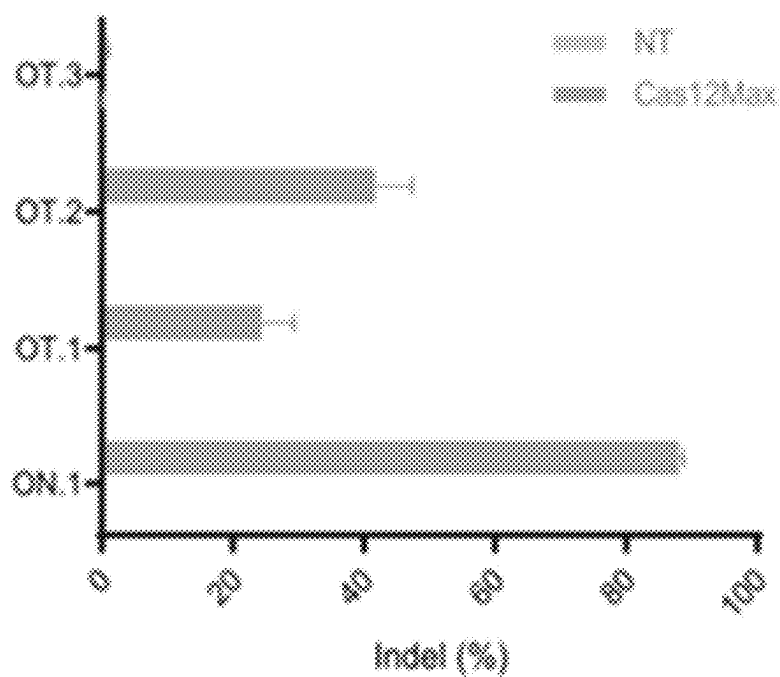


FIG. 7B

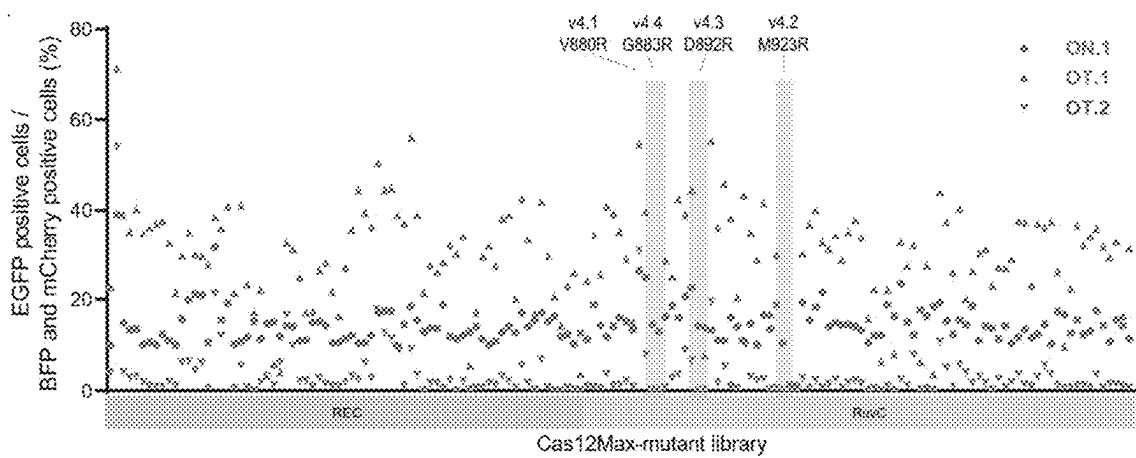


FIG. 8A

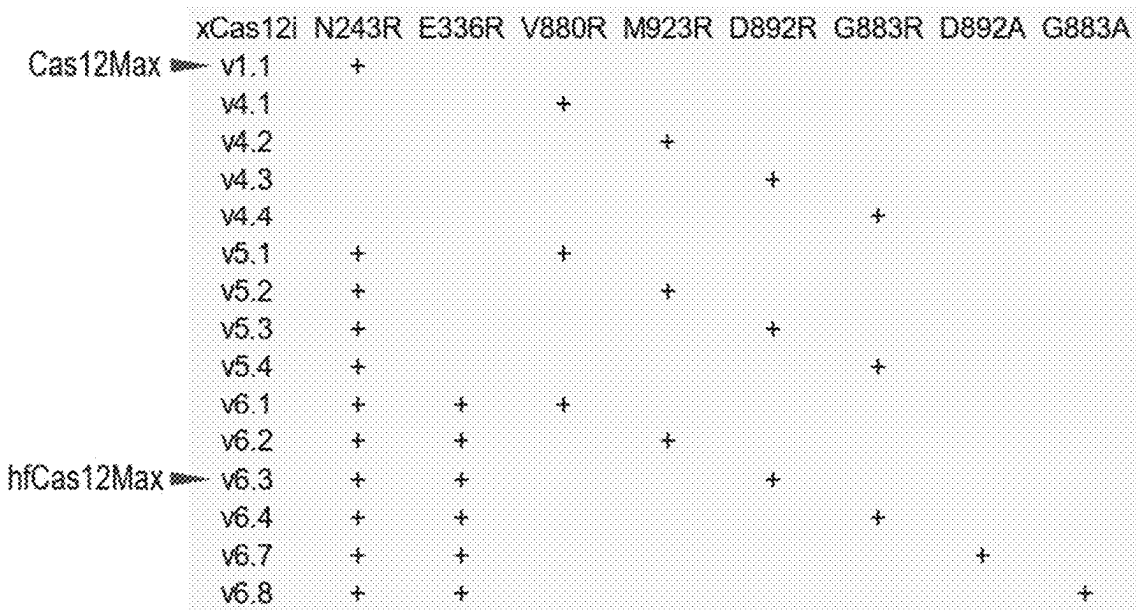


FIG. 8B

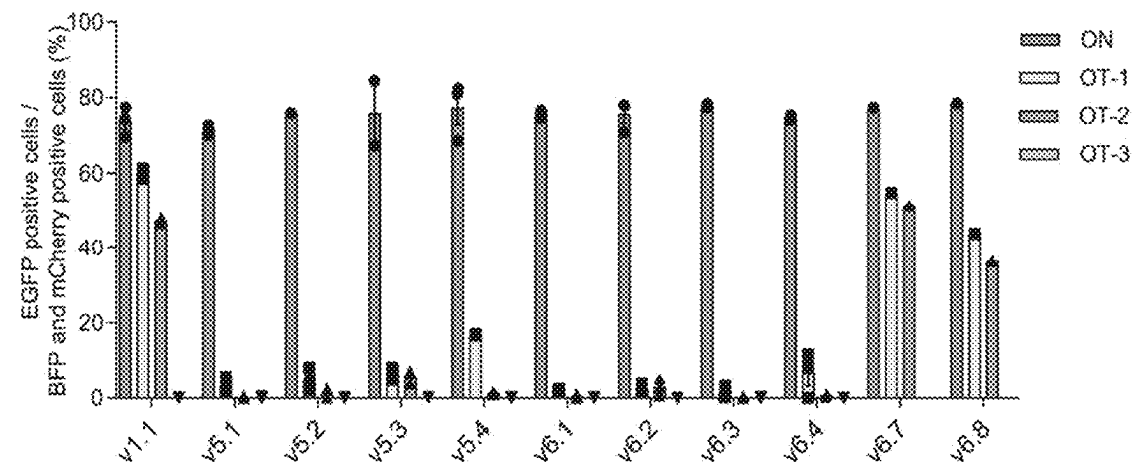


FIG. 8C

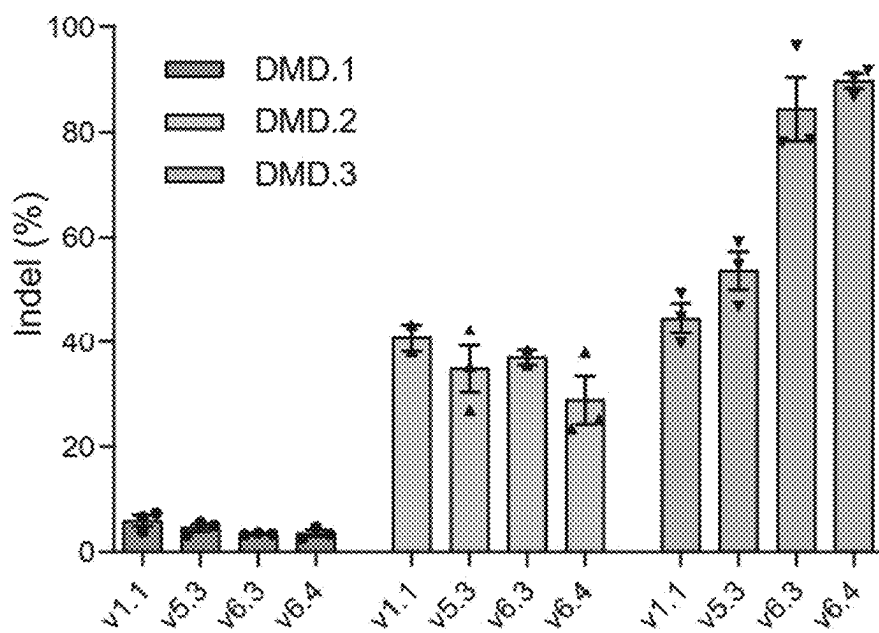


FIG. 8D

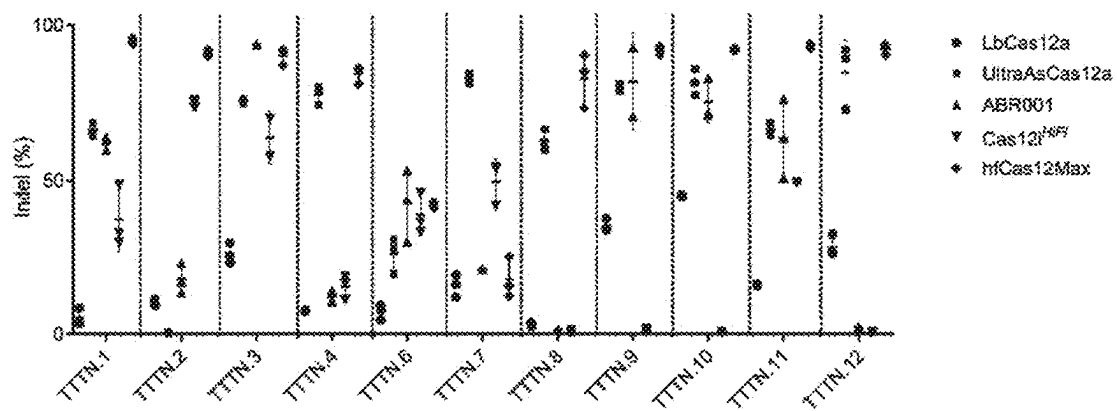


FIG. 9

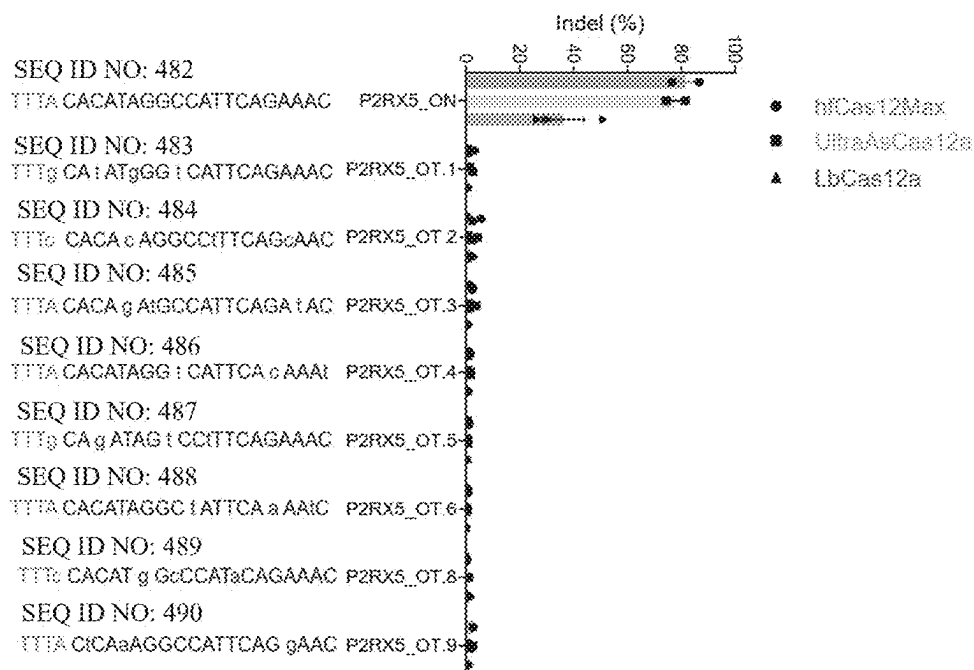


FIG. 10A

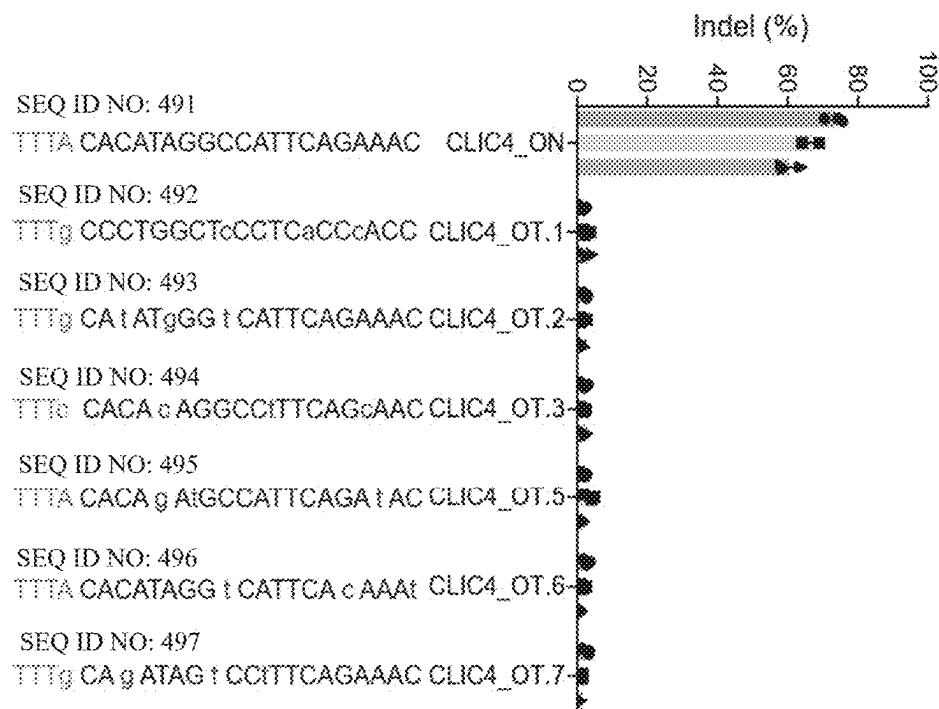


FIG. 10B

SEQ ID NO:

498		VRD	YDQLSYEGV	Y				
499	Cas121	SQVRDKSY	QLSYNGVKL					
500	Cas122	TENRGNQF	QLSYEGLAY		D700			
501	xCas12	VIVRDEVY	QLSYEGVPY					
502		GDRFLGYDQ	NQ	NTY				
503	Cas121	NYKVVLGY	ANIVRKNTY		D650			
504	Cas122	IGDRFCGY	QNQTASHAY					
505	xCas12	VGDRFLGF	QNQSENHTY					
506		-G	K	I	GE	DL	TT	
507	Cas121	YGPVLIK	NISDTTKKG					E875
508	Cas122	- - -	KFIRG	GDLSTTNNA				
509	xCas12	-GCKMIVC	DDL	PVADGK				
510		KQ	WVCNAD	VAA	NIAL			
511	Cas121	KQFWVCNA	LVAAYNVGL					D1049
512	Cas122	KQVWVCNA	HVAAAANIAL					
513	xCas12	EDMWQYHA	EIAAVNIAM					

FIG. 11A

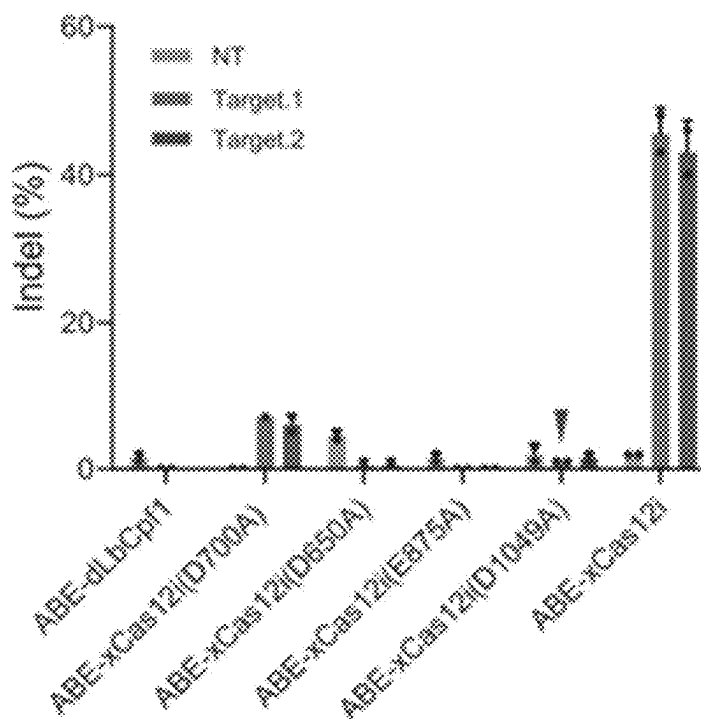
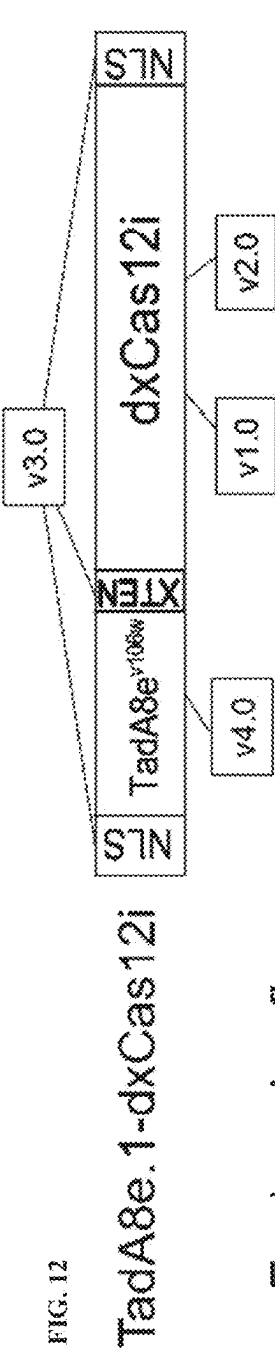


FIG. 11B



Engineering flow

v1.0		v2.0		v3.0		v4.0					
		v2.2	N'	Linker	C'	v2.2	N'	Linker	TadA8e		
v1.1	E161R	v2.1	N243R/E161R	v3.1	BpNLS	XTEN	3xSV40 NLS	v4.1	BpNLS	XTEN	W106V
v1.2	N243R	v2.2	N243R/E336R	v3.2	BpNLS	XTEN	BpNLS-dMycNLS	v4.2	dMyc-NLS	XTEN	W106V
v1.3	E336R	v2.3	N243R/D474R	v3.3	BpNLS	XTEN	CHRONO NLS	v4.3	BpNLS	BpNLS	W106V
v1.4	D474R	v2.4	N243R/Q862R	v3.4	BpNLS	XTEN	SGCS-NLS	v4.4	dMyc-NLS	BpNLS	W106V
v1.5	S245R			v3.5	BpNLS	XTEN	v8pNLS				
v1.6	N269R			v3.6	BpNLS	XTEN	BpNLS				
v1.7	S155R			v3.7	BpNLS	BpNLS	NLS				
v1.8	S268R			v3.8	dMyc-NLS	BpNLS	NLS				
v1.9	A276R										
v1.10	D287R										
v1.11	D381R										

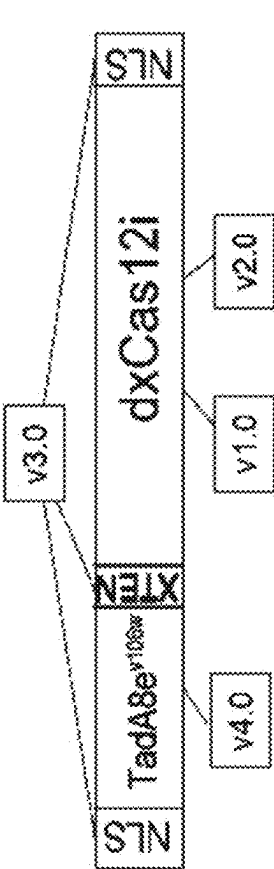


FIG. 13A

TadA8e.1-dxCas12i

Engineering flow

v1.0		v2.0		v3.0		v4.0	
v1.1	E161P	v2.1	N243P/E181R	v3.1	BpNLS	XTEH	TadA8e
v1.2	N243R	v2.2	N243R/E336R	v3.2	BpNLS	XTEH	W108V
v1.3	E336P	v2.3	N243R/D474P	v3.3	BpNLS	XTEH	W108V
v1.4	D474P	v2.4	N243R/Q362P	v3.4	BpNLS	XTEH	W108V
v1.5	S245R			v3.5	BpNLS	XTEH	
v1.6	H136R			v3.6	BpNLS	XTEH	
v1.7	S155R			v3.7	BpNLS	XTEH	
v1.8	S268P			v3.8	BpNLS	XTEH	
v1.9	A276R						
v1.10	D287P						
v1.11	D381P						

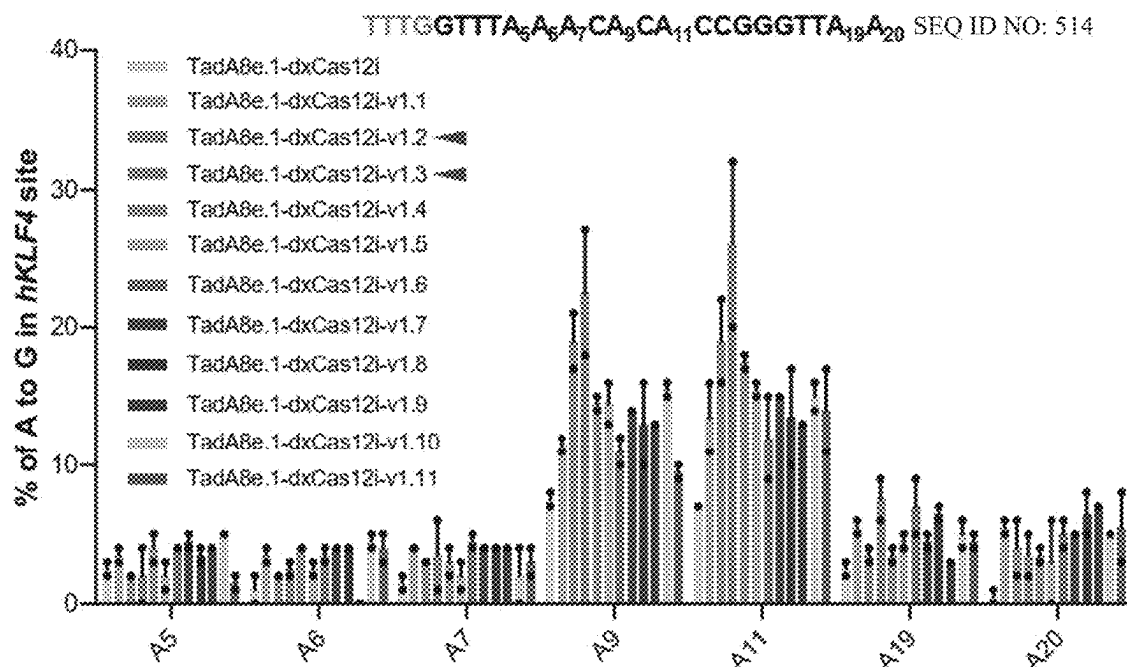


FIG. 13B

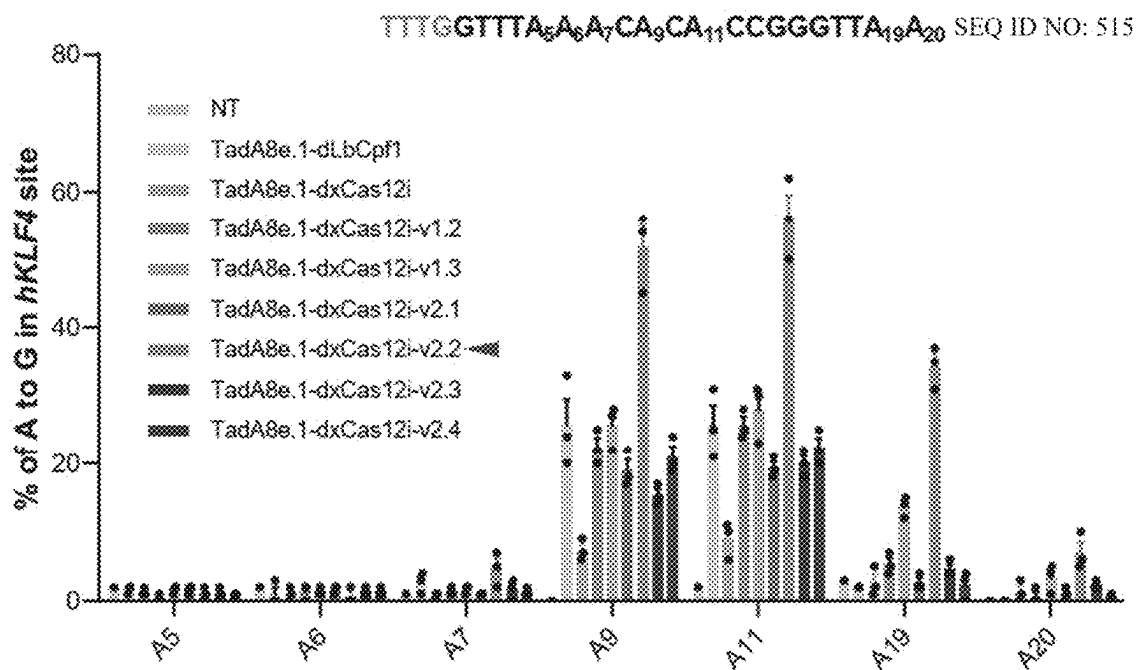


FIG. 13C

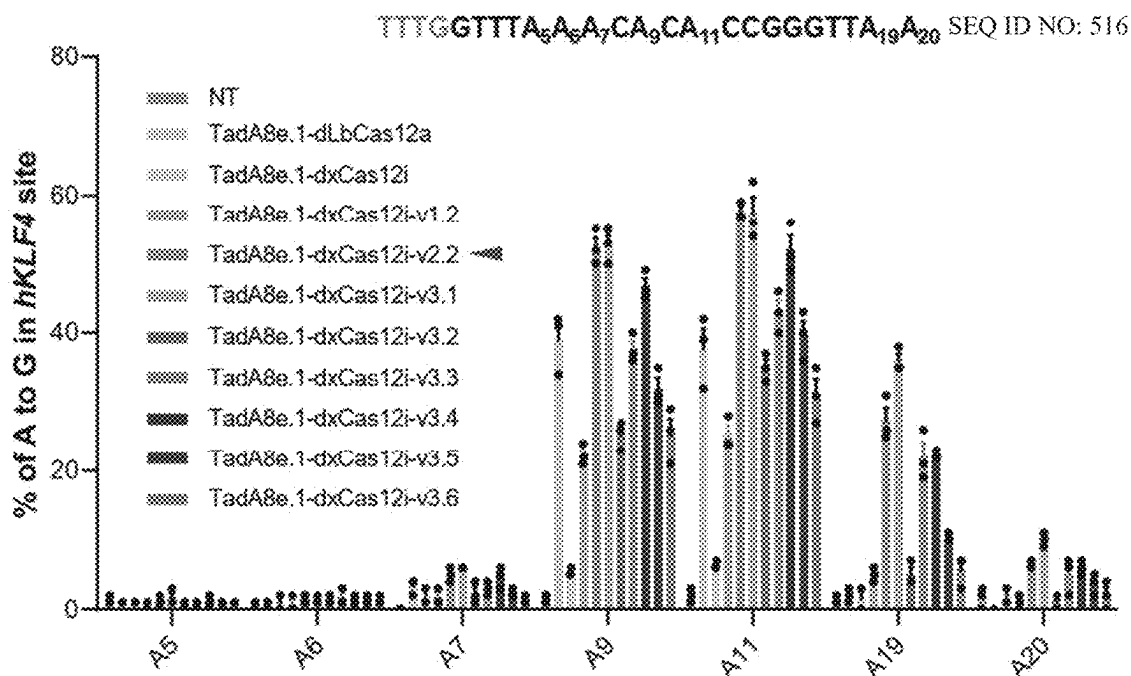


FIG. 13D

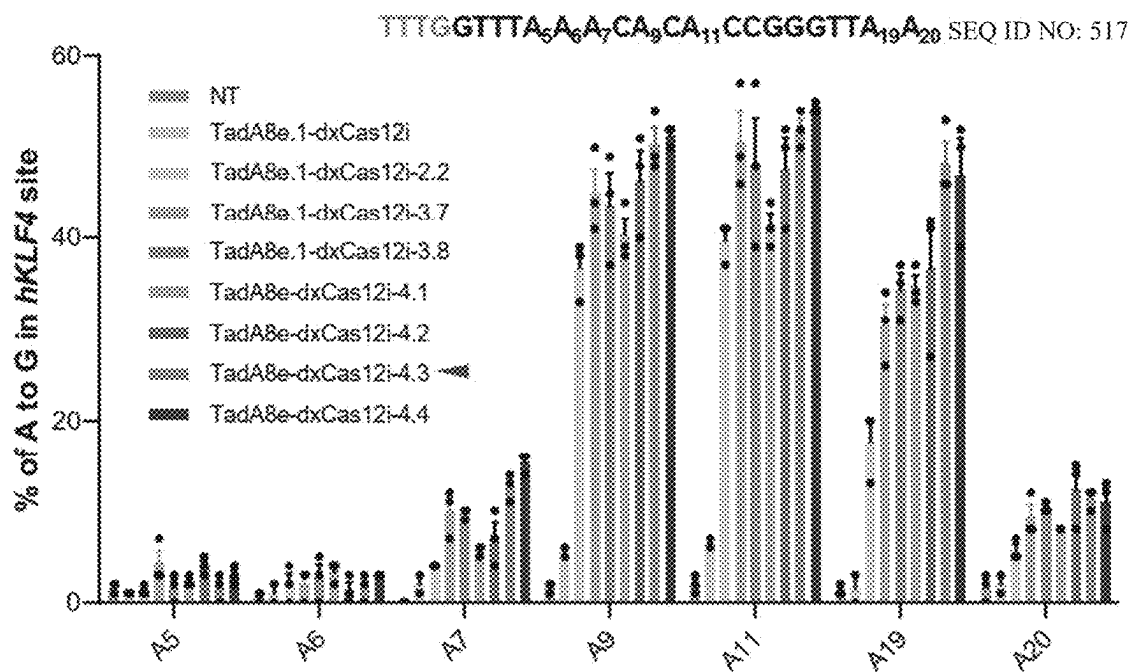


FIG. 13E

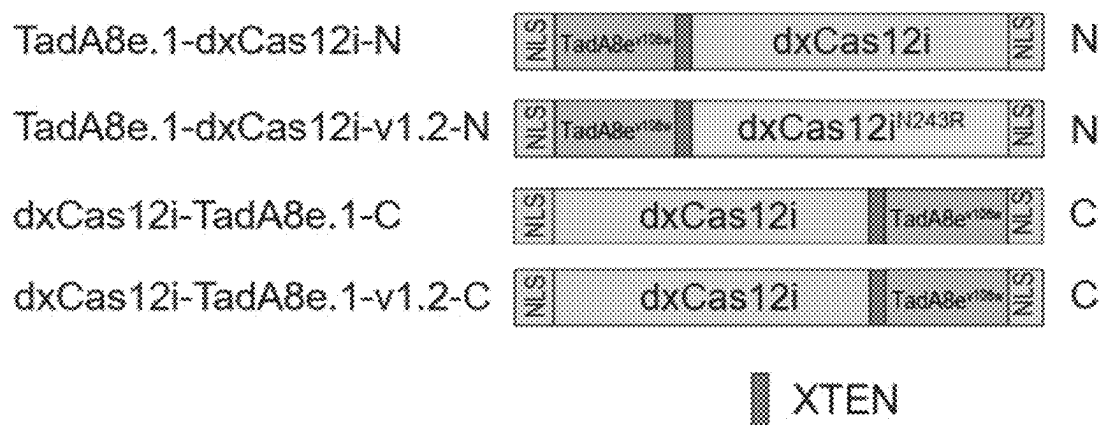


FIG. 14A

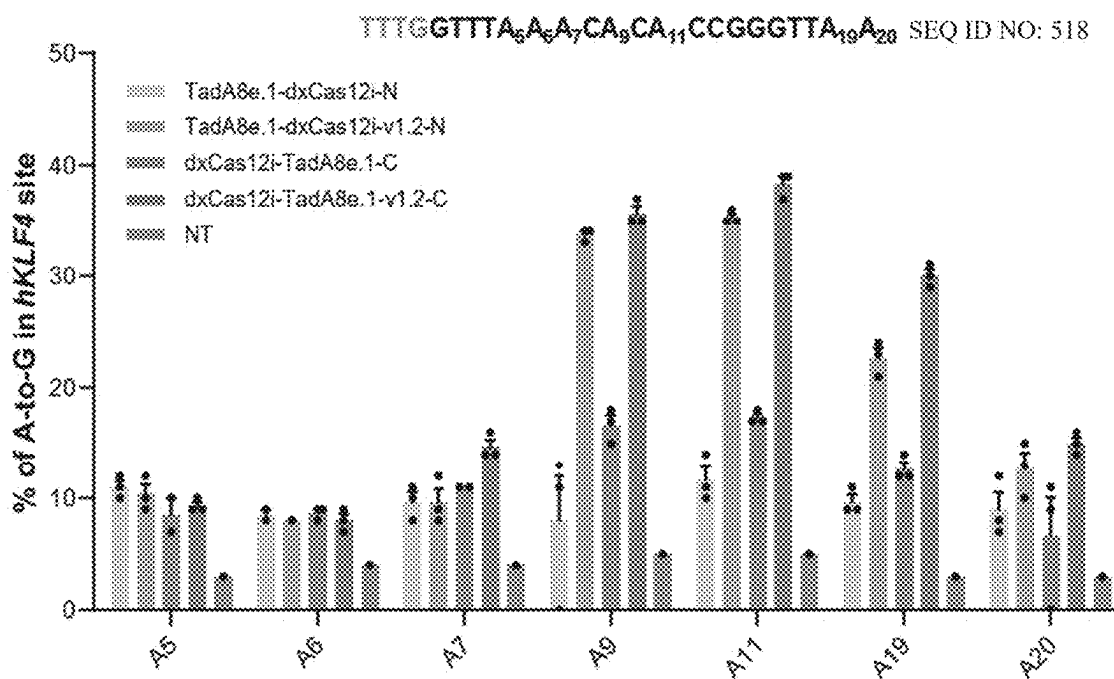


FIG. 14B

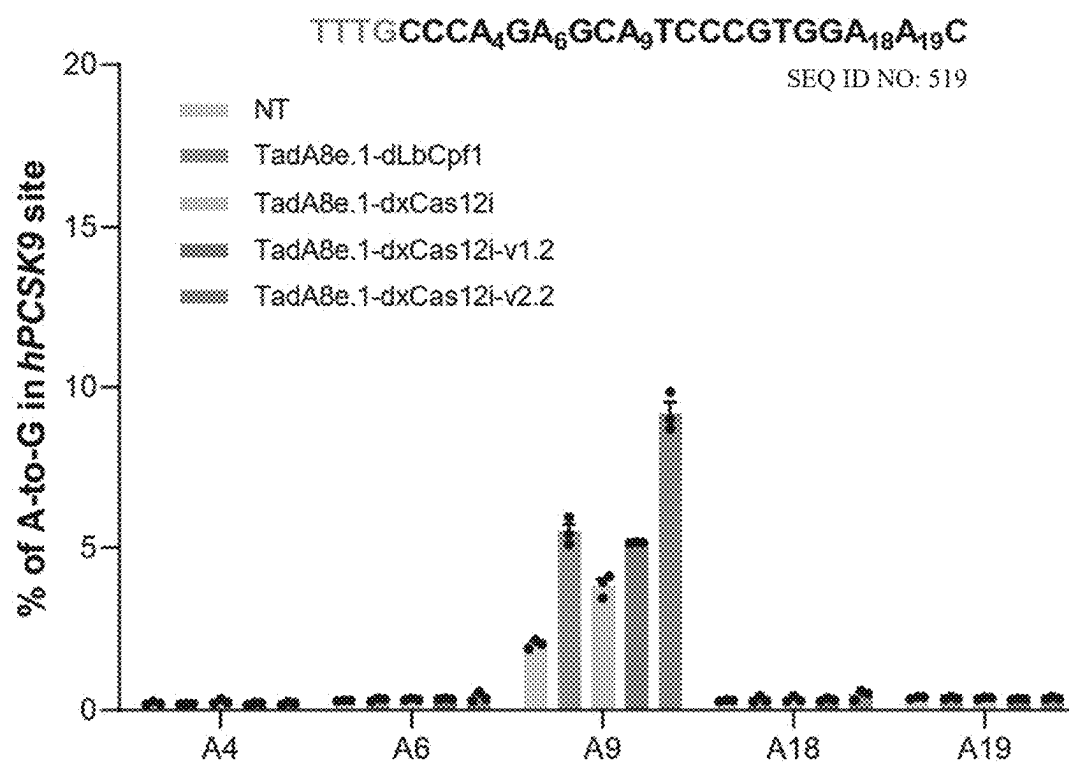


FIG. 15A

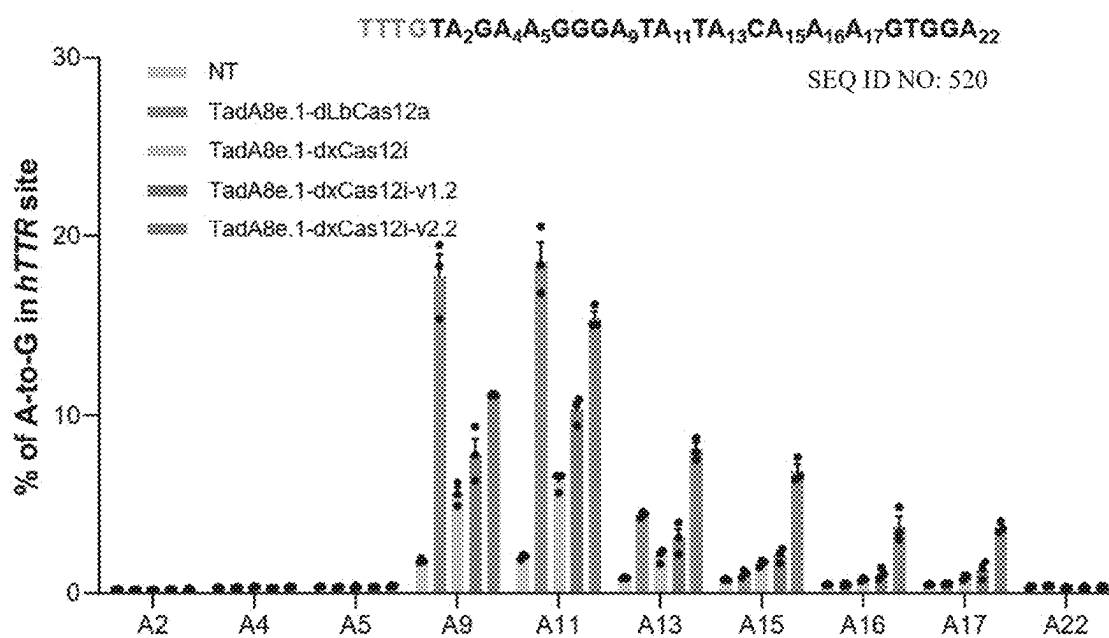


FIG. 15B

FIG. 16A

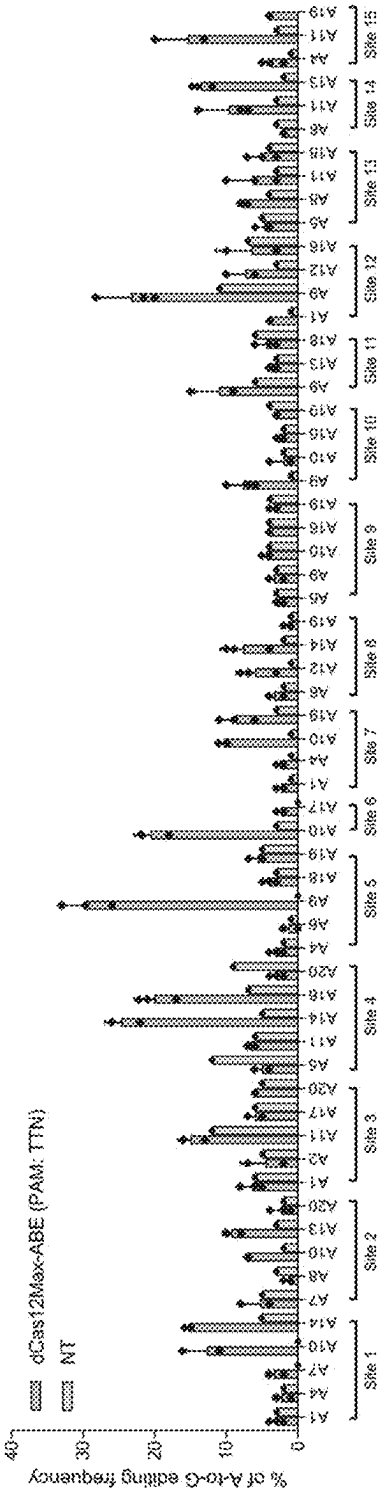
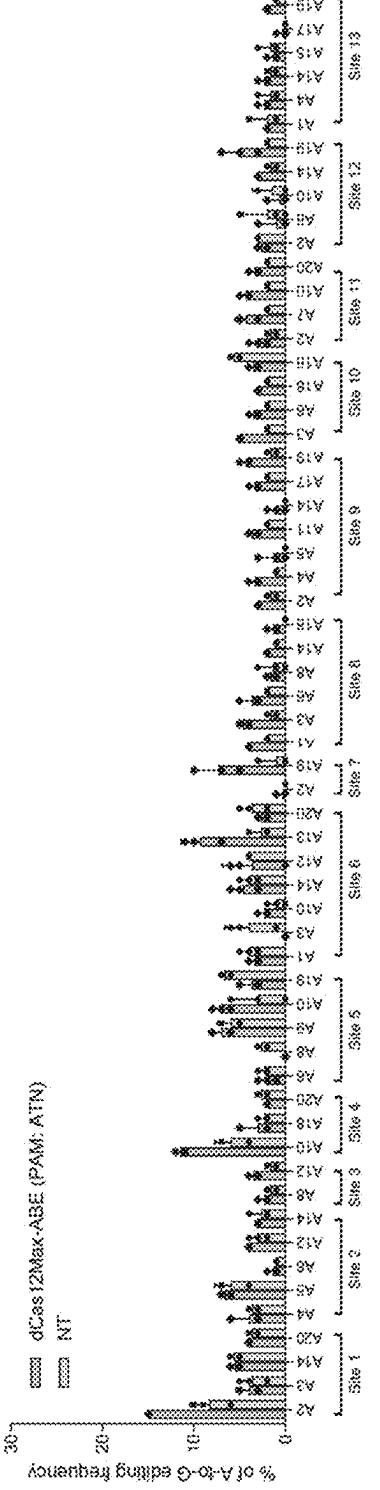
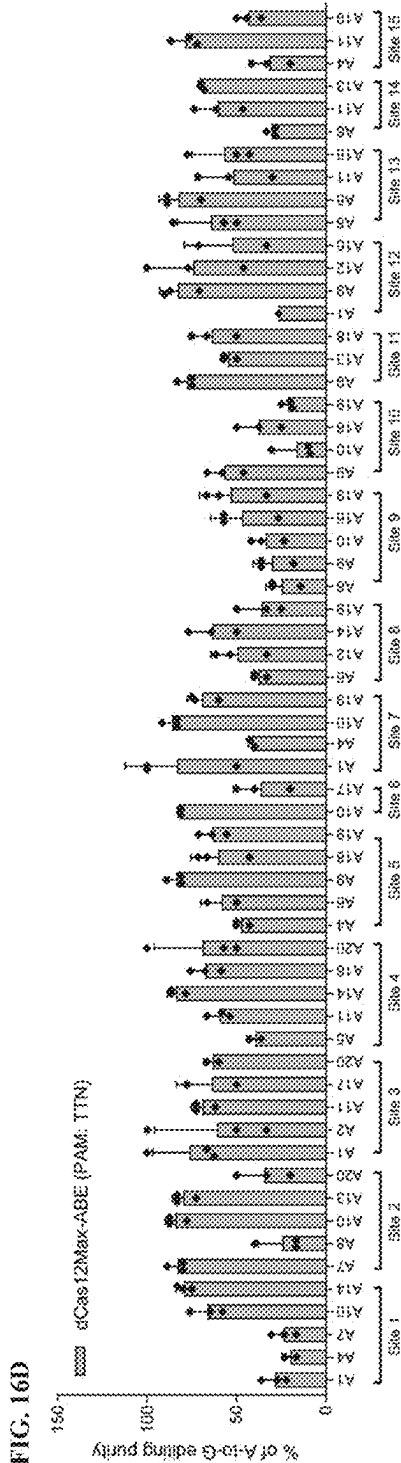
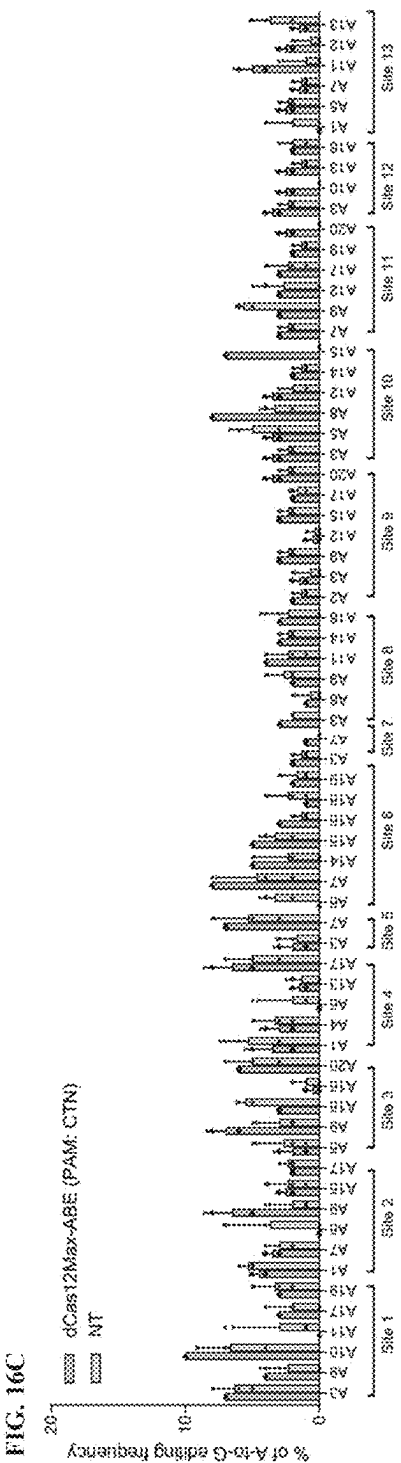


FIG. 16B





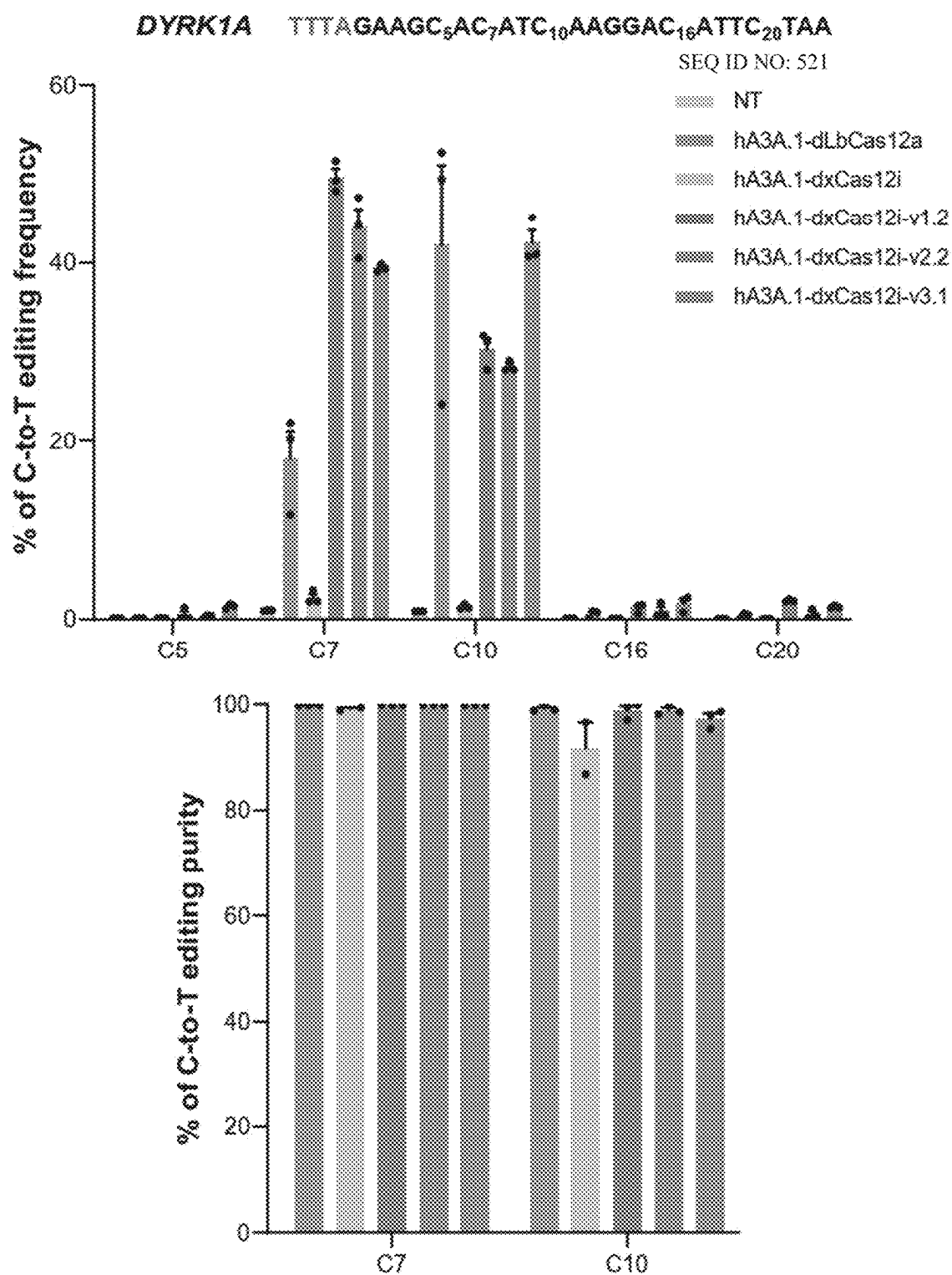


FIG. 17A

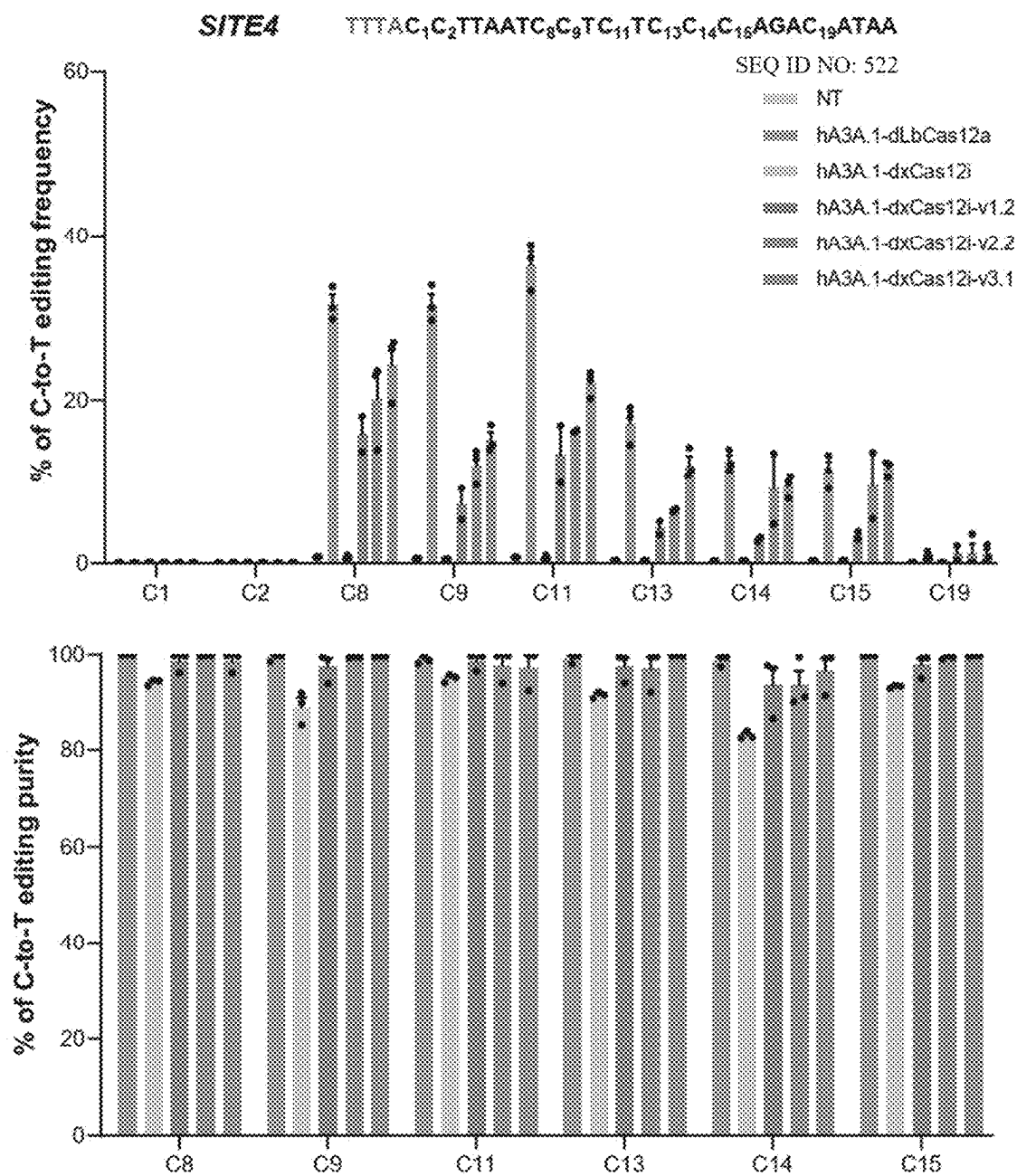


FIG. 17B

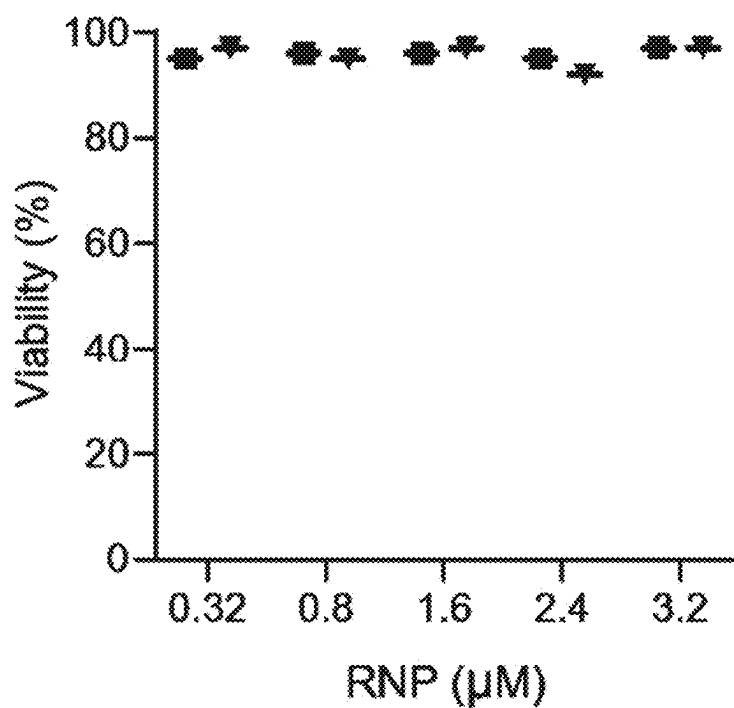


FIG. 18A

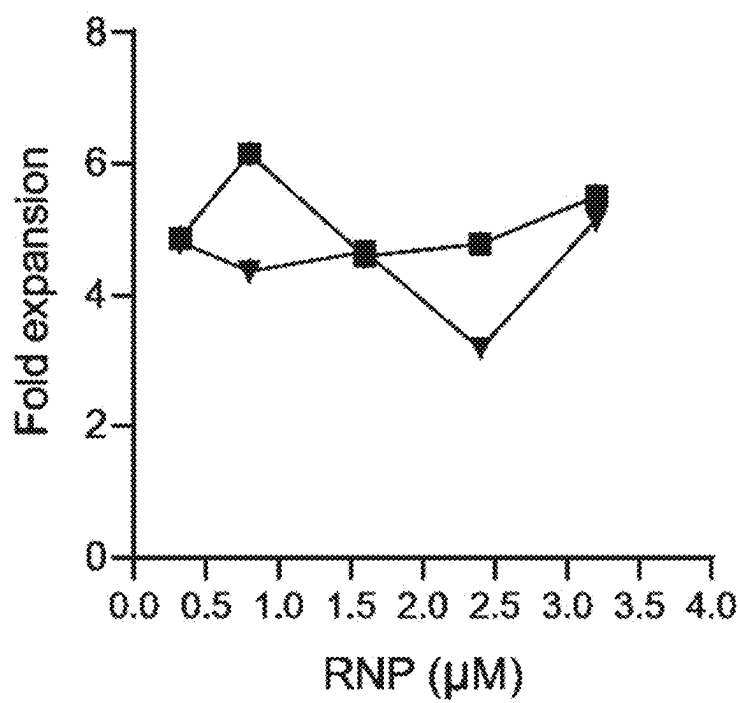


FIG. 18B

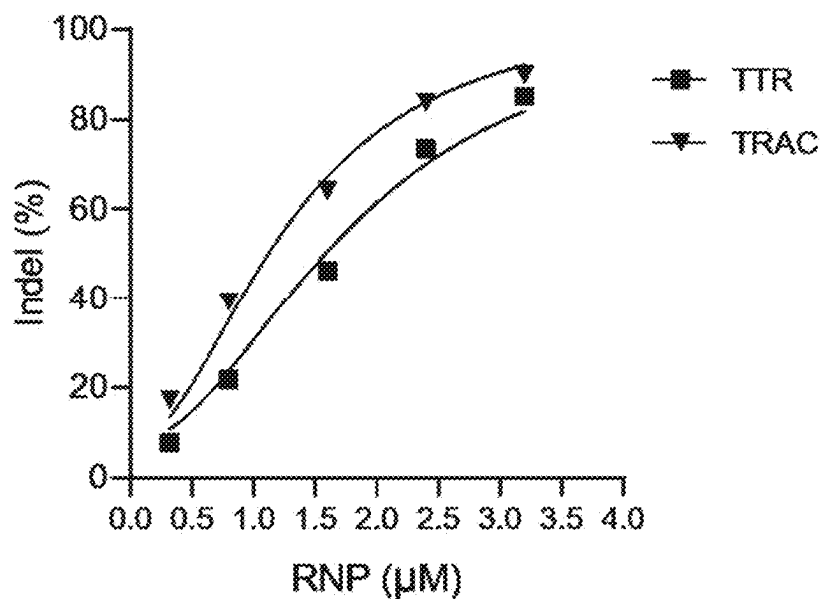


FIG. 18C

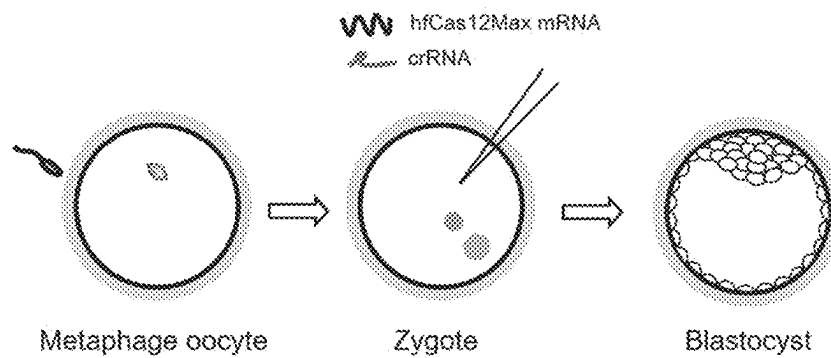


FIG. 19A

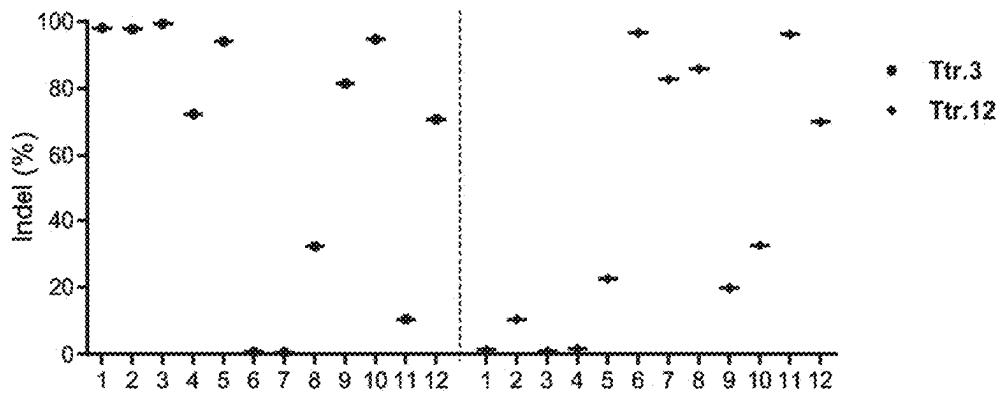


FIG. 19B

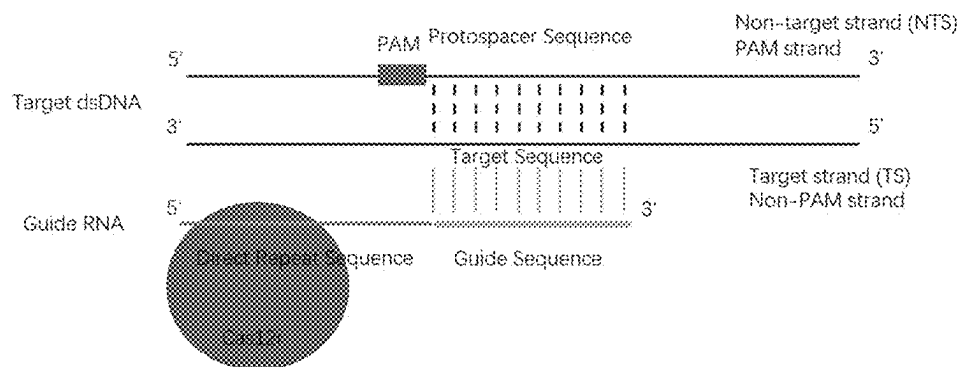


FIG. 20

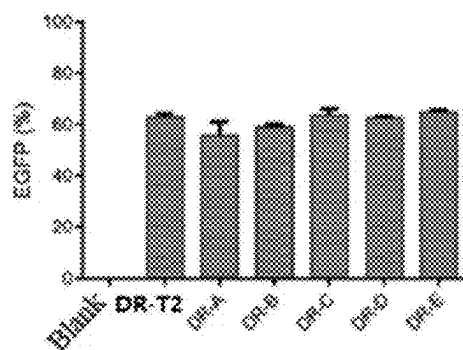


FIG. 21

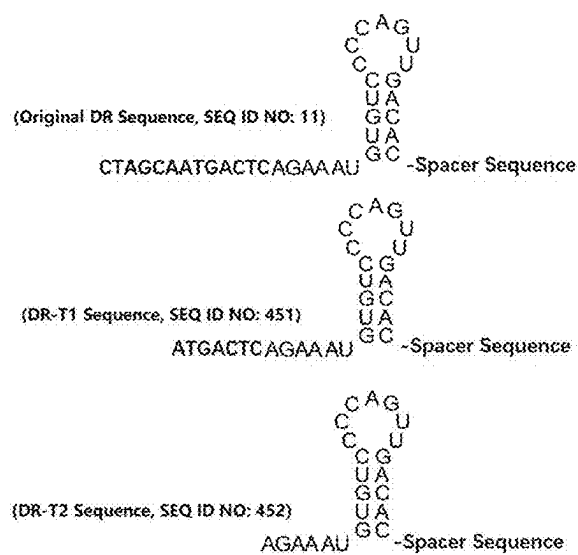


FIG. 22

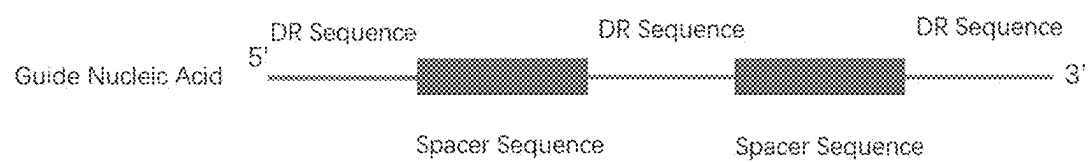


FIG. 23

NOVEL CRISPR-CAS12I SYSTEMS AND USES THEREOF

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims the benefits of and priorities to PCT Patent Application No. PCT/CN2022/089074, filed on Apr. 25, 2022, entitled “NOVEL CRISPR-CAS12I SYSTEMS”, PCT Patent Application No. PCT/CN2022/129376, filed on Nov. 2, 2022, entitled “NOVEL CRISPR-CAS12I SYSTEMS AND USES THEREOF”, and PCT Patent Application No. PCT/CN2023/073420, filed on Jan. 20, 2023, entitled “NOVEL CRISPR-CAS12I SYSTEMS AND USES THEREOF” the entire contents of which, including any sequence listing and drawings, are incorporated herein by reference in their entireties.

REFERENCE TO AN ELECTRONIC SEQUENCE LISTING

[0002] The disclosure contains an electronic sequence listing (“HEP001PCT3. xml” created on Apr. 25, 2023, by software “WIPO Sequence” according to WIPO Standard ST. 26), which is incorporated herein by reference in its entirety. According to WIPO Standard ST. 26, symbol “t” is used to denote both T in DNA and U in RNA (See “Table 1: List of nucleotides symbols”, the definition of symbol “t” is “thymine in DNA/uracil in RNA (t/u)”). Thus, in a sequence listing prepared according to ST. 26, wherever a sequence is an RNA, the T in the sequence shall be deemed as U.

BACKGROUND

[0003] The clustered regularly interspaced short palindromic repeats-Cas (CRISPR-Cas) systems, including type II Cas9 and type V Cas12 systems, which serve in the adaptive immunity of prokaryotes against viruses, have been developed into genome editing tools. Compared with type II systems, the type V systems including V-A to V-K showed more functional diversity. Amongst them, Cas12i has a relatively smaller size compared to SpCas9 and Cas12a. Cas12i is characterized by the capability of autonomously processing precursor crRNA (pre-crRNA) to form short mature crRNA.

[0004] Cas12i mediates cleavage of dsDNA with a single RuvC domain, by preferentially nicking the non-target strand and then cutting the target strand. These intrinsic features of Cas12i enable multiplex high-fidelity genome editing.

[0005] Citation or identification of any document in the disclosure is not an admission that such a document is available as prior art to the disclosure. Each of the references mentioned or cited in the disclosure is incorporated by reference in its entirety.

SUMMARY

[0006] It is against the above background that the disclosure provides certain advantages and advancements over the prior art. Although the disclosure is not limited to specific advantages or functionalities, in one aspect, the disclosure provides a Cas12i polypeptide comprising an amino acid substitution at E336, V880, G883, D892, and/or M923 of SEQ ID NO: 458.

[0007] In another aspect, the disclosure provides a system comprising:

[0008] (1) the Cas12i polypeptide of the disclosure or a polynucleotide encoding the Cas12i polypeptide; and

[0009] (2) a guide nucleic acid or a polynucleotide encoding the guide nucleic acid, the guide nucleic acid comprising:

[0010] (i) a direct repeat (DR) sequence capable of forming a complex with the Cas12i polypeptide; and

[0011] (ii) a spacer sequence capable of hybridizing to a target sequence of a target DNA, thereby guiding the complex to the target DNA.

[0012] In yet another aspect, the disclosure provides a polynucleotide encoding the Cas12i polypeptide of the disclosure. In yet another aspect, the disclosure provides a vector comprising the polynucleotide of the disclosure.

[0013] In yet another aspect, the disclosure provides a ribonucleoprotein (RNP) comprising the Cas12i polypeptide of the disclosure and a guide nucleic acid optionally as defined in the disclosure.

[0014] In yet another aspect, the disclosure provides a lipid nanoparticle (LNP) comprising the Cas12i polypeptide of the disclosure or the system of the disclosure.

[0015] In yet another aspect, the disclosure provides a method for modifying a target DNA, comprising contacting the target DNA with the system of the disclosure, the vector of the disclosure, the ribonucleoprotein of the disclosure, or the lipid nanoparticle of the disclosure, wherein the spacer sequence is capable of hybridizing to a target sequence of the target DNA, wherein the target DNA is modified by the complex.

[0016] In yet another aspect, the disclosure provides a cell modified by the method of the disclosure.

[0017] In yet another aspect, the disclosure provides a pharmaceutical composition comprising (1) the system of the disclosure, the vector of the disclosure, the ribonucleoprotein of the disclosure, the lipid nanoparticle of the disclosure, or the cell of the disclosure; and (2) a pharmaceutically acceptable excipient.

[0018] In yet another aspect, the disclosure provides a method for diagnosing, preventing, or treating a disease in a subject in need thereof, comprising administering to the subject the system of the disclosure, the vector of the disclosure, the ribonucleoprotein of the disclosure, the lipid nanoparticle of the disclosure, the cell of the disclosure, or the pharmaceutical composition of the disclosure, wherein the disease is associated with a target DNA, wherein the spacer sequence is capable of hybridizing to a target sequence of the target DNA, wherein the target DNA is modified by the complex, and wherein the modification of the target DNA diagnose, prevents, or treats the disease.

[0019] In yet another aspect, the disclosure provides a method of detecting a target DNA, comprising contacting the target DNA with the system of the disclosure, wherein the target DNA is modified by the complex, and wherein the modification detects the target DNA.

[0020] The details of one or more embodiments of the disclosure are set forth in the description below. Other features or advantages of the disclosure will be apparent from the following drawings and detailed description of several embodiments, and also from the appended claims. It is understood that any aspect or embodiment of the disclosure can be combined with any other aspect or embodiment of the disclosure to constitute another embodiment explicitly or implicitly disclosed herein unless otherwise indicated.

OVERVIEW

[0021] Cas12i, as a subtype of Class 2, Type V CRISPR associated protein (Cas12), is capable of binding to or function on a target nucleic acid (e.g., a dsDNA) as guided by a guide nucleic acid (e.g., a guide RNA (gRNA, used interchangeably with single guide RNA or sgRNA in the disclosure)) comprising a guide sequence targeting the target nucleic acid. In some embodiments, the target nucleic acid is eukaryotic.

[0022] Without wishing to be bound by theory, in some embodiments, the guide nucleic acid comprises a scaffold sequence (used interchangeable with a direct repeat sequence in the disclosure) responsible for forming a complex with the Cas12i, and a guide sequence (used interchangeable with a spacer sequence in the disclosure) that is intentionally designed to be responsible for hybridizing to a target sequence of the target nucleic acid, thereby guiding the complex comprising the Cas12i and the guide nucleic acid to the target nucleic acid.

[0023] Referring to FIG. 20, an exemplary target dsDNA (e.g., a target gene) is depicted to comprise a 5' to 3' single DNA strand and a 3' to 5' single DNA strand.

[0024] An exemplary guide nucleic acid is depicted to comprise a guide sequence and a scaffold sequence. The guide sequence is designed to hybridize to a part of the 3' to 5' single DNA strand, and so the guide sequence “targets” that part. And thus, the 3' to 5' single DNA strand is referred to as a “target strand (TS)” of the target dsDNA, while the opposite 5' to 3' single DNA strand is referred to as a “nontarget strand (NTS)” of the target dsDNA. That part of the target strand based on which the guide sequence is designed and to which the guide sequence may hybridize is referred to as a “target sequence”, while the opposite part on the nontarget strand corresponding to that part is referred to as the “protospacer sequence”, which is 100% (fully) reversely complementary to the target sequence.

[0025] Generally, a nucleic acid sequence (e.g., a DNA sequence, an RNA sequence) is written in 5' to 3' direction/orientation.

[0026] For example, for a DNA sequence of ATGC, it is usually understood as 5'-ATGC-3' unless otherwise indicated. Its reverse sequence is 5'-CGTA-3', its fully complement sequence is 5'-TACG-3', and its fully reverse complement sequence is 5'-GCAT-3'.

[0027] Generally, the double-strand sequence of a dsDNA may be represented with the sequence of its 5' to 3' single DNA strand conventionally written in 5' to 3' direction/orientation unless otherwise indicated.

[0028] For example, for a dsDNA having a 5' to 3' single DNA strand of 5'-ATGC-3' and a 3' to 5' single DNA strand of 3'-TACG-5', the dsDNA may be simply represented as 5'-ATGC-3'.

[0029] 5'-----ATGC-----3'

[0030] 3'-----TACG-----5'

[0031] It should be noted that either the 5' to 3' single DNA strand or the 3' to 5' single DNA strand of a dsDNA can be a nontarget strand from which a protospacer sequence is selected or a target strand to which the guide sequence is designed to hybridize.

[0032] Generally, for a gene as a dsDNA, the 5' to 3' single DNA strand is the sense strand of the gene, and the 3' to 5' single DNA strand is the antisense strand of the gene. But it should be noted that either the sense strand or the antisense strand of a gene can be a nontarget strand from which a

protospacer sequence is selected or a target strand to which the guide sequence is designed to hybridize.

[0033] To hybridize to a target dsDNA, in one embodiment, the guide sequence of a guide nucleic acid (e.g., a guide RNA) is designed to have a RNA sequence of 5'-AUGC-3' that is fully reversely complementary to the 3' to 5' strand of the target dsRNA, which would be set forth in ATGC in the electric sequence listing but annotated as RNA; and in another embodiment, the guide sequence of a guide nucleic acid (e.g., a guide RNA) is designed to have a RNA sequence of 5'-GCAU-3' that is fully reversely complementary to the 5' to 3' strand of the target dsRNA, which would be set forth in GCAT in the electric sequence listing but annotated as RNA.

[0034] In the case that the guide sequence of a guide nucleic acid is fully reversely complementary to the target sequence and the target sequence is fully reversely complementary to the protospacer sequence, the guide sequence is identical to the protospacer sequence except for the U in the guide sequence if it is an RNA sequence and correspondingly the T in the protospacer sequence. According to WIPO standard ST. 26, symbol “t” is used to denote both T in DNA and U in RNA (See “Table 1: List of nucleotides symbols”, the definition of symbol “t” is “thymine in DNA/uracil in RNA (t/u)”). Thus, in the sequence listing of the disclosure prepared according to ST. 26, such a guide sequence could be set forth in the same sequence as a corresponding protospacer sequence. For convenience, a single SEQ ID NO in the sequence listing can be used to denote both such guide sequence and protospacer sequence, although such a single SEQ ID NO may be marked as either DNA or RNA in the sequence listing. When a reference is made to such a SEQ ID NO that sets forth a protospacer/guide sequence, it refers to either a protospacer sequence that is a DNA sequence or a guide sequence that may be an RNA sequence depending on the context, no matter whether it is marked as DNA or RNA in the sequence listing.

Term

[0035] Unless otherwise specified, all technical and scientific terms used in the disclosure have the meaning commonly understood by one of ordinary skill in the art to which the disclosure belongs. Throughout the specification, several terms are employed that are defined in the following paragraphs. Other definitions are also found within the body of the specification.

[0036] As used herein, the terms “nucleic acid”, “nucleic acid molecule”, or “polynucleotide” are used interchangeably. They refer to a polymer of deoxyribonucleotides or ribonucleotides or their mixtures in either single- or double-stranded form, and unless otherwise stated, encompass known analogs of natural nucleotides that can function in a similar manner as naturally occurring nucleotides. The terms encompass nucleic acid-like structures with synthetic backbones, as well as amplification products. DNAs and RNAs are both polynucleotides. The polymer may include natural nucleosides (i.e., adenosine, thymidine, guanosine, cytidine, uridine, deoxyadenosine, deoxythymidine, deoxyguanosine, and deoxycytidine), nucleoside analogs (e.g., 2-aminoadenosine, 2-thiothymidine, inosine, pyrrolo-pyrimidine, 3-methyl adenosine, C5-propynylcytidine, C5-propynyluridine, C5-bromouridine, C5-fluorouridine, C5-iodouridine, C5-methylcytidine, 7-deazaadenosine, 7-deazaguanosine, 8-oxoadenosine, 8-oxoguanosine, O (6)-methylguanine, and

2-thiocytidine), chemically modified bases, biologically modified bases (e.g., methylated bases), intercalated bases, modified sugars (e.g., 2'-fluororibose, ribose, 2'-deoxyribose, arabinose, and hexose), or modified phosphate groups (e.g., phosphorothioates and 5'-N-phosphoramidite linkages).

[0037] As used herein, the term “polypeptide” and “protein” are used interchangeably to refer to polymers of amino acids of any length. The polymer may be linear or branched, it may comprise modified amino acids, and it may be interrupted by non-amino acids. The terms also encompass an amino acid polymer that has been modified; for example, by disulfide bond formation, glycosylation, lipidation, acetylation, phosphorylation, or any other manipulation, such as conjugation with a labeling component.

[0038] As used herein, a “fusion protein” refers to a protein created through the joining of two or more originally separate proteins, or portions thereof. In some embodiments, a linker may be present between each protein.

[0039] As used herein, the term “heterologous,” in reference to polypeptide domains, refers to the fact that the polypeptide domains do not naturally occur together (e.g., in the same polypeptide). For example, in fusion proteins generated by the hand of man, a polypeptide domain from one polypeptide may be fused to a polypeptide domain from a different polypeptide. The two polypeptide domains would be considered “heterologous” with respect to each other, as they do not naturally occur together.

[0040] As used herein, the term “nuclease” refers to a polypeptide capable of cleaving the phosphodiester bonds between the nucleotide subunits of nucleic acids; the term “endonuclease” refers to a polypeptide capable of cleaving the phosphodiester bond within a polynucleotide chain.

[0041] As used herein, the term “Cas12i” is used interchangeably with Cas12i protein or Cas12i polypeptide in the disclosure and used in its broadest sense and includes parental or reference Cas12i proteins (e.g., Cas12i protein comprising any of SEQ ID NOs: 1-10), derivatives or variants thereof, and functional fragments such as nucleic acid-binding fragments thereof, including endonuclease deficient (dead) Cas12i polypeptides, and Cas12i nickases.

[0042] As used herein, the term “guide nucleic acid” refers to a nucleic acid-based molecule capable of forming a complex with a CRISPR-Cas protein (e.g., a Cas12i of the disclosure) (e.g., via a scaffold sequence of the guide nucleic acid), and comprises a sequence (e.g., guide sequences) that are sufficiently complementary to a target nucleic acid to hybridize to the target nucleic acid and guide the complex to the target nucleic acid, which include but are not limited to RNA-based molecules, e.g., guide RNA. As used herein, the term “crRNA” is used interchangeably with guide RNA (gRNA), single guide RNA (sgRNA), or RNA guide. As used in the disclosure, the term “guide sequence” is used interchangeably with the term “spacer sequence”, and the term “scaffold sequence” is used interchangeably with the term “direct repeat sequence”. The guide nucleic acid may be a DNA molecule, an RNA molecule, or a DNA/RNA mixture molecule. By “DNA/RNA mixture molecule” it refers to a nucleic acid comprising both one or more modified or unmodified ribonucleotides and one or more modified or unmodified deoxyribonucleotides, whether consecutive or not. However, by “DNA molecule” or “RNA molecule” it may also refer to a DNA molecule containing one or more modified or unmodified ribonucleotides,

whether consecutive or not, or an RNA molecule containing one or more modified or unmodified deoxyribonucleotides, whether consecutive or not.

[0043] As used herein, the term “complex” refers to a grouping of two or more molecules. In some embodiments, the complex comprises a polypeptide and a nucleic acid interacting with (e.g., binding to, coming into contact with, adhering to) one another. As used herein, the term “complex” can refer to a grouping of a guide nucleic acid and a polypeptide (e.g., a Cas12i polypeptide). As used herein, the term “complex” can refer to a grouping of a guide nucleic acid, a polypeptide, and a target nucleic acid.

[0044] As used herein, the term “activity” refers to a biological activity. In some embodiments, the activity includes enzymatic activity, e.g., catalytic ability of an effector. For example, the activity can include nuclease activity, e.g., DNA nuclease activity, dsDNA endonuclease activity, guide sequence-specific (on-target) dsDNA endonuclease activity, guide sequence-independent (off-target) dsDNA endonuclease activity.

[0045] As used herein, the term “spacer sequence-specific (on-target) dsDNA cleavage” may be termed as “dsDNA cleavage” for short unless otherwise indicated.

[0046] As used herein, the term “cleavage” refers to the breakage of the covalent backbone of a DNA molecule. Cleavage can be initiated by a variety of methods including, but not limited to, enzymatic or chemical hydrolysis of a phosphodiester bond. Both single-stranded cleavage and double-stranded cleavage are possible, and double-stranded cleavage can occur as a result of two distinct single-stranded cleavage events. DNA cleavage can result in the production of either blunt ends or cohesive ends.

[0047] As used herein, the meanings of “cleaving a nucleic acid” or “modifying a nucleic acid” may overlap. Modifying a nucleic acid includes not only modification of a mononucleotide but also insertion or deletion of a nucleic acid fragment.

[0048] As used herein, the term “on-target” refers to binding, cleavage, and/or editing of an intended or expected region of DNA, for example, by Cas12i of the disclosure.

[0049] As used herein, the term “off-target” refers to binding, cleavage, and/or editing of an unintended or unexpected region of DNA, for example, by Cas12i of the disclosure. In some embodiments, a region of DNA is an off-target region when it differs from the region of DNA intended or expected to be bound, cleaved and/or edited by 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 or more nucleotides.

[0050] As used herein, if a DNA sequence, for example, 5'-ATGC-3' is transcribed to an RNA sequence, with each dT (deoxythymidine, or “T” for short) in the primary sequence of the DNA sequence replaced with a U (uridine) and each dA (deoxyadenosine, or “A” for short), dG (deoxyguanosine, or “G” for short), and dC (deoxycytidine, or “C” for short) replaced with A (adenosine), G (guanosine), and C (cytidine), respectively, for example, 5'-AUGC-3', it is said in the disclosure that the DNA sequence “encodes” the RNA sequence.

[0051] As used herein, the term “protospacer adjacent motif” or “PAM” refers to a short sequence (or a motif) adjacent to a protospacer sequence on the nontarget strand of a dsDNA recognized by CRISPR complexes.

[0052] As used herein, the term “adjacent” includes instances wherein there is no nucleotide between the protospacer sequence and the PAM and also instances wherein

there are a small number (e.g., 1, 2, 3, 4, or 5) of nucleotides between the protospacer sequence and the PAM. As used herein, A “immediately adjacent (to)” B, A “immediately 5' to” B, and A “immediately 3' to” B mean that there is no nucleotide between A and B.

[0053] As described herein, the guide sequence is so designed to be capable of hybridizing to a target sequence. As used herein, the term “hybridize”, “hybridizing”, or “hybridization” refers to a reaction in which one or more polynucleotide sequences react to form a complex that is stabilized via hydrogen bonding between the bases of the one or more polynucleotide sequences. The hydrogen bonding may occur by Watson Crick base pairing, Hoogsteen binding, or in any other sequence specific manner. A polynucleotide sequence capable of hybridizing to a given polynucleotide sequence is referred to as the “complement” of the given polynucleotide sequence. As used herein, the hybridization of a guide sequence and a target sequence is so stabilized to permit a Cas12i polypeptide that is complexed with a guide nucleic acid comprising the guide sequence or a function domain (e.g., a deaminase domain) associated (e.g., fused) with the Cas12i polypeptide to act (e.g., cleave, deaminize) at or near the target sequence or its complement (e.g., a sequence of a target DNA or its complement).

[0054] For the purpose of hybridization, in some embodiments, the guide sequence is reversely complementary to a target sequence. As used herein, the term “complementary” refers to the ability of nucleobases of a first polynucleotide sequence, such as a guide sequence, to base pair with nucleobases of a second polynucleotide sequence, such as a target sequence, by traditional Watson-Crick base-pairing. Two complementary polynucleotide sequences are able to non-covalently bind under appropriate temperature and solution ionic strength conditions. In some embodiments, a first polynucleotide sequence (e.g., a guide sequence) comprises 100% (fully) complementarity to a second nucleic acid (e.g., a target sequence). In some embodiments, a first polynucleotide sequence (e.g., a guide sequence) is complementary to a second polynucleotide sequence (e.g., a target sequence) if the first polynucleotide sequence comprises at least about 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99% complementarity to the second nucleic acid. As used herein, the term “substantially complementary” refers to a polynucleotide sequence (e.g., a guide sequence) that has a certain level of complementarity to a second polynucleotide sequence (e.g., a target sequence) such that the first polynucleotide sequence (e.g., a guide sequence) can hybridize to the second polynucleotide sequence (e.g., a target sequence) with sufficient affinity to permit a Cas12i polypeptide that is complexed with the first polynucleotide sequence or a nucleic acid comprising the first polynucleotide sequence or a function domain associated (e.g., fused) with the Cas12i polypeptide to act (e.g., cleave, deaminize) on the target sequence or its complement (e.g., a sequence of a target DNA or its complement). In some embodiments, a guide sequence that is substantially complementary to a target sequence has 100% or less than 100% complementarity to the target sequence. In some embodiments, a guide sequence that is substantially complementary to a target sequence has at least about 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99% complementarity to the target sequence.

[0055] As used herein, the term “identity” refers to the overall relatedness between polymeric molecules, e.g., between nucleic acid molecules (e.g., DNA molecules and/or RNA molecules) and/or between polypeptide molecules. In some embodiments, polymeric molecules are considered to be “substantially identical” to one another if their sequences are at least 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, or 99% identical. Calculation of the percent identity of two nucleic acid or polypeptide sequences, for example, can be performed by aligning the two sequences for optimal comparison purposes (e.g., gaps can be introduced in one or both of a first and a second sequences for optimal alignment and non-identical sequences can be disregarded for comparison purposes). In certain embodiments, the length of a sequence aligned for comparison purposes is at least 30%, at least 40%, at least 50%, at least 60%, at least 70%, at least 80%, at least 90%, at least 95%, or substantially 100% of the length of a reference sequence. The nucleotides at corresponding positions are then compared. The comparison of sequences and determination of percent identity between two sequences can be accomplished using a mathematical algorithm. As is well known in the art, amino acid or nucleic acid sequences may be compared using any of a variety of algorithms, including those available in commercial computer programs such as BLASTN for nucleotide sequences and BLASTP, gapped BLAST, and PSI-BLAST for amino acid sequences. In some embodiments, the sequence identity is calculated by global alignment, for example, using the Needleman-Wunsch algorithm and an online tool at ebi.ac.uk/Tools/psa/emboss_needle/. In some embodiments, the sequence identity is calculated by local alignment, for example, using the Smith-Waterman algorithm and an online tool at ebi.ac.uk/Tools/psa/emboss_water/.

[0056] As used herein, the term “variant” refers to an entity that shows significant structural identity with a reference entity (e.g., a wild-type sequence) but differs structurally from the reference entity in the presence or level of one or more chemical moieties as compared with the reference entity. In many embodiments, a variant also differs functionally from its reference entity. In general, whether a particular entity is properly considered to be a “variant” of a reference entity is based on its degree of structural identity with the reference entity. As will be appreciated by those skilled in the art, any biological or chemical reference entity has certain characteristic structural elements. A variant, by definition, is a distinct chemical entity that shares one or more such characteristic structural elements. To give but a few examples, a polypeptide may have a characteristic sequence element comprising a plurality of amino acids having designated positions relative to one another in linear or three-dimensional space and/or contributing to a particular biological function; a nucleic acid may have a characteristic sequence element comprising a plurality of nucleotide residues having designated positions relative to one another in linear or three-dimensional space. For example, a variant polypeptide may differ from a reference polypeptide as a result of one or more differences in amino acid sequence and/or one or more differences in chemical moieties (e.g., carbohydrates, lipids, etc.) covalently attached to the polypeptide backbone. In some embodiments, a variant polypeptide shows an overall sequence identity with a reference polypeptide (e.g., a nuclease described herein) that is at least 60%, 65%, 70%, 75%, 80%, 85%, 86%, 87%, 88%, 89%,

90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99%. Alternatively or additionally, in some embodiments, a variant polypeptide does not share at least one characteristic sequence element with a reference polypeptide. In some embodiments, the reference polypeptide has one or more biological activities. In some embodiments, a variant polypeptide shares one or more of the biological activities of the reference polypeptide, e.g., nuclease activity. In some embodiments, a variant polypeptide lacks one or more of the biological activities of the reference polypeptide. In some embodiments, a variant polypeptide shows a reduced level of one or more biological activities (e.g., nuclease activity, e.g., off-target nuclease activity) as compared with the reference polypeptide. In some embodiments, a polypeptide of interest is considered to be a “variant” of a parent or reference polypeptide if the polypeptide of interest has an amino acid sequence that is identical to that of the parent but for a small number of sequence alterations at particular positions. Typically, fewer than 20%, 15%, 10%, 9%, 8%, 7%, 6%, 5%, 4%, 3%, 2%, or 1% of the residues in the variant are substituted as compared with the parent or reference polypeptide. In some embodiments, a variant has 10, 9, 8, 7, 6, 5, 4, 3, 2, or 1 substituted residue as compared with a parent or reference polypeptide. Often, a variant has a very small number (e.g., fewer than 5, 4, 3, 2, or 1) of substituted functional residues (i.e., residues that participate in a particular biological activity). In some embodiments, a variant has not more than 5, 4, 3, 2, or 1 additions or deletions, and often has no additions or deletions, as compared with the parent or reference polypeptide. Moreover, any additions or deletions are typically fewer than about 25, about 20, about 19, about 18, about 17, about 16, about 15, about 14, about 13, about 12, about 11, about 10, about 9, about 8, about 7, about 6, and commonly are fewer than about 5, about 4, about 3, or about 2 residues. In some embodiments, the parent or reference polypeptide is a wild type. A variant of a polynucleotide or polypeptide may be naturally occurring such as an allelic variant, or it may be a variant that is not known to occur naturally. Non-naturally occurring variants of polynucleotides and polypeptides may be made by mutagenesis techniques, by direct synthesis, and by other recombinant methods known to skilled artisans.

[0057] As used herein, the terms “non-naturally occurring” and “engineered” are used interchangeably and refer to artificial participation. When these terms are used to describe a nucleic acid or a polypeptide, it is meant that the nucleic acid or polypeptide is at least substantially freed from at least one other component of its association in nature or as found in nature.

[0058] Conservative substitutions of non-critical amino acids of a protein may be made without affecting the normal functions of the protein. Conservative substitutions refer to the substitution of amino acids with chemically or functionally similar amino acids. In some embodiments, a conservative amino acid substitution refers to an amino acid substitution that does not alter the relative charge or size characteristics of the protein in which the amino acid substitution was made. In some embodiments, a “conservative substitution” refers to a substitution of an amino acid made among amino acids within the following groups: i) methionine, isoleucine, leucine, valine, ii) phenylalanine, tyrosine, tryptophan, iii) lysine, arginine, histidine, iv) alanine, glycine, v) serine, threonine, vi) glutamine, asparagine and vii) glutamic acid, aspartic acid.

[0059] As used herein, the term “wild type” has the meaning commonly understood by those skilled in the art to mean a typical form of an organism, a strain, a gene, or a feature that distinguishes it from a mutant or variant when it exists in nature. It can be isolated from sources in nature and not intentionally modified.

[0060] As used herein, the description of “a variant (e.g., a Cas12i polypeptide) comprising an amino acid mutation (e.g., substitution) at a given position (e.g., E336) of a given polypeptide (e.g., SEQ ID NO: 458)” or similar description means that the polypeptide as set forth in the amino acid sequence of the given polypeptide serves as a parent or reference polypeptide, and the variant is a variant of the parent or reference polypeptide and comprises an amino acid mutation at a position of the amino acid sequence of the variant corresponding to the given position of the amino acid sequence of the given polypeptide. The position of the amino acid mutation in the amino acid sequence of the variant may be the same as the given position of the given polypeptide, for example, when the variant comprises just an amino acid substitution as compared with the given polypeptide and has the same length as the given polypeptide. The position of the amino acid mutation in the amino acid sequence of the variant may also be different from the given position of the given polypeptide, for example, when the variant comprises a N-terminal truncation as compared with the given polypeptide and the first N-terminal amino acid of the variant is not corresponding to the first N-terminal amino acid of the given polypeptide but to an amino acid within the given polypeptide, but the position of the amino acid mutation can be determined by alignment of the variant and the given polypeptide to identify the corresponding amino acids in their sequences as understood by a skilled in the art. For example, if the variant has a N-terminal truncation of 20 amino acids as compared with the given polypeptide, then the variant comprising an amino acid mutation at E336 of a given polypeptide means that the variant comprises an amino acid mutation at E316 of the variant since E316 in the variant is corresponding to E336 in the given polypeptide as determined by alignment of the variant and the given polypeptide.

[0061] As used herein, the description of “a variant (e.g., a Cas12i polypeptide) comprising a given amino acid substitution (e.g., E336R) relative to a given polypeptide (e.g., SEQ ID NO: 458)” means that the polypeptide as set forth in the amino acid sequence of the given polypeptide serves as a parent or reference polypeptide that does not comprise the given amino acid substitution, and the variant is a variant of the parent or reference polypeptide and comprises an amino acid substitution having the same type of substitution as the given amino acid substitution and at a position in the amino acid sequence of the variant corresponding to the position of the given amino acid substitution. For example, a Cas12i polypeptide comprising an amino acid substitution E336R relative to SEQ ID NO: 458 refers to the fact that the amino acid sequence of SEQ ID NO: 458 comprises amino acid E at position 336, and the Cas12i polypeptide comprises amino acid R at a position corresponding to position 336 of the amino acid sequence of SEQ ID NO: 458. The corresponding relationship of positions in two amino acid sequences as determined by alignment is explained in the previous paragraph.

[0062] As used herein, the terms “upstream” and “downstream” refer to relative positions within a single nucleic

acid (e.g., DNA) sequence in a nucleic acid. “Upstream” and “downstream” relate to the 5' to 3' direction, respectively, in which transcription occurs. For a first sequence and a second sequence present on the same strand of a single nucleic acid written in 5' to 3' direction, the first sequence is upstream of the second sequence when the 3' end of the first sequence is on the left side of the 5' end of the second sequence, and the first sequence is downstream of the second sequence when the 5' end of the first sequence is on the right side of the 3' end of the second sequence. For example, a promoter is usually at the upstream of a sequence under the regulation of the promoter; and on the other hand, a sequence under the regulation of a promoter is usually at the downstream of the promoter.

[0063] As used herein, the term “regulatory element” refers to a DNA sequence that controls or impacts one or more aspects of transcription and/or expression is intended to include promoters, enhancers, silencers, termination signals, internal ribosome entry sites (IRES), and other expression control elements (e.g., transcription termination signals such as polyadenylation signals and poly-U sequences). Regulatory elements include those that direct constitutive expression of a nucleotide sequence in many types of host cells and those that direct expression of a nucleotide sequence only in certain host cells (e.g., tissue-specific regulatory sequences). Regulatory elements may also direct expression in a time-dependent manner, e.g., in a cell cycle-dependent or developmental stage-dependent manner, which may or may not be tissue or cell type specific.

[0064] As used herein, the term “operably linked” refers to a juxtaposition wherein the components described are in a relationship permitting them to function in their intended manner. A regulatory element “operably linked” to a functional element is associated in such a way that transcription, expression, and/or activity of the functional element is achieved under conditions compatible with the regulatory element. In some embodiments, “operably linked” regulatory elements are contiguous (e.g., covalently linked) with the functional elements of interest; in some embodiments, regulatory elements act in trans to or otherwise at a distance from the functional elements of interest.

[0065] As used herein, the term “cell” is understood to refer not only to a particular individual cell, but to the progeny or potential progeny of the cell. Because certain modifications may occur in succeeding generations due to either mutation or environmental influences, such progeny may not, in fact, be identical to the parent cell, but are still included within the scope of the term.

[0066] As used herein, the term “in vivo” means inside the body of an organism, and the terms “ex vivo” or “in vitro” means outside the body of an organism.

[0067] As used herein, the term “treat”, “treatment”, or “treating” is an approach for obtaining beneficial or desired results including clinical results. For purposes of the disclosure, the beneficial or desired clinical results include, but are not limited to, one or more of the following: alleviating one or more symptoms resulting from a disease, diminishing the extent of a disease, stabilizing a disease (e.g., preventing or delaying the worsening of a disease), preventing or delaying the spread (e.g., metastasis) of a disease, preventing or delaying the recurrence of a disease, reducing recurrence rate of a disease, delay or slowing the progression of a disease, ameliorating a disease state, providing a remission (partial or total) of a disease, decreasing the dose of one or

more other medications required to treat a disease, delaying the progression of a disease, increasing the quality of life, and prolonging survival. Also encompassed by the term is a reduction of pathological consequence of a disease (such as cancer). The methods of the disclosure contemplate any one or more of these aspects of treatment.

[0068] As used herein, the term “disease” includes the terms “disorder” and “condition” and is not limited to those specific diseases that have been medically or clinically defined.

[0069] As used herein, reference to “not” a value or parameter generally means and describes “other than” a value or parameter. For example, the method is not used to treat cancer of type X means the method may be used to treat cancer of types other than X.

[0070] As used herein, the singular forms “a”, “an”, and “the” include plural referents unless the context clearly dictates otherwise. That is, articles “a/an” and “the” are used herein to refer to one or more than one (i.e., at least one) grammatical object of the article. For example, “an element” means one element or more than one element, e.g., two elements.

[0071] As used herein, the term “and/or” in a phrase such as “A and/or B” is intended to mean either or both of the alternatives, including both A and B, A or B, A (alone), and B (alone). Likewise, the term “and/or” in a phrase such as “A, B, and/or C” is intended to encompass each of the following embodiments: A, B, and C; A, B, or C; A or C; A or B; B or C; A and C; A and B; B and C; A (alone); B (alone); and C (alone).

[0072] As used herein, when the term “about” is ahead of a series of numbers (for example, about 1, 2, 3), it is understood that each of the series of numbers is modified by the term “about” (that is, about 1, about 2, about 3). The term “about X-Y” used herein has the same meaning as “about X to about Y.”

[0073] As used herein, the terms “about” and “approximately,” in reference to a number, is used herein to include numbers that fall within a range of 20%, 15%, 10%, 9%, 8%, 7%, 6%, 5%, 4%, 3%, 2%, or 1% in either direction (greater than or less than) of the number unless otherwise stated or otherwise evident from the context (except where such number would exceed 100% of a possible value).

[0074] As used herein, a numerical range includes the end values of the range, and each specific value within the range, for example, “16 to 100 nucleotides” includes 16 nucleotides and 100 nucleotides, and each specific value between 16 and 100, e.g., 17, 23, 34, 52, 78.

[0075] As used herein, the terms “comprise”, “include”, “contain”, and “have” are to be understood as implying that a stated element or a group of elements is included, but not excluding any other element or a group of elements, unless the context requires otherwise. In certain embodiments, the terms “comprise”, “include”, “contain”, and “have” are used synonymously.

[0076] As used herein, the phrase “consist essentially of” is intended to include any element listed after the phrase “consist essentially of” and is limited to other elements that do not interfere with or contribute to the activities or actions specified in the disclosure of the listed elements. Thus, the phrase “consist essentially of” is intended to indicate that the listed elements are required, but no other elements are

optional, and may or may not be present depending on whether they affect the activities or actions of the listed elements.

[0077] As used herein, the phrase “consist of” means including but limited to any element after the phrase “consist of”. Thus, the phrase “consist of” indicates that the listed elements are required, and that no other elements can be present.

[0078] As used herein, the term “comprises” also encompasses the terms “consists essentially of” and “consists of”. It is understood that the “comprising” embodiments of the disclosure described herein also include “consisting essentially of” and “consisting of” embodiments.

[0079] Throughout the specification, reference to “one embodiment”, “embodiment”, “a specific embodiment”, “a related embodiment”, “an embodiment”, “another embodiment”, or “a further embodiment” or a combination thereof means that specific elements, features, structures, or characteristics described in connection with the embodiment are included in at least one embodiment of the disclosure. Accordingly, the appearances of the foregoing phrases in various places throughout the specification are not necessarily all referring to the same embodiments. Furthermore, specific elements, features, structures, or characteristics may be combined in any suitable manner in one or more embodiments.

[0080] It is further noted that the claims may be drafted to exclude any optional element. As such, this statement is intended to serve as antecedent basis for use of such exclusive terminology as “solely”, “only”, and the like in connection with the recitation of claim elements, or use of a “negative” limitation.

BRIEF DESCRIPTION OF THE DRAWINGS

[0081] An understanding of the features and advantages of the disclosure will be obtained by reference to the following detailed description that sets forth illustrative embodiments, in which the principles of the disclosure may be utilized, and the accompanying drawings of which:

[0082] FIG. 1 shows that hfCas12Max, an engineered variant of xCas12i, mediated high-efficient and high-specificity genome editing, and dCas12i base editor exhibited high base editing activity in mammalian cells. FIG. 1A, xCas12i mediated EGFP activation efficiency determined by flow cytometry. NC represents non-specific (non-targeting) control. FIG. 1B, Schematics of protein engineering strategy for mutants with high efficiency and high fidelity (specificity) using an activatable EGFP reporter screening system with on-targeted and off-targeted crRNA. FIG. 1C and FIG. 1D, Cas12Max exhibited significantly increased cleavage activity than xCas12i at reporter plasmids (FIG. 1C) or various genomic target sites (FIG. 1D). Each dot represents the mean indel frequency at one targeted site (n=3). FIG. 1E, NGS analysis showed that hfCas12Max retained comparable activity at TTR. 2-ON targets to Cas12Max and almost no activity at 6 OT sites. FIG. 1F, Both Cas12Max and hfCas12Max exhibited a broader PAM recognition profile including 5'-TN and 5'-TNN PAM than other Cas proteins. FIG. 1G, Comparison of indel activity of Cas12Max, hfCas12Max, LbCas12a, Ultra AsCas12a, SpCas9 and KKH-saCas9 at TTR locus. hfCas12Max retained comparable activity to Cas12Max, and higher gene-editing efficiency than other Cas proteins. Each dot represents one of three repeats of single target site. FIG. 1H, Schematics of

different versions of dxCas12i adenine base editors. FIG. 1I, Comparison of A-to-G editing frequency and product purity at the KLF4 site of TadA8e. 1-dxCas12i-v1.2, v2.2 and v4.3, v4.3 showed a high editing activity of 80%. TadA8e-dxCas12i-v4.3, named as ABE-dCas12Max. TadA8e. 1 represents TadA8e-V106W. FIG. 1J, Schematics of different versions of dxCas12i cytosine base editors. FIG. 1K, Comparison of C-to-T editing frequency and product purity at the RUNX1 site of hA3A. 1-dxCas12i, -v1.2 v2.2 and v3.1, and also hA3A. 1-dLbCas12a, v3.1 showed a high editing activity of 50%. hA3A. 1-dxCas12i-v3.1, named as CBE-dCas12Max. hA3A. 1 represents human APOBEC3A-W104A.

[0083] FIG. 2 shows that hfCas12Max mediated high-efficiency gene editing ex vivo and in vivo. FIG. 2A, Schematics of hfCas12Max gene editing in primary human cells. FIG. 2B, Viability and indel activity of human CD3+T cells following delivery of hfCas12Max RNPs with three different TRAC targeting gRNAs at 1.6 μ M and 3.2 μ M respectively (n=2 or 3). NC represents blank control, untreated with RNP. FIG. 2C, Representative flow cytometric analysis of edited CD3+ T cell 5 days after RNP delivery. NC represents blank control, untreated with RNP. FIG. 2D, Schematics of in vivo non-liposome delivery containing IVT-mRNA, LNP packaging process. FIG. 2E, Editing efficiency of LNP packaging with hfCas12Max mRNA and Ttr targeting gRNA at increased concentrations in N2a cells (n=8). FIG. 2F, Schematics of Ttr locus. FIG. 2G, Indel rates of LNP packaging with hfCas12Max mRNA and Ttr targeting gRNA at three doses (0.1, 0.3 and 0.5 mpk) in C57 mouse (n=6). FIG. 2H, The A-to-G editing percentage of LNP packaging with dCas12i-ABE mRNA and Ttr targeting gRNA at 3 mpk in C57 mouse (n=2).

[0084] FIG. 3 shows screening for functional Cas12i in HEK293T cells. FIG. 3A, Transfection of plasmids coding Cas12i and gRNA mediate EGFP activation. FIG. 3B, Five of ten Cas12i nucleases mediated EGFP-activated efficiency in HEK293T cells.

[0085] FIG. 4 shows identification and characterization of type V-I systems. FIG. 4A, Nuclease domain organization of SpCas9, LbCas12a, and xCas12i. FIG. 4B, Effective spacer sequence length for xCas12i. FIG. 4C, PAM scope comparison of LbCas12a and xCas12i. xCas12i exhibited a higher dsDNA cleavage activity at 5'-TTN PAM than LbCas12a. FIG. 4D, Flow diagram for detection of genome cleavage activity by transfection of an all-in-one plasmid containing xCas12i and gRNA into HEK293T cells, followed by FACS and NGS analysis. FIG. 4E-FIG. 4F, xCas12i mediated robust genome cleavage (up to 90%) at Ttr locus in N2a cells and TTR and PCSK9 locus in HEK293T cells.

[0086] FIG. 5 shows screening for engineered xCas12i mutants with single point mutation and various dsDNA cleavage activity. FIG. 5A, The relative dsDNA cleavage activity of over 500 rationally engineered xCas12i mutants. v1.1 represents xCas12i with N243R, named as Cas12Max.

[0087] FIG. 6 shows additional xCas12i-N243 mutants mediated high-efficiency editing. FIG. 6A, Of all the saturated mutants of xCas12i-N243, xCas12i-N243R showed the mostly increased EGFP-activated fluorescence. FIG. 6B-FIG. 6C, xCas12i mutant with N243R increased 1.2, 5, and 20-fold activity at DMD. 1, DMD. 2 and DMD. 3 locus, respectively. FIG. 6D, Both Cas12Max (xCas12i-N243R)

and Cas12Max-E336R (xCas12i-N243R+E336R) elevated EGFP-activated fluorescence at different PAM recognition sites.

[0088] FIG. 7 shows that Cas12Max induced off-target dsDNA cleavage activity at sites with mismatches using the reporter system (FIG. 7A) and targeted deep sequence (FIG. 7B).

[0089] FIG. 8 shows that hfCas12Max (xCas12i-N243R+E336R+D892R) mediates high-efficiency and high-specificity editing. FIG. 8A, Rational protein engineering screening of over 200 mutants for highly-fidelity (specificity) Cas12Max. Four mutants show significantly decreased cleavage activity at both OT (off-target) sites and retained cleavage activity at ON. 1 (on-target) site. FIG. 8B, Different versions of xCas12i mutants. FIG. 8C, v6.3 reduced off-target at OT. 1, OT. 2 and OT. 3 sites and retained indel activity at TTR-ON targets, compared to v1.1. FIG. 8D, v6.3 exhibited comparable indel activity at DMD. 1, DMD. 2, and higher at DMD. 3 locus, than v1.1. v1.1, i.e., Cas12Max. v6.3, named as hfCas12Max.

[0090] FIG. 9 shows comparison of the gene-editing efficiency of hfCas12Max with LbCas12a, Ultra AsCas12a, ABR001, and Cas12i^{HFE} at TTR locus.

[0091] FIG. 10 shows that hfCas12Max mediated high-efficient and high-specific editing. FIG. 10A-FIG. 10B, Off-target efficiency of hfCas12Max, LbCas12a, and UltraAsCas12a at in-silico predicted off-target sites, determined by targeted deep sequencing. Sequences of on-target and predicted off-target sites are shown, PAM sequences are in blue and mismatched bases are in red.

[0092] FIG. 11 shows conserved cleavage sites of Cas12i. FIG. 11A, Sequence alignment of xCas12i, Cas12i1 and Cas12i2 shows that D650, D700, E875 and D1049 are conserved cleavage sites at RuvC domain. FIG. 11B, Introducing point mutations of D700A, D650A, E875A, or D1049A result in abolished activity of xCas12i.

[0093] FIG. 12 and FIG. 13 shows engineering for highly efficient dxCas12i-ABE. FIG. 12 and FIG. 13A, Engineering schematic of TadA8e. 1-dxCas12i. Four parts for engineering are indicated. FIG. 13B, TadA8e. 1-dxCas12i-v1.2 and v1.3 exhibit significantly increased A-to-G editing activity among various variants at KLKF4 site of genome. FIG. 13C, Increased A-to-G editing activity of TadA8e-dxCas12i-v2.2 by combining v1.2 and v1.3. FIG. 13D, Unchanged or even decreased editing activity from various dCas12-ABEs carrying different NLS at N-terminal. FIG. 13E, Increased A-to-G editing activity of TadA8e-dxCas12i-v4.3 by combining v2.2, changed-NLS linker and high-activity Tade8e.

[0094] FIG. 14 shows additional strategies for highly efficient dxCas12i-ABE. FIG. 14A, Schematics of different versions of dxCas12i ABEs. FIG. 14B, dxCas12i-ABE-N by TadA at the C-terminus of the dxCas12i slightly increased editing activity.

[0095] FIG. 15 shows comparison of editing frequencies induced by various dCas12-ABEs at different genomic target sites. FIG. 15A-FIG. 15B, Comparison of A-to-G editing frequencies induced by indicated TadA8e. 1-dxCas12i-v1.2, v2.2, and TadA8e. 1-dLbCas12a at PCSK9 and TTR genomic locus.

[0096] FIG. 16 shows characterization of dxCas12i-ABE in HEK293T cells. A-C, dCas12Max-ABE base editing of the target sites with TTN (FIG. 16A), ATN (FIG. 16B), and

CTN (FIG. 16C) PAMs. FIG. 16D, dCas12Max-ABE base editing product purity at each target site with TTN PAM in FIG. 16A.

[0097] FIG. 17 shows comparison of editing frequencies induced by various dCas12-CBEs at different genomic target sites. FIG. 17A-FIG. 17B, Comparison of C-to-T editing frequencies and product purity induced by indicated hA3A. 1-dxCas12i, v1.2, v2.2, and hA3A. 1-dLbCas12a at DYRK1A and SITE4 genomic locus. hA3A. 1 represents human APOBEC3A-W104A.

[0098] FIG. 18 shows that hfCas12Max mediated high editing efficiency in HEK293 cells. FIG. 18A-FIG. 18C, Unchanged viability and proliferation and increased indel activity of HEK293 cells following delivery of hfCas12Max RNPs with TTR or TRAC targeting gRNA at increasing concentration (n=1).

[0099] FIG. 19 shows that hfCas12Max mediated high editing efficiency in mouse blastocyst. FIG. 19A, Schematics of hfCas12Max gene editing in mouse blastocyst. hfCas12Max mRNA and Ttr targeting gRNA were injected into mouse zygotes, and the injected zygotes were cultured into blastocyst stage for genotyping analysis by targeted deep sequencing. FIG. 19B, Indel rates of hfCas12Max targeting Ttr. 3 and Ttr. 12 in mouse blastocyst (n=12).

[0100] FIG. 20 is a schematic illustrating an exemplary target dsDNA, an exemplary guide nucleic acid having one DR sequence 5' to one spacer sequence, and an exemplary Cas12i.

[0101] FIG. 21 shows the dsDNA cleavage activity of xCas12i when using various DR sequence variants.

[0102] FIG. 22 is a schematic illustrating the secondary structures of direct repeat sequences of the guide RNAs of the disclosure.

[0103] FIG. 23 shows another exemplary guide nucleic acid having three DR sequences and two spacer sequences, and each of the two spacer sequences is flanked by two DR sequences.

[0104] The figures herein are for illustrative purposes only and are not necessarily drawn to scale.

DETAILED DESCRIPTION OF THE EXAMPLE EMBODIMENTS

Overview

[0105] The disclosure provides Cas12i polypeptides with high spacer sequence-specific (on-target) dsDNA cleavage activity and/or low spacer sequence-independent (off-target) dsDNA cleavage activity based on parent or reference Cas12i polypeptides, and fusions and uses thereof.

[0106] In some embodiments, the parent or reference Cas12i polypeptide may be: (i) any one of SEQ ID NOs: 1-10 (Cas12i3 to Cas12i12) of the disclosure and Cas12i polypeptides (such as, Cas12i1 and Cas12i2) in PCT/CN2022/089074, PCT/CN2022/129376, PCT/CN2023/073420, WO2019090173A1, WO2019178033A1, WO2019222555A1, WO2020018142A1, WO2020180699A1, WO2020252378A1, WO2021007563A1, WO2021041569A1, WO2021046442A1, WO2021050534A1, WO2021113522A1, WO2021202800A1, WO2021243267A3, WO2021257730A3, WO2022040224A1, WO2022094313A1, WO2022094309A1, WO2022094329A1, WO2022094323A8, WO2022150608A1,

WO2022159585A1, WO2022162623A1, WO2022174099A3, WO2022192381A1, WO2022256619A3, WO2022256642A3, WO2023010084A3, WO2023018858A1, WO2023034475A1, WO2023039534A2, and WO2023039472A2, (ii) a naturally-occurring ortholog, paralog, or homolog of any one of (i); (iii) a Cas12i polypeptide having a sequence identity of at least about 40%, 50%, 60%, 70%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99% to any one of (i) and (ii); or (iv) any mutant or variant of (i) to (iii). The parent or reference Cas12i polypeptide may be a wild type or not.

Representative Cas12i Polypeptides and Characterization of Cas12i Polypeptide

[0107] In some aspects of the disclosure, the Cas12i polypeptide of the disclosure has or retains or has improved endonuclease activity against a target DNA for on-target DNA cleavage. Still for the purpose of on-target DNA cleavage, the Cas12i polypeptide of the disclosure may not only have on-target endonuclease activity but also substantially lack off-target endonuclease activity such that it can have specificity for a target DNA. On the other hand, the Cas12i polypeptide of the disclosure can be engineered to substantially lack endonuclease activity (either on-target or off-target) but retain its ability of complexing with a guide nucleic acid and thus being guided to a target DNA, so as to indirectly guide a functional domain associated with the Cas12i polypeptide to the target DNA. Therefore, the characterization of the Cas12i polypeptide of the disclosure is not limited to its ability of on-target DNA cleavage.

[0108] In some embodiments, the Cas12i polypeptide has spacer sequence-specific (on-target) dsDNA cleavage activity.

[0109] In some embodiments, the Cas12i polypeptide substantially retains the spacer sequence-specific (on-target) dsDNA cleavage activity of SEQ ID NO: 458 or SEQ ID NO: 1.

Increased On-Target Cleavage

[0110] As representatives of the disclosure, in an aspect, the disclosure provides an Cas12i polypeptide comprising an amino acid substitution at E336, V880, G883, D892, and/or M923 of SEQ ID NO: 458. The polypeptide as set forth in the amino acid sequence of SEQ ID NO: 458 (Cas12Max; xCas12i-N243R) serves as a parent or reference polypeptide, based on which the Cas12i polypeptide of the disclosure is engineered.

[0111] In some embodiments, the Cas12i polypeptide has an increased spacer sequence-specific (on-target) dsDNA cleavage activity compared to that of SEQ ID NO: 458 or SEQ ID NO: 1 when both are used in combination with a same guide nucleic acid, e.g., an increase by at least about 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 100%, 110%, 120%, 130%, 140%, 150%, 160%, 170%, 180%, 190%, 200%, 210%, 220%, 230%, 240%, 250%, 260%, 270%, 280%, 290%, 300%, or more.

[0112] In some embodiments, the Cas12i polypeptide has a sequence identity of at least about 80%, at least about 85%, at least about 90%, at least about 95%, or at least about 99% to SEQ ID NO: 458. In some embodiments, the Cas12i polypeptide has a sequence identity of at least about 80%, at least about 85%, at least about 90%, at least about 95%, or at least about 99% to any one of SEQ ID NOs: 1-10.

[0113] Typically, amino acid substitution is a substitution with a non-polar amino acid residue (such as, Glycine (Gly/G), Alanine (Ala/A), Valine (Val/V), Cysteine (Cys/C), Proline (Pro/P), Leucine (Leu/L), Isoleucine (Ile/I), Methionine (Met/M), Tryptophan (Trp/W), Phenylalanine (Phe/F), a polar amino acid residue (such as, Serine (Ser/S), Threonine (Thr/T), Tyrosine (Tyr/Y), Asparagine (Asn/N), Glutamine (Gln/Q)), a positively charged amino acid residue (such as, Lysine (Lys/K), Arginine (Arg/R), Histidine (His/H)), or a negatively charged amino acid residue (such as, Aspartic Acid (Asp/D), Glutamic Acid (Glu/E)).

[0114] In some embodiments, the amino acid substitution is a substitution with a positively charged amino acid residue (such as, Lysine (Lys/K), Arginine (Arg/R), Histidine (His/H)). In some embodiments, the amino acid substitution is a substitution with Arginine (Arg/R).

[0115] In some embodiments, the amino acid substitution is a substitution with a non-polar amino acid residue (such as, Glycine (Gly/G), Alanine (Ala/A), Valine (Val/V), Cysteine (Cys/C), Proline (Pro/P), Leucine (Leu/L), Isoleucine (Ile/I), Methionine (Met/M), Tryptophan (Trp/W), Phenylalanine (Phe/F)). In some embodiments, the amino acid substitution is a substitution with Alanine (Ala/A).

[0116] In some aspects, the disclosure provides Cas12i polypeptide comprises one indicated amino acid substitution based on the parent or reference Cas12i polypeptide.

[0117] In some embodiments, the Cas12i polypeptide comprises an amino acid substitution at one position selected from the group consisting of E336, V880, G883, D892, and M923 of SEQ ID NO: 458. In some embodiments, the amino acid substitution is a substitution with a positively charged amino acid residue (such as, Lysine (Lys/K), Arginine (Arg/R), Histidine (His/H)), and optionally a substitution with Arginine (Arg/R).

[0118] In some embodiments, the Cas12i polypeptide comprises an amino acid substitution E336R relative to SEQ ID NO:458. In some embodiments, the Cas12i polypeptide comprises the amino acid sequence of SEQ ID NO:467 (xCas12i-N243R+E336R), or an amino acid sequence having a sequence identity of at least about 80% (e.g., at least about 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, 99.1%, 99.2%, 99.3%, 99.4%, 99.5%, 99.6%, 99.7%, 99.8%, 99.9%, or 100%) to the amino acid sequence of SEQ ID NO: 467.

[0119] In some other aspects, the disclosure provides Cas12i polypeptide comprises one indicated amino acid substitution based on the parent or reference Cas12i polypeptide.

[0120] In some embodiments, the Cas12i polypeptide comprises one amino acid substitution selected from the group consisting of V880R, G883R, D892R, and M923R relative to SEQ ID NO: 458.

[0121] In some aspects, the disclosure provides Cas12i polypeptide comprises two indicated amino acid substitutions based on the parent or reference Cas12i polypeptide.

[0122] In some embodiments, the Cas12i polypeptide comprises two amino acid substitutions at any two positions

of E336, V880, G883, D892, and M923 of SEQ ID NO: 458. In some embodiments, each of the two amino acid substitutions is independently a substitution with a positively charged amino acid residue (such as, Lysine (Lys/K), Arginine (Arg/R), Histidine (His/H)). In some embodiments, each of the two amino acid substitutions is independently a substitution with Arginine (Arg/R).

[0123] In some embodiments, the Cas12i polypeptide comprises amino acid substitutions E336R and one amino acid substitution selected from the group consisting of V880R, G883R, D892R, and M923R relative to SEQ ID NO: 458.

[0124] In some embodiments, the Cas12i polypeptide comprises amino acid substitutions E336R and D892R relative to SEQ ID NO: 458. In some embodiments, the Cas12i polypeptide comprises the amino acid sequence of SEQ ID NO: 459 (hfCas12Max; xCas12i-N243R+E336R+D892R), or an amino acid sequence having a sequence identity of at least about 80% (e.g., at least about 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, 99.1%, 99.2%, 99.3%, 99.4%, 99.5%, 99.6%, 99.7%, 99.8%, 99.9%, or 100%) to the amino acid sequence of SEQ ID NO: 459.

[0125] In some aspects, the disclosure provides Cas12i polypeptide further comprise an indicated amino acid substitutions based on the parent or reference Cas12i polypeptide or the Cas12i polypeptide, e.g., for increased spacer-sequence specific dsDNA cleavage activity.

[0126] In some embodiments, the Cas12i polypeptide further comprises an additional amino acid substitution at a position selected from the group consisting of K109, L112, D125, I27, F144, L147, A148, L151, L157, V195, Y226, F252, I258, M293, W305, A308, I309, S312, A314, D315, V316, A318, L324, I327, A348, L352, Y365, L372, L376, L379, L383, I405, L424, I427, A436, F439, A443, V447, A457, H458, P459, T460, S463, S814, F859, A864, H867, Y977, S1031, A1053, and F1068 of SEQ ID NO: 458. In some embodiments, the additional amino acid substitution is a substitution with a positively charged amino acid residue (such as, Lysine (Lys/K), Arginine (Arg/R), Histidine (His/H)), and optionally a substitution with Arginine (Arg/R).

Decreased Off-Target Cleavage

[0127] In some embodiments, the Cas12i polypeptide substantially lacks spacer sequence-independent (off-target) dsDNA cleavage activity.

[0128] In some embodiments, the Cas12i polypeptide substantially lacks the spacer sequence-independent (off-target) dsDNA cleavage activity of SEQ ID NO: 458 or SEQ ID NO: 1.

[0129] In some embodiments, the Cas12i polypeptide has a decreased spacer sequence-independent (off-target) dsDNA cleavage activity compared to that of SEQ ID NO: 458 or SEQ ID NO: 1 when both are used in combination with a same guide nucleic acid, e.g., a decrease by at least about 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, or 100%.

Endonuclease Deficient (Dead) Cas12i Polypeptide

[0130] In some aspects, the disclosure provides a Cas12i polypeptide that is endonuclease deficient, which means the Cas12i polypeptide is substantially incapable of functioning

as an endonuclease to cleave (either double strands or a single strand of) a dsDNA or a ssDNA, either against a target DNA or against a non-target DNA (For convenience of experiment design, performance, and evaluation, the defect of endonuclease activity is usually indicated by substantial loss of spacer sequence-specific dsDNA cleavage activity against a target DNA). Such a Cas12i polypeptide is named as “dead Cas12i (dCas12i)” and may be generated based on the parent or reference Cas12i polypeptide, for example, by mutating one or more functional domains of the parent or reference Cas12i polypeptide that is/are responsible for endonuclease activity.

[0131] In some embodiments, the Cas12i polypeptide is further engineered to substantially lack spacer sequence-specific (on-target) dsDNA cleavage activity.

[0132] In some embodiments, the Cas12i polypeptide substantially lacks the spacer sequence-specific (on-target) dsDNA cleavage activity of SEQ ID NO: 458 or SEQ ID NO: 1.

[0133] In some embodiments, the Cas12i polypeptide has a decreased spacer sequence-specific (on-target) dsDNA cleavage activity compared to that of SEQ ID NO: 458 or SEQ ID NO: 1 when both used in combination with a same guide nucleic acid, e.g., a decrease by at least about 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, or 100%.

[0134] In some embodiments, the Cas12i polypeptide comprise a further amino acid substitution at a position selected from the group consisting of D650, D700, E875, and D1049 of SEQ ID NO: 458. In some embodiments, the amino acid substitution is a substitution with a non-polar amino acid residue (such as, Glycine (Gly/G), Alanine (Ala/A), Valine (Val/V), Cysteine (Cys/C), Proline (Pro/P), Leucine (Leu/L), Isoleucine (Ile/I), Methionine (Met/M), Tryptophan (Trp/W), Phenylalanine (Phe/F)). In some embodiments, the amino acid substitution is a substitution with Alanine (Ala/A).

[0135] In some embodiments, the Cas12i polypeptide comprises amino acid substitutions E336R and D1049A relative to SEQ ID NO: 458. In some embodiments, the Cas12i polypeptide comprises the amino acid sequence of SEQ ID NO: 466 (xCas12i-N243R+E336R+D1049A), or an amino acid sequence having a sequence identity of at least about 80% (e.g., at least about 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, 99.1%, 99.2%, 99.3%, 99.4%, 99.5%, 99.6%, 99.7%, 99.8%, 99.9%, or 100%) to the amino acid sequence of SEQ ID NO: 466.

Cas12i Nickase

[0136] In some aspects, the disclosure provides a Cas12i polypeptide that is not completely endonuclease deficient but the endonuclease activity is not against the double strand of a dsDNA but against one strand (the sense or nonsense strand; or the target or nontarget strand) of a dsDNA or a ssDNA, which means the Cas12i polypeptide is substantially incapable of functioning as a dsDNA endonuclease to cleave double strands of a dsDNA, either against a target DNA or against a non-target DNA, but is substantially capable of functioning as a ssDNA endonuclease to cleave a ssDNA or “nick” one strand of a dsDNA. Such a Cas12i polypeptide is named as “nickase” and may be generated based on the parent or reference Cas12i polypeptide, for example, by mutating one or more functional domains of the

parent or reference Cas12i polypeptide that is/are responsible for endonuclease activity.

[0137] In some embodiments, the Cas12i polypeptide is further engineered to be a nickase.

[0138] In some embodiments, the Cas12i polypeptide comprise an amino acid substitution at a position selected from the group consisting of W896, S924, and S925 of SEQ ID NO: 458.

[0139] In some embodiments, the Cas12i polypeptide comprise an amino acid substitution selected from the group consisting of W896R, W896P, W896K, S924R, S924F, S924D, S924E, S924H, S925R, and S925T relative to SEQ ID NO: 458.

Fusion Protein

[0140] In some aspects, the disclosure provides a fusion protein comprising the Cas12i polypeptide and a functional domain. In some embodiments, the functional domain is a heterologous functional domain. Such a function protein may also be regarded as a Cas12i polypeptide further comprising a functional domain fused to the Cas12i polypeptide.

[0141] In some embodiments, the Cas12i polypeptide further comprises a functional domain fused to the Cas12i polypeptide.

[0142] In some embodiments, the functional domain is selected from the group consisting of a nuclear localization signal (NLS), a nuclear export signal (NES), a base editing domain, for example, a deaminase or a catalytic domain thereof, a base excising domain, an uracil glycosylase inhibitor (UGI) or a catalytic domain thereof, an uracil glycosylase (UNG) or a catalytic domain thereof, a methylpurine glycosylase (MPG) or a catalytic domain thereof, a methylase or a catalytic domain thereof, a demethylase or a catalytic domain thereof, an transcription activating domain (e.g., VP64 or VPR), an transcription inhibiting domain (e.g., KRAB moiety or SID moiety), a reverse transcriptase or a catalytic domain thereof, an exonuclease (e.g., T5E (SEQ ID NO: 449)) or a catalytic domain thereof, a destabilized domain (e.g., destabilized domains (DD) of *E. coli* dihydrofolate reductase (ecDHFR)), a histone residue modification domain, a nuclease catalytic domain (e.g., FokI), a transcription modification factor, a light gating factor, a chemical inducible factor, a chromatin visualization factor, a targeting polypeptide for providing binding to a cell surface portion on a target cell or a target cell type, a reporter (e.g., fluorescent) polypeptide or a detection label (e.g., GST, HRP, CAT, GFP, HcRed, DsRed, CFP, YFP, BFP), a localization signal, a polypeptide targeting moiety, a DNA binding domain (e.g., MBP, Lex A DBD, Gal4 DBD), an epitope tag (e.g., His, myc, V5, FLAG, HA, VSV-G, Trx, etc), a transcription release factor, an HDAC, a moiety having ssRNA cleavage activity, a moiety having dsRNA cleavage activity, a moiety having ssDNA cleavage activity, a DNA or RNA ligase, a functional domain exhibiting activity to modify a target DNA, selected from the group consisting of: methyltransferase activity, DNA repair activity, DNA damage activity, dismutase activity, alkylation activity, dealkylation activity, depurination activity, oxidation activity, deoxidation activity, pyrimidine dimer forming activity, integrase activity, transposase activity, recombinase activity, polymerase activity, ligase activity, helicase activity, photolyase activity, glycosylase activity, acetyl transferase activity,

deacetylase activity, kinase activity, phosphatase activity, ubiquitin ligase activity, deubiquitination activity, adenylation activity, deadenylation activity, SUMOylation activity, deSUMOylation activity, ribosylation activity, deribosylation activity, myristoylation activity, demyristoylation activity, glycosylation activity (e.g., from O-GlcNAc transferase), deglycosylation activity, and a catalytic domain thereof, and a functional fragment (e.g., a functional truncation) thereof, and any combination thereof.

[0143] In some embodiments, the NLS comprises or is SV40 NLS (SEQ ID NO: 444), bpSV40 NLS (BP NLS, bpNLS, SEQ ID NO: 443 or 462), or NP NLS (*Xenopus laevis* Nucleoplasmin NLS, nucleoplasmin NLS, SEQ ID NO: 445).

Base Editing

[0144] In some embodiments, the base editing domain is capable of substituting a base of a nucleotide with a different base.

[0145] In some embodiments, the base editing domain is capable of deaminating a base of a nucleotide.

[0146] In some embodiments, the base editing domain comprises a deaminase domain capable of deaminating a base (e.g., an adenine, a guanine, a cytosine, a thymine, an uracil) of a nucleotide. In some embodiments, the deaminase domain is capable of deaminating an adenine (A) to a hypoxanthine (I). In some embodiments, the deamination of the adenine to the hypoxanthine converts the adenosine (A) or deoxyadenosine (dA) containing the adenine to a guanosine (G) or deoxyguanosine (dG). In some embodiments, the deaminase domain is capable of deaminating a cytosine (C) to an uracil (U). In some embodiments, the deamination of the cytosine to the uracil converts the cytidine (C) or deoxycytidine (dC) containing the cytosine to a uridine (U) or a deoxythymidine (dT).

[0147] In some embodiments, the base editing domain is capable of excising a base (e.g., an adenine, a guanine, a cytosine, a thymine, an uracil) of a nucleotide.

[0148] In some embodiments, the base editing domain comprises a base excising domain capable of excising a base of a nucleotide.

[0149] In some embodiments, the base editing domain comprises a deaminase domain and a base excising domain.

[0150] In some embodiments, the deaminase domain is tRNA adenosine deaminase (TadA), or the deaminase domain thereof, or a functional variant or fragment thereof, e.g., TadA8e (SEQ ID NO: 3), TadA8.17, TadA8.20, TadA9, TadA8E^{V106W}, TadA8E^{V106W+D108Q}, TadA-CDa, TadA-CDb, TadA-CDc, TadA-CDd, TadA-CDe, TadA-dual, T_{AD}AC-1.2, T_{AD}AC-1.14, T_{AD}AC-1.17, T_{AD}AC-1.19, T_{AD}AC-2.5, T_{AD}AC-2.6, T_{AD}AC-2.9, T_{AD}AC-2.19, T_{AD}AC-2.23, TadA8e-N46L, TadA8e-N46P.

[0151] In some embodiments, the deaminase domain is an apolipoprotein B mRNA-editing complex (APOBEC) family deaminase, an activation induced deaminase (AID), a cytidine deaminase 1 from *Petromyzon marinus* (pmCDA1), or the deaminase domain thereof, or a functional variant or fragment thereof, e.g., APOBEC1, APOBEC2, APOBEC3A, APOBEC3B, APOBEC3C, APOBEC3D, APOBEC3F, APOBEC3G, APOBEC3H.

[0152] In some embodiments, the deaminase or catalytic domain thereof is an adenine deaminase (e.g., TadA, such as, TadA8e, TadA8.17, TadA8.20, TadA9) or a catalytic domain

thereof, for example, TadA8e-V106W (SEQ ID NO: 439), TadA8e-W106V (SEQ ID NO: 461).

[0153] In some embodiments, the deaminase or catalytic domain thereof is a cytidine deaminase (e.g., APOBEC, such as, APOBEC3, for example, APOBEC3A, APOBEC3B, APOBEC3C; DddA) or a catalytic domain thereof, for example, hAPOBEC3-W104A (SEQ ID NO: 440).

[0154] In some embodiments, the UGI is human UGI domain (such as, SEQ ID NO: 441).

[0155] In some embodiments, the Cas12i polypeptide comprises amino acid substitutions E336R and D1049A relative to SEQ ID NO: 458, and a base editing domain, for example, a deaminase or a catalytic domain thereof.

[0156] In some embodiments, the Cas12i polypeptide comprises the amino acid sequence of SEQ ID NO: 463 (dCas12Max-ABE), or an amino acid sequence having a sequence identity of at least about 80% (e.g., at least about 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, 99.1%, 99.2%, 99.3%, 99.4%, 99.5%, 99.6%, 99.7%, 99.8%, 99.9%, or 100%) to the amino acid sequence of SEQ ID NO: 463.

[0157] In some embodiments, the Cas12i polypeptide comprises the amino acid sequence of SEQ ID NO: 464 (dCas12Max-CBE), or an amino acid sequence having a sequence identity of at least about 80% (e.g., at least about 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, 99.1%, 99.2%, 99.3%, 99.4%, 99.5%, 99.6%, 99.7%, 99.8%, 99.9%, or 100%) to the amino acid sequence of SEQ ID NO: 464.

[0158] In some embodiments, the functional domain comprises a reverse transcriptase (RT) or a catalytic domain thereof. In some embodiments, the guide nucleic acid further comprises or is used in combination with a reverse transcription donor RNA (RT donor RNA) comprising a primer binding site (PBS) and a template sequence. For details of prime editing with Class 2, Type V Cas proteins, references is made to WO2022256440A3, which is incorporated herein by reference in its entirety.

System

[0159] The Cas12i polypeptide of the disclosure may be used in combination with and guided by a guide nucleic acid to a target DNA to function on the target DNA. In another aspect, the disclosure provides a system comprising:

- [0160]** (1) the Cas12i polypeptide of the disclosure or a polynucleotide (e.g., a DNA, an RNA) encoding the Cas12i polypeptide; and
- [0161]** (2) a guide nucleic acid or a polynucleotide (e.g., a DNA or an RNA) encoding the guide nucleic acid, the guide nucleic acid comprising:
 - [0162]** (i) a direct repeat (DR) sequence capable of forming a complex with the Cas12i polypeptide; and
 - [0163]** (ii) a spacer sequence capable of hybridizing to a target sequence of a target DNA, thereby guiding the complex to the target DNA.

[0164] In some embodiments, the system is a non-naturally occurring or engineered system.

[0165] In some embodiments, the system is a complex comprising the Cas12i polypeptide complexed with the guide nucleic acid. In some embodiments, the complex further comprises the target DNA hybridized with the target sequence.

[0166] In another aspect, the disclosure provides a guide nucleic acid comprising:

- [0167]** (1) a direct repeat (DR) sequence capable of forming a complex with the Cas12i polypeptide of the disclosure, and
- [0168]** (2) a spacer sequence capable of hybridizing to a target sequence of a target DNA, thereby guiding the complex to the target DNA.

[0169] In some embodiments, the guide nucleic acid is a guide RNA (gRNA). In some embodiments, the guide nucleic acid comprises a crRNA. In some embodiments, the guide nucleic acid does not comprise a tracrRNA.

[0170] In some embodiments, the direct repeat sequence is 5' to the spacer sequence.

Design of Protospacer Sequence/Target Sequence; Target Site

[0171] For the purpose of the disclosure, in some embodiments, the protospacer sequence or target sequence is located such that the target DNA is specifically modified by the Cas12i polypeptide.

[0172] To facilitate the evaluation of selected protospacer sequences or target sequence and designed guide sequences in mouse models, in some embodiments, the protospacer sequence or target sequence is located such that a mouse target DNA is specifically modified by the Cas12i polypeptide. In some embodiments, the protospacer sequence or target sequence is located such that both a human target DNA and a mouse target DNA are specifically modified by the Cas12i polypeptide. That is, the protospacer sequence or target sequence is selected to be cross-reactive to both human and mouse species.

[0173] In some embodiments, the protospacer sequence is a stretch of contiguous nucleotides identified from the nontarget strand of the target DNA by identifying the stretch of contiguous nucleotides immediately 3' to the PAM on the nontarget strand. In some embodiments, the PAM is 5'-TTN, 5'-TTN, or 5'-GCC, wherein N is A, T, G, or C. In some embodiments, the PAM is 5'-TTN, wherein N is A, T, G, or C. The protospacer sequence is the reversely complementary sequence of the target sequence.

[0174] In some embodiments, the protospacer sequence is a stretch of about or at least about 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, or more contiguous nucleotides of the target DNA, or a stretch of contiguous nucleotides of the target DNA in a numerical range between any two of the preceding values, e.g., a stretch of from about 16 to about 50, or from about 17 to about 22 contiguous nucleotides. In some embodiments, the protospacer sequence is a stretch of about 20 contiguous nucleotides of the target DNA.

[0175] In some embodiments, the protospacer sequence comprises about or at least about 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, or more contiguous nucleotides of the target DNA, or contiguous nucleotides in a numerical range between any two of the preceding values, e.g., from about 16 to about 50, or from about 17 to about 22 contiguous nucleotides of the target DNA. In some embodiments, the protospacer sequence comprises about 20 contiguous nucleotides of the target DNA.

[0176] In some embodiments, the target sequence is a stretch of contiguous nucleotides identified from the target strand of the target DNA. The target sequence is the reversely complementary sequence of the protospacer sequence.

[0177] In some embodiments, the target sequence is a stretch of about or at least about 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, or more contiguous nucleotides on the target strand of the target DNA, or a stretch of contiguous nucleotides on the target strand of the target DNA in a numerical range between any two of the preceding values, e.g., a stretch of from about 16 to about 50, or from about 17 to about 22 contiguous nucleotides. In some embodiments, the target sequence is a stretch of about 20 contiguous nucleotides on the target strand of the target DNA.

[0178] In some embodiments, the target sequence comprises about or at least about 16 contiguous nucleotides of the target DNA, e.g., about or at least about 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, or more contiguous nucleotides of the target DNA, or in a numerical range between any two of the preceding values, e.g., from about 16 to about 50, or from about 17 to about 22 contiguous nucleotides of the target DNA. In some embodiments, the target sequence comprises about 20 contiguous nucleotides of the target DNA.

[0179] In some embodiments, the target sequence comprises about or at least about 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, or more contiguous nucleotides on the target strand of the target DNA, or contiguous nucleotides in a numerical range between any two of the preceding values, e.g., from about 16 to about 50, or from about 17 to about 22 contiguous nucleotides on the target strand of the target DNA. In some embodiments, the target sequence comprises about 20 contiguous nucleotides on the target strand of the target DNA.

[0180] In some embodiments, the reversely complementary sequence of the target sequence is immediately 3' to a protospacer adjacent motif (PAM); optionally, wherein the PAM is 5'-TN, 5'-TTN, or 5'-GCC, wherein N is A, T, G, or C, wherein N is A, T, G, or C.

[0181] In some embodiments, the nontarget strand is the sense strand of the target DNA.

[0182] In some embodiments, the nontarget strand is the antisense strand of the target DNA.

[0183] In some embodiments, the target strand is the sense strand of the target DNA.

[0184] In some embodiments, the target strand is the antisense strand of the target DNA.

[0185] In some embodiments, the protospacer sequence or target sequence is located within Exon 1 of the target DNA.

[0186] In some embodiments, the protospacer sequence or target sequence is located within about 50, 100, 150, 200, 250, 300, or more 5' end nucleotides of Exon 1 of the target DNA.

[0187] In some embodiments, the target DNA comprises a pathogenic mutation.

[0188] In some embodiments, the target DNA comprises a premature stop codon (e.g., TAG).

[0189] In some embodiments, the target DNA is a dsDNA, such as, a eukaryotic dsDNA, e.g., a gene in a eukaryotic cell.

[0190] In some embodiments, the target DNA is human target DNA, non-human primate target DNA, or mouse target DNA.

[0191] In some embodiments, the target DNA is in a eukaryotic cell, for example, a human cell, a non-human primate cell, or a mouse cell.

Design of Guide Sequence According to Protospacer/Target Sequence

[0192] In some embodiments, the spacer sequence is about or at least about 16 nucleotides in length, e.g., about or at least about 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, or more nucleotides in length, or in a length of a numerical range between any two of the preceding values, e.g., in a length of from about 16 to about 50 nucleotides, or from about 17 to about 22 nucleotides. In some embodiments, the spacer sequence is about 20 nucleotides in length.

[0193] In some embodiments, (1) the guide sequence is at least about 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, or 100% (fully), optionally about 100% (fully), reversely complementary to the target sequence; (2) the guide sequence contains no more than 5, 4, 3, 2, or 1 mismatch or contains no mismatch with the target sequence; or (3) the guide sequence comprises no mismatch with the target sequence in the first 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, or 70 nucleotides at the 5' end of the guide sequence. In some embodiments, the guide sequence is about 100% (fully), reversely complementary to the target sequence.

Selection of Protospacer/Target/Guide Sequence; Effect of System

[0194] In some embodiments, the protospacer sequence, the target sequence, or the guide sequence is selected such that the target DNA is modified by the system of the disclosure. In some embodiments, the modification decreases or eliminates the transcription of the target DNA and/or translation of a transcript (e.g., mRNA) of the target DNA.

[0195] In some embodiments, the level of the transcript (e.g., mRNA) of the target DNA is decreased in a cell model (e.g., HEK293T cell model) or an animal model (e.g., a mouse model, a non-human primate model) by at least about 10%, about 15%, about 20%, about 25%, about 30%, about 35%, about 40%, about 45%, about 50%, about 55%, about 60%, about 65%, about 70%, about 75%, about 80%, about 85%, or more, upon administration of the system of the disclosure to the cell model or the animal model, compared to the level of the transcript (e.g., mRNA) of the target DNA in the same cell model or animal model that does not receive the administration.

[0196] In some embodiments, the level of the transcript (e.g., mRNA) of the target DNA is increased in a cell model (e.g., HEK293T cell model) or an animal model (e.g., a mouse model, a non-human primate model) by at least about 10%, about 15%, about 20%, about 25%, about 30%, about 35%, about 40%, about 45%, about 50%, about 55%, about 60%, about 65%, about 70%, about 75%, about 80%, about 85%, or more, upon administration of the system of the disclosure to the cell model or the animal model, compared to the level of the transcript (e.g., mRNA) of the target DNA in the same cell model or animal model that does not receive the administration.

[0197] In some embodiments, the level of the expression product (e.g., protein) of the target DNA is decreased in a cell model (e.g., HEK293T cell model) or an animal model (e.g., a mouse model, a non-human primate model) by at least about 10%, about 15%, about 20%, about 25%, about 30%, about 35%, about 40%, about 45%, about 50%, about 55%, about 60%, about 65%, about 70%, about 75%, about 80%, about 85%, or more, upon administration of the system of the disclosure to the cell or the animal model, compared to the level of the expression product (e.g., protein) of the target DNA in the same cell model or animal model that does not receive the administration.

[0198] In some embodiments, the level of the expression product (e.g., protein) of the target DNA is increased in a cell model (e.g., HEK293T cell model) or an animal model (e.g., a mouse model, a non-human primate model) by at least about 10%, about 15%, about 20%, about 25%, about 30%, about 35%, about 40%, about 45%, about 50%, about 55%, about 60%, about 65%, about 70%, about 75%, about 80%, about 85%, or more, upon administration of the system of the disclosure to the cell or the animal model, compared to the level of the expression product (e.g., protein) of the target DNA in the same cell model or animal model that does not receive the administration. In some embodiments, the expression product is a functional mutant of the expression product of the target DNA.

Overall Structure of Guide Nucleic Acid

[0199] In some embodiments, the guide nucleic acid is a single molecule.

[0200] In some embodiments, the guide nucleic acid comprises one spacer sequence capable of hybridizing to one target sequence.

[0201] In some embodiments, the guide nucleic acid comprises a plurality (e.g., 2, 3, 4, 5 or more) of the spacer sequences capable of hybridizing to a plurality of the target sequences, respectively.

[0202] In some embodiments, the guide nucleic acid comprises, from 5' to 3', the direct repeat sequence, the spacer sequence, the direct repeat sequence, the spacer sequence, and the direct repeat sequence.

[0203] In some embodiments, the guide nucleic acid comprises one scaffold sequence and one guide sequence.

[0204] In some embodiments, the guide nucleic acid comprises one scaffold sequence 5' to one guide sequence. In some embodiments, the guide nucleic acid comprises one scaffold sequence 3' to one guide sequence.

[0205] In some embodiments, the guide nucleic acid comprises one or more scaffold sequence and/or one or more guide sequence, provided that the guide nucleic acid does not comprise one scaffold sequence and one guide sequence.

[0206] In some embodiments, the guide nucleic acid comprises, from 5' to 3', one scaffold sequence, one guide sequence, and one scaffold sequence, wherein scaffold sequences are the same or different.

[0207] In some embodiments, the guide nucleic acid comprises, from 5' to 3', one guide sequence, one scaffold sequence, and one guide sequence, wherein guide sequences are the same or different.

[0208] In some embodiments, the guide nucleic acid comprises, from 5' to 3', one scaffold sequence, one guide sequence, one scaffold sequence, and one guide sequence, wherein scaffold sequences are the same or different, and wherein guide sequences are the same or different.

[0209] In some embodiments, the guide nucleic acid comprises, from 5' to 3', one guide sequence, one scaffold sequence, one guide sequence, and one scaffold sequence, wherein scaffold sequences are the same or different, and wherein guide sequences are the same or different.

[0210] In some embodiments, the guide nucleic acid comprises, from 5' to 3', one scaffold sequence, one guide sequence, one scaffold sequence, one guide sequence, and one scaffold sequence, wherein scaffold sequences are the same or different, and wherein guide sequences are the same or different.

[0211] In some embodiments, the guide nucleic acid comprises, from 5' to 3', one guide sequence, one scaffold sequence, one guide sequence, one scaffold sequence, and one guide sequence, wherein scaffold sequences are the same or different, and wherein guide sequences are the same or different.

[0212] In some embodiments, the guide nucleic acid comprises, from 5' to 3', one scaffold sequence, one guide sequence, one scaffold sequence, one guide sequence, one scaffold sequence, and one guide sequence, wherein scaffold sequences are the same or different, and wherein guide sequences are the same or different.

[0213] In some embodiments, the guide nucleic acid comprises, from 5' to 3', one guide sequence, one scaffold sequence, one guide sequence, one scaffold sequence, one guide sequence, and one scaffold sequence, wherein scaffold sequences are the same or different, and wherein guide sequences are the same or different.

[0214] In some embodiments, the guide nucleic acid comprises a linker or no linker between any adjacent scaffold sequence and guide sequence. In some embodiments, the guide nucleic acid comprises no linker between any adjacent scaffold sequence and guide sequence.

Multiple Guide Nucleic Acid

[0215] The system of the disclosure may comprise or encode one guide nucleic acid or comprise or encode multiple (e.g., 2, 3, 4, or more) guide nucleic acids, e.g., for the purpose of improving the editing efficiency of the system on target DNA.

[0216] In some embodiments, the system further comprises one or more additional guide nucleic acids, or the first polynucleotide sequence further comprises one or more additional sequences encoding one or more additional guide nucleic acids, each of the additional guide nucleic acids comprising:

[0217] (1) an additional scaffold sequence capable of forming a complex with the Cas12i polypeptide, and

[0218] (2) an additional guide sequence capable of hybridizing to an additional target sequence on a target

strand of the target DNA or an additional target sequence on the transcript thereof, thereby guiding the complex to the target DNA or the transcript.

[0219] In some embodiments, the additional protospacer sequence is on the same strand as the protospacer sequence.

[0220] In some embodiments, the additional protospacer sequence is on the different strand from the protospacer sequence.

[0221] In some embodiments, the additional protospacer sequence is the same or different from the protospacer sequence.

[0222] In some embodiments, the additional target sequence is the same or different from the target sequence.

[0223] In some embodiments, the additional guide sequence is the same or different from the guide sequence.

[0224] In some embodiments, the additional scaffold sequence is the same or different from the scaffold sequence. In some embodiments wherein the system comprises the same Cas12i polypeptide and multiple guide nucleic acids, the scaffold sequences of the multiple guide nucleic acids may be the same or different (e.g., different by no more than 5, 4, 3, 2, or 1 nucleotide) to be compatible to the same Cas12i polypeptide. In some embodiments wherein that the system comprises different Cas12i polypeptides and multiple guide nucleic acids, the scaffold sequences of the multiple guide nucleic acids may be different to be compatible to the different Cas12i polypeptides.

[0225] In some embodiments, the additional guide nucleic acid and the guide nucleic acid are operably linked to or under the regulation of the same regulatory element (e.g., promoter) or separate regulatory elements (e.g., promoters).

Nature and Modification of Guide Nucleic Acid

[0226] In some embodiments, the guide nucleic acid (e.g., the guide nucleic acid, the additional guide nucleic acid) is an RNA. In some embodiments, the guide nucleic acid is an unmodified guide RNA. In some embodiments, the guide nucleic acid is a modified guide RNA. In some embodiments, the guide nucleic acid comprises a modification. In some embodiments, the guide nucleic acid is a modified RNA containing a modified ribonucleotide. In some embodiments, the guide nucleic acid is a modified RNA containing a modified deoxyribonucleotide. In some embodiments, the guide nucleic acid is a modified RNA containing a modified deoxyribonucleotide. In some embodiments, the guide nucleic acid comprises a modified or unmodified deoxyribonucleotide and a modified or unmodified ribonucleotide.

Scaffold Sequence

[0227] For the purpose of the disclosure, the scaffold sequence is compatible with the Cas12i polypeptide of the disclosure and is capable of complexing with the Cas12i polypeptide. The scaffold sequence may be a naturally occurring scaffold sequence identified along with the Cas12i polypeptide, or a variant thereof maintaining the ability to complex with the Cas12i polypeptide. Generally, the ability to complex with the Cas12i polypeptide is maintained as long as the secondary structure of the variant is substantially identical to the secondary structure of the naturally occurring scaffold sequence. A nucleotide deletion, insertion, or substitution in the primary sequence of the scaffold sequence may not necessarily change the secondary structure of the scaffold sequence (e.g., the relative locations and/or sizes of

the stems, bulges, and loops of the scaffold sequence do not significantly deviate from that of the original stems, bulges, and loops). For example, the nucleotide deletion, insertion, or substitution may be in a bulge or loop region of the scaffold sequence so that the overall symmetry of the bulge and hence the secondary structure remains largely the same. The nucleotide deletion, insertion, or substitution may also be in the stems of the scaffold sequence so that the lengths of the stems do not significantly deviate from that of the original stems (e.g., adding or deleting one base pair in each of two stems correspond to 4 total base changes).

[0228] In some embodiments, the direct repeat sequence or the additional scaffold sequence has substantially the same secondary structure as the secondary structure of any one of SEQ ID NOs: 11 and 451-457.

[0229] In some embodiments, the direct repeat sequence or the additional scaffold sequence:

[0230] (i) comprises the polynucleotide sequence of any one of SEQ ID NOs: 11 and 451-457; or

[0231] (ii) comprises a polynucleotide sequence having a sequence identity of at least about 80% (e.g., at least about 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, 99.1%, 99.2%, 99.3%, 99.4%, 99.5%, 99.6%, 99.7%, 99.8%, 99.9%, or 100%) to any one of SEQ ID NOs: 11 and 451-457.

[0232] In some embodiments, the scaffold sequence or the additional scaffold sequence comprises a sequence having a sequence identity of at least about 80% (e.g., at least about 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100%) to the sequence of any one of SEQ ID NOs: 11 and 451-457; or a sequence having at most 1, 2, 3, 4, 5, 6, 7, 8, 9, or 10 nucleotide differences, whether consecutive or not, compared to the sequence of any one of SEQ ID NOs: 11 and 451-457.

[0233] In some embodiments, the scaffold sequence or the additional scaffold sequence comprises the sequence of SEQ ID NO: 452.

Regulation of Guide Nucleic Acid

[0234] In some embodiments, the polynucleotide encoding the guide nucleic acid is a DNA, a RNA, or a DNA/RNA mixture. By “DNA/RNA mixture” it refers to a nucleic acid comprising both one or more modified or unmodified ribonucleotides and one or more modified or unmodified deoxyribonucleotides, whether consecutive or not. However, by “DNA” or “RNA” it may also refer to a DNA containing one or more modified or unmodified ribonucleotides, whether consecutive or not, or an RNA containing one or more modified or unmodified deoxyribonucleotides, whether consecutive or not.

[0235] In some embodiments, the guide nucleic acid is operably linked to or under the regulation of a promoter.

[0236] In some embodiments, the promoter is a ubiquitous, tissue-specific, cell-type specific, constitutive, or inducible promoter.

[0237] Suitable promoters are known in the art and include, for example, a Cbh promoter, a Cba promoter, a pol I promoter, a pol II promoter, a pol III promoter, a T7 promoter, a U6 promoter, a H1 promoter, a retroviral Rous sarcoma virus LTR promoter, a cytomegalovirus (CMV) promoter, a SV40 promoter, a dihydrofolate reductase promoter, a β -actin promoter, an elongation factor 1 α short (EFS) promoter, a Bglucuronidase (GUSB) promoter, a cytomegalovirus (CMV) immediate-early (Ie) enhancer and/

or promoter, a chicken β -actin (CBA) promoter or derivative thereof such as a CAG promoter, CB promoter, a (human) elongation factor 1 α -subunit (EF1 α) promoter, a ubiquitin C (UBC) promoter, a prion promoter, a neuron-specific enolase (NSE), a neurofilament light (NFL) promoter, a neurofilament heavy (NFH) promoter, a platelet-derived growth factor (PDGF) promoter, a platelet-derived growth factor B-chain (PDGF- β) promoter, a synapsin (Syn) promoter, a synapsin 1 (Syn1) promoter, a methyl-CpG binding protein 2 (MeCP2) promoter, a Ca2+/calmodulin-dependent protein kinase II (CaMKII) promoter, a metabotropic glutamate receptor 2 (mGluR2) promoter, a neurofilament light (NFL) promoter, a neurofilament heavy (NFH) promoter, a β -globin minigene n β 2 promoter, a preproenkephalin (PPE) promoter, an enkephalin (Enk) promoter, an excitatory amino acid transporter 2 (EAAT2) promoter, a glial fibrillary acidic protein (GFAP) promoter, and a myelin basic protein (MBP) promoter.

Regulation of Cas12i Polypeptide

[0238] In some embodiments, the polynucleotide encoding the Cas12i polypeptide is a DNA, a RNA, or a DNA/RNA mixture. By “DNA/RNA mixture” it refers to a nucleic acid comprising both one or more modified or unmodified ribonucleotides and one or more modified or unmodified deoxyribonucleotides, whether consecutive or not. However, by “DNA” or “RNA” it may also refer to a DNA containing one or more modified or unmodified ribonucleotides, whether consecutive or not, or an RNA containing one or more modified or unmodified deoxyribonucleotides, whether consecutive or not.

[0239] In some embodiments, the polynucleotide encoding the Cas12i polypeptide is operably linked to or under the regulation of a promoter.

[0240] In some embodiments, the promoter is a ubiquitous, tissue-specific, cell-type specific, constitutive, or inducible promoter.

[0241] Suitable promoters are known in the art and include, for example, a Cbh promoter, a Cba promoter, a pol I promoter, a pol II promoter, a pol III promoter, a T7 promoter, a U6 promoter, a H1 promoter, a retroviral Rous sarcoma virus LTR promoter, a cytomegalovirus (CMV) promoter, a SV40 promoter, a dihydrofolate reductase promoter, a β -actin promoter, an elongation factor 1a short (EFS) promoter, a β glucuronidase (GUSB) promoter, a cytomegalovirus (CMV) immediate-early (Ie) enhancer and/or promoter, a chicken β -actin (CBA) promoter or derivative thereof such as a CAG promoter, CB promoter, a (human) elongation factor 1 α -subunit (EF1 α) promoter, a ubiquitin C (UBC) promoter, a prion promoter, a neuron-specific enolase (NSE), a neurofilament light (NFL) promoter, a neurofilament heavy (NFH) promoter, a platelet-derived growth factor (PDGF) promoter, a platelet-derived growth factor B-chain (PDGF- β) promoter, a synapsin (Syn) promoter, a human synapsin (hSyn) promoter, a synapsin 1 (Syn1) promoter, a methyl-CpG binding protein 2 (MeCP2) promoter, a Ca2+/calmodulin-dependent protein kinase II (CaMKII) promoter, a metabotropic glutamate receptor 2 (mGluR2) promoter, a neurofilament light (NFL) promoter, a neurofilament heavy (NFH) promoter, a β -globin minigene n β 2 promoter, a preproenkephalin (PPE) promoter, an enkephalin (Enk) promoter, an excitatory amino acid transporter 2 (EAAT2) promoter, a glial fibrillary acidic protein (GFAP) promoter, a myelin basic protein (MBP) promoter,

a OTOF promoter, a GRK1 promoter, a CRX promoter, a NRL promoter, a MECP2 promoter, a mMECP2 promoter, a hMECP2 promoter, an APP promoter, and a RCVRN promoter.

Delivery

[0242] Various ways of delivery can be applied to the Cas12i polypeptide of the disclosure or the system of the disclosure as needed in practices.

[0243] In yet another aspect, the disclosure provides a polynucleotide encoding the Cas12i polypeptide of the disclosure.

[0244] In yet another aspect, the disclosure provides a delivery system comprising (1) the Cas12i polypeptide of the disclosure, the polynucleotide of the disclosure, or the system of the disclosure; and (2) a delivery vehicle.

[0245] In yet another aspect, the disclosure provides a vector comprising the polynucleotide of the disclosure. In some embodiments, the vector encodes a guide nucleic acid as defined in the disclosure. In some embodiments, the vector is a plasmid vector, a recombinant AAV (rAAV) vector (vector genome), or a recombinant lentivirus vector.

[0246] In yet another aspect, the disclosure provides a recombinant AAV (rAAV) particle comprising the rAAV vector genome of the disclosure. A simple introduction of AAV for delivery may refer to “Adeno-associated Virus (AAV) Guide” (addgene.org/guides/aav/).

[0247] Adeno-associated virus (AAV), when engineered to delivery, e.g., a protein-encoding sequence of interest, may be termed as a (r) AAV vector, a (r) AAV vector particle, or a (r) AAV particle, where “r” stands for “recombinant”. And the genome packaged in AAV vectors for delivery may be termed as a (r) AAV vector genome, vector genome, or vg for short, while viral genome may refer to the original viral genome of natural AAVs.

[0248] The serotypes of the capsids of rAAV particles can be matched to the types of target cells. For example, Table 2 of WO2018002719A1 lists exemplary cell types that can be transduced by the indicated AAV serotypes (incorporated herein by reference).

[0249] In some embodiments, the rAAV particle comprising a capsid with a serotype suitable for delivery into ear cells (e.g., inner hair cells). In some embodiments, the rAAV particle comprising a capsid with a serotype of AAV1, AAV2, AAV3A, AAV3B, AAV4, AAV5, AAV6, AAV7, AAVrh74, AAV8, AAV9, AAV10, AAV11, AAV12, AAV13, AAV-DJ, or AAV. PHP. eB, a member of the Clade to which any of the AAV1-AAV13 belong, or a functional variant (e.g., a functional truncation) thereof, encapsidating the rAAV vector genome. In some embodiments, the serotype of the capsid is AAV9 or a functional variant thereof.

[0250] General principles of rAAV particle production are known in the art. In some embodiments, rAAV particles may be produced using the triple transfection method (described in detail in U.S. Pat. No. 6,001,650).

[0251] The vector titers are usually expressed as vector genomes per ml (vg/ml). In some embodiments, the vector titer is above 1×10^9 , above 5×10^{10} , above 1×10^{11} , above 5×10^{11} , above 1×10^{12} , above 5×10^{12} , or above 1×10^{13} vg/ml.

[0252] Instead of packaging a single strand (ss) DNA sequence as a vector genome of a rAAV particle, systems and methods of packaging an RNA sequence as a vector genome into a rAAV particle is recently developed and

applicable herein. See PCT/CN2022/075366, which is incorporated herein by reference in its entirety.

[0253] When the vector genome is RNA as in, for example, PCT/CN2022/075366, for simplicity of description and claiming, sequence elements described herein for DNA vector genomes, when present in RNA vector genomes, should generally be considered to be applicable for the RNA vector genomes except that the deoxyribonucleotides in the DNA sequence are the corresponding ribonucleotides in the RNA sequence (e.g., dT is equivalent to U, and dA is equivalent to A) and/or the element in the DNA sequence is replaced with the corresponding element with a corresponding function in the RNA sequence or omitted because its function is unnecessary in the RNA sequence and/or an additional element necessary for the RNA vector genome is introduced.

[0254] As used herein, a coding sequence, e.g., as a sequence element of rAAV vector genomes herein, is construed, understood, and considered as covering and covers both a DNA coding sequence and an RNA coding sequence. When it is a DNA coding sequence, an RNA sequence can be transcribed from the DNA coding sequence, and optionally further a protein can be translated from the transcribed RNA sequence as necessary. When it is an RNA coding sequence, the RNA coding sequence per se can be a functional RNA sequence for use, or an RNA sequence can be produced from the RNA coding sequence, e.g., by RNA processing, or a protein can be translated from the RNA coding sequence.

[0255] For example, a Cas13 coding sequence encoding a Cas13 polypeptide covers either a Cas13 DNA coding sequence from which a Cas13 polypeptide is expressed (indirectly via transcription and translation) or a Cas13 RNA coding sequence from which a Cas13 polypeptide is translated (directly).

[0256] For example, a gRNA coding sequence encoding a gRNA covers either a gRNA DNA coding sequence from which a gRNA is transcribed or a gRNA RNA coding sequence (1) which per se is the functional gRNA for use, or (2) from which a gRNA is produced, e.g., by RNA processing.

[0257] In some embodiments for rAAV RNA vector genomes, 5'-ITR and/or 3'-ITR as DNA packaging signals may be unnecessary and can be omitted at least partly, while RNA packaging signals can be introduced.

[0258] In some embodiments for rAAV RNA vector genomes, a promoter to drive transcription of DNA sequences may be unnecessary and can be omitted at least partly.

[0259] In some embodiments for rAAV RNA vector genomes, a sequence encoding a polyA signal may be unnecessary and can be omitted at least partly, while a polyA tail can be introduced.

[0260] Similarly, other DNA elements of rAAV DNA vector genomes can be either omitted or replaced with corresponding RNA elements and/or additional RNA elements can be introduced, in order to adapt to the strategy of delivering an RNA vector genome by rAAV particles.

[0261] In yet another aspect, the disclosure provides a ribonucleoprotein (RNP) comprising the Cas12i polypeptide of the disclosure and a guide nucleic acid optionally as defined in the disclosure.

[0262] In yet another aspect, the disclosure provides a lipid nanoparticle (LNP) comprising an RNA (e.g., mRNA)

encoding the Cas12i polypeptide of the disclosure and a guide nucleic acid of the disclosure.

Method of Modification

[0263] The CRISPR-Cas12i system of the disclosure comprising the Cas12i polypeptide of the disclosure has a wide variety of utilities, including modifying (e.g., cleaving, deleting, inserting, translocating, inactivating, or activating) a target DNA in a multiplicity of cell types. The CRISPR-Cas12i systems have a broad spectrum of applications requiring high cleavage activity and low collateral activity, e.g., drug screening, disease diagnosis and prognosis, and treating various genetic disorders.

[0264] The methods and/or the systems of the disclosure can be used to modify a target DNA, for example, to modify the translation and/or transcription of one or more genes of the cells. For example, the modification may lead to increased transcription/translation/expression of a gene. In other embodiments, the modification may lead to decreased transcription/translation/expression of a gene.

[0265] In yet another aspect, the disclosure provides a method for modifying a target DNA, comprising contacting the target DNA with the system of the disclosure, the vector of the disclosure, the ribonucleoprotein of the disclosure, or the lipid nanoparticle of the disclosure, wherein the spacer sequence is capable of hybridizing to a target sequence of the target DNA, wherein the target DNA is modified by the complex.

[0266] In some embodiments, the target DNA is in a cell.

[0267] In some embodiments, the modification comprises one or more of cleavage, base editing, repairing, and exogenous sequence insertion or integration of the target DNA.

Cells

[0268] The methods of the disclosure can be used to introduce the systems of the disclosure into a cell and cause the cell to alter the production of one or more cellular products, such as antibody, starch, ethanol, or any other desired products. Such cells and progenies thereof are within the scope of the disclosure.

[0269] In yet another aspect, the disclosure provides a cell comprising the system of the disclosure. In some embodiments, the cell is a eukaryote. In some embodiments, the cell is a human cell.

[0270] In yet another aspect, the disclosure provides a cell modified by the system of the disclosure or the method of the disclosure. In some embodiments, the cell is a eukaryote. In some embodiments, the cell is a human cell. In some embodiments, the cell is modified in vitro, in vivo, or ex vivo.

[0271] In some embodiments, the cell is a stem cell. In some embodiments, the cell is not a human embryonic stem cell. In some embodiments, the cell is not a human germ cell.

[0272] In some embodiments, the cell is a prokaryotic cell.

[0273] In some embodiments, the cell is a eukaryotic cell (e.g., an animal cell, a vertebrate cell, a mammalian cell, a non-human mammalian cell, a non-human primate cell, a rodent (e.g., mouse or rat) cell, a human cell, a plant cell, or a yeast cell) or a prokaryotic cell (e.g., a bacteria cell).

[0274] In some embodiments, the cell is from a plant or an animal.

[0275] In some embodiments, the plant is a dicotyledon. In some embodiments, the dicotyledon is selected from the

group consisting of soybean, cabbage (e.g., Chinese cabbage), rapeseed, *Brassica*, watermelon, melon, potato, tomato, tobacco, eggplant, pepper, cucumber, cotton, alfalfa, eggplant, grape.

[0276] In some embodiments, the plant is a monocotyledon. In some embodiments, the monocotyledon is selected from the group consisting of rice, corn, wheat, barley, oat, sorghum, millet, grasses, Poaceae, *Zizania*, *Avena*, *Coix*, *Hordeum*, *Oryza*, *Panicum* (e.g., *Panicum miliaceum*), *Secale*, *Setaria* (e.g., *Setaria italica*), *Sorghum*, *Triticum*, *Zea*, *Cymbopogon*, *Saccharum* (e.g., *Saccharum officinarum*), *Phyllostachys*, *Dendrocalamus*, *Bambusa*, *Yushania*.

[0277] In some embodiments, the animal is selected from the group consisting of pig, ox, sheep, goat, mouse, rat, alpaca, monkey, rabbit, chicken, duck, goose, fish (e.g., zebra fish).

[0278] In some embodiments, the cell is a eukaryotic cell, such as a mammalian cell, including a human cell (a primary human cell or an established human cell line). In some embodiments, the cell is a non-human mammalian cell, such as a cell from a non-human primate (e.g., monkey), a cow/bull/cattle, sheep, goat, pig, horse, dog, cat, rodent (such as rabbit, mouse, rat, hamster, etc.). In some embodiments, the cell is from fish (such as salmon), bird (such as poultry bird, including chick, duck, goose), reptile, shellfish (e.g., oyster, clam, lobster, shrimp), insect, worm, yeast, etc. In some embodiments, the cell is from a plant, such as monocot or dicot. In certain embodiment, the plant is a food crop such as barley, cassava, cotton, groundnuts or peanuts, maize, millet, oil palm fruit, potatoes, pulses, rapeseed or canola, rice, rye, sorghum, soybeans, sugar cane, sugar beets, sunflower, and wheat. In certain embodiment, the plant is a cereal (barley, maize, millet, rice, rye, sorghum, and wheat). In certain embodiment, the plant is a tuber (cassava and potatoes). In certain embodiment, the plant is a sugar crop (sugar beets and sugar cane). In certain embodiment, the plant is an oil-bearing crop (soybeans, groundnuts or peanuts, rapeseed or canola, sunflower, and oil palm fruit). In certain embodiment, the plant is a fiber crop (cotton). In certain embodiment, the plant is a tree (such as a peach or a nectarine tree, an apple or pear tree, a nut tree such as almond or walnut or pistachio tree, or a citrus tree, e.g., orange, grapefruit or lemon tree), a grass, a vegetable, a fruit, or an algae. In certain embodiment, the plant is a nightshade plant; a plant of the genus *Brassica*; a plant of the genus *Lactuca*; a plant of the genus *Spinacia*; a plant of the genus *Capsicum*; cotton, tobacco, asparagus, carrot, cabbage, broccoli, cauliflower, tomato, eggplant, pepper, lettuce, spinach, strawberry, blueberry, raspberry, blackberry, grape, coffee, cocoa, etc.

Pharmaceutical Composition

[0279] In yet another aspect, the disclosure provides a pharmaceutical composition comprising (1) the system of the disclosure, the vector of the disclosure, the ribonucleoprotein of the disclosure, the lipid nanoparticle of the disclosure, or the cell of the disclosure; and (2) a pharmaceutically acceptable excipient.

[0280] In some embodiments, the pharmaceutical composition comprises the rAAV particle in a concentration selected from the group consisting of about 1×10^{10} vg/mL, 2×10^{10} vg/mL, 3×10^{10} vg/mL, 4×10^{10} vg/mL, 5×10^{10} vg/mL, 6×10^{10} vg/mL, 7×10^{10} vg/mL, 8×10^{10} vg/mL,

9×10^{10} vg/mL, 1×10^{11} vg/mL, 2×10^{11} vg/mL, 3×10^{11} vg/mL, 4×10^{11} vg/mL, 5×10^{11} vg/mL, 6×10^{11} vg/mL, 7×10^{11} vg/mL, 8×10^{11} vg/mL, 9×10^{11} vg/mL, 1×10^{12} vg/mL, 2×10^{12} vg/mL, 3×10^{12} vg/mL, 4×10^{12} vg/mL, 5×10^{12} vg/mL, 6×10^{12} vg/mL, 7×10^{12} vg/mL, 8×10^{12} vg/mL, 9×10^{12} vg/mL, 1×10^{13} vg/mL, or in a concentration of a numerical range between any of two preceding values, e.g., in a concentration of from about 9×10^{10} vg/mL to about 8×10^{11} vg/mL.

[0281] In some embodiments, the pharmaceutical composition is an injection.

[0282] In some embodiments, the volume of the injection is selected from the group consisting of about 1 microliter, 10 microliters, 50 microliters, 100 microliters, 150 microliters, 200 microliters, 250 microliters, 300 microliters, 350 microliters, 400 microliters, 450 microliters, 500 microliters, 550 microliters, 600 microliters, 650 microliters, 700 microliters, 750 microliters, 800 microliters, 850 microliters, 900 microliters, 950 microliters, 1000 microliters, and a volume of a numerical range between any of two preceding values, e.g., in a concentration of from about 10 microliters to about 750 microliters.

Method of Treatment

[0283] In yet another aspect, the disclosure provides a method for diagnosing, preventing, or treating a disease in a subject in need thereof, comprising administering to the subject (e.g., a therapeutically effective dose) the system of the disclosure, the vector of the disclosure, the ribonucleoprotein of the disclosure, the lipid nanoparticle of the disclosure, the cell of the disclosure, or the pharmaceutical composition of the disclosure, wherein the disease is associated with a target DNA, wherein the spacer sequence is capable of hybridizing to a target sequence of the target DNA, wherein the target DNA is modified by the complex, and wherein the modification of the target DNA diagnose, prevents, or treats the disease.

[0284] In some embodiments, the disease is selected from the group consisting of Angelman syndrome (AS), Alzheimer's disease (AD), transthyretin amyloidosis (ATTR), transthyretin amyloid cardiomyopathy (ATTR-CM), cystic fibrosis (CF), hereditary angioedema, diabetes, progressive pseudohypertrophic muscular dystrophy, Duchenne muscular dystrophy (DMD), Becker muscular dystrophy (BMD), spinal muscular atrophy (SMA), alpha-1-antitrypsin deficiency, Pompe disease, myotonic dystrophy, Huntington's disease (HTT), fragile X syndrome, Friedreich ataxia, amyotrophic lateral sclerosis (ALS), frontotemporal dementia, hereditary chronic kidney disease, hyperlipidemia, Leber congenital amaurosis (LCA), sickle cell disease, thalassemia (e.g., β -thalassemia), Parkinson's disease (PD), myelodysplastic syndrome (MDS), retinitis pigmentosa (RP), age-related macular degeneration (AMD), Hepatitis B, nonalcoholic fatty liver disease (NAFLD), Acquired Immune Deficiency Syndrome, corneal dystrophy (CD), hypercholesterolemia, familial hypercholesterolemia (FH), heart disease (e.g., hypertrophic cardiomyopathy (HCM)), and cancer.

[0285] In some embodiments, the target DNA encodes a mRNA, a tRNA, a ribosomal RNA (rRNA), a microRNA (miRNA), a non-coding RNA, a long non-coding (lnc) RNA, a nuclear RNA, an interfering RNA (iRNA), a small interfering RNA (siRNA), a ribozyme, a riboswitch, a satellite RNA, a microswitch, a microzyme, or a viral RNA.

[0286] In some embodiments, the target DNA is a eukaryotic DNA.

[0287] In some embodiments, the eukaryotic DNA is a mammal DNA, such as a non-human mammalian DNA, a non-human primate DNA, a human DNA, a plant DNA, an insect DNA, a bird DNA, a reptile DNA, a rodent (e.g., mouse, rat) DNA, a fish DNA, a nematode DNA, or a yeast DNA.

[0288] In some embodiments, the target DNA is in a eukaryotic cell, for example, a human cell, a non-human primate cell, or a mouse cell.

[0289] In some embodiments, the administering comprises local administration or systemic administration.

[0290] In some embodiments, the administering comprises intrathecal administration, intramuscular administration, intravenous administration, transdermal administration, intranasal administration, oral administration, mucosal administration, intraperitoneal administration, intracranial administration, intracerebroventricular administration, or stereotaxic administration.

[0291] In some embodiments, the administration is injection or infusion.

[0292] In some embodiments, the subject is a human, a non-human primate, or a mouse.

[0293] In some embodiments, the level of the transcript (e.g., mRNA) of the target DNA is decreased in the subject by at least about 10%, about 15%, about 20%, about 25%, about 30%, about 35%, about 40%, about 45%, about 50%, about 55%, about 60%, about 65%, about 70%, about 75%, about 80%, about 85%, or more compared to the level of the transcript (e.g., mRNA) of the target DNA in the subject prior to the administration.

[0294] In some embodiments, the level of the transcript (e.g., mRNA) of the target DNA is increased in the subject by at least about 10%, about 15%, about 20%, about 25%, about 30%, about 35%, about 40%, about 45%, about 50%, about 55%, about 60%, about 65%, about 70%, about 75%, about 80%, about 85%, or more compared to the level of the transcript (e.g., mRNA) of the target DNA in the subject prior to the administration.

[0295] In some embodiments, the level of the expression product (e.g., protein) of the target DNA is decreased in the subject by at least about 10%, about 15%, about 20%, about 25%, about 30%, about 35%, about 40%, about 45%, about 50%, about 55%, about 60%, about 65%, about 70%, about 75%, about 80%, about 85%, or more compared to the level of the expression product (e.g., protein) of the target DNA in the subject prior to the administration.

[0296] In some embodiments, the level of the expression product (e.g., protein) of the target DNA is increased in the subject by at least about 10%, about 15%, about 20%, about 25%, about 30%, about 35%, about 40%, about 45%, about 50%, about 55%, about 60%, about 65%, about 70%, about 75%, about 80%, about 85%, or more compared to the level of the expression product (e.g., protein) of the target DNA in the subject prior to the administration. In some embodiments, the expression product is a functional mutant of the expression product of the target DNA.

[0297] In some embodiments, the median survival of the subject suffering from the disease but receiving the administration is 5 days, 10 days, 20 days, 30 days, 2 months, 3 months, 4 months, 5 months, 6 months, 7 months, 8 months, 9 months, 10 months, 11 months, 12 months, 1.5 year, 2 years, 2.5 years, 3 years, 4 years, 5 years, 6 years, 7 years,

8 years, 9 years, 10 years or more longer than that of a subject or a population of subjects suffering from the disease and not receiving the administration.

[0298] The therapeutically effective dose may be either via a single dose, or multiple doses. One skilled in the art understands that the actual dose may vary greatly depending upon a variety of factors, such as the vector choices, the target cells, organisms, tissues, the general conditions of the subject to be treated, the degrees of transformation/modification sought, the administration routes, the administration modes, the types of transformation/modification sought, etc.

[0299] For example, the therapeutically effective dose of the rAAV particle may be about 1.0E+8, 2.0E+8, 3.0E+8, 4.0E+8, 6.0E+8, 8.0E+8, 1.0E+9, 2.0E+9, 3.0E+9, 4.0E+9, 6.0E+9, 8.0E+9, 1.0E+10, 2.0E+10, 3.0E+10, 4.0E+10, 6.0E+10, 8.0E+10, 1.0E+11, 2.0E+11, 3.0E+11, 4.0E+11, 6.0E+11, 8.0E+11, 1.0E+12, 2.0E+12, 3.0E+12, 4.0E+12, 6.0E+12, 8.0E+12, 1.0E+13, 2.0E+13, 3.0E+13, 4.0E+13, 6.0E+13, 8.0E+13, 1.0E+14, 2.0E+14, 3.0E+14, 4.0E+14, 6.0E+14, 8.0E+14, 1.0E+15, 2.0E+15, 3.0E+15, 4.0E+15, 6.0E+15, 8.0E+15, 1.0E+16, 2.0E+16, 3.0E+16, 4.0E+16, 6.0E+16, 8.0E+16, or 1.0E+17 vg, or within a range of any two of the those point values. vg stands for vector genomes of rAAV particles for administration.

Method of Detection

[0300] In yet another aspect, the disclosure provides a method of detecting a target DNA, comprising contacting the target DNA with the system of the disclosure, wherein the target DNA is modified by the complex, and wherein the modification detects the target DNA. In some embodiments, the modification generates a detectable signal, e.g., a fluorescent signal.

Kits

[0301] In yet another aspect, the disclosure provides a kit comprising the Cas12i polypeptide of the disclosure, the system of the disclosure, the polynucleotide of the disclosure, the vector of the disclosure, the RNP of the disclosure, the LNP of the disclosure, the delivery system of the disclosure, the cell of the disclosure, or the pharmaceutical composition of the disclosure, or any one, two, or all components of the same.

[0302] In some embodiments, the kit further comprises an instruction to use the component(s) contained therein, and/or instructions for combining with additional component(s) that may be available or necessary elsewhere.

[0303] In some embodiments, the kit further comprises one or more buffers that may be used to dissolve any of the component(s) contained therein, and/or to provide suitable reaction conditions for one or more of the component(s). Such buffers may include one or more of PBS, HEPES, Tris, MOPS, Na₂CO₃, NaHCO₃, NaB, or combinations thereof. In some embodiments, the reaction condition includes a proper pH, such as a basic pH. In some embodiments, the pH is between 7-10.

[0304] In some embodiments, any one or more of the kit components may be stored in a suitable container or at a suitable temperature, e.g., 4 Celsius degree.

[0305] Further embodiments are illustrated in the following Examples which are given for illustrative purposes only and are not intended to limit the scope of the disclosure.

EXAMPLES

Material and Methods

[0306] Unless otherwise specified, the experimental methods used in the Examples are conventional.

[0307] Unless otherwise specified, the materials, reagents, etc., used in the Examples are commercially available.

[0308] Unless otherwise specified, the following materials and experimental methods were used in the Examples.

Plasmid Vector Construction.

[0309] Human codon-optimized Cas12i, TadA8e and human APOBEC3A genes were synthesized by the GenScript Co., Ltd., and cloned to generate pCAG-NLS-Cas12i-NLS_pA_pU6_BpiI_pCMV_mCherry_pA by Gibson Assembly. crRNA oligos were synthesized by HuaGene Co., Ltd., annealed and ligated into BpiI site to produce the pCAG-NLS-Cas12i-NLS_pA_pU6_crRNA_pCMV_mCherry_pA.

Cell Culture, Transfection, and Flow Cytometry Analysis.

[0310] The mammalian cell lines used in this study were HEK293T and N2A. Cells were cultured in Dulbecco's modified Eagle's medium (DMEM) supplemented with 10% FBS, penicillin/streptomycin and GlutMAX. Transfections were performed using Polyetherimide (PEI). For variant/mutant screening, HEK293T cells were cultured in 24-well plates, and after 12 hours 2 µg of the plasmids (1 µg of an expression plasmid and 1 µg of a reporter plasmid) were transfected into these cells with 4 µL PEI. 48 hours after transfection, BFP, mCherry, and EGFP fluorescence were analyzed using a Beckman CytoFlex flow-cytometer. For assay of mutations in target sites of endogenous genes, 1 µg of expression plasmid was transfected into HEK293T or N2A cells, which were then sorted using a BD FACS Aria III, BD LSRFortessa X-20 flow cytometer, 48 hours after transfection.

Detection of Gene Editing Frequency.

[0311] Six thousand sorted cells were lysed in 20 µl of lysis buffer (Vazyme). Targeted sequence primers were synthesized and used in nested PCR amplification by Phanta Max Super-Fidelity DNA Polymerase (Vazyme). Targeted deep sequence analysis was used to determine indel frequencies. A-to-G or C-to-T editing frequencies were calculated by targeted deep sequence analysis or Sanger sequencing and EditR. A-to-G editing purity were calculated as A-to-G editing efficiency/(A-to-T editing efficiency+A-to-C editing efficiency+A-to-G editing efficiency). C-to-T editing purity were calculated as C-to-T editing efficiency/(C-to-A editing efficiency+C-to-G editing efficiency+C-to-T editing efficiency).

Pem-Seq.

[0312] PEM-seq in HEK293 cells was performed as previously described. Briefly, all-in-one plasmids containing LbCas12a, Ultra-AsCas12a, hfCas12Max, ABR001 or Cas12i2HiFi with targeting TTR. 2 crRNA were transfected into HEK293 cells by PEI respectively, and after 48 hrs, positive cells were harvested for DNA extraction. The 20 µg genomic DNA was fragmented with a peak length of 300-700 bp by Covaris sonication. DNA fragments was tagged

with biotin by a one-round biotinylated primer extension at 5'-end, and then primer removal by AMPure XP beads and purified by streptavidin beads. The single-stranded DNA on streptavidin beads is ligated with a bridge adapter containing 14-bp RMB, and PCR product was performed nested PCR for enriching DNA fragment containing the bait DSB and tagged with illumine adapter sequences. The prepared sequencing library was sequencing on an Hi-seq 2500, with a 2x150 bp.

RNP Delivery and Ex Vivo Editing.

[0313] RNP was complexed by mixing purified hfCas12Max proteins with chemically synthesized RNA oligonucleotides (Genscript) at a 1:2 molar ratio in 1xPBS. RNP was incubated at room temperature for >15 min prior to electroporation with Lonza® 4D-Nucleofector™. 0.2x 10⁶ cells were resuspended in 20 µL of Lonza buffer and mixed with 5 µL RNP with different concentrations electroporated according to Lonza specifications. HEK293 or CD3+ T cells were harvested 72 hrs post-electroporation for targeted deep sequence analysis.

LNP Delivery and In Vivo Editing.

[0314] LNPs were formulated with ALC0315, cholesterol, DMG-PEG2k, DSPC in 100% ethanol, carrying in vitro transcription (IVT) mRNA and chemically synthesized RNA oligonucleotides (Genscript) with a 1:1 weight ratio. LNPs were formed according to the manufacturer's protocol, by microfluidic mixing the lipid with RNA solutions using a Precision Nano-systems NanoAssemblr Benchtop Instrument. LNPs diluted in PBS were transfected into N2a cells at 0.1, 0.3, 0.5, 1 µg RNA, or delivered into C57 mouse with different dose by through tail intravenous injection. Cells were harvested 48 hrs post-transfection for lysis and targeted deep sequence analysis. For in vivo editing, liver tissue was collected from the left or median lateral lobe of each mouse 7 days post-injection for DNA extraction and targeted deep sequence analysis.

Zygote Injection and Embryo Culturing.

[0315] Super ovulated female C57 mice (7-8 weeks old) by injecting 5 IU of pregnant mare serum gonadotropin (PMSG), followed by 5 IU of human chorionic gonadotropin (hCG) 48 hrs later were mated to B6D2F1 males, and fertilized embryos were collected from oviducts 20 hrs post hCG injection. For zygote injection, hfCas12Max mRNA (100 ng/µL) and gRNA (100 ng/µL) were mixed and injected into the cytoplasm of fertilized eggs in a droplet of HEPES-CZB medium containing 5 mg/ml cytochalasin B (CB) using a FemtoJet microinjector (Eppendorf) with constant flow settings. The injected zygotes were cultured in KSOM medium with amino acids at 37° C. under 5% CO₂ in air to blastocysts and harvested for targeted deep sequence analysis.

Example 1 Identification of Cas12i Proteins and Evaluation of their dsDNA Cleavage Activity

[0316] In order to identify more Cas12i proteins, the applicant developed and employed a bioinformatics pipeline to annotate Cas12i proteins, CRISPR arrays, DR sequences, and predicted PAM preferences, and identified 10 Cas12i proteins and associated sequences in Table 1 below.

TABLE 1

Name Cas12i protein	Name Cas12i protein	SEQ ID NO:			
		Cas12i amino acid sequence	Corresponding DR sequence	Cas12i coding sequences	Codon-optimized Cas12i coding sequence
SiCas12i	Cas12i12 (xCas12i)	1	11	21	31
Si2Cas12i	Cas12i3	2	12	22	32
WiCas12i	Cas12i7	3	13	23	33
Wi2Cas12i	Cas12i8	4	14	24	34
Wi3Cas12i	Cas12i9	5	15	25	35
SaCas12i	Cas12i11	6	16	26	36
Sa2Cas12i	Cas12i4	7	17	27	37
Sa3Cas12i	Cas12i5	8	18	28	38
WaCas12i	Cas12i6	9	19	29	39
Wa2Cas12i	Cas12i10	10	20	30	40

[0317] To evaluate the guide sequence-specific dsDNA cleavage activity (“dsDNA cleavage activity” for short as used in the disclosure) of these Cas12i proteins in mammalian cells, the applicant designed a dual plasmid fluorescent reporter system, which detected the increased enhanced green fluorescent protein (EGFP) signal intensity activated by Cas-mediated dsDNA cleavage or double strand breaks (FIG. 3A). This system relied on the co-transfection of an expression plasmid encoding mCherry, a nuclear localization signal (NLS)-tagged Cas protein, and a guide RNA (gRNA, or crRNA), and a reporter plasmid encoding BFP and activatable EGxxFP cassette, which is EGxx-target site-xxFP. EGFP activation was carried out by Cas mediated DSB and single-strand annealing (SSA)-mediated repair.

[0318] Specifically, referring to FIG. 3A, the reporter plasmid comprised a polynucleotide encoding, from 5' to 3', BFP-P2A-activatable EGxxFP (SEQ ID NO: 41) (EGxx-insertion sequence (SEQ ID NO: 42) (containing, from 5' to 3', a protospacer adjacent motif (PAM)) of [TTC] for Cas12i protein, a protospacer sequence (SEQ ID NO: 43) (which is

the reverse complementary sequence of a target sequence (SEQ ID NO: 44)), and a protospacer adjacent motif (PAM))

of [GGG] for Cas9 protein-xxFP), followed by a bGH polyA (SEQ ID NO: 448) coding sequence, operably linked to human CMV promoter (SEQ ID NO: 447). The protospacer sequence (SEQ ID NO: 43) contained a premature stop

codon [TAG] that prevented the expression of EGFP and hence emission of green fluorescent signals. The BFP coding sequence expresses BFP to indicate the successful transfection and expression of the reporter plasmid into host cells through blue fluorescence.

[0319] Most of the known Cas12i proteins recognize a 5'-T-rich PAM 5' to protospacer sequence in dsDNA, while Cas9 recognizes a 3'-G-rich PAM 3' to protospacer sequence in dsDNA. The co-existence of the 5' PAM of [TTC] for Cas12i protein and the 3' PAM of [GGG] for Cas9 protein flanking the protospacer sequence (SEQ ID NO: 43) allows the simultaneous evaluation and comparison of dsDNA cleavage activity of Cas12i protein and Cas9 protein.

Activatable EGxxxxFP coding sequence, SEQ ID NO: 41

atgagcgagctgattaaggagaacatgcacatgaagctgtataggaggccacgtggacaaccatcacttcaagtgacatccgagggcggaaggaag
ccctacgagggcaccagaccatgagaatcaaggtggtcgagggcgccctctcccttcgccttcgacatcctggctactagcttccctctacggcagc
aagaccttcatcaaccacacccaggcatccccgacttcttcaagcagtccttccctgagggcttcacatgggagagagtcaccacatacaggagcggg
gcggtgctgaccgctaccaggacaccagcctccaggacggctgcctcatctacaacgtcaagatcagaggggtgaacttcacatccaacggccctg
tgatgcagaagaaacactcggtgggagccttcaccgagacactgtaccgcgtgacggcgccctggaaggcagaaacgacatggccctgaagctcgt
ggcggggagccatctgatcgcaaacatcaagaccacatatagataccaagaacccgctaagaacctcaagatgcctggcgctctacatgtggactacagac
tggaaagaatcaaggaggccaacaacgagacatacgtcgagcagcagcaggtggcagtgccagatactgcgacccctctagcaaacctggggcacaagc
tgaatgaattcgagggcgagggcgagcctgctgacctgcggcgagctggaggagaaccccgccccatggtgagcaaggcgagggagctgttcaccgggg
tggtgcccactcctggtcgagctggagggcgagctaaacggccacaagttcagcgtgtccggcgagggcgagggcgatgccacctacggcaagctgacct
gaagttcatctgcaccacggcaagctgcccgtgcccggccaccctcgtgaccacctgacctacggcgtgcagtgcttcagccgctaccccgacca
catgaagcagcagcacttcttcaagtcggccatgcccgaaggctacgtccaggagcgccacatcttcttcaaggacgagggcaactacaagaccccgccc
gaggtgaagttcgagggcgacaccctggtgaaccgcatcgagctgaaggcgatcgacttcaaggaggacggcaacatcctggggcacaagctggagtag
aactacaacagccacaacgtctatcatatcatggccgacaagcagaagaacggcatcaaggtgaacttcaag

-continued

atccgccacaacatcgaggacggcagcgtgcagctcgccgaccactaccagcagaacaccGGATCCGTGCTCTTCCTATTACAGTAGG

AGCATACGGGAGACAAGCTTTGccacctaaggcaagctgacctgaagtctcaccacccggaagctgcccgtgccctggcccacct

cgtagaccacctcgacctacggcgtgcagtgcttcagccgctaccccgaccacatgaagcagcagcacttcttcaagtcggccatgcccgaaggctacgtc

caggagcgcaccatcttcttaaggacgacggcaactacaagaccgcgcgaggtgaagttcgagggcgacacctgggtgaa

ccgcatcgagctgaaggcgatcgacttcaaggaggacggcaacatcctggggcacaagctggagtacaactacaacagccacaacg

tctatatcatggccgacaagcagaagaacggcatcaaggtgaacttcaagatccgccacaacatcgaggacggcagcgtgcagctcgc

cgaccactaccagcagaacacccccatcgggcagcggccccgtgctgctgcccgacaaccactacctgagcaccagtcggccctgagcaaa

gaccccaacgagaagcgcgatcacatggctcctgctggagttcgtgaccgcgcgggatcactctcgcatggacg

agctgtacaagtaa

Insertion sequence, SEQ ID NO: 42
GGATCCGTGCTCTTCCTATTACAGTAGGAGCATACGGGAGACAAGCTTTG

Protospacer sequence (Reverse complementary sequence of the target sequence),
20bp, SEQ ID NO: 43
CCATTACAGTAGGAGCATAC

Target sequence, 20 nt, SEQ ID NO: 44
GTATGCTCCTACTGTAATGG

EGxxxxFP-targeting spacer sequence, 20 nt, SEQ ID NO: 45
CCATTACAGTAGGAGCATAC

Non-targeting ("NT") spacer sequence, 20 nt SEQ ID NO: 46
GGTCTTCGATAAGAACCT

PAM for Cas12i TTC

PAM for Cas9 GGG

[0320] Also referring to FIG. 3A, the expression plasmid comprised from 5' to 3' i) a Cas12i coding sequence codon optimized for expression in mammalian cells (one of SEQ ID NOs: 31-40) encoding a Cas12i protein (one of SEQ ID NOs: 1-10) flanked by a SV40 NLS (SEQ ID NO: 444) coding sequence on its 5' end and a NP NLS (SEQ ID NO: 445) coding sequence on its 3' end, followed by a bGH polyA (SEQ ID NO: 448) coding sequence, operably linked to CAG promoter (SEQ ID NO: 500), ii) a sequence encoding a guide RNA (gRNA) composed of 5'-DR sequence-spacer sequence-3' operably linked to human U6 promoter (SEQ ID NO: 446); and iii) a coding sequence for mCherry followed by a bGH polyA (SEQ ID NO: 448) coding sequence operably linked to human CMV promoter (SEQ ID NO: 447). The mCherry coding sequence expresses mCherry to indicate the successful transfection and expression of the expression plasmid into host cells through red fluorescence.

[0321] In the event that both the target sequence on the target strand and the protospacer sequence on the nontarget strand of the target dsDNA are successfully cleaved by a Cas12i protein guided by a gRNA to generate a double-strand break (DSB), the subsequent DNA repairing such as single-strand annealing (SSA)-mediated repair triggered by the DSB would restore the EGFP coding sequence to express EGFP with green fluorescence emission indicative of dsDNA cleavage activity.

[0322] For test groups, the spacer sequence comprised in the gRNA ("crEGFP", one of SEQ ID NOs: 51-60) for use

with each corresponding tested Cas12i protein (one of SEQ ID NOs: 1-10) is a EGxxxxFP-targeting spacer sequence (SEQ ID NO: 45) designed to target and hybridize to the target sequence (SEQ ID NO: 44), and the DR sequence in the gRNA (one of SEQ ID NOs: 51-60) is a DR sequence (one of SEQ ID NOs: 11-20) corresponding to each tested Cas12i protein (one of SEQ ID NOs: 1-10), as shown in Table 2.

TABLE 2

Cas12i protein	SEQ ID NO:		
	DR sequence	Spacer sequence	Guide RNA
SiCas12i (xCas12i)	11	45	51
Si2Cas12i	12	45	52
WiCas12i	13	45	53
Wi2Cas12i	14	45	54
Wi3Cas12i	15	45	55
SaCas12i	16	45	56
Sa2Cas12i	17	45	57
Sa3Cas12i	18	45	58
WaCas12i	19	45	59
Wa2Cas12i	20	45	60

[0323] For negative control ("NT") for each tested Cas12/9 protein (Cas12i, SpCas9, LbCas12a), a non-targeting spacer sequence ("NT", SEQ ID NO: 46) incapable of hybridizing to the target sequence (SEQ ID NO: 44) was used in place of the EGxxxxFP-targeting spacer sequence

(SEQ ID NO: 45), while the other elements of each tested CRISPR-Cas12/9 system remained.

[0324] For positive control, CRISPR-SpCas9 and CRISPR-LbCas12a systems each comprising a Cas protein and a guide RNA as shown in Table 3 below were used in place of the tested CRISPR-Cas12i systems in Tables 1 and 2 above, using the same EGxxxxFP-targeting spacer sequence (SEQ ID NO: 45). Note that the gRNA for the CRISPR-SpCas9 system was composed of 5'-spacer sequence-scaffold sequence-3', and the gRNA for the CRISPR-LbCas12a system was composed of 5'-DR sequence-spacer sequence-3'.

TABLE 3		
Control Cas protein	Control Cas amino acid sequence	Guide RNA
SpCas9	SEQ ID NO: 47	SEQ ID NO: 48
LbCas12a	SEQ ID NO: 49	SEQ ID NO: 50

[0325] HEK293T cells were cultured in 24-well tissue culture plates according to standard methods for 12 hours, before the reporter and expression plasmids were co-transfected into the cells using standard polyethyleneimine (PEI) transfection. The transfected cells were then cultured at 37° C. under 5% CO₂ for 48 hours. Then the cultured cells were analyzed by flow cytometry for BFP, EGFP, and mCherry fluorescent signals. A “blank” control group was also set up, where only the reporter plasmid was transfected, and no expression plasmid was introduced.

[0326] The dsDNA cleavage activities of the tested Cas proteins were calculated as the percentage of EGFP positive cells in BFP & mCherry dual-positive cells (“EGFP⁺”, indicating dsDNA cleavage at the indicated target site on the reporter plasmid; “mCherry⁺ BFP⁺”, indicating successful co-transfection and co-expression of the expression and reporter plasmids). The higher the % EGFP⁺/mCherry⁺ BFP⁺ is, the higher the dsDNA cleavage activity would be.

[0327] Using this dual plasmid fluorescent reporter system, it was observed that five Cas12i proteins (Cas12i3

(SEQ ID NO: 2), Cas12i7 (SEQ ID NO: 3), Cas12i10 (SEQ ID NO: 10), Cas12i11 (SEQ ID NO: 6), and Cas12i12 (SEQ ID NO: 1, also referred to as SiCas12i or xCas12i in the disclosure)) exhibited targeted gRNA induced significant activation of EGFP expression indicative of significant dsDNA cleavage (FIG. 1A, FIG. 3B), and among them, Cas12i12 even exhibited a higher dsDNA cleavage activity than both LbCas12a and SpCas9 as determined by Fluorescence Activated Cell Sorter (FACS) analysis (FIG. 1A, FIG. 3B). The xCas12i (1080 aa) was smaller in size compared to SpCas9 (1368 aa) and LbCas12a (1228 aa) (FIG. 4A).

Example 2 Evaluation of Effective Spacer Sequence Length for xCas12i

[0328] Using the dual plasmid fluorescent reporter system in Example 1, to test the effective spacer sequence length for xCas12i, 22 spacer sequences of different lengths ranging from 10 to 50 nt (SEQ ID NOs: 45 and 61-81 as shown in Table 4 below) were designed to target and hybridize to the reverse complementary sequence of a protospacer sequence (SEQ ID NO: 43, or one of SEQ ID NOs: 61-81) comprised in the insertion sequence (SEQ ID NO: 42) of the GFxxxxFP reporter plasmid in Example 1, wherein the 20 nt spacer sequence in Table 4 is exactly the EGxxxxFP-targeting spacer sequence (SEQ ID NO: 45) in Example 1. To evaluate the additional spacer sequence lengths, the EGxxxxFP targeting spacer sequence (SEQ ID NO: 45) in the guide RNA encoded in the expression plasmid was replaced with the spacer sequence in respective length (one of SEQ ID NOs: 61-81) in Table 4, while the other elements of the dual plasmid fluorescent reporter system remained. To save drafting, the sequences in Table 4 refer to both the protospacer sequence (a DNA sequence) and the corresponding spacer sequence (an RNA sequence) with any “T” in the sequence when referring to a protospacer sequences standing for “T” and when referring to such a spacer sequence standing for “U”, although the assigned SEQ ID NOs: 61-81 in the sequence listing are annotated as RNA.

TABLE 4		
	Protospacer sequence/Spacer sequence	SEQ ID NO:
10-nt	CCATTACAGT	61
12-nt	CCATTACAGTAG	62
14-nt	CCATTACAGTAGGA	63
15-nt	CCATTACAGTAGGAG	64
16-nt	CCATTACAGTAGGAGC	65
17-nt	CCATTACAGTAGGAGCA	66
18-nt	CCATTACAGTAGGAGCAT	67
19-nt	CCATTACAGTAGGAGCATA	68
20-nt	CCATTACAGTAGGAGCATAC	45
21-nt	CCATTACAGTAGGAGCATACG	69
22-nt	CCATTACAGTAGGAGCATACGG	70
23-nt	CCATTACAGTAGGAGCATACGGG	71

TABLE 4-continued

	Protospacer sequence/Spacer sequence	SEQ ID NO:
24-nt	CCATTACAGTAGGAGCATACGGGA	72
26-nt	CCATTACAGTAGGAGCATACGGGAGA	73
27-nt	CCATTACAGTAGGAGCATACGGGAGAC	74
28-nt	CCATTACAGTAGGAGCATACGGGAGACA	75
30-nt	CCATTACAGTAGGAGCATACGGGAGACAAG	76
32-nt	CCATTACAGTAGGAGCATACGGGAGACAAGCT	77
35-nt	CCATTACAGTAGGAGCATACGGGAGACAAGCTTTG	78
40-nt	CCATTACAGTAGGAGCATACGGGAGACAAGCTTTGGCCAC	79
45-nt	CCATTACAGTAGGAGCATACGGGAGACAAGCTTTGGCCACCTACG	80
50-nt	CCATTACAGTAGGAGCATACGGGAGACAAGCTTTGGCCACCTACGGCAAG	81

[0329] By using the experimental procedure in Example 1, it was observed that a spacer sequence length range of at least 16 nucleotides is effective for xCas12i’s spacer sequence-specific cleavage activity, and among that range, 17-22 nt is optimal (FIG. 4B).

Example 3 Evaluation of PAM Recognition for xCas12i

[0330] Considering the 5'-TTN PAM preference of Cas12i, the applicant performed a NTTN PAM identification assay (wherein N is A, T, C, or G) using the dual plasmid fluorescent reporter system in Example 1, in which various 5' PAM was used in place of the original 5' PAM of TTT, while the other elements of the dual plasmid fluorescent reporter system remained.

[0331] By using the experimental procedure in Example 1, it was observed that xCas12i showed a consistently high frequency of EGFP activation at target sites with 5'-NTTN

[0332] PAM sequences, wherein N is A, T, C, or G, while LbCas12a had comparable activity at just 5'-TTTN PAM, respectively (FIG. 4C), showing the much more broad PAM site recognition of xCas12i.

Example 4 Tolerance of Variation in DR Sequence of xCas12i System

[0333] To test whether the original direct repeat (DR) sequence (SEQ ID NO: 11, 36 nt) identified together with xCas12i could tolerate variation, the applicant truncated the original DR sequence to generate two functional fragments DR-T1 (30 nt) and DR-T2 (23 nt) of SEQ ID NOs: 451 and 452, respectively, without destroying the secondary structure of the original DR sequence (FIG. 22), and then designed five DR variants of DR-T2 to generate DR-A, DR-B, DR-C, DR-D, and DR-E sequences of SEQ ID NOs: 453-457, respectively, each containing 5% to 30% mutations in the stem-loop regions without destroying the secondary structure of the original DR sequence. That is, the secondary structures of the 7 DR variants were substantially the same as that of the original DR sequence.

DR-T1, 30 nt	SEQ ID NO: 451
ATGACTCAGAAATGTGTCCCCAGTTGACAC	
DR-T2 sequence, 23 nt	SEQ ID NO: 452
AGAAATGTGTCCCCAGTTGACAC	
DR-A sequence, 23 nt	SEQ ID NO: 453
AGAAATCCGTCCTTAGTTGACGG	
DR-B sequence, 22 nt	SEQ ID NO: 454
AGACATGTGTCCCCAGTGACAC	
DR-C sequence, 23 nt	SEQ ID NO: 455
AGAAATGTTTCCCCAGTTGAAAC	
DR-D sequence, 23 nt	SEQ ID NO: 456
AGAAATGTGTTCCCGTTAACAC	
DR-E sequence, 23 nt	SEQ ID NO: 457
AGAAATTTGTCCCCAGTTGACAA	

[0334] By using the dual plasmid fluorescent reporter system for xCas12i in Example 1 with the original DR sequence (SEQ ID NO: 11) replaced with each of the DR variants (DR-T1, DR-T2, DR-A, DR-B, DR-C, DR-D, and DR-E), while the other element of the reporter system remained, the results (FIG. 21) show that xCas12i still exhibited high dsDNA cleavage activity mediated by gRNAs with various DR sequence variants. It can be seen that under the condition that the secondary structure of the DR sequence is maintained (i.e., the secondary structures of the DR variants are substantially the same as that of the original DR sequence), the CRISPR-SiCas12i system tolerated mismatching or deletion on DR sequence without substantial loss of dsDNA cleavage activity, indicating wide adaptability to variations in the DR sequence. These data also demonstrated that the two truncations of the original xCas12i DR sequence of SEQ ID NO: 11 (36 nt), i.e., DR-T1 (SEQ ID NO: 451, 30 nt) and DR-T2 (SEQ ID NO: 452, 23 nt), could still mediate high dsDNA cleavage activity of xCas12i.

Example 5 Evaluation of dsDNA Cleavage Activity
of xCas12i at Endogenous Gene

[0335] To further verify the dsDNA cleavage activity of xCas12i at an endogenous gene (genome cleavage) in mammalian cells, the applicant transfected the expression plasmid (FIG. 3A, FIG. 4D) in Example 1 encoding NLS tagged xCas12i with gRNAs targeting 37 sites from human TTR gene and human PCSK9 gene in HEK293T (human embryonic kidney 293 cells) or mouse Ttr gene in N2a cells (Neuro2a cells, a fast-growing mouse neuroblastoma cell line). The EGxxxxFP targeting spacer sequence (SEQ ID NO: 45) in Example 1 was replaced with respective gene-targeting spacer sequence (SEQ ID NOs: 82-119 and 121-125 in Table 5), the DR-T1 sequence (SEQ ID NO: 451) was used in place of the original DR sequence (SEQ ID NO: 11) (and also in the Examples below unless otherwise specified), while the other elements of the CRISPR-xCas12i system in

Example 1 remained. The dsDNA cleavage activity, i.e., indel (insertion and/or deletion) formation, at these loci was measured 48 hours after transfection using FACS and targeted deep sequencing. To save drafting, the sequences in Table 5 refer to both the protospacer sequence (a DNA sequence) and the corresponding spacer sequence (an RNA sequence) with any “T” in the sequence when referring to a protospacer sequences standing for “T” and when referring to a such spacer sequence standing for “U”, although the assigned SEQ ID NOs: 82-119 and 121-125 in the sequence listing are annotated as DNA.

[0336] It was observed that xCas12i mediated a high frequency, up to 90%, of indel formation at most sites from Ttr, TTR and PCSK9, with a mean indel formation rate of over 50% (FIG. 4E-F). These data indicate that xCas12i exhibits a robust genome editing efficiency in mammalian cells, suggesting that it has excellent potential for therapeutic genome editing applications.

TABLE 5

Sequences for testing genome cleavage at target loci					
Genomic loci	Guide RNA	PAM	Protospacer sequence/ Spacer sequences	SEQ ID NO of protospacer/ sequence	Figure
DMD	DMD_sg1	TTTG	CAAAAACCCAAATATTTTA	82	FIG. 1D, DIG. 6B-C and FIG. 8D
	DMD_sg2	TTTA	GCTCCTACTCAGACTGTTAC	83	FIG. 1D, DIG. 6B-C and FIG. 8D
	DMD_sg3	GTTG	TGTCACCAGAGTAACAGTC	84	FIG. 1D, DIG. 6B-C and FIG. 8D
Ttr	Ttr_sg1	TTTG	CCTCGCTGGACTGGTATTTG	85	FIG. 4E-F
	Ttr_sg2	TTTG	TGTCCTGAAGCTGGCCCCGCG	86	FIG. 1D and FIG. 4E-F
	Ttr_sg3	CTTC	CCTTCGACTCTTCCTCCTTTG	87	FIG. 1D and FIG. 4E-F, 19B
	Ttr_sg4	CTTC	CTCCTTTGCTCGCTGGACTG	88	FIG. 4E-F
	Ttr_sg5	TTTG	ACCATCAGAGGACATTTGGA	89	FIG. 4E-F
	Ttr_sg6	TTTG	GATTCTCCAGCACCTGGGC	90	FIG. 4E-F
	Ttr_sg7	TTTA	CAGCCACGTCTACAGCAGGG	91	FIG. 4E-F
	Ttr_sg8	TTTT	ACAGCCACGTCTACAGCAGG	92	FIG. 4E-F
	Ttr_sg9	TTTT	GAACACTTTTACAGCCACGT	93	FIG. 4E-F
	Ttr_sg10	GTTC	AAAAAGACCTCTGAGGGATC	94	FIG. 4E-F
	Ttr_sg11	TTTG	AACACTTTTACAGCCACGTC	95	FIG. 4E-F
	Ttr_sg12	TTTG	TAGAAGGAGTGATCAGAGTA	96	FIG. 1D, FIG. 2F-H and FIG. 4E-F, 19B
	Ttr_sg13	CTTG	GCATTTCCCCGTTCCATGAA	97	FIG. 4E-F
	Ttr_sg14	CTTC	TCATCTGTGGTGAGCCCCGTG	98	FIG. 4E-F
	Ttr_sg15	TTTG	GTGTCCAGTTCTACTCTGTA	99	FIG. 4E-F
	Ttr_sg16	CTTC	CAGTACGATTTGGTGTCCAG	100	FIG. 4E-F
	Ttr_sg17	CTTC	TACAACTTCTCATCTGTGG	101	FIG. 4E-F
	Ttr_sg18	TTTT	CACAGCCAACGACTCTGGCC	102	FIG. 4E-F
	Ttr_sg19	TTTC	ACAGCCAACGACTCTGGCCA	103	FIG. 4E-F
	Ttr_sg20	GTTG	CTGACGACAGCCGTGGTCTG T	104	FIG. 4E-F
	Ttr_sg21	GTTC	AAAAAGACCTCTGAGGGATCC T	105	FIG. 4E-F
TTR	TTR_sg1	GTTC	AGAAAGGCTGCTGATGACACC T	106	FIG. 1D and FIG. 4E-F
	TTR_sg2	TTTG	TAGAAGGGATATACAAAGTGG A	107	FIG. 1D and FIG. 4E-F, 16
	TTR_sg3	ATTC	CACCACGGCTGTCGTACACAA T	108	FIG. 4E-F
	TTR_sg5	TTTG	AATCCAAGTGTCTCTGATGGT	109	FIG. 4E-F
	TTR_sg6	TTTC	AATGTGGCCGTGCATGTGTTCA	110	FIG. 4E-F
	TTR_sg7	GTTC	TAGATGCTGTCCGAGGCAGTC C	111	FIG. 4E-F
	TTR_sg8	ATTC	GCATGGGCTCACAACTGAGGA G	112	FIG. 4E-F

TABLE 5-continued

Sequences for testing genome cleavage at target loci					
Genomic loci	Guide RNA	PAM	Protospacer sequence/ Spacer sequences	SEQ ID NO of protospacer/ sequence	Figure
	TTR_sg10	TTTG	TATACAAAGTGGAAATAGACA C	113	FIG. 4E-F
	TTR_sg11	CTTA	CTGGAAGGCACTTGGCATCTC C	114	FIG. 1D and FIG. 4E-F
	TTR_sg12	CTTG	GCATCTCCCCATTCCATGAGCA	115	FIG. 1D and FIG. 4E-F
	TTR_sg14	ATTC	ACAGCCAACGACTCCGGCCCC C	116	FIG. 4E-F
PCSK9	PCSK9_sg5	GTTG	CCTGGCACCTACGTGGTGG	117	FIG. 4E-F
	PCSK9_sg6	CTTC	CATGGCCTTCTTCTCTGGC	118	FIG. 1D and FIG. 4E-F
	PCSK9_sg7	CTTC	TTCTGGCTTCCTGGTGAAG	119	FIG. 4E-F
	PCSK9_sg9	CTTG	AAGTTGCCCCATGTCGACTA	121	FIG. 1D and FIG. 4E-F
	PCSK9_sg10	TTTG	CCCAGAGCATCCCGTGAAC	122	FIG. 1D and FIG. 4E-F
TRAC	TRAC_sg1	TTTA	CAGATACGAACCTAACTTT	123	FIG. 2B-C
	TRAC_sg2	TTTA	GAGTCTCTCAGCTGGTACAC	124	FIG. 2B-C
	TRAC_sg3	TTTG	TCTGTGATATACATCAGA	125	FIG. 2B-C

Example 6 Development of xCas12i Mutants and Evaluation of their dsDNA Cleavage Activity

[0337] To vary xCas12i's dsDNA cleavage activity and/or expand its scope of PAM site recognition, the applicant engineered xCas12i protein via mutagenesis and screened for mutants with various dsDNA cleavage activity and broader PAM using a dual plasmid fluorescent reporter system similar to the dual plasmid fluorescent reporter system in Example 1, except that the EGxxxxFP-targeting guide RNA (SEQ ID NO: 51; "crON/crRNA On-target") coding sequence operably linked to U6 promoter was not located on the expression plasmid together with the xCas12i (or its mutant) coding sequence (SEQ ID NO: 31) but located on the reporter plasmid together with the BFP-P2A-EGxxxxFP coding sequence (SEQ ID NO: 41) (referring to "On-Target Reporter" in FIG. 1B). Combined with predictive structural analysis of xCas12i, the applicant performed an arginine (R) scanning mutagenesis approach in the domains including PI domain (amino acid residue position 173-291), REC-I domain (amino acid residue position 427-473), and RuvC-II domain (amino acid residue position 800-1082) of xCas12i, generating a library of 599 xCas12i mutants with a single non-R amino acid substitution with R. The xCas12i (SEQ ID NO: 1) coding sequence on the expression plasmid was replaced with a sequence encoding each of the xCas12i mutants in Table 6, the DR-T1 sequence (SEQ ID NO: 451) was used in place of the original DR sequence (SEQ ID NO: 11), while the other elements of the reporter system remained. The applicant then individually transfected the expression plasmid and the reporter plasmid into HEK293T cells and analyzed them by FACS (FIG. 1B).

[0338] For negative control ("NT"), a non-targeting spacer sequence ("NT", SEQ ID NO: 46) incapable of hybridizing to the target sequence (SEQ ID NO: 44) was used in place of the EGxxxxFP-targeting spacer sequence (SEQ ID NO: 45) and used in combination with xCas12i (SEQ ID NO: 1),

while the other elements of the reporter system remained. For positive control ("WT"), the original xCas12i (SEQ ID NO: 1) was used.

TABLE 6

Mutants of xCas12i and dsDNA cleavage activity thereof	
Mutant	dsDNA cleavage activity
K109R	0.034
N110R	0.778
Y111R	0.634
L112R	0.041
M113R	0.062
S114R	0.837
N115R	0.312
I116R	0.836
D117R	0.499
S118R	1.481
D119R	1.337
F121R	1.356
V122R	0.737
W123R	1.010
V124R	0.119
D125R	0.040
C126R	0.051
K128R	0.844
F129R	0.802
A130R	0.064
K131R	0.728
D132R	0.839
F133R	0.990
A134R	0.076
Y135R	0.863
Q136R	1.067
M137R	0.128
E138R	1.010
L139R	0.194
G140R	0.957
F141R	0.429
H142R	0.941
E143R	1.240
F144R	0.007

TABLE 6-continued

Mutants of xCas12i and dsDNA cleavage activity thereof	
Mutant	dsDNA cleavage activity
T145R	0.951
V146R	1.106
L147R	0.038
A148R	0.013
E149R	0.319
T150R	0.686
L151R	0.038
L152R	0.097
A153R	1.000
N154R	0.307
S155R	1.577
I156R	0.531
L157R	0.041
V158R	1.990
L159R	0.085
N160R	0.860
E161R	2.115
S162R	2.096
T163R	1.054
K164R	0.760
A165R	3.151
N166R	1.548
W167R	0.775
A168R	0.058
W169R	0.161
G170R	0.572
T171R	0.211
V172R	0.564
S173R	0.202
A174R	0.398
L175R	0.170
Y176R	0.215
G177R	0.135
G178R	1.920
G179R	0.737
D180R	1.025
K181R	0.172
E182R	0.235
D183R	0.279
S184R	0.987
T185R	1.685
L186R	0.641
K187R	0.193
S188R	0.234
K189R	1.010
I190R	0.070
L191R	0.118
L192R	0.910
A193R	1.566
F194R	0.194
V195R	0.019
D196R	1.317
A197R	0.791
L198R	0.204
N199R	1.354
N200R	1.417
H201R	0.183
E202R	1.102
L203R	1.344
K204R	0.817
T205R	0.973
K206R	0.871
E208R	0.279
I209R	0.108
L210R	0.346
N211R	0.499
Q212R	0.650
V213R	0.114
C214R	0.166
E215R	0.329
S216R	0.591

TABLE 6-continued

Mutants of xCas12i and dsDNA cleavage activity thereof	
Mutant	dsDNA cleavage activity
L217R	0.465
K218R	0.294
Y219R	0.375
Q220R	0.371
S221R	1.150
Y222R	0.417
Q223R	0.574
D224R	0.301
M225R	0.099
Y226R	0.000
V227R	0.177
D228R	0.168
F229R	0.190
S231R	0.284
V232R	0.559
V233R	1.253
D234R	0.217
E235R	1.727
N236R	1.242
G237R	0.470
N238R	0.069
K239R	0.988
K240R	0.908
S241R	1.828
P242R	0.167
N243R	3.606
G244R	0.060
S245R	1.293
M246R	0.124
P247R	0.240
I248R	0.962
V249R	0.114
T250R	0.140
K251R	1.434
F252R	0.009
E253R	0.321
T254R	0.927
D255R	1.182
D256R	0.595
L257R	1.162
I258R	0.044
S259R	0.531
D260R	0.293
N261R	0.484
Q262R	0.498
K264R	0.671
A265R	0.725
M266R	0.250
I267R	0.933
S268R	0.959
N269R	0.401
F270R	0.131
T271R	0.450
K272R	0.383
N273R	1.652
A274R	0.207
A275R	0.713
A276R	0.309
K277R	0.282
A278R	0.471
A279R	0.683
K280R	0.556
K281R	0.671
P282R	0.575
I283R	0.390
P284R	0.274
Y285R	0.287
L286R	0.745
D287R	1.084
L289R	0.400
K290R	0.403

TABLE 6-continued

Mutants of xCas12i and dsDNA cleavage activity thereof	
Mutant	dsDNA cleavage activity
E291R	0.363
M293R	0.019
V294R	0.665
S295R	1.172
L296R	0.752
C297R	0.061
D298R	0.719
Y300R	0.168
N301R	0.359
V302R	1.517
Y303R	0.324
A304R	0.067
W305R	0.026
A306R	0.187
A307R	0.265
A308R	0.030
I309R	0.009
T310R	0.163
N311R	0.120
S312R	0.037
N313R	0.246
A314R	0.030
D315R	0.046
V316R	0.007
T317R	0.143
A318R	0.037
N320R	0.098
T321R	0.156
L324R	0.035
T325R	0.209
F326R	0.183
I327R	0.031
G328R	0.879
E329R	0.249
Q330R	0.159
N331R	0.538
S332R	1.136
K335R	0.577
E336R	1.463
L337R	0.613
S338R	1.505
V339R	1.183
L340R	0.419
Q341R	0.766
T342R	0.322
T343R	0.710
T344R	0.646
N345R	0.218
E346R	0.554
K347R	0.684
A348R	0.048
K349R	0.461
D350R	0.474
I351R	0.146
L352R	0.023
N353R	0.553
K354R	0.681
N356R	0.542
D357R	0.472
N358R	0.554
L359R	0.398
I360R	0.580
Q361R	0.676
E362R	1.430
V363R	0.696
Y365R	0.016
T366R	0.973
P367R	0.195
A368R	0.709
K370R	0.648
H371R	0.068

TABLE 6-continued

Mutants of xCas12i and dsDNA cleavage activity thereof	
Mutant	dsDNA cleavage activity
L372R	0.006
G373R	0.430
D375R	1.408
L376R	0.006
A377R	1.097
N378R	1.113
L379R	0.008
F380R	0.087
D381R	1.502
T382R	1.517
L383R	0.006
K384R	0.941
E385R	1.424
K386R	0.980
D387R	1.050
I388R	0.317
N389R	0.895
N390R	1.066
I391R	0.685
E392R	0.996
N393R	0.662
E394R	0.871
E395R	1.144
E396R	1.214
K397R	0.918
Q398R	1.043
N399R	1.050
V400R	1.222
I401R	0.754
N402R	0.934
D403R	1.712
C404R	0.689
I405R	0.048
E406R	1.758
Q407R	1.735
Y408R	0.064
V409R	1.004
D410R	0.771
D411R	1.447
C412R	1.852
L415R	0.650
N416R	1.541
N418R	1.292
P419R	0.171
I420R	0.058
A421R	0.910
A422R	0.674
L423R	0.092
L424R	0.013
K425R	0.745
H426R	0.742
I427R	0.005
S428R	0.075
Y430R	0.359
Y431R	0.856
E432R	0.670
D433R	0.605
F434R	0.161
S435R	0.981
A436R	0.033
K437R	0.880
N438R	0.309
F439R	0.010
L440R	1.379
D441R	0.671
G442R	0.051
A443R	0.033
K444R	0.547
L445R	0.107
N446R	0.410
V447R	0.004

TABLE 6-continued

Mutants of xCas12i and dsDNA cleavage activity thereof	
Mutant	dsDNA cleavage activity
L448R	1.369
T449R	0.514
E450R	0.887
V451R	1.883
V452R	0.735
N453R	0.895
Q455R	1.190
K456R	0.887
A457R	0.004
H458R	0.008
P459R	0.008
T460R	0.009
I461R	0.801
W462R	0.358
S463R	0.020
E464R	1.127
I800R	0.596
S801R	0.204
L802R	0.398
K803R	0.436
M804R	0.130
I805R	0.325
S806R	1.214
D807R	0.899
F808R	0.261
K809R	0.905
G810R	0.954
V811R	0.178
V812R	0.187
Q813R	0.161
S814R	0.023
Y815R	0.284
F816R	0.299
S817R	1.290
V818R	1.410
S819R	1.130
G820R	0.407
C821R	0.801
V822R	0.699
D823R	0.911
D824R	0.939
A825R	0.884
S826R	0.707
K827R	0.654
K828R	0.917
A829R	0.954
H830R	0.593
D831R	0.318
S832R	1.010
M833R	1.088
L834R	0.835
F835R	1.280
T836R	1.402
F837R	1.270
M838R	0.961
C839R	1.700
A840R	1.412
A841R	0.245
E842R	1.540
E843R	1.710
K844R	1.520
T846R	1.620
N847R	1.180
K848R	1.230
E850R	0.867
E851R	0.977
K852R	0.337
T853R	0.928
N854R	1.031
A856R	1.262
A857R	0.384

TABLE 6-continued

Mutants of xCas12i and dsDNA cleavage activity thereof	
Mutant	dsDNA cleavage activity
S858R	1.117
F859R	0.000
I860R	0.146
L861R	0.770
Q862R	1.882
K863R	1.427
A864R	0.000
Y865R	1.179
L866R	1.417
H867R	0.000
G868R	1.613
C869R	0.131
K870R	1.510
M871R	1.334
I872R	0.163
V873R	0.306
C874R	0.519
E875R	0.100
D876R	2.637
D877R	2.492
L878R	0.132
P879R	0.132
V880R	1.458
A881R	0.236
D882R	0.356
G883R	1.303
K884R	1.624
T885R	0.464
G886R	1.856
K887R	1.606
A888R	2.077
Q889R	0.720
N890R	0.151
A891R	2.265
D892R	1.417
M894R	1.386
D895R	0.539
W896R	0.265
C897R	0.873
A898R	0.192
A900R	1.324
L901R	0.376
A902R	0.621
K903R	1.115
K904R	1.106
V905R	0.203
N906R	1.606
D907R	0.238
G908R	0.244
C909R	0.499
V910R	1.406
A911R	0.222
M912R	1.106
S913R	1.471
I914R	1.000
C915R	1.663
Y916R	1.356
A918R	1.882
P920R	0.831
A921R	0.338
Y922R	0.446
M923R	1.044
S924R	0.351
S925R	1.276
H926R	1.440
Q927R	1.933
D928R	0.164
P929R	0.179
F930R	0.203
V931R	1.547
H932R	0.229

TABLE 6-continued

Mutants of xCas12i and dsDNA cleavage activity thereof	
Mutant	dsDNA cleavage activity
M933R	1.827
Q934R	2.147
D935R	1.413
K936R	1.489
K937R	1.442
T938R	1.452
S939R	1.413
V940R	1.333
L941R	0.988
P943R	0.812
F945R	1.055
M946R	1.207
E947R	0.885
V948R	1.231
N949R	1.893
K950R	1.640
D951R	2.347
S952R	1.500
I953R	0.382
D955R	1.221
Y956R	1.768
H957R	0.681
V958R	0.541
A959R	1.635
G960R	1.840
L961R	0.152
L965R	0.443
N966R	1.933
S967R	1.529
K968R	1.241
S969R	1.548
D970R	1.451
A971R	1.848
G972R	1.152
T973R	0.641
S974R	1.180
V975R	1.097
Y976R	1.148
Y977R	0.007
Q979R	1.421
A980R	1.057
A981R	0.341
L982R	1.146
H983R	1.372
F984R	0.580
C985R	1.076
E986R	1.137
A987R	1.220
L988R	0.954
G989R	1.420
V990R	1.094
S991R	1.211
P992R	1.128
E993R	1.154
L994R	1.148
V995R	1.109
K996R	1.038
N997R	1.211
K998R	1.171
K999R	1.348
T1000R	1.128
H1001R	1.209
A1002R	1.171
A1003R	1.241
E1004R	1.460
L1005R	0.665
G1006R	1.031
M1009R	0.980
G1010R	1.172
S1011R	0.558
A1012R	1.098

TABLE 6-continued

Mutants of xCas12i and dsDNA cleavage activity thereof	
Mutant	dsDNA cleavage activity
M1013R	1.207
L1014R	1.044
M1015R	0.535
P1016R	0.088
W1017R	1.744
G1019R	0.387
G1020R	0.396
V1022R	1.260
Y1023R	0.814
I1024R	0.296
A1025R	0.062
S1026R	0.971
K1027R	0.978
K1028R	1.550
L1029R	0.444
T1030R	0.824
S1031R	0.000
D1032R	1.230
A1033R	0.563
K1034R	1.301
S1035R	0.790
V1036R	0.627
K1037R	1.750
Y1038R	0.666
C1039R	1.430
G1040R	1.077
E1041R	0.920
D1042R	0.928
M1043R	0.930
W1044R	0.870
Q1045R	1.560
Y1046R	0.708
H1047R	1.430
A1048R	0.739
D1049R	0.699
E1050R	0.788
I1051R	0.678
A1052R	0.114
A1053R	0.035
V1054R	0.122
N1055R	0.108
I1056R	0.078
A1057R	0.285
M1058R	0.354
Y1059R	0.762
E1060R	0.623
V1061R	0.947
C1062R	0.699
C1063R	1.137
Q1064R	0.948
T1065R	0.781
G1066R	0.906
A1067R	0.994
F1068R	0.010
G1069R	1.067
K1070R	0.969
K1071R	0.833
Q1072R	0.879
K1073R	0.464
K1074R	0.286
S1075R	0.971
D1076R	0.777
E1077R	0.709
L1078R	0.915
P1079R	0.860
G1080R	0.996
WT	1.000
NT	0.0084

[0339] Based on the fluorescence intensity of cells with activated EGFP, it was observed that almost 200 xCas12i

mutants showed an increased dsDNA cleavage activity relative to xCas12i (WT; SEQ ID NO: 1) (FIG. 5A, Table 6), and among them, one mutant, xCas12i-N243R, referred to as Cas12Max, showed about 3.6-fold improvement (FIG. 5A). In addition, about 50 xCas12i mutants has no more than 5% dsDNA cleavage activity relative to WT xCas12i (SEQ ID NO: 1) (FIG. 5A, Table 6).

[0340] The applicant then performed saturation mutagenesis of N243 and observed that the mutation to R indeed showed the highest dsDNA cleavage activity (FIG. 6A).

[0341] The applicant next targeted DMD or Ttr sites using the fluorescent reporter system (replacing the insertion sequence (SEQ ID NO: 42) with an insertion sequence containing DMD or Ttr protospacer and corresponding 5' PAM as listed in Table 5) and observed that Cas12Max displayed a markedly increased frequency of EGFP activation, relative to xCas12i (WT) (FIG. 1C, FIG. 6B-C). In addition, it was observed that the incorporation of E336R into Cas12Max (resulting in mutant xCas12i-N243R+E336R; SEQ ID NO: 467) further increased the dsDNA cleavage activity of Cas12Max at all three sites with different PAMs (TTC-site 1, TTG-site 2, ATG-site 3) (FIG. 6D).

[0342] To further test the efficacy of Cas12Max in targeting genomic loci, the applicant designed a total of eight gRNAs to target sites TTR and PCSK9 in HEK293T cells and three more to target Ttr in N2a cells (Table 5), in which DR-T2 (SEQ ID NO: 452) was used. Consistent with the previous results, Cas12Max exhibited a significantly increased frequency of indels compared to WT xCas12i (FIG. 1D).

Example 7 Further Development of Mutants Based on Cas12Max and Evaluation of their Off-Target dsDNA Cleavage Activity

[0343] To examine the specificity of Cas12Max, the applicant transfected a construct designed to express it with a gRNA targeting TTR (with TTR-targeting (on-target) spacer sequence of SEQ ID NO: 130), and performed indel frequency analysis of on- and off-target (OT) sites predicted by Cas-OFFinder.

TABLE 7

	Off-target protospacer sequence (with 5' PAM of TTTG)	SEQ ID NO:
TTR off-target.3 (OT.3)	<u>CAGCAGGCTTCT</u> TACAAAGTGG	127
TTR off-target.2 (OT.2)	TAAAGGGATATACAA <u>TATGT</u> A	128
TTR off-target.1 (OT.1)	TAGAAGGGATATAGAAAGT <u>TATC</u>	129
	On-target protospacer sequence/ spacer sequence (with 5' PAM of TTTG)	
TTR on-target.1 (ON.1)	TAGAAGGGATATACAAAGTGG	130

[0344] A dual plasmid fluorescent reporter system for evaluation of off-target dsDNA cleavage activity (off-target reporter system; referring to “Off-Target Reporter” in FIG. 1B) was established, which was similar to the dual plasmid fluorescent reporter system in Example 6 for evaluation of (on-target) dsDNA cleavage activity, except that the insertion sequence of the EGxxxFP coding sequence contains an TTR off-target protospacer sequence (one of SEQ ID NOs: 127-129) containing one or more mismatches (bold, under-

lined) with a TTR-targeting spacer sequence (SEQ ID NO: 130) in the gRNA, rather than containing a TTR on-target protospacer sequence (SEQ ID NO: 130; which is the same as SEQ ID NO: 107 in Example 5); DR-T1 sequence (SEQ ID NO: 451) was used. To save drafting, the on-target protospacer sequence/spacer sequence in Table 7 refer to both the protospacer sequence (a DNA sequence) and the corresponding spacer sequence (an RNA sequence) with any “T” in the sequence when referring to a protospacer sequences standing for “T” and when referring to a such spacer sequence standing for “U”, although the assigned SEQ ID NO: 130 in the sequence listing is annotated as DNA.

[0345] Using the on-target and off-target reporter systems (FIG. 7A) or targeted deep sequence analysis on endogenous gene (FIG. 7B), the applicant observed that Cas12Max efficiently edited the target site (“ON. 1”), while resulting in indel formation at 2 (“OT. 1” and “OT. 2”) of the 3 predicted off-target sites (“OT. 1”, “OT. 2”, and “OT. 3”), indicating off-target dsDNA cleavage activity.

[0346] To eliminate the off-target activity of Cas12Max, the applicant selected those mutants in Example 5 with a single mutation in the REC and RuvC domains and undiminished on-target cleavage activity (comparable to xCas12i (WT)), and then tested their off-target dsDNA cleavage activity by using two off-target reporter systems above with TTR OT1 and OT2, respectively (FIG. 1B). It was observed that four xCas12i mutants (xCas12i-V880R (v4.1), xCas12i-M923R (v4.2), xCas12i-D892R (v4.3), and xCas12i-G883R (v4.4); FIG. 8B) maintained a high level of on-target dsDNA cleavage activity and showed substantially no off-target dsDNA cleavage activity at both TTR OT1 and OT2 (FIG. 8A).

[0347] The applicant further combined one or more of these four amino acid substitutions with N243R or N243R+E336R (FIG. 8B). As shown in FIG. 8C, all the four mutants v5.1, v5.2, v5.3, and v5.4 with two amino acid substitutions of N243R and one of V880R, G883R, D892R, and M923R, respectively, had comparable or higher on-target cleavage activity and greatly reduced off-target cleavage activity

compared with Cas12Max; and all the four mutants v6.1, v6.2, v6.3, and v6.4 with three amino acid substitutions of N243R and E336R and one of V880R, G883R, D892R, and M923R, respectively, had comparable or higher on-target cleavage activity and greatly reduced off-target cleavage activity compared with Cas12Max.

[0348] In particular, it was observed that the mutant v6.3 (N243R+E336R+D892R) showed the best overall performance of both on-target and off-target cleavage activities

(FIG. 8B-C). Targeted deep sequencing analysis of endogenous TTR. 2 site and its off-target sites in HEK293T showed that v6.3 (N243R+E336R+D892R) significantly reduced off-target indel frequencies at all the six OT sites and retained on-target indel frequency at ON site, compared to Cas12Max (FIG. 1E). In addition, relative to Cas12Max (v1.1), v6.3 (N243R+E336R+D892R) retained comparable or even higher on-target activity at DMD. 1, DMD. 2 and DMD. 3 sites (FIG. 8D). Therefore, the applicant named v6.3 as high-fidelity Cas12Max (hfCas12Max).

TABLE 8

Mutant	Version	ON	OT-1	OT-2	OT-3
N243R	v1.1	73.80	60.17	47.50	0.11
N243R + V880R	v5.1	71.60	3.82	0.24	0.15
N243R + M923R	v5.2	76.10	4.90	0.92	0.15
N243R + D892R	v5.3	75.85	6.66	5.46	0.21
N243R + G883R	v5.4	77.30	16.80	1.36	0.15
N243R + E336R + V880R	v6.1	75.70	2.04	0.44	0.15
N243R + E336R + M923R	v6.2	75.57	2.41	2.90	0.05
N243R + E336R + D892R (hfCas12Max)	v6.3	77.73	1.55	0.25	0.13
N243R + E336R + G883R	v6.4	74.75	6.65	0.64	0.03
N243R + E336R + D892A	v6.7	77.30	54.80	51.50	
N243R + E336R + G883A	v6.8	78.50	44.00	36.40	
NT		0.028	0.048	0.067	0.014

[0349] Additionally, to investigate hfCas12Max's PAM preference, the applicant performed a 5'-NNN PAM recognition assay by designing reporter plasmids with the same target sequence but different PAM, similar to Example 3. Besides showing a consistent or higher cleavage activity at sites with a 5'-TTN PAM, hfCas12Max and Cas12Max showed a similarly high cleavage activity for targets with 5'-TNN, 5'-ATN, 5'-GTN, and 5'-CTN PAM sites, compared with the commonly used Cas12 (LbCas12a, Ultra-

AsCas12a) and recently reported improved Cas12i2 (ABR001, Cas12i2^{Hifi}) (FIG. 1F). Taken together, these results demonstrate that hfCas12Max exhibits high-efficiency editing activity with highly flexible 5'-TN (TTN/ATN/GTN/CTN) or 5'-TNN (TAN/TCN/TGN/TTN) PAM recognition, along with some discrete 5' PAM, such as, 5'-GCC, which advantageously expands the application scope of this tool.

Example 8 Verification and Comparison of hfCas12Max's On- and Off-Target dsDNA Cleavage Activity at TTR Gene

[0350] To comprehensively evaluate the performance of hfCas12Max in human cells, the applicant designed a large number of target sites in the exons of TTR for various Cas nucleases. DR-T2 (SEQ ID NO: 452) was used in this and subsequent Example unless otherwise specified.

[0351] In total, cleavage activity was monitored at 43 sites for hfCas12Max with 5'-TTN PAMs, 43 sites for ABR001 (engineered Cas12i2 from Arbor Biotechnologies) with TTN PAMs, 43 sites for Cas12i2-1 with TTN PAMs, 45 sites for SpCas9 with NGG PAMs, 12 sites for LbCas12a with TTTN PAMs, 12 sites for Ultra AsCas12a with TTTN PAMs, and 20 sites for KKH-saCas9 with NNNRRT PAMs (Table 9). Indel analysis showed that hfCas12Max exhibited a higher average on-target dsDNA cleavage activity than all the other Cas nucleases and Cas12Max (FIG. 1G, FIG. 9). To save drafting, the sequences in Table 9 refer to both the protospacer sequence (aDNA sequence) and the corresponding spacer sequence (an RNA sequence) with any "T" in the sequence when referring to a protospacer sequences standing for "T" and when referring to a such spacer sequence standing for "U", although the assigned SEQ ID NOs: 131-381 in the sequence listing are annotated as DNA.

TABLE 9

Sequences for testing genome cleavage at target loci (FIG. 1G, FIG. 9)					
Genomic loci	Cas	SITE	5'/3' PAM	Protospacer sequence/ Spacer Sequence	SEQ ID NO of protospacer sequence/ spacer sequence
TTR	LbCas12a	TTTN.1	TTTG	TGCTGAGGCTGGCCCTACGGTG	131
		TTTN.2	TTTG	ACCATCAGAGGACACTTGGATTG	132
		TTTN.3	TTTC	TGAACACATGCACGGCCACATTG	133
		TTTN.4	TTTG	CCTCTGGGTAAGTTGCCAAGAA	134
		TTTN.5	TTTG	GCAACTTACCCAGAGGCAAAATGG	135
		TTTN.6	TTTC	ACACCTTATAGGAAAACCAAGTGA	136
		TTTN.7	TTTC	CTATAAGGTGTGAAAGTCTGGAT	137
		TTTN.8	TTTT	CCTATAAGGTGTGAAAGTCTGGA	138
		TTTN.9	TTTG	TAGAAGGGATATACAAAGTGGAA	139
		TTTN.10	TTTG	TATATCCCTTCTACAAATTCCTC	140
		TTTN.11	TTTC	CACTTTGTATATCCCTTCTACAA	141
		TTTN.12	TTTG	GTGTCTATTTCCACTTTGTATAT	142
UltraCas12a	UltraCas12a	TTTN.1	TTTG	TGCTGAGGCTGGCCCTACGGTG	143
		TTTN.2	TTTG	ACCATCAGAGGACACTTGGATTG	144
		TTTN.3	TTTC	TGAACACATGCACGGCCACATTG	145
		TTTN.4	TTTG	CCTCTGGGTAAGTTGCCAAGAA	146
		TTTN.5	TTTG	GCAACTTACCCAGAGGCAAAATGG	147
		TTTN.6	TTTC	ACACCTTATAGGAAAACCAAGTGA	148
		TTTN.7	TTTC	CTATAAGGTGTGAAAGTCTGGAT	149
		TTTN.8	TTTT	CCTATAAGGTGTGAAAGTCTGGA	150
		TTTN.9	TTTG	TAGAAGGGATATACAAAGTGGAA	151
		TTTN.10	TTTG	TATATCCCTTCTACAAATTCCTC	152
		TTTN.11	TTTC	CACTTTGTATATCCCTTCTACAA	153
		TTTN.12	TTTG	GTGTCTATTTCCACTTTGTATAT	154

TABLE 9-continued

Sequences for testing genome cleavage at target loci (FIG. 1G, FIG. 9)					
Genomic loci	Cas	SITE	5' /3' PAM	Protospacer sequence/ Spacer Sequence	SEQ ID NO of protospacer sequence/ spacer sequence
KKH-SaCas9		NNGRRT.1	ACAGAT	CCACCTATGAGAGAAGACAG	155
		NNGRRT.2	AGGAAT	GGCTGTCGTCACCAATCCCA	156
		NNGRRT.3	AGGAGT	GACGACAGCCGTGGTGAAT	157
		NNGRRT.4	ATTGAT	CTGAACACATGCACGGCCAC	158
		NNGRRT.5	CCAAGT	CACCCAGGGCACCAGTGAAT	159
		NNGRRT.6	CGGAGT	AATGGTGTAGCGGCGGGGC	160
		NNGRRT.7	GCAAAT	CTTTGGCAACTTACCCAGAG	161
		NNGRRT.8	GTGAGT	TGTCTGAGGCTGGCCCTACG	162
		NNGRRT.9	TACGGT	TTTGTGTCTGAGGCTGGCCC	163
		NNGRRT.10	TGGAAT	ATTGGTGACGACAGCCGTGG	164
		NNGRRT.11	TGGGAT	AGGAGAAGTCCCTCATTCCT	165
		NNGRRT.12	TTTGGT	CCAAGTGCCTTCCAGTAAGA	166
SpCas9		NGG.1	AGG	ACACAAATACCAGTCCAGCA	167
		NGG.2	AGG	CCAGTCCAGCAAGGCAGAGG	168
		NGG.3	AGG	GAAGTCCACTCATTCCTGGC	169
		NGG.4	AGG	AAAGTCTAGATGCTGTCCG	170
		NGG.5	AGG	CCCAGAGGCAAATGGCTCCC	171
		NGG.6	AGG	TTCTTTGGCAACTTACCCAG	172
		NGG.7	AGG	ACTGAGGAGGAATTTGTAGA	173
		NGG.8	AGG	CCCATTCCATGAGCATGCAG	174
		NGG.9	AGG	GCATGGGCTCACAACTGAGG	175
		NGG.10	AGG	AATAGGAGTAGGGGCTCAGC	176
		NGG.11	AGG	GACGACAGCCGTGGTGAAT	177
		NGG.12	AGG	GGCTGTCGTCACCAATCCCA	178
		NGG.13	AGG	GTCACCAATCCCAAGGAATG	179
		NGG.14	CGG	TGTGTCTGAGGCTGGCCCTA	180
		NGG.15	CGG	AGCCTTTCTGAACACATGCA	181
		NGG.16	CGG	CAGAGGACACTTGGATTAC	182
		NGG.17	CGG	CATTGATGGCAGGACTGCCT	183
		NGG.18	CGG	CTTCTCTACACCCAGGGCAC	184
		NGG.19	CGG	AATGGTGTAGCGGCGGGGC	185
		NGG.20	CGG	CCCCTACTCCTATTCCACCA	186
		NGG.21	CGG	GCAGGGCGGCAATGGTGTAG	187
		NGG.22	CGG	GGAGTAGGGGCTCAGCAGGG	188
		NGG.23	CGG	GTATTCACAGCCAACGACTC	189
		NGG.24	GGG	TCACAGAAACACTCACCGTA	190
		NGG.25	GGG	AAAGGCTGCTGATGACACCT	191
		NGG.26	GGG	CTTGGATTACCCGGTGCCT	192
		NGG.27	GGG	GCCGTGGTGAATAGGAGTA	193
		NGG.28	GGG	GCGGCAATGGTGTAGCGGCG	194
		NGG.29	GGG	GGAGAAGTCCCTCATTCCTT	195
		NGG.30	GGG	GGCGGCAATGGTGTAGCGGC	196
		NGG.31	GGG	TCACCAATCCCAAGGAATGA	197
		NGG.32	TGG	GCAACTTACCCAGAGGCAAA	198
		NGG.33	TGG	AAGTGCCTTCCAGTAAGATT	199
		NGG.34	TGG	ACCTCTGCATGCTCATGGAA	200
		NGG.35	TGG	TACTCACCTCTGCATGCTCA	201
		NGG.36	TGG	TGTAGAAGGGATATACAAAG	202
		NGG.37	TGG	AGGAGAAGTCCCTCATTCCT	203
		NGG.38	TGG	ATTGGTGACGACAGCCGTGG	204
		NGG.39	TGG	GCGGCGGGGGCGGAGTCGT	205
		NGG.40	TGG	GGGATTGGTGACGACAGCCG	206
		NGG.41	TGG	GGGGCTCAGCAGGGCGGCAA	207
Cas12Max		TTTN.1	TTTG	TGTCGTAGGCTGGCCCTACGGTG	208
		TTTN.2	TTTG	ACCATCAGAGGACACTTGGATTC	209
		TTTN.3	TTTC	TGAACACATGCACGGCCACATTG	210
		TTTN.4	TTTG	CCTCTGGGTAAGTTGCCAAAGAA	211
		TTTN.5	TTTG	GCAACTTACCCAGAGGCAAAATGG	212
		TTTN.6	TTTC	ACACCTTATAGGAAAACCACTGA	213
		TTTN.7	TTTC	CTATAAGGTGTGAAAGTCTGGAT	214
		TTTN.8	TTTT	CCTATAAGGTGTGAAAGTCTGGA	215
		TTTN.9	TTTG	TAGAAGGGATATACAAAGTGGAA	216
		TTTN.10	TTTG	TATATCCCTTCTACAAATTCCTC	217
		TTTN.11	TTTC	CACCTTGTATATCCCTTCTACAA	218
		TTTN.12	TTTG	GTGTCTATTTCCACTTTGTATAT	219
		VTTN.1	CTTA	CTGGAAGGCATTTGGCATCT	220
		VTTN.2	CTTA	TAGGAAAACCAAGTGAGTCTG	221
		VTTN.3	CTTC	TCATCGTCTGCTCCTCCTCT	222
		VTTN.4	ATTC	TTGGCAGGATGGCTTCTCAT	223
		VTTN.5	ATTC	ACCGGTGCCCTGGGTGTAGA	224
		VTTN.6	GTTc	AGAAAGGCTGCTGATGACAC	225

TABLE 9-continued

Sequences for testing genome cleavage at target loci (FIG. 1G, FIG. 9)					
Genomic loci	Cas	SITE	5'/3'PAM	Protospacer sequence/ Spacer Sequence	SEQ ID NO of protospacer sequence/ spacer sequence
hfCas12Max		VTTN.7	GTTC	TAGATGCTGTCCGAGGCAGT	226
		VTTN.8	CTTC	TCTACACCCAGGGCACCAGG	227
		VTTN.9	GTTC	TTTGGCAACTTACCCAGAGG	228
		VTTN.10	CTTC	CAGTAAGATTGGTGTCTAT	229
		VTTN.11	ATTG	CATGAGCATGCAGAGGTGAG	230
		VTTN.12	ATTG	CTCCTCAGTTGTGAGCCCAT	231
		VTTN.13	CTTC	TACAAATTCCTCCTCAGTTG	232
		VTTN.14	ATTG	ACAGCCAACGACTCCGGCCC	233
		VTTN.15	ATTG	CACCACGGCTGTCGTACCA	234
		VTTN.16	ATTG	CTTGGGATTGGTGACGACAG	235
		VTTN.17	CTTC	TCTCATAGGTGGTATTCA _q A	236
		VTTN.18	CTTG	CTGGACTGGTATTGTGTCT	237
		VTTN.19	CTTG	GCAGGATGGCTTCTCATCGT	238
		VTTN.20	ATTG	ATGGCAGGACTGCCTCGGAC	239
		VTTN.21	CTTG	GATTACCCGGTGCCCTGGGT	240
		VTTN.22	CTTG	GCATCTCCCCATTCCATGAG	241
		VTTN.23	GTTC	TGAGCCCATGCAGCTCTCCA	242
		VTTN.24	ATTG	CCGCCCTGCTGAGCCCTAC	243
		VTTN.25	GTTC	GCTGTGAATACCACCTATGA	244
		VTTN.26	CTTG	GGATTGGTGACGACAGCCGT	245
		VTTN.27	ATTG	GTGACGACAGCCGTGGTGGA	246
		VTTN.28	ATTT	GTGTCTGAGGCTGGCCCTAC	247
		VTTN.29	CTTT	GACCATCAGAGGACACTTGG	248
		VTTN.30	ATTT	GCCTCTGGGTAAGTTGCCAA	249
		VTTN.31	CTTT	GGCAACTTACCCAGAGGCAA	250
		VTTN.32	ATTT	GGTGTCTATTTCCACTTTGT	251
		VTTN.33	CTTT	GTATATCCCTTCTACAAATT	252
		TTTN.1	TTTG	TGTCTGAGGCTGGCCCTACGGTG	253
		TTTN.2	TTTG	ACCATCAGAGGACACTTGGATT	254
		TTTN.3	TTTC	TGAACACATGCACGGCCACATTG	255
		TTTN.4	TTTG	CCTCTGGGTAAGTTGCCAAAGAA	256
		TTTN.6	TTTC	ACACCTTATAGGAAAACCAAGTGA	257
		TTTN.7	TTTC	CTATAAGGTGTGAAAGTCTGGAT	258
		TTTN.8	TTTT	CCTATAAGGTGTGAAAGTCTGGA	259
		TTTN.9	TTTG	TAGAAGGGATATACAAAGTGGAA	260
		TTTN.10	TTTG	TATATCCCTTCTACAAATTCCTC	261
		TTTN.11	TTTC	CACCTTGTATATCCCTTCTACAA	262
		TTTN.12	TTTG	GTGTCTATTTCCACTTTGTATAT	263
		VTTN.1	CTTA	CTGGAAGGCACCTGGCATCT	264
		VTTN.2	CTTA	TAGGAAAACCAAGTGTCTG	265
		VTTN.3	CTTC	TCATCGTCTGCTCCTCCTCT	266
		VTTN.4	ATTG	TTGGCAGGATGGCTTCTCAT	267
		VTTN.5	ATTG	ACCGGTGCCCTGGGTGTAGA	268
		VTTN.6	GTTC	AGAAAGGCTGCTGATGACAC	269
		VTTN.7	GTTC	TAGATGCTGTCCGAGGCAGT	270
		VTTN.9	GTTC	TTTGGCAACTTACCCAGAGG	271
		VTTN.10	CTTC	CAGTAAGATTGGTGTCTAT	272
		VTTN.11	ATTG	CATGAGCATGCAGAGGTGAG	273
		VTTN.12	ATTG	CTCCTCAGTTGTGAGCCCAT	274
		VTTN.13	CTTC	TACAAATTCCTCCTCAGTTG	275
		VTTN.14	ATTG	ACAGCCAACGACTCCGGCCC	276
		VTTN.15	ATTG	CACCACGGCTGTCGTACCA	277
		VTTN.16	ATTG	CTTGGGATTGGTGACGACAG	278
		VTTN.17	CTTC	TCTCATAGGTGGTATTACACA	279
		VTTN.18	CTTG	CTGGACTGGTATTGTGTCT	280
		VTTN.19	CTTG	GCAGGATGGCTTCTCATCGT	281
		VTTN.20	ATTG	ATGGCAGGACTGCCTCGGAC	282
		VTTN.21	CTTG	GATTACCCGGTGCCCTGGGT	283
		VITN.22	CTTG	GCATCTCCCCATTCCATGAG	284
		VTTN.23	GTTC	TGAGCCCATGCAGCTCTCCA	285
		VTTN.24	ATTG	CCGCCCTGCTGAGCCCTAC	286
		VTTN.25	GTTC	GCTGTGAATACCACCTATGA	287
		VTTN.26	CTTG	GGATTGGTGACGACAGCCGT	288
		VTTN.27	ATTG	GTGACGACAGCCGTGGTGGA	289
		VTTN.28	ATTT	GTGTCTGAGGCTGGCCCTAC	290
		VTTN.29	CTTT	GACCATCAGAGGACACTTGG	291
		VTTN.30	ATTT	GCCTCTGGGTAAGTTGCCAA	292
		VTTN.31	CTTT	GGCAACTTACCCAGAGGCAA	293
		VTTN.32	ATTT	GGTGTCTATTTCCACTTTGT	294
		VTTN.33	CTTT	GTATATCCCTTCTACAAATT	295

TABLE 9-continued

Sequences for testing genome cleavage at target loci (FIG. 1G, FIG. 9)					
Genomic loci	Cas	SITE	5'/3'PAM	Protospacer sequence/ Spacer Sequence	SEQ ID NO of protospacer sequence/ spacer sequence
ABR001		TTTN.1	TTTG	TGTCTGAGGCTGGCCCTACGGTG	296
		TTTN.2	TTTG	ACCATCAGAGGACACTTGGATTG	297
		TTTN.3	TTTC	TGAACACATGCACGGCCACATTG	298
		TTTN.4	TTTG	CCTCTGGGTAAGTTGCCAAAGAA	299
		TTTN.6	TTTC	ACACCTTATAGGAAAACCAAGTGA	300
		TTTN.7	TTTC	CTATAAGGTGTGAAAGTCTGGAT	301
		TTTN.8	TTTT	CCTATAAGGTGTGAAAGTCTGGA	302
		TTTN.9	TTTG	TAGAAGGGATATACAAAGTGGAA	303
		TTTN.10	TTTG	TATATCCCTTCTACAAATTCCTC	304
		TTTN.11	TTTC	CACCTTGTATATCCCTTCTACAA	305
		TTTN.12	TTTG	GTGTCTATTTCCACTTTGTATAT	306
		VTTN.1	CTTA	CTGGAAGGCACCTGGCATCT	307
		VTTN.2	CTTA	TAGGAAAACCAAGTGTCTG	308
		VTTN.3	CTTC	TCATCGTCTGCTCCTCCTCT	309
		VTTN.4	ATTC	TTGGCAGGATGGCTTCTCAT	310
		VTTN.5	ATTC	ACCGGTGCCCTGOGTGTAGA	311
		VTTN.6	G TTC	AGAAAGGCTGCTGATGACAC	312
		VTTN.7	G TTC	TAGATGCTGTCCGAGGCAGT	313
		VTTN.9	G TTC	TTTGGCAACTTACCCAGAGG	314
		VTTN.10	CTTC	CAGTAAGATTTGGTGTCTAT	315
		VTTN.11	ATTC	CATGAGCATGCAGAGGTGAG	316
		VTTN.12	ATTC	CTCCTCAGTTGTGAGCCCAT	317
		VTTN.13	CTTC	TACAAATTCCTCCTCAGTTG	318
		VTTN.14	ATTC	ACAGCCAACCACTCCGGCCC	319
		VTTN.15	ATTC	CACCACGGCTGTCTGCACCA	320
		VTTN.16	ATTC	CTTGGGATTGGTGACGACAG	321
		VTTN.17	CTTC	TCTCATAGGTGGTATTACACA	322
		VTTN.18	CTTG	CTGGAATCGTATTGTGTCT	323
		VTTN.19	CTTG	GCAGGATGGCTTCTCATCGT	324
		VTTN.20	ATTG	ATGGCAGGACTGCCTCGGAC	325
		VTTN.21	CTTG	GATTACCCGGTGCCCTGGGT	326
		VTTN.22	CTTG	GCATCTCCCCATTCCATGAG	327
		VTTN.23	GTTG	TGAGCCCATGCAGCTCTCCA	328
		VTTN.24	ATTG	CCGCCCTGTGAGCCCTTAC	329
		VTTN.25	GTTG	GCTGTGAATACCACCTATGA	330
		VTTN.26	CTTG	GGATTGGTGACGACAGCCGT	331
		VTTN.27	ATTG	GTGACGACAGCCGTGGTGGA	332
		VTTN.28	ATTT	GTGTCTGAGGCTGGCCCTAC	333
		VTTN.29	CTTT	GACCATCAGAGGACACTTGG	334
		VTTN.30	ATTT	GCCTCTGGGTAAGTTGCCAA	335
		VTTN.31	CTTT	GGCAACTTACCCAGAGGCAA	336
		VTTN.32	ATTT	GGTGTCTATTTCACATTGT	337
		VTTN.33	CTTT	GTATATCCCTTCTACAAAT	338
Cas12i2 _{H1F1}		TTTN.1	TTTG	TGTCTGAGGCTGGCCCTACGGTG	339
		TTTN.2	TTTG	ACCATCAGAGGACACTTGGATTG	340
		TTTN.3	TTTC	TGAACACATGCACGGCCACATTG	341
		TTTN.4	TTTG	CCTCTGGGTAAGTTGCCAAAGAA	342
		TTTN.6	TTTC	ACACCTTATAGGAAAACCAAGTGA	343
		TTTN.7	TTTC	CTATAAGGTGTGAAAGTCTGGAT	344
		TTTN.8	TTTT	CCTATAAGGTGTGAAAGTCTGGA	345
		TTTN.9	TTTG	TAGAAGGGATATACAAAGTGGAA	346
		TTTN.10	TTTG	TATATCCCTTCTACAAATTCCTC	347
		TTTN.11	TTTC	CACCTTGTATATCCCTTCTACAA	348
		TTTN.12	TTTG	GTGTCTATTTCCACTTTGTATAT	349
		VTTN.1	CTTA	CTGGAAGGCACCTGGCATCT	350
		VTTN.2	CTTA	TAGGAAAACCAAGTGTCTG	351
		VTTN.3	CTTC	TCATCGTGTGCTCCTCCTCT	352
		VTTN.4	ATTC	TTGGCAGGATGGCTTCTCAT	353
		VTTN.5	ATTC	ACCGGTGCCCTGGGTGTAGA	354
		VTTN.6	G TTC	AGAAAGGCTGCTGATGACAC	355
		VTTN.7	G TTC	TAGATGCTGTCCGAGGCAGT	356
		VTTN.9	G TTC	TTTGGCAACTTACCCAGAGG	357
		VTTN.10	CTTC	CAGTAAGATTTGGTGTCTAT	358
		VTTN.11	ATTO	CATGAGCATGCAGAGGTGAG	359
		VTTN.12	ATTC	CTCCTCAGTTGTGAGCCCAT	360
		VTTN.13	CTTC	TACAAATTCCTCCTCAGTTG	361
		VTTN.14	ATTC	ACAGCCAACCACTCCGGCCC	362
		VTTN.15	ATTC	CACCACGGCTGTCTGCACCA	363
		VTTN.16	ATTC	CTTGGGATTGGTGACGACAG	364
		VTTN.17	CTTC	TCTCATAGGTGGTATTACACA	365

TABLE 9-continued

Sequences for testing genome cleavage at target loci (FIG. 1G, FIG. 9)					
Genomic loci	Cas	SITE	5'/3'PAM	Protospacer sequence/ Spacer Sequence	SEQ ID NO of protospacer sequence/ spacer sequence
		VTTN.18	CTTG	CTGGACTGGTATTGTGTCT	366
		VTTN.19	CTTG	GCAGGATGGCTTCTCATCGT	367
		VTTN.20	ATTG	ATGGCAGGACTGCCTCGGAC	368
		VTTN.21	CTTG	GATTCACCGGTGCCCTGGGT	369
		VTTN.22	CTTG	GCATCTCCCCATTCCATGAG	370
		VTTN.23	GTTG	TGAGCCCATGCAGCTCTCCA	371
		VTTN.24	ATTG	CCGCCCTGCTGAGCCCTAC	372
		VTTN.25	GTTG	GCTGTGAATACCACCTATGA	373
		VTTN.26	CTTG	GGATTGGTGACGACAGCCGT	374
		VTTN.27	ATTG	GTGACGACAGCCGTGGTGGA	375
		VTTN.28	ATTT	GTGTCTGAGGCTGGCCCTAC	376
		VTTN.29	CTTT	GACCATCAGAGGACACTTGG	377
		VTTN.30	ATTT	GCCTCTGGGTAAGTTGCCAA	378
		VTTN.31	CTTT	GGCAACTTACCCAGAGGCAA	379
		VTTN.32	ATTT	GGTGTCTATTTCACCTTGT	380
		VTTN.33	CTTT	GTATATCCCTTCTACAAATT	381

[0352] To further evaluate the specificity of hfCas12Max on endogenous genes in human cells, the applicant determined indel frequencies of P2RX5 and NLRC4 on-target and their corresponding in silico predicted off-target sites. Targeted deep sequence analysis showed that hfCas12Max had a higher on-target editing efficiency and similarly almost no indel activity at potential off target sites, compared to Ultra AsCas12a and LbCas12a (FIG. 10A-B; protospacer sequences/spacer sequences of SEQ ID NOs: 382-390 (not including the 5' PAM TTTN in blue) from upside to downside in FIG. 10A; protospacer sequences/spacer sequences of SEQ ID NOs: 391-397 (not including the 5' PAM TTTN in blue) from upside to downside in FIG. 10B. To save drafting, the sequences in black in FIGS. 10A and 10B refer to both the protospacer sequence (a DNA sequence) and the corresponding spacer sequence (an RNA sequence) with any “T” in the sequence when referring to a protospacer sequences standing for “T” and when referring to a such spacer sequence standing for “U”, although the assigned SEQ ID NOs: 382-397 in the sequence listing are annotated as DNA. To sufficiently detect off-target of hfCas12Max and to compare to other Cas proteins, the applicant used PEM-seq to quantify germline events (uncut or perfect rejoining) and editing events including indels and translocations events of TTR. 2 libraries.

[0353] Overall, these results demonstrate that hfCas12Max has high efficiency and specificity and is superior to SpCas9 and other Cas12 nucleases.

Example 9 Development and Evaluation of Base Editor Based on Dead xCas12i

[0354] The applicant further explored the base editing of xCas12i by generating a nuclease-deactivated xCas12i

mutant (dead xCas12i, dxCas12i). This was done by introducing single mutations (D650A, D700A, E875A, or D1049A) in the conserved active site of xCas12i based on alignment to Cas12i1 and Cas12i2 (FIG. 11A). The dsDNA cleavage activity (Indel %) of each of the four dxCas12i mutants (xCas12i-D650A, xCas12i-D700A, xCas12i-E875A, and xCas12i-D1049A) was measured in comparison to dead LbCpf1 (dLbCpf1-D832A) and xCas12i (WT), with N-terminally fusion of TadA8e^{P106W} (SEQ ID NO: 439, TadA8e. 1), and the results confirmed that all the four dxCas12i mutants had none or little dsDNA cleavage activity (FIG. 11B). xCas12i-D1049A had the lowest overall dsDNA cleavage activity and thus used in further base editor designs.

[0355] Then, initial versions of adenine base editor (ABE) and cytidine base editor (CBE) were constructed based on dxCas12i-D1049A (FIGS. 1H and 1J). dxCas12i-D1049A was C-terminally fused to TadA8e^{P106W} (SEQ ID NO: 439, TadA8e. 1) via a GS linker containing a XTEN linker (SEQ ID NO: 442) to form an initial version of ABE named TadA8e. 1-dxCas12i. dxCas12i-D1049A was C-terminally fused to human APOBEC3A^{H104A} (SEQ ID NO: 440, hA3A. 1) via a GS linker containing a XTEN linker (SEQ ID NO: 442), and fused to one UGI (SEQ ID NO: 441), to form an initial version of CBE named hA3A. 1-dxCas12i (FIGS. 1H and 1J). For the ABE, it further contained a N-terminal SV40 NLS (SEQ ID NO: 444) and a C-terminal BP NLS (SEQ ID NO: 443) flanking the fusion of the TadA8e^{P106W} and the dxCas12i-D1049A. For the CBE, it further contained a N-terminal BP NLS (SEQ ID NO: 443) and a C-terminal BP NLS (SEQ ID NO: 443) flanking the fusion of the hA3A. 1, the dxCas12i-D1049A, and the UGI.

TadA8e^{P106W},
SEVEFSHEYWMRHALTLAKRARDEREVPVGAVLVLLNNRVIGEGWNRAIGLHDPTAHAEIMALRQGGVL
MQNYRLIDATLYVTFEPCVMCAGAMIHSRIGRVVFGWRNSKRGAGSLMNVLNYPGMNHRVEITEGILA
DECAALLCDFYRMPRQVFNAQKKAQSSIN

SEQ ID NO: 439

-continued

TadA8e^{#10617},
SEVEFSHEYWMRHALTLAKRARDEREVPVGAVLVLNNRVIGEGWNRAIGLHDPTAHAEIMALRQGGLV
MQNYRLIDATLYVTTFEPCVMCAGAMIHSRIGRVVFGVRNSKRGAGSLMNVNLYPGMNRVEITEGILA
DECAALLCDFYRMPRQVFNAQKKAQSSIN
hAPOBEC3^{#1044},
MEASPASGPRHLMDPHIFTSNFNNGIGRHKTYLCYEVERLDNGTSVKMDQHRGFLHNQAKNLLCGFYG
RHAELRFLDLVPSLQLDPAQIYRVTFISYSPCFSAGCAGEVRAFLQENTHVRLRIFAARIFDYDPLYKEAL
QMLRDAGAQVSIMTYDEFKHCWDTFVDHQGCPFPQPDGLDEHSQALSGRRLRAILQNQGN
UGI,
TNLSDIEKETGKQLVIQESILMLPEEVEEVIGNKPESDILVHTAYDESTDENVMLLTSDAPEYKPWALVIQD
SNGENKIKML
XTEN linker,
SGSETPGTSESATPES
bpNLS (also known as BP NLS or bpSV40 NLS) 1, (doi: 10.1038/nature20565.),
KRTADGSEFESPKKKRKV
bpNLS 2,
KRTADGSESEPKKKRKV
SV40 NLS, from Betapolyomavirus macacae,
PKKKRKV
NP NLS (also known as *Xenopus laevis* Nucleoplasmin NLS or nucleoplasmin
NLS), (doi: 10.1126/science.abj6856.), also a bipartite NLS,
KRPAATKKAGQAKKKK
human U6 promoter, 241 bp,
gagggcctatttcccatgattccttcatatttgcataacgatacaaggctgttagagagataattggaattaatttgactgtaaacacaaagat
attagtacaaaatacgtgacgtagaaagtaataatttcttgggtagtttgcagttttaaattatgttttaaatggactatcatatgcttacc
gtaacttgaaagtatttcgatttcttggctttatatcttggtaaaggac
human CMV promoter, 204 bp,
gtgatgcggttttggcagtagcatcaatggcggttagcaggtttgactcacggggtttccaagctctccacccattgacgtcaatgggagttt
gttttggcaccaaaatcaacgggactttccaaaatgtcgtaacaactccgccccattgacgcaaatgggcggtaggcgtgtacgggtgggaggtct
atataagcagagct
bGH polyA signal, 208 bp,
ctgtgccttctagttgccagccatctgtgttttgcctctccctgccttcttgaccctggaaggtgccactcccactgtcctttcctaataa
aatgaggaaaattgcacgcattgtctgagtaggtgtcattctattctgggggtgggggtggggcaggacagcaaggggaggattgggaagagaa
tagcaggcatgctgggga
T5 EXO,
MSKSWGKFIEEEAEMASRRNLMIVDGTNLGFRFKHNSKKPFASSYSTIQSLAKSYSARTTIVLGDKG
KSVFRLEHLPEYKGNRDEKYAQRTEEEKALDEQFFEYLKDAFELCKTTFTFTIRGVEADMAAYIVKLI
GHLVDHVWLITDGDWDTLLTDKVSRSFSTTRREYHLRDMYEHNVDDVEQFISLKAIMGDLGDNIRGV
EGIGAKRGYNIIREFGNVLDIIDQLPLPGKQKYIQNLNASEELLERNLILVDLPTYCVDALAAGQDVLDPK
TKDILEIAEQ

-continued

CAG promoter (human CMV enhancer+ chicken β -actin promoter) (containing a hybrid intron),
 SEQ ID NO: 450
 cgttacataacttacggtaaatggcccgcttggtgacggcccaacgacccccgccattgacgtcaatagtaacgccaatagggactttccatt
 gacgtcaatgggtggagtatttacggtaaaactgcccacttgccagtagcatcaagtgtatcatatgccaaagtacgccccctattgacgtcaatgac
 ggtaaatggcccgctggcattGtgcccagtagcatgaccttatgggactttctacttgccagtagcatctacgtattagtcacgtcgtattaccat
 ggtcgagggtgagccccacgtttctgcttcactctccccatctccccccctccccacccccattttgtattttatttttttaattttttgtg
 cagcgatggggggggggggggggggggggggcgcgccagggcgggggggggggcgagggggggggggggggggcgagggcgagaggtgctggcgga
 gccaatcagagcgggcgcgctccgaaagtttcttttatggcgaggcgggcgggcgggcgccctataaaaagcgaagcgcgggggggggagtcg
 ctgacgctgctctcgcccgctgccccgctcgcgcgcgcctcgcgcccgcccgcccggtctgactgacgcggttactccacaggtgagcgg
 gcgggacggcccttctcctccgggtgttaattagctgagcaagaggaaggggttaagggttggttggtgggtattaatgtttaattacc
 tggagcacctgctgaaatcactttttttcag

[0356] The initial versions of ABE and CBE showed low base editing activity with frequencies of about 8% A-to-G and about 2% C-to-T, respectively (FIG. 1I, 1K). To address this, the applicant conducted a series of designs, including introduction of single and combined mutations for high cleavage activity into the PI and Rec domains of dxCas12i (FIG. 12 and FIG. 13A), which resulted in significantly increased A-to-G editing activity.

[0357] As shown in FIG. 1I, TadA8e. 1-dxCas12i-v1.2 (N243R) achieved significantly higher A-to-G base editing efficiency than TadA8e. 1-dxCas12i (initial version) at sites A9, A11, A19, and A20 of the KLF4 locus, indicating that the introduction of a mutation (e.g., N243R) that has been demonstrated to improve on-target dsDNA cleavage activity can also improve the A-to-G base editing of the base editor comprising the dCas12i and a deaminase domain. Further, TadA8e. 1-dxCas12i-v2.2 (N243R+E336R) achieved significantly higher A-to-G base editing efficiency than TadA8e. 1-dxCas12i-v1.2 (N243R) at sites A7, A9, A11, A19, and A20 of KLF4, further confirming that the introduction of a mutation (e.g., E336R) that has been demonstrated to improve on-target dsDNA cleavage activity can also improve the A-to-G base editing of the base editor comprising the dCas12i and a deaminase domain.

[0358] TadA8e. 1-dxCas12i-v2.2 (D1049A+N243R+E336R) achieved 50% activity at A9 and A11 sites of the KLF4 locus, markedly higher than the 30% activity of TadA8e. 1-dLbCas12a (FIG. 1I, FIG. 13B-C). At target sites within PCSK9 and TTR, TadA8e. 1-dxCas12i-v2.2 showed a similarly increased efficiency to mediate A-to-G transitions, and higher than TadA8e. 1-dLbCas12a at PCSK9 site (FIG. 15).

[0359] To test whether the orientation of deaminase fusion affects the base editing efficiency, the applicant constructed dxCas12i-ABE by fusing the TadA8e. 1 to N or C terminus of dxCas12i and found that TadA8e. 1 at C terminus of dxCas12i showed slightly higher activity than N terminus (FIG. 14).

[0360] The applicant then further engineered the NLS, linker, and TadA8e. 1 protein (return back to TadA8e (SEQ ID NO: 461; TadA8e^{H106V})) (FIG. 12; FIG. 13A) to produce v3.1-v3.8 and v4.1-v4.4, where TadA8e-dxCas12i-v4.3 exhibited a nearly 80% A-to-G editing efficiency and >95% editing purity, which is significantly higher than TadA8e.

1-dxCas12i-v2.2, indicating that the base editing efficiency can also be improved by specific selections of the NLS, linker, and deaminase domain (FIG. 1H-1I, FIG. 13D-13E). The applicant named TadA8e-dxCas12i-v4.3 as dCas12Max-ABE (SEQ ID NO: 463), which contains, from N-terminal to C-terminal, Methionine (M), bpNLS 1 (SEQ ID NO: 443), TadA8e-W106V (SEQ ID NO: 461), bpNLS 1-containing GS linker (SEQ ID NO: 465), xCas12i-N243R+E336R+D1049A (SEQ ID NO: 466), and npNLS (SEQ ID NO: 445).

[0361] To further characterize the base editing activity of dCas12Max-ABE, the applicant performed 21 sites with TTN PAM, 13 sites with ATN PAMs and 13 sites with CTN PAMs (Table 10). It was observed that dCas12Max-ABE exhibited significant A-to-G activity at sites with TTN PAM (FIG. 16).

[0362] Similarly for CBE, hA3A. 1-dxCas12i-v1.2 (N243R), hA3A. 1-dxCas12i-v2.2 (N243R+E336R), and hA3A. 1-dxCas12i-v3.1 (N243R+E336R-bpNLS) showed consistently elevated C-to-T editing efficiency along with >95% editing purity at RUNX1, DYRK1A, and SITE4 locus, even higher than hA3A. 1-dLbCas12a at RUNX1 and DYRK1A (FIG. 1J-K and FIG. 17). The applicant named hA3A. 1-dxCas12i-v3.1 (N243R+E336R-bpNLS) as dCas12Max-CBE (SEQ ID NO: 464), which contains, from N-terminal to C-terminal, Methionine (M), bpNLS 1 (SEQ ID NO: 443), hAPOBEC3_{H104A} (SEQ ID NO: 440), bpNLS 1-containing GS linker (SEQ ID NO: 465), xCas12i-N243R+E336R+D1049A (SEQ ID NO: 466), a short GS linker, SV40 NLS (SEQ ID NO: 444), a short GS linker, UGI (SEQ ID NO: 441), a short GS linker, and bpNLS 2 (SEQ ID NO: 462).

[0363] These results together demonstrate that both the engineered dxCas12i-based ABE and CBE exhibited high base editing activity in mammalian cells.

[0364] To save drafting, the sequences in Table 10 refer to both the protospacer sequence (a DNA sequence) and the corresponding spacer sequence (an RNA sequence) with any "T" in the sequence when referring to a protospacer sequences standing for "T" and when referring to a such spacer sequence standing for "U", although the assigned SEQ ID NOs: 398-438 in the sequence listing are annotated as DNA.

TABLE 10

Sequence of target loci for A to G frequency at different sites (FIG. 16)					
Genomic loci	ABE	SITE	5'/3'PAM	Protospacer/ Spacer Sequence	SEQ ID NO of Protospacer/ Spacer Sequence
TTR	TTN	site1	CTTC	AGCACCACCACGTAGGTGCC	398
		site2	CTTC	CTGGTGAAGATGAGTGGCGA	399
		site3	CTTG	AAGTTGCCCATGTGCGACTA	400
		site4	GTTG	CCCATGTGCGACTACATCGA	401
		site5	TTTG	CCCAGAGCATCCCGTGGAAC	402
		site6	TTTC	CCGGTGGTCACTCTGTATGC	403
		site7	GTTG	AGCACGCGCAGGCTGCGCAT	404
		site8	GTTA	GCGGCACCTCATAGGTGAG	405
		site9	GTTG	GGGCCACCAATGCCCAGGAC	406
		site10	ATTG	GTGGCCCCAACTGTGATGAC	407
		site11	ATTG	GTGCCTCCAGCGACTGCAGC	408
		site12	ATTC	ACCCCTGCACCAGGCATTGC	409
		site13	GTTT	CCTGAGGACCAGCGGGTACT	410
		site14	GTTG	GTGGCAGTGGACACGGGTCC	411
		site15	GTTG	TCTACGGCGTAGGCCCCAG	412
	ATN	site1	AATC	CAAGTGTCTCTGATGGTCA	413
		site2	GATG	GTCAAAGTTCTAGATGTGT	414
		site3	GATG	CTGTCCGAGGCAGTCTTGCC	415
		site4	AATG	TGGCCGTGCATGTGTTGAGA	416
		site5	CATG	TGTTTCAGAAAGGCTGCTGAT	417
		site6	GATG	ACACCTGGGAGCCATTGACC	418
		site7	GATT	CACCGGTGCCCTGGGTGTAG	419
		site8	CATC	AGAGGACACTTGGATTACCC	420
		site9	CATC	TAGAACTTTGACCATCAGAG	421
		site10	GATG	GCAGGACTGCCTCGGACAGC	422
		site11	CATT	GATGGCAGGACTGCCTCGGA	423
		site12	CATG	CACGGCCACATTGATGGCAG	424
		site13	CATC	AGCAGCCTTTCTGAACACAT	425
	CTN	site1	CCTC	TGATGGTCAAAGTTCTAGAT	426
		site2	TCTG	ATGGTCAAAGTTCTAGATGC	427
		site3	GCTG	TCCGAGGCAGTCTGCGCATC	428
		site4	GCTG	ATGACACCTGGGAGCCATTT	429
		site5	CCTG	GGAGCCATTTGCCTCTGGGT	430
		site6	CCTC	TGGGTAAGTTGCCAAAGAAC	431
		site7	ACTT	GGATTACCCGGTGCCCTGGG	432
		site8	ACTT	TGACCATCAGAGGACACTTG	433
		site9	TCTA	GAACCTTGACCATCAGAGGA	434
		site10	CCTC	GGACAGCATCTAGAACTTTG	435
		site11	ACTG	CCTCGGACAGCATCTAGAAC	436
		site12	GCTC	CCAGGTGTCATCAGCAGCCT	437
		site13	ACTT	ACCCAGAGGCAAATGGCTCC	438

Example 10 Evaluation of RNP Delivery of hfCas12Max in T Cells

[0365] To explore the therapeutic potential application of hfCas12Max, the applicant delivered hfCas12Max RNP targeting TRAC in CD3+ T cells (FIG. 2A). Beforehand, the applicant tested hfCas12Max RNP targeting TTR and TRAC in HEK293 cells, and it was found that the gene editing efficiency was increased following increasing dose of RNPs, with unaffected cellular viability and proliferation (FIG. 18A-C). The applicant achieved about 90% dsDNA cleavage activity and >95% viability at 3.2 μ M dose for TRAC (FIG. 18A-C) in HEK293 cells. Three spacer sequences (TRAC_sg. 1, TRAC_sg. 2, and TRAC_sg. 3) were designed to target TRAC (Table 5), and both TRAC_sg. 2 and TRAC_sg. 3 generated ~90% editing at both 1.6 and 3.2 μ M doses along with ~80% viability (FIG. 2B) in CD3+ T cells. Flow cytometric analysis showed that TRAC expression was detected to be reduced to a level of 2.54% and 3.72% in CD3+ T cells post 5 days post electroporation treated with RNPs comprising TRAC_sg. 2 or TRAC_sg. 3, respectively, compared to 96.6% in untreated cells (FIG.

2C). The guide RNA used in this Example was composed of 5' DR-T1-spacer sequence-DR-T2-spacer sequence-3'.

Example 11 Evaluation of LNP Delivery of hfCas12Max In Vivo

[0366] To assess the feasibility of the hfCas12Max or the base editor of in vivo gene editing, the applicant delivered a guide RNA and a mRNA encoding hfCas12Max or the base editor by LNP packaging to the liver of C57 mouse via tail intravenous injection (FIG. 2D). The applicant targeted the exon 3 in the murine transthyretin (Ttr) gene (Ttr_sg12 in Table 5) by gene editing (dsDNA cleavage) and base editing (FIG. 2E). Robust editing efficiencies were detected at four concentration and nearly 100% at 1 μ g dose in N2a cells (FIG. 2F). Similarly, targeted deep sequence analysis indicated that the editing efficiencies of murine liver were approximately 70% at the dose of 0.3 and 0.5 milligrams per kilogram (mpk), equivalent to saturation (FIG. 2G). Further, through the LNP packaging delivery, TadA8e-dxCas12i-v4.3 (dCas12Max-ABE) achieved approximately 25% A-to-G efficiency at A13 in Ttr locus in murine liver at 3

mpk dose (FIG. 2H). The guide RNA used in this Example was composed of 5' DR-T1-spacer sequence-DR-T2-spacer sequence-3'.

[0367] In addition, the applicant injected hfCas12Max mRNA with two gRNAs (with spacer sequences of Ttr_sg3 and Ttr_sg12 in Table 5) targeting Ttr gene into murine zygotes, which were cultured to blastocyst stage for genotyping analysis (FIG. 19A). Targeted deep sequence analysis showed that most zygotes were edited and some up to 100% (FIG. 19B). These results indicate that hfCas12Max mediated robust ex vivo and in vivo gene editing, showing significant potential for disease modeling and therapies.

[0368] Mis-folding and aggregation of transthyretin (TTR) is associated with amyloid diseases, including transthyretin-related wild-type amyloidosis (ATTRwt), transthyretin-related hereditary amyloidosis (ATTRm), familial amyloid polyneuropathy (FAP), and familial amyloid cardiomyopathy (FAC). Gene silencing of TTR to reduce TTR protein production may have therapeutic effects in TTR-associated amyloid diseases. The high-efficiency cleavage of TTR target sites in mice in this Example demonstrates that the CRISPR-Cas12i system of the disclosure has very promising prospects for the treatment of TTR-related amyloid diseases, such as ATTR (e.g., ATTRwt or ATTRm).

Example 12: Screening of xCas12i Mutant with Nickase Activity

[0369] To screen xCas12i mutant with nickase activity (i.e., having ssDNA cleavage activity and substantially lacking dsDNA cleavage activity), xCas12i mutant in Tables 11-14 were designed and tested for their nickase activity and dsDNA cleavage activity, by using the reporter system for dsDNA cleavage activity in Example 1 and a reporter system for nickase activity established based on the reporter system for dsDNA cleavage activity in Example 1 wherein the insertion sequence was replaced with an insertion sequence containing, from 5' to 3', a 5' PAM, a protospacer sequence (SEQ ID NO: 43), a linker, a target sequence (SEQ ID NO: 44), and a reverse complementary sequence of the 5' PAM.

[0370] When the xCas12i mutant has only nickase activity, it does not generate green fluorescence with the reporter system for dsDNA cleavage activity but can generate green fluorescence with the reporter system for nickase activity. When the xCas12i mutant has dsDNA cleavage activity, it can generate green fluorescence with both the reporter systems for nickase activity and dsDNA cleavage activity. So the reporter system for nickase activity indicates the sum of the dsDNA cleavage activity and nickase activity. The nickase activity is calculated as green fluorescence from the reporter system for nickase activity minus green fluorescence from the reporter system for dsDNA cleavage activity. Nickase preference was calculated as nickase activity/dsDNA cleavage activity.

[0371] It was observed that xCas12i-W896R, xCas12i-S924R, and xCas12i-S925R exhibited significant nickase activity relative to xCas12i (WT) and substantially lacked dsDNA cleavage activity compared with xCas12i (WT).

TABLE 11

Mutant	Nickase (ssDNA cleavage) activity (%)	dsDNA cleavage activity (%)	Nickase activity/dsDNA cleavage activity
NT	0.000	0.020	0.000
Blank	0.000	0.020	0.000

TABLE 11-continued

Mutant	Nickase (ssDNA cleavage) activity (%)	dsDNA cleavage activity (%)	Nickase activity/dsDNA cleavage activity
xCas12i	-0.300	76.100	-0.004
xCas12i-W896R	30.130	4.970	6.062
xCas12i-S924R	22.300	26.800	0.832
xCas12i-S925R	6.650	5.350	1.243

[0372] Further mutagenesis was conducted at W896, S924, or S925 of xCas12i to generate the mutants in Tables 12-14. It was observed that eight xCas12i mutants, W896R, W896P, W896K, S924F, S924D, S924E, S924H, and S925T, achieved both significant nickase preference (Nickase activity/dsDNA cleavage activity >1.0) and high nickase activity (higher than 20%).

TABLE 12

xCas12i-W896 mutants			
Mutant	Nickase (ssDNA cleavage) activity (%)	dsDNA cleavage activity (%)	Nickase activity/dsDNA cleavage activity
W896G	-3.100	72.900	-0.043
W896A	6.500	75.700	0.086
W896V	-0.300	64.300	-0.005
W896L	13.900	61.300	0.227
W896I	-0.600	74.700	-0.008
W896M	0.500	76.800	0.007
W896F	5.800	74.100	0.078
W896W	-0.400	80.300	-0.005
W896P	32.170	8.030	4.006
W896S	0.000	72.000	0.000
W896T	0.600	67.200	0.009
W896C	2.200	72.800	0.030
W896Y	2.300	67.700	0.034
W896N	0.700	63.700	0.011
W896Q	1.500	69.800	0.021
W896D	-1.900	49.200	-0.039
W896E	11.900	58.400	0.204
W896K	37.500	14.700	2.551
W896H	3.100	68.000	0.046

TABLE 13

xCas12i-S924 mutants			
Mutant	Nickase (ssDNA cleavage) activity (%)	dsDNA cleavage activity (%)	Nickase activity/dsDNA cleavage activity
S924G	0.100	70.900	0.001
S924A	18.000	53.400	0.337
S924V	11.100	53.500	0.207
S924L	2.800	54.500	0.051
S924I	14.900	41.800	0.356
S924M	8.100	49.600	0.163
S924F	26.600	15.500	1.716
S924W	3.530	8.670	0.407
S924P	15.500	10.100	1.535
S924S	-5.000	82.200	-0.061
S924T	2.800	78.200	0.036
S924C	2.700	70.700	0.038
S924Y	11.000	11.000	1.000
S924N	8.400	71.800	0.117
S924Q	23.400	29.200	0.801

TABLE 13-continued

xCas12i-S924 mutants			
Mutant	Nickase (ssDNA cleavage) activity (%)	dsDNA cleavage activity (%)	Nickase activity/dsDNA cleavage activity
S924D	29.000	12.700	2.283
S924E	22.800	15.400	1.481
S924K	14.600	41.600	0.351
S924H	36.000	25.300	1.423

TABLE 14

xCas12i-S925 mutants			
Mutant	Nickase (ssDNA cleavage) activity (%)	dsDNA cleavage activity (%)	Nickase activity/dsDNA cleavage activity
S925G	28.700	40.900	0.702
S925A	-0.600	12.700	-0.047
S925V	3.000	3.560	0.843
S925L	6.650	5.750	1.157
S925I	9.000	5.800	1.552
S925M	5.350	5.150	1.039
S925F	7.530	6.870	1.096
S925W	3.330	9.770	0.341
S925P	4.700	9.700	0.485
S925S	-0.300	76.300	-0.004
S925T	32.000	21.200	1.509
S925C	7.600	8.000	0.950
S925Y	7.780	5.820	1.337
S925N	1.300	12.300	0.106
S925Q	6.230	5.970	1.044
S925D	9.320	6.180	1.508
S925E	11.690	6.610	1.769
S925K	6.700	10.800	0.620
S925H	6.100	10.600	0.575

[0373] In the Examples, it was demonstrated that the Type V-I Cas12i system enables versatile and efficient genome editing in mammalian cells. Among others, xCas12i that shows high editing efficiency at TTN-PAM sites was identified. By semi-rational design and protein engineering of its PI, REC, RuvC domains, a high-efficiency, high-fidelity variant, hxCas12Max, was obtained which contains N243R, E336R, and D892R substitutions. It was demonstrated that the introduction of N243R in the PI domain and E336R at REC domain significantly increased editing activity and expanded PAM recognition. Interestingly, D892R or G883R substitutions in the RuvC domain reduced off-target and retained on-target cleavage activity. The D892R substituted hxCas12Max was obviously more sensitive to mismatch, which suggests that D892R or G883R improved gRNA binding specificity. According to sequence alignment and predicted structure of xCas12i to Cas12i2, asparagine 892 is located on NUC domain, together with RuvC domain to

form a cleft, in which crRNA: DNA heteroduplex was located. The variant with D892R did not alter the on-target but eliminated off-target activity, probably due to arginine substitution of asparagine affecting the binding of non-target gRNA. The data of the disclosure suggests that a semi-rational engineering strategy with arginine substitutions based on the EGFP-activated reporter system could be used as a general approach to improve the activity of CRISPR editing tools.

[0374] Through engineering, the Cas12i system of the disclosure has achieved high editing activity, high specificity, and a broad PAM range, comparable to SpCas9, and better than other Cas12 systems. Given its smaller size, short crRNA guide, and self-processing features, the type V-I Cas12i system is suitable for in vivo multiplexed gene-editing applications, including AAV or LNP. Indeed, the data of the disclosure indicates type V-I Cas12i system mediates the robust ex vivo or in vivo genome-editing efficiencies via ribonucleoprotein (RNP) delivery and lipid nanoliposomes (LNP) delivery, respectively, demonstrating the great potential for therapeutic genome-editing applications.

[0375] In addition, it was confirmed that the type V-I Cas12i system can be used in base editing applications. For base editor, the dCas12i system shows high A-to-G editing at A9-A11 sites and even A19 site of KLF locus, and C-to-T editing at C7-C10 sites, which is similar to the dCas12a system but is distinct from the dCas9/nCas9 system. Comparable to dCas12a, dCas12i-BE exhibited higher base editing activity at KLF4, PCSK9, and DYRK1A loci, suggesting it may have more potential as a base editor. This suggests that the dCas12i system of the disclosure is useful for broad genome engineering applications, including epigenome editing, genome activation, and chromatin imaging.

[0376] In summary, the Cas12i system of the disclosure, which has robust editing activity and high specificity, is a versatile platform for genome editing or base editing in mammalian cells and could be useful in the future for in vivo or ex vivo therapeutic applications.

[0377] Various modifications and variations of the described products, methods, and uses of the disclosure will be apparent to those skilled in the art without departing from the scope and spirit of the disclosure. Although the disclosure has been described in connection with specific embodiments, it will be understood that it is capable of further modifications and that the disclosure as claimed should not be unduly limited to such specific embodiments. Indeed, various modifications of the described modes for carrying out the disclosure that are obvious to those skilled in the art are intended to be within the scope of the disclosure. This application is intended to cover any variations, uses, or adaptations of the disclosure following, in general, the principles of the disclosure and including such departures from the present disclosure come within known customary practice within the art to which the disclosure pertains and may be applied to the essential features herein before set forth.

SEQUENCE LISTING

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Sequence total quantity: 467
SEQ ID NO: 1             moltype = AA  length = 1080
FEATURE                  Location/Qualifiers
source                   1..1080
                        mol_type = protein

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organism = unidentified

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EAHVTDASLC	AINWFRPMSK	RYSKYATTF	NMLELFKEYS	GHEPDAYSKN	YLMNIDSDR	120
FVWVDCRKA	KDFAYQMELG	FHEFTVLAET	LLANSILVLN	ESTKANWAWG	TVSALYGGGD	180
KEDSTLKSKI	LLAFVDALNN	HELKTREIL	NOVCESLKYQ	SYQDMYVDFR	SVVDENGKK	240
SPNGSMPIVT	KFETDDLSD	NQRKAMISNF	TKNAAAKAK	KPIPYLDRK	EHMVSLCDEY	300
NVYAWAAIT	NSNADVARN	TRNLTFIGEQ	NSRRKELSVL	QTTTNEKAKD	ILNKINDNLI	360
QEVRYTPAPK	HLGRDLANLF	DTLKEKDINN	IENEEKQNV	INDCIEQYVD	DCRSLNRNPI	420
AALLKHISRY	YEDFSAKNFL	DGAKLNVLTE	VVNRQKAHPT	IWSEKAYTWI	SKFDKNRRQA	480
NSSLVGWVVP	PEEVHKEKIA	GQQSMMWVTL	TLLDDGKWVK	HHIPFSDSR	YSEVYAYNPN	540
LPYLDGGIIPR	QSKFGNKPTT	NLTAESQALL	ANSKYKANK	SFLRAKENAT	HNVRVSPNTS	600
LCIRLLKDSA	GNQMFQINHG	VLPQMGINHK	ITVGKPNYKI	EVGDRFLGFD	QNQSENHTYA	660
VLQRVSESSH	DTLHFNWDV	KVLEKGVTS	DVIVRDEVYD	QLSYEGVPYD	SSKFAEWRDK	720
RRRFVLENL	IQLEEGKTFL	TEFDKLNKDS	LYRWNNMYLK	LLRKAIRAGG	KEFAKIAKTE	780
IPELAVERRG	YEDFSAKNFL	SLKMIAFPKG	VVQSYFSVSG	CVDDASKKAH	DSMLFTMCA	840
ABEKRNTKRE	EKTNRASFI	LQKAYLHGCK	MIVCEDDLPV	ADGKTGKAQN	ADRMDCARA	900
LAKKVNDDGV	AMSCYRAIP	AYMSSHQDPF	VHMQDKKTSV	LRPRFMEVKN	DSIRDYHVAG	960
LRRMLNSKSD	ALHFCEALGV	SPELVKNKKT	HAAELGKHM	SAMLMWVRGG		1020
RVYIASKKLT	SDAKSVKCYG	EDMWQYHADE	IAAVNIAMYE	VCCQTGAFGK	KQKKSDELPG	1080

SEQ ID NO: 2 moltype = AA length = 1061

FEATURE Location/Qualifiers

source 1..1061

 mol_type = protein

 organism = unidentified

SEQUENCE: 2

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EDHVTADSLC	AINWFRPMSK	RYIKYATTF	KMLELFKEYS	GHEPDYTSKN	YLMNIVSDR	120
FVWVDCRKA	KDFANQMLSL	FHEFTTLSET	LLANSILVLN	ESTKANWAWG	AVSALYGGGD	180
KEDSTLKSKI	LLAFVDALNN	HELKTREIL	NHVCESLKYQ	SYQDMYVDFR	SVVDDKGNKK	240
SPNGSMPIVT	KFESDDLIGD	NQRKTMISSE	TKNAAAKASK	KPIPYLDILK	DHMISLCEEY	300
NVYAWAAIT	NSNADVARN	TRNLTFIGEQ	NTRRKELSVL	QTSTNEKAKD	ILNKINDNLI	360
PEVRYTPAPK	HLGRDLANLF	EMPEKEDINQ	IGNEEEKQNV	INDCIEQYVD	DCRSLNRNPV	420
AALLKHISGY	YEDFSAKNFL	DGAKLNVLTE	VVNRQKAHPT	ICSEKAYTWI	SKIDKNRRQA	480
NSSLVGWVVP	PEEVHKEKIA	GQQSMMWVTL	TLLDDGKWVK	HHIPFADSRY	YSEVYAYNPN	540
LPYLEGGIIPR	QSKFGNKPTT	NLTAESQALL	ANSKHKANK	TFLRAKENIT	HNVRVSPNTS	600
LCIRPLKDSA	GNQMFQINHG	MLPQMGINHR	ITVGKPNYKI	EVGDRFLGFD	QNQSENHTYA	660
VLQRVSESSH	DTLHFNWDV	KVLEKGVTS	DVIVRDEVYD	QLSYEGVPYD	SPKFTEWREK	720
RRKFVLENN	QSKFGNKPTT	TEFDKLNKDS	LYRWNNMYMK	LLRKAIRAGG	KEFAKITKAE	780
IPELGVMRFG	PMNLGSLSQV	SLKMIAAFKG	VIQSYFSVSG	CIDDASKKAH	DSMLFAFLCS	840
ADEKRNTKRE	EKTNRASFI	LQKAYSHGCK	MIVCEDDLPI	ADGKVGKAQN	ADRMDCARS	900
LAKKVNDDGV	AMSCYRAIP	AYMSSHQDPF	THMQDKKTSV	LRPRFMEVGK	DSIRDHHVAG	960
LRRMLNSKGN	TGTSVYYREA	ALRFCEALGV	LPELVKNKKT	HASELGKHM	SAMLMWVRGG	1020
RIYVASKKLT	SDAKSIKCYG	EDMWQYHADE	IAAINIAMYE	V		1061

SEQ ID NO: 3 moltype = AA length = 1075

FEATURE Location/Qualifiers

source 1..1075

 mol_type = protein

 organism = unidentified

SEQUENCE: 3

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DSELVCAIGW	FRLVDKVTWS	ENEIKQENLV	RQYETYSKE	ASEVIKTYLS	SPSSDKYVWI	120
DCRQKFLRFQ	RDLGTRNLSE	DFECMLFEQY	LRLTKGELDG	HTAMSNMFGT	KTKEKDRATKL	180
RYAARMKEWL	EANEETWEQ	YHQALQDKLD	ANTLEEAVDN	YKKGAGGSP	FFSYTLLNRG	240
QIDKKTHEQQ	LKKFNKVLKT	KSKNLNFPNK	EKLKQYLETA	IGIPVDAQVY	GQMFNNGVSE	300
VQPKTTRNMS	FSMEKLELLN	ELKSLNKTDG	FERANEVLNG	FFDSELHTTE	DKFNITSRYL	360
GGDRNNRPLK	LYELWKKKEG	DREEGIQQFS	QAIQDKMGQI	PVKNVRLRYI	EFRETVSAD	420
FEAAAKANQL	EEKITRTKAH	PVVISNRYWT	FGSSALVGN	MPADKMHKQ	YAGQSPKMWL	480
EAEHLHYDGK	VKHLPLPFYNA	RFFEEVYCYH	PSVAEVTPEK	TKQFGYAIK	DIPADVSVVL	540
KDNPKYKATK	RFLRAISNPV	ANTVDVNKPT	VCSFMIKREN	DEYKLVINRK	IGVDRPKRIK	600
VGRKVMGYDR	NQTASDTYWI	GELVPHGTTG	AYRIGESVQ	YIKSGPVLSS	TQGVNDSTTD	660
QLIYNGMPSS	SERFKAWKKS	RMSFIRKLIR	QLNAEGLESK	GQDVVPENPS	SFDVVRGETLY	720
VFNSTNYMKAL	VSKHRKAKKP	VEGILEEIEA	LTSKAKDSCS	LMRLSSLSDA	AMQGIASLKS	780
LINSYFNKNG	CKTIEDKEKF	NPDLYVKLVE	VBQKRTNKRK	EKVGRIGAGS	EQALLANGVD	840
VVIGEADLGE	VKGKSKKQK	SRNMDWCAQ	VAERLEYKLT	FHCIGYFGVN	PMYTSHQDPF	900
EHRRVADHLV	MRARFEVNV	SNVSEWHMRN	FSNYLRADSG	TGLYKQATL	DFLKHYDLEE	960
HADDEKQNI	KFYDFRKILE	DKQLTSVIVP	KRGGRIMAT	NPVTSSTPVP	TYAGKTYNRC	1020
NADEVAAANI	AISVLAPHSK	KEEKDKIPI	ISKPKSKNT	PKARKNLKTS	QLPQK	1075

SEQ ID NO: 4 moltype = AA length = 1003

FEATURE Location/Qualifiers

source 1..1003

 mol_type = protein

 organism = unidentified

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SEQUENCE: 4

MASKHVVRPF	NGKVTATGKR	LAYLEETPHY	LEKAAGGVST	LFAALGSYLD	ATTISNLINK	60
NQDLAVVIFR	YHVVPKGEAH	TLPVGTDMVS	RFVADYGMEP	NEFQRAYLDS	PIDQEKYCWQ	120
DNRDVGCWLG	EQLGVSEADM	RAIAVTFYNN	QMLYDCVKGT	SGSNAVSLLF	SGSKKSDYSM	180
KGVIAGKAAS	VLAKYRPATY	QDARKMILEA	NGFTSVKDLV	TSYGITGRSS	ALQIFMEGIE	240
SGPISSKTL	ARIKFTEDS	ERNRKNLVP	HAGAIRNWLI	EQAGSSVENY	QMAWCEVYGN	300
VSADWNAAKVE	SNFNFVAEKV	KALTELSNIQ	KSTPDLGKAL	KLFEYLTTC	QDEFIAIPYH	360
FSVMEEVRME	MATGREFNDA	YDDALNSLDM	ESKQPIQPLC	KFLIERGGSI	SFDTFKSAAK	420
YLKTQSKIAG	RYPHFPVKGN	QGFTFGSKNI	WAAINDPMME	YADGRIAGGS	AMMWVTATLL	480
DGKKWVRHHI	PFANTRYFEE	VYASKKGLPV	LPCARDGKHS	FKLGNNLSVE	RVEKVKEGGR	540
TKATKAQERI	LSNLTHNVQF	DSSTTFIIRR	QEESEFVICVN	HRHPAPLMKK	EMEVGDKIIG	600
IDQNVTAPTT	YAIIVERVASG	GIERNGKQYK	VTAMGAISSV	QKTRGGEVDV	LSYMGVELSD	660
SKNGFQSLWN	KCLDFVTKHG	TENDVKYNN	TAVWANKLYV	WHKMYFRLK	QLMRRAKDLK	720
PPRDHLQHL	FHPNLSPLQR	HSLSLTSLEA	TKIVRNCIHS	YFSLGLKTL	DERKAADINL	780
LEVLEKLYAG	LVERRKERTK	LTAGLLVRLC	NEHGISFAAI	EGDLPVVVGE	KSKAANNTQQ	840
DWTARELEKR	LSMAEVVGI	KVIAVLPHYT	SHQDPFVYSK	NTKKMRCRWN	WRTTKTFTDR	900
DALSIRRLS	KPETGTNLYY	QKGLKAFPAK	HGLDLAEMKK	RKDAQWYLER	IQDKNFLVPM	960
NGGRVYLSV	KLAKGETIDM	GGELIYLND	DQVAALNVLL	VKI		1003

SEQ ID NO: 5 moltype = AA length = 1018
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source 1..1018
 mol_type = protein
 organism = unidentified

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EEIETTSTFC	SFRVLVGHEN	FTMPLTNTMI	KHFQKTFNIN	PSEQQAIYLS	SGFDSKRYRW	120
QDTSFVSRNF	ANKCRLTNQE	FQEFAEQALL	NMCFIGCSGS	PGATNAVSIQ	FGTGEKSDYQ	180
RKSQIAKIAA	DTLENHKPST	YESARLMVLN	TLGHKTIEDC	VNDYGAIGAK	SAFRLFMESK	240
ETGPIITSEQL	TTKIKKFRED	HKNSIKKQL	PHVEKVRNAL	LSQFKEQYLP	SAWAEAWCNI	300
MGEFNSKLSN	NNNFIDQKTK	MVNDNCDNIK	SNPQLDKAVN	MLDEWKYKNW	DDNSAIHPYH	360
IGDLKKLMAI	FNINNEGTFD	ERFSASWEQF	STSLEYGEKP	PVRDLLAHII	KNMNDLTYTD	420
VINAAKFLKL	QDNIRNKYPH	PFVMPNKGCT	FGKDNLWGEI	NDPTAKIKST	EEVAGQRPMM	480
WLTAKLLDNG	KWVEHHIPFA	SSRYFAEVVY	TNPALPTLPI	ARDGKHSYKL	TKTIDANTAK	540
TLVNNPRDKA	AKLIARTKAN	TTHNVKWIKP	TYRIQKENNQ	FVITINHRHP	CITPPKEIIL	600
GDRIILSFDQN	ETAPTAFSIL	EKTTKGTEFC	GHHIKVLKTG	MLEAKIKTSK	KSIDAFPTYM	660
PMEDDHASGF	PTLLNICEKF	ISENGDEKDK	SFSSRKLFPK	RSLYFPFHGSH	FDLLKKMIRK	720
AKNDPKKLKL	VRIHINEILF	NSNLSPIKLH	SLSIHSMENT	KKVIAAISCY	MNVHEWKTID	780
EQKNADITLY	NAKEKLYNNL	VNRKRKERVK	TAGMLIRLAR	ENNRCPFMVGE	AELPTQQQKG	840
SKKNNNSKQD	WAKMDIQRRC	EDMCCEVVGK	WNGVTPHNST	HQNPFIYKNT	SGQMQRCRYS	900
LVKKSEMTDK	MAEKIRNLIH	AEPVGTAYY	REGILEFAKH	HGLDLGMMKK	RRDAKYDNL	960
PDEFLLPTRG	GRIYLSNQ	GGNETIVING	KKYFVNQADQ	VAAVNIGLLY	LLPKKNQS	1018

SEQ ID NO: 6 moltype = AA length = 1047
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source 1..1047
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 organism = unidentified

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SGAVGSPDLW	SAVINWFRIVP	LAEAGDARVG	QASLKNLFRG	YAGHEPDEEA	SIYMESRVDD	120
KRAHWVDCRA	MFRMALEECG	LEEAQLASDV	FALASREVIV	FKDGEINGWG	IASLLFGEGE	180
KADSQKQVAL	LRSVRLALEG	DYATYEELSG	LMLAKTGASS	GSDLLDEYKR	SEKGGSSSGR	240
HPFFDEVFRR	GGRVKQERE	RLKSCDTAI	QKQGQALPLS	HVASWRQWFL	RRVTLLRNRR	300
QESFAVCITN	ALMDLQPKNL	RNVHYVTNPK	SEKDKGVLEL	RVDVKNNEGP	DVAGAQAQVD	360
AYMARLAPDL	RFSVMPRHG	SLKDLALWA	KLGRDEAIEE	YLEGYEGPFS	KRPIAGILQI	420
IHAHRGKVG	DSLRLAARLN	RAMDRLERKR	AHACAAGNKG	YVYGKSSMVG	RINPQSLEV	480
GRKSGRSPMM	WVTLDLVGD	RFAQHHLFPQ	SARFFSEVYC	HGDGLPATRV	PGMVRNRRNG	540
LAIGNGLGEG	GLSALRAGSD	RRKRANKRTL	RALENITHNV	EIDPSTSFTL	REDGIIISHR	600
IEKIEPKLVA	FGDRALGFDL	NQTGAHTFAV	LQKVDSGGLD	VGHSRVIVL	TGTVRSICKG	660
NQASGGRDYD	LLSYDGPEDR	DGAFTAWRSD	RQAFLSAIR	ELPTPAEGEK	DYKADLLSQM	720
ASLDHYRRLY	AYNRKCLGIY	IGALRRATRR	QAVAAFKDEI	LSIANHRCGP	LMRGSLSVNG	780
MESLANLKG	ATAYLSKFKD	SKSEDLSSKD	EEMADLYRAC	ARRMTGKRKE	RYRRAASEIV	840
RLANEHGCLEF	VFGEKELPTT	SKGNKSKQNG	RNTDWSARAI	VKAVKEACEG	CGLGFKPVVK	900
EYSSLTDPFE	RDGDRPALR	KRFAKVAAPD	SELPPRLTKA	VGSYVKNALK	ADKAEKKQTC	960
YQGAIEFCS	RHGIDVRKAT	DKAIRKAVRG	SSDLLVPFDG	GRTFLLSTRL	SPESRKVEWA	1020
GRTLYEFPD	MVAANIACR	GLEPRKA				1047

SEQ ID NO: 7 moltype = AA length = 1029
FEATURE Location/Qualifiers
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 organism = unidentified

SEQUENCE: 7

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YNDELYAWQD	TLVLRARQIIA	ETGLTEDQFR	APAHACFKNG	NIIGCAGGPG	ASNAISGIFG	180
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LILFMEDTKT	GAPPSNEFDY	KLKLLKEDAE	RVGRKGIPH	RDVIASVLRN	QTGADIEYNS	300
KAWCESYCCA	VSEYNSKMSN	NVRFAFEKSL	DLTKLDETIR	ETPKISEAML	VFENYMARID	360
ADLRFIIVSKH	HLGNLAKFRQ	TMMHVSASEF	EEAFKAMWAD	YLAGLEYGEK	PAICELVRYV	420
LTHGNDLPVE	AFYAACKFLS	LDDKIKNRYP	HPFVPGNKGY	TFGAKNLWAE	INDPFKPIRQ	480
GNPEVAGQRP	MMWATADLLD	NNKWLHHP	FASSRYFEEV	YYTDPSPPTA	QKARDGKHGY	540
RLGKVLDEAA	RERLKANNRQ	RKAAKAIERI	KANCEHNVAV	DPTTTFMLQL	DSEGNVKMTI	600
NHRHIAIRAP	KEIGVGDVRI	GIDQNETAPT	TYAILERTEN	PRDLEYNGKY	YRVVKMGSVT	660
SPNVSKYRTV	DALTYDGVSL	SDDASGAVNF	VVLCREFFAA	HGDDEGRKYL	ERTLGWSSSL	720
YSFHGNYFKC	LTQMRRRSAR	SGGDLTVYRA	HLQQILFQHN	LSPLRMHSLS	LRSMESTMKV	780
ISCMKSYMSL	CGWKTADARI	ANDRSLFEAA	RKLYTSLVNR	RTERVVRVTA	ILMRLCLEHN	840
VRFIHMEDEL	PVAETGSKKK	SNGAKMHWCA	RELAVRLSQM	AEVTSVKFTG	VSPHYTSHQD	900
PFVHSKTSKV	MRARWSWRNR	ADFTDKDAER	IRTILGGDDA	GTKAYYRSAL	AEFASRYGLD	960
MEQMRKRRDA	QWYQERLPET	FIIPQRGGRV	YLSHDLGSG	QKVDGIYGR	AFVNHADAVA	1020
ALNVALVRL						1029

SEQ ID NO: 8 moltype = AA length = 1029
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 mol_type = protein
 organism = unidentified

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QDNRLLAQNI	ASQLNISETQ	FQEIHAHLS	NNLYIGQTAS	NAAANFISQV	TGTGQKAPKA	180
ARLDVLFQTN	QALAKTQPTT	FGQLQQIIVQ	ACGESTTDAV	LAKFGNKGAA	TSLQLALKTD	240
PNTTLDQKKY	EALQKKFAED	ETKYRNKVDI	PHKTQLRNL	LNTSNQFCNW	HTKPAIEAFK	300
CAIADIQSKV	SNNLRIMQEK	AKLYEAFRNV	DPQVQIAVQA	LENHMTLEE	PYAPYAHSG	360
SVKDFYEDLN	NGSNLDEAIQ	TIVHSDNFN	RKPDNPWLRI	IAPLHSSHSA	SQIMEAVKYL	420
SSKQDYELRK	PFPFVATNLP	ATYGKFNIPG	TLNPPDLSLH	GRLNGSHSNM	WLTALLLDGR	480
DWKNHHLCPA	SSRYFEEVYF	TNPSLPTTDK	VRSPKCGFTL	KSVLDSEAKD	RIRNAPKSRT	540
KAVKAIERIK	ANSTHNVAWN	PETSPQMQR	NDEFYITINH	RIEMEKIPGQ	KKTDDGFTIH	600
PKGLFAILKE	GDRILSQDLN	QTAATHCAVY	EVAKPDQNTF	NHHGIHLKLI	ATEELKMLPK	660
TKKSTIPDAL	SYQGIHAHDR	ENGLQQLKDA	CGAFISPRLD	PKQKATWDNS	VSKKENLYPF	720
ITAYMKLLKK	VMKAGRQELK	LFRTHLDHIL	FKHNLSPKL	HGVSIMIGLES	SRATKSVINS	780
FFNLQNAKTE	QQIALDRLE	FEAGKTLINN	QTRRRQERVR	LETSLTMRLA	HKYNAKAI	840
EGELPHSSTG	TSQYQNNVRL	DWSAKKSAKL	KTESANCAGI	AICQIDPCHT	SHQNPFPHTP	900
TNPDLRPRFA	QVKKGKMFQY	QLNGLQRLLN	PRSKSSTAIY	YRQAVQSFC	HHNLTERDIT	960
SAKFPSDLEK	KIKDDTYLIP	QRGGRIYISS	FPVTSARCP	TSNHYFGGGQ	PECNADAVAA	1020
VNIMLKVHP						1029

SEQ ID NO: 9 moltype = AA length = 1011
FEATURE Location/Qualifiers
source 1..1011
 mol_type = protein
 organism = unidentified

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ICANWFKIV	EKTDCIADD	ALLNQFAQY	GEKPNEKVQ	FLTASYNKDK	YVWVDCRQKF	120
YTLQKDLGVQ	NLENDLECLI	REDLLPVGSD	KEVNGWHSIS	KLPGCGEKED	RTIKAKILNG	180
LWERIEKEDI	LTEEDARNEL	LHSAGVLTPK	EFRKVYKGAA	GGRDCYHTLL	VDGRNFTFNL	240
KTLIKQTKDK	LKEKSVDVEI	PNKEALRLYL	EKRIGRSFEQ	KPWSEMYKTA	LSAVMPKQNTL	300
NYCFAIDRHA	QYTKIQTLLQ	PYDSAITALN	GFFESECFTG	SDVFPVISP	LGKTLKKLYN	360
YKDVESGISE	IVEDDNSLR	SGVNVNLLRY	IFTLKDMSFA	EDFIKAAEYN	VVFERYNRQK	420
VHPTVKGNQS	FTFGNSALSG	KVIPPCKCLS	NLPGQMWLAI	NLLDQGEWKE	HHIPPHSARF	480
YEEIYATSDN	QNNPVDLRTK	RFGCSLNKTF	SAADIEKVKE	SAKKKHGKAA	KRILRAKNTN	540
TAVNWVDCGF	MLEKTEVNF	ITVNYKLDPQ	KLKGFEPVIG	TKILAYDQNG	TAPDAYAILE	600
ICDDSEAFDY	KGKIKCLST	GDLASKSLTK	QTEVDQLAYK	GVDKTSNFKY	KWKQQRRLFV	660
KSLNIPDALK	SFENINKEYL	YGFNNSYLKL	LKQILRGKFG	PILVDIRPEL	IEMCQIGSI	720
MRLSSLNHD	LDATQSLKSL	LHSYFDLKV	EEIKTEELRE	KADKEVFKLL	QQVIQKQK	780
RKEKVNRVTD	AILTLAADEQ	VQVIVGEGDL	CVSTKGTKKR	QNNRTIDWCA	RAVVEKLEKA	840
CKLHGLHFKE	IPPHYTSHQD	CFEHNKDIE	PKEVMKCRFN	SSENVAPWMI	KKFANYLKCE	900
TKYYVQGMQD	PLEHYGLVEY	KHIKKGKIS	IGDFQKLIK	ALEKVGEKEI	VFPCKGGRIY	960
LSTYCLTNES	KPIVFNGRR	YVINADHVAA	INVIGCLNLF	NARAKVAEKT	P	1011

SEQ ID NO: 10 moltype = AA length = 1093
FEATURE Location/Qualifiers
source 1..1093
 mol_type = protein
 organism = unidentified

SEQUENCE: 10

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NTQDELQLA	YAVRMFRLVP	KDFSKNENNI	PPDMLISKLA	SYTNINQSPT	NVLSYVNSNY	120
DPEKYKWIDS	RNEAISLSKE	IGIKLDELAD	YATTMLWEDW	LPLNKDVTNG	WGTTSGLFGA	180
GKKEDRTQKV	QMLNALLLGL	KNNPPKDYKQ	YSTILLKAPD	AKSWEEAVKI	YKGECSGRTS	240

SEQ ID NO: 11	moltype = RNA	length = 36	
FEATURE	Location/Qualifiers		
source	1..36		
	mol_type = other RNA		
	organism = unidentified		
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SEQ ID NO: 12	moltype = RNA	length = 36	
FEATURE	Location/Qualifiers		
source	1..36		
	mol_type = other RNA		
	organism = unidentified		
SEQUENCE: 12			
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SEQ ID NO: 13	moltype = RNA	length = 36	
FEATURE	Location/Qualifiers		
source	1..36		
	mol_type = other RNA		
	organism = unidentified		
SEQUENCE: 13			
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SEQ ID NO: 14	moltype = RNA	length = 36	
FEATURE	Location/Qualifiers		
source	1..36		
	mol_type = other RNA		
	organism = unidentified		
SEQUENCE: 14			
ctcaaagtgt caaaagaatg	tccttgctaa	tgggac	36
SEQ ID NO: 15	moltype = RNA	length = 37	
FEATURE	Location/Qualifiers		
source	1..37		
	mol_type = other RNA		
	organism = unidentified		
SEQUENCE: 15			
tcccaaagtg gcaaaagaat	ctccctgtta	atggggag	37
SEQ ID NO: 16	moltype = RNA	length = 36	
FEATURE	Location/Qualifiers		
source	1..36		
	mol_type = other RNA		
	organism = unidentified		
SEQUENCE: 16			
gtctaactgc catagaatcg	tgctgcaat	tggcac	36
SEQ ID NO: 17	moltype = RNA	length = 35	
FEATURE	Location/Qualifiers		
source	1..35		
	mol_type = other RNA		
	organism = unidentified		
SEQUENCE: 17			
tcggggcacc aaaataatct	ccttggtaat	ggggag	35
SEQ ID NO: 18	moltype = RNA	length = 36	
FEATURE	Location/Qualifiers		
source	1..36		
	mol type = other RNA		

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                                organism = unidentified
SEQUENCE: 18
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SEQ ID NO: 19                moltype = RNA  length = 36
FEATURE                      Location/Qualifiers
source                      1..36
                                mol_type = other RNA
                                organism = unidentified

SEQUENCE: 19
gtaacagtg ctaagtaatg tgtcttccaa tgacac                               36

SEQ ID NO: 20                moltype = RNA  length = 24
FEATURE                      Location/Qualifiers
source                      1..24
                                mol_type = other RNA
                                organism = unidentified

SEQUENCE: 20
gagagaatgt gtgcaaagtc acac                                           24

SEQ ID NO: 21                moltype = DNA  length = 3243
FEATURE                      Location/Qualifiers
source                      1..3243
                                mol_type = other DNA
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SEQUENCE: 21
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aatatgtttt tgcaaaacttt caagcgactt aattctatct ctcttaataca ttttgatctc   120
ttaattttgc tttatgtctgc cattaccaac aagaaggcag aagaatataa gtctgaaaaa   180
gaagctcatg taaccgctga tagcctttgt gctatcaatt ggttcocgtcc tatgtccaag   240
cgttacagca aatacgcaac tacaactttc aatatgcttg aattgttcaa agaatactct   300
gggcatgaac catagtgctta ttccaagaat tatcttatgt ccaatattga ctcagacagg   360
tttgtctggg ttgattgccc taaatttgcc aaagattttg cgtatcaaat ggaacttggt   420
ttccatgaat ttacagtctt ggcagaaacc ttgttgccaa atagtattct tgtactcaac   480
gaatcaacta aggcaaaattg ggcattggggc accgtttctg cactttacgg tggaggcgat   540
aaggaaagatt ctacgctgaa gtcgaaaatc cttttggctt ttgttgatgc actcaataac   600
cacgaactta aaactaagcg tgaaattctc aatcaagttt gtgaatcact aaaatatcaa   660
tcataccaag acatgtatgt tgatttccgt tctgttgggt acgaaaatgg aaacaagaag   720
tctcccaatg gctcaatgcc aatcgccacc aagtttgaaa cagatgattt gatttctgat   780
aatcaacgca aagtaatgat ttctaatttc acaaagaatg ctgctgctaa agcgggctaaa   840
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aatgtttatg cttggggcagc agctatcact aactctaatt ccgatgtaac agctaggaat   960
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caaaactcaa caaacgaaaa agcaaaaagat atcttgtaata agattaatga caatcttatt  1080
caagaagtaa ggtatacccc tgcccccaag cacttggggc gtgatcttgc caatcttttt  1140
gatactctga aagaaaaaga tatcaataat attgaaaacg aagaagagaa gcagaatgta  1200
attaatgatt gcattgagca atatgttgat gattgcccgt cactgaaccg caatcccat  1260
gctgctttgc tcaagcacat tagccgatac tatgaagatt tttcagccaa gaatttcttg  1320
gatggtgccg agttgaatgt ctgactgaa gttgtaaatc gtcaaaaggc acatccaact  1380
atlttggtctg aaaaggctta taacttggtt tccaagtttg acaagaatag gcgacaagca  1440
aactcttctt tggttgatg ggttggtcca ccagaagaag tccataaaga gaagattgct  1500
gggtcaacaaa gcatgattgtg ggtcactttg actctgcttg atgatggcaa gtgggtaaag  1560
caccatattc ctttttcaga ttccagatat tattctgaag tctatgecta caatccaaat  1620
ttgccatatt ttgatgggtg tattccacgc cagtcaaatg ttggcaataa accaaccact  1680
aatctgactg ctgaaagtca agcgttactt gcaaacagca agtataaaaa ggcaataaag  1740
tcatttctcc gtgccaaagg aaatgctact cacaatgtcc gtgttagtcc aaacacttcc  1800
ttgtgcattc gtttgctcaa ggatagtgct ggtaaatcaa tgtttgataa gattggcaat  1860
gttctgtttg gaatgcagat caaccataaa atcacccgtt gcaagcccaa ctacaagatc  1920
gaagttgggtg ataggttctt tgggttcgac cagaacccaa gtgaaaacca cacttatgct  1980
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aagggttcttg aaaaggggcaa agtaacaagt gatgtcatcg ttagagatga ggtctatgac  2100
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aggagaaggt ttgttttgga aaacttgctc atccagttgg aagaaggcaa aacattcttg  2220
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tgtgttgatg acgcatccaa gaaggcacat gattccatgc tcttcaactt catggtgtgca  2520
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ttgcggagaa tgctgaacag caagagtgtg gcaggcactt ccgtttacta tctgcaggca  2940
gctttgcatt tctgcgaagc gttgggcgtg tctccagaat tagtcaagaa caaaaagact  3000

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catgctgccc aattaggaaa gcatatgggt tctgccatgt tgatgccttg gcgggggtggc 3060
aggggtttata ttgccagcaa gaagttgact tcggatgcta aaagtgtaaa atactgtgga 3120
gaagatatgt ggcagtatca tgctgatgag attgctgctg tcaatatcgc aatgtatgaa 3180
gtttgctgccc agacaggtgc gtttggcaag aagcaaaaaga agagtgatga actaccggga 3240
taa 3243

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SEQ ID NO: 22      moltype = DNA length = 3244
FEATURE           Location/Qualifiers
source            1..3244
                  mol_type = other DNA
                  organism = unidentified

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SEQUENCE: 22
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cttggtttgt ctttatgctg ctatcaccaa caaaaaagct gaagaatata agtcagaaaa 180
agaagatcat gtaaccgctg atagcctttg cgccatcaat tggttccgct ctatgtocaa 240
gcgttatatc aaatcacgcaa ccactacttt taagatgctt gaattgttta aggagtactc 300
tggatcatgaa ccagatactt attccaagaa ttatctcatg tccaatatcg tctcagatag 360
gtttgtttgg gttgattgct gcaaatttgc caaagatttt gccaatcaaa tggaaacttag 420
tttccacgaa tttaccactt tgcagagac tttgttggca aatagtatcc ttgtactcaa 480
tgagtcaacc aaggcaaat gggcatgggg tgctgtttca gcactttatg gtggaggcga 540
caaagaagat tctacgctga agtccaaaaa ccttttggtt tttgttgatg ctctcaataa 600
tcttgaactt aaaactaggc gggaaaattct caatcatggt tgtgaatcac taaaatatca 660
atcataccaa gatattgtat ttgattttcg atctgtcgtt gatgataagg gaaacaagaa 720
gtctcccaat ggtccaatgc caatcgtcac taagtttgaa tcagatgatt tgattggtga 780
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gaagcccatc ccatatctag acattctaaa agaccacatg atttcttgt gcgaggaata 900
caatgtctat gcttgggcag cagctattac caattccaat gctgatgtaa ctgctagaaa 960
cactcgcaat ctgacattca tcgggggaaca aaatacccga aggaagaac tatcggtttt 1020
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ataa 3244

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SEQ ID NO: 23      moltype = DNA length = 3225
FEATURE           Location/Qualifiers
source            1..3225
                  mol_type = other DNA
                  organism = unidentified

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SEQUENCE: 23
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gaaatgttgg ataagttttt caccacgcta gcaaaaggtc agcgtgtttt tgcggatctg 120

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ggactgtgca tttacggcag ccttacttta gaaatggtaa agcggcttga gccagaatcc 180
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gagaatgaaa ttaacaaga gaacctgggt agacaatatg agacgtattc aggaaaaaga 300
gcgtctgagg ttaacaagc ttacctaaag tctccaagtt cagacaagta tgtgtggata 360
gactgccgac aaaagtttct taggtttcaa agggatctgg gaacacgtaa tctgtctgaa 420
gactttgagt gcatgctttt tgaacagtac ctcagactca caaagggaga gcttgatggg 480
cataccgcta tgtccaacat gtttgaaca aaaacaaaag aagatcgccg cacaaaactg 540
agatatgccg caaggatgaa agaattggct gaggctaacg aagaaattac ttgggaacaa 600
tatcaccaa gcttgcaaga taatttagac gccaaacttt tagaggaggg tgttgataat 660
tacaaaggca aagcggggagg ctctaattca ttttttagtt acacgctttt aaacagaggt 720
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SEQ ID NO: 24      moltype = DNA length = 3012
FEATURE
source             Location/Qualifiers
                   1..3012
                   mol_type = other DNA
                   organism = unidentified

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ttgtttgctg cccttggttc ttatcttgat gcaaccacaa taagcaattt aattaataaa 180
aatcaagatt tagccgttgt aatatttcgt tatcatgtgg ttcccaaaag tgaggctcat 240
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aatgagtttc agagagctta ttgggacagt ccgattgacc aagaaaagta ttgttggcag 360
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 organism = unidentified

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SEQ ID NO: 26

moltype = DNA length = 3144

FEATURE

Location/Qualifiers

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1..3144

mol_type = other DNA

organism = unidentified

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FEATURE            Location/Qualifiers
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                    mol_type = other DNA
                    organism = unidentified

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FEATURE            Location/Qualifiers
source              1..3090
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SEQ ID NO: 29 moltype = DNA length = 3036
 FEATURE Location/Qualifiers
 source 1..3036
 mol_type = other DNA
 organism = unidentified

SEQUENCE: 29

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ctgtatgggt	ggatcgcccc	aaaaatgatt	ccacaagacc	tggggatcaa	tgaacaagta	180
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SEQ ID NO: 30 moltype = DNA length = 3282
 FEATURE Location/Qualifiers
 source 1..3282
 mol_type = other DNA
 organism = unidentified

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aaagacttct	ccaagaatga	aaacaacata	ccccccgata	tgctcaatag	caagcttgct	300
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gatccagaaa	agtataagtg	gatcgactca	cgcaacgaag	ccatctcatt	gtccaaagaa	420
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SEQ ID NO: 31 moltype = DNA length = 3240
 FEATURE Location/Qualifiers
 source 1..3240
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 31

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SEQ ID NO: 32 moltype = DNA length = 3183
 FEATURE Location/Qualifiers
 source 1..3183
 mol_type = other DNA

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organism = synthetic construct
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SEQ ID NO: 33      moltype = DNA length = 3225
FEATURE
source             Location/Qualifiers
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                   mol_type = other DNA
                   organism = synthetic construct

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FEATURE           Location/Qualifiers
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source 1..3054
 mol_type = other DNA
 organism = synthetic construct

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gaaaaggcca	tcaagacacc	catcaaaaac	atcctggagt	acatctggaa	caacaagaac	1260
gtccctgtgg	gcaatggggg	cgggatcgct	aaatacaacc	agctgaagga	cagattagca	1320
gggatcaagg	ccaatccac	agtggaatgc	aacagaggca	tgacatttgg	caacagcgcc	1380
atgggtggcg	aagtgtatgc	ctccaaccgg	atcagcacca	gcaccaagaa	caagggccag	1440
atcttgggcc	agatgcacaa	cgaccggcct	gtgggcagca	acaacatgat	ttggctggaa	1500
atgaccctcc	tgaaccaacg	caagtggcag	aagcacccaca	tccccacaca	caacaacaaa	1560
tttttcgagg	aagtgcacgc	cttcaaccct	gaactgaagc	agagcgtgaa	cgtgagaaac	1620
agaatgtaca	gaagccagaa	ctactcacag	ctgcctacca	gcctgaccga	cgccctgcag	1680
ggaaatccta	accccaagat	cttcaagaga	cagtacagag	ccctgaacaa	catgaccgct	1740
aatgtgatcg	agccaaagct	gtccttcac	gtgaacaaga	aagatgggag	attcgagatc	1800
agcatcatcc	acaacgtgga	agtgatccga	gccagacggg	acgtgctggt	cggcgactac	1860
ctgggtggga	tcggaccaaa	ccagacggct	tctaatacct	acgcccctcat	gcaggtggtg	1920
cagcctaaca	cccccgacag	ccatgagttc	agaaaccagt	gggtcaagtt	catcgagagc	1980
ggcaagatcg	agagctcaac	actgaactcc	cggggtgagt	acatcgacca	gctgagccac	2040
gatggcgtcg	acctgcagga	gattaaaggt	tctgagtggg	ttcctgcccgc	cgaaaaattc	2100
ctgaacaagc	taggagctat	caacaagagc	ggcaccccca	tcagcatctc	caacaccagc	2160
aaacggggcct	acacattcaa	tagcatctat	ttcaaaatcc	tgctgaatta	tctgagagcc	2220
aacgacgtgg	acctgaatct	ggtgcgggaa	gagatcctgc	ggatcgccaa	cggcagattc	2280
agccctatgc	ggctggggtc	tctgtcctgg	accacactaa	aaatgctggg	caatttcccg	2340
aacctaatcc	acagctactt	cgaccactgt	ggctttaagg	aaatgctgta	gagagaaagc	2400
aaggacaaga	ccatgtacga	tctgctgatg	cacaccatca	ccaagctgac	caacaagcgg	2460
gcccagcgca	ccagcagaat	cgttggaagc	ctgatgaacg	tggtccacaa	gtacaagatc	2520
ggcacaagcg	tggtccacgt	ggtggtggaa	ggctctctga	gcaaaaccga	caagagcagc	2580
tccaagggca	acaatcgga	taccacagac	tggtgcagcc	gggcccgtgt	gaagaagctt	2640
gaagatattg	gcgtgtctta	cggcttcaac	ctgaaagccg	tgagcgccca	ctacaccagc	2700
caccaggacc	ctctggttca	tagagccgat	tacgatgatc	ctaagttggc	cctgagatgc	2760
agatactctt	cttacagcag	agctgatttt	gagaagtggt	gcaaaaaatc	tttcgcccgc	2820
gtgatcagat	gggcccacga	caagaagagc	aacacctgct	acaaggtggg	agccgtagag	2880
ttcttcaaga	actacaaaat	ccttgaggac	aagatcacca	aaaagctgac	catcaaaagag	2940
ttcctggaaa	ttatgtgcgc	tgagagccac	taccctaagt	agtacgacga	cattctgatc	3000
cctagaaggg	gcggcagaat	ctacctcaca	actaagaagc	tgctgtccga	tagcaccac	3060
cagagagagt	ctgtgcatag	ccataccgcc	gtggtgaaga	tgaacggcaa	ggaatactat	3120
agcagcgagc	ccgatgaggt	ggctgctatc	aatatctgcc	tgacgactg	ggtggtcccc	3180
ctgaattgga	caaatcactg	cctgctgcc	ggatggtgta	gcgaccacct	gaagggaatgc	3240
gtgcaatgtc	acacccctga	tcctgtgaga	atcagcatg			3279

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SEQ ID NO: 41 moltype = DNA length = 1979
 FEATURE Location/Qualifiers
 source 1..1979
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 41

atgagcgcgc	tgattaagga	gaacatgcac	atgaagctgt	atatggaggg	caccgtggac	60
aaccatcact	tcaagtgcac	atccgagggc	gaaggcaagc	cctacgaggg	caccagacc	120
atgagaatca	agggtgtcga	ggcgggccct	ctccccttcg	ccttcgacat	cctggctact	180
agcttcctct	acggcgagca	gaccttcac	aaccacaccc	agggcatccc	cgacttcttc	240
aagcagtcct	tccctgaggg	cttcacatgg	gagagagtca	ccacatacga	ggacgggggc	300
gtgctgaccg	ctacccagga	caccagcctc	caggacggct	gcctcatcta	caacgtcaag	360
atcagagggg	tgaacttcac	atccaacggc	cctgtgatgc	agaagaaaac	actcggctgg	420
gaggccttca	ccgagacact	gtaccccgct	gacggcggcc	tggaaggcag	aaacgacatg	480
gccctgaagc	tcgtgggcgg	gagccatctg	atcgcaaaac	tcaagaccac	atatagatcc	540
aagaaacccg	ctaagaacct	caagatgcct	ggcgtctact	atgtggacta	cagactggaa	600
agaatcaagg	aggccaacaa	cgagacatac	gtcagcagc	acgaggtggc	agtggccaga	660
tactgcgacc	tccctagcaa	actggggcac	aagctgaatg	aattcgaggg	caggggcagc	720
ctgctgacct	gcggcgacgt	ggaggagaa	cccgcccca	tggtgagcaa	ggcgagggag	780
ctgttcacgg	gggtgggtgc	catcctggtc	gagctggacg	gcgacgtaaa	cgccacacaag	840
ttcagcgtgt	ccggcgaggg	cgagggcgat	gccacctacg	gcaagctgac	cctgaagtgc	900
atctgcacca	ccggcaagct	gcccgcgccc	tgggccaccc	tcgtgaccac	cctgacctac	960
ggcgtgcagt	gcttcagcgc	ctaccccgac	cacatgaagc	agcacgactt	cttcaagtcc	1020
gccatgcccc	aaggctacgt	ccaggagcgc	accatcttct	tcaaggacga	cggaactac	1080
aagacccgcg	cgaggtgaa	gttcgagggc	gacacccctg	tgaaccgcat	cgagctgaag	1140
ggcatcgact	tcaaggagga	cggaacatc	ctggggcaca	agctggagta	caactacaac	1200
agccacaacg	tctatatcat	ggccgacaag	cagaagaacg	gcatcaaggt	gaacttcaag	1260
atccgccaca	acatcgagga	cgccagcgtg	cagctcgccg	accactacca	gcagaacacc	1320
ggatccgtgt	ctttccatt	acagtaggag	catcggggag	acaagctttg	gccacctacg	1380
gcaagctgac	cctgaagtgc	atctgcacca	ccggcaagct	gcccgtgccc	tgccccaccc	1440
tcgtgaccac	cctgacctac	ggcgtgcagt	gcttcagccg	ctaccccgac	cacatgaagc	1500
agcacgactt	cttcaagtcc	gccatgcccg	aaggctacgt	ccaggagcgc	accatcttct	1560
tcaaggacga	cggaacttac	aagaccgcgc	ccgaggtgaa	gttcgagggc	gacacccctg	1620
tgaaaccgat	cgagctgaag	ggcatcgact	tcaaggagga	cggaacatc	ctggggcaca	1680
agctggagta	caactacaac	agccacaacg	tctatatcat	ggccgacaag	cagaagaacg	1740
gcatcaaggt	gaacttcaag	atccgccaca	acatcgagga	cgccagcgtg	cagctcgccg	1800
accactacca	gcagaacacc	cccatcggcg	acggcccccgt	gctgctgccc	gacaaccact	1860
acctgagcac	ccagtccgcc	ctgagcaaa	gaaagcggat	caacatggtc		1920
tgctggaggt	cgtgaccgcc	ggcgggatca	ctctcggcat	ggacgagctg	tacaagtaa	1979

SEQ ID NO: 42 moltype = DNA length = 50
 FEATURE Location/Qualifiers
 source 1..50
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 42

ggatccgtgt	ctttccatt	acagtaggag	catcggggag	acaagctttg	50
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SEQ ID NO: 43 moltype = DNA length = 20
 FEATURE Location/Qualifiers
 source 1..20
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 43

ccattacagt	aggagcatatc	20
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SEQ ID NO: 44 moltype = DNA length = 20
 FEATURE Location/Qualifiers
 source 1..20
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 44

gtatgctcct	actgtaatgg	20
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SEQ ID NO: 45 moltype = RNA length = 20
 FEATURE Location/Qualifiers
 source 1..20
 mol_type = other RNA
 organism = synthetic construct

SEQUENCE: 45

ccattacagt	aggagcatatc	20
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SEQ ID NO: 46 moltype = RNA length = 20
 FEATURE Location/Qualifiers
 source 1..20
 mol_type = other RNA

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organism = synthetic construct

SEQUENCE: 46
 ggtcttcgat aagaagacct 20

SEQ ID NO: 47
 FEATURE Location/Qualifiers
 source 1..1368
 mol_type = protein
 organism = Streptococcus pyogenes

SEQUENCE: 47
 MDKKYSIGLD IGTNSVGWAV ITDEYKVP SK KFKVLGNTDR HSIKKNLIGA LLFDSGETAE 60
 ATRLKRTARR RYTRRKNRIC YLQEIFSNEM AKVDDSFPHR LEESFLVEED KKHRRHPIFG 120
 NIVDEVAYHE KYPTIYHLRK KLVDSDDKAD LRLIYLALAH MIKFRGHFLI EGDNLNPDNSD 180
 VDKLFIQLVQ TYNQLFEENP INASGVDAKA ILSARLSKSR RLENLIAQLP GEKKNGLFGN 240
 LIALSLGLTP NFKSNFDLAE DAKLQLSKDT YDDDLNLLA QIGDQYADLF LAAKNLSDAI 300
 LLSDILRVMT BITKAPLSAS MIKRYDEHHQ DLTLKALVR QQLPEKYKEI FFDQSKNGYA 360
 GYIDGGASQE EFYKFIKPIE EKMKGTEELL VKLNREDLLR KQRTFDNGSI PHQIHLGELH 420
 AILRRQEDFY PFLKDNREKI EKILTFRIPY YVGPLARGNS RFAWMTRKSE ETITPWNFEE 480
 VVDKGASQAS FIERMTNFDK NLPNEKVLKP HSLLYEYFTV YNELTKVKYV TEGMRKPAFL 540
 SGEQKKAIVD LLFPTNRKVT VKQLKEDYFK KIECFDSVEI SGVEDRPNAS LGTYHDLLEI 600
 IKDKDFLDNE ENEDILEDIV LTLTLFEDRE MIEERLKTYA HLPDDKVMKQ LKRRRYTGWG 660
 RLSRKLINGR RDKQSGKTIL DFLKSDGFAN RNFMQLIHDD SLTPKEDIQK AQVSGQGDSL 720
 HEHIANLAGS PAIKKGILQT VKVDELVKV MGRHKPENIV IEMARENQTT QKGQKNSRER 780
 MKRIEEGIKE LGSQILKEHP VENTQLQNEK LYLYYLQNGR DMYVDQELDI NRLSDYVDH 840
 IVPQSFLKDD SIDNKVLTRS DKNRGKSDNV PSEEVVKMK NYWRQLLNAK LITQRKFDNL 900
 TKAERGGLSE LDKAGFIKQ LVETRQITKH VAQILDSRMN TKYDENDKLI REVKVITLKS 960
 KLVSDFRKDF QFYKVRINN YHHAHDAYLN AVVGTAIIK YPKLESEFVY GDYKVYDVRK 1020
 MIAKSEQEIG KATAKYFFYS NIMNFPKTEI TLANGEIRK PLIETNGETG EIVWDKGRDF 1080
 ATVRKVLSP QVNIVKKEV QTGGFSKESI LPKRNSDKLI ARKKDWDPKK YGGFDSPTVA 1140
 YSVLVVAVKE KGSKKLKSV KELLGITIME RSSPEKNPID FLEAKGYKEV KDLIIKLKP 1200
 YSLFELENGR KRMLASAGEL QKGNELALPS KYVNFYLYAS HYEKLKGSPE DNEQKQLFVE 1260
 QHKHYLDEII EQISEFSKRV ILADANLDKV LSAYNKHARDK PIREQAENII HLFITLNLGA 1320
 PAAFKYFDTT IDRKYRSTK EVLDATLIHQ SITGLYETRI DLSQLGGD 1368

SEQ ID NO: 48
 FEATURE Location/Qualifiers
 source 1..96
 mol_type = other RNA
 organism = synthetic construct

SEQUENCE: 48
 ccattacagt aggagcatatc gtttttagagc tagaaatagc aagttaaaaa aaggctagtc 60
 cgttatcaac ttgaaaaagt ggcaccgagt cggtgc 96

SEQ ID NO: 49
 FEATURE Location/Qualifiers
 source 1..1228
 mol_type = protein
 note = Lachnospiraceae bacterium MA2020
 organism = unidentified

SEQUENCE: 49
 MSKLEKFTNC YLSKTLRFK AIPVGKTQEN IDNKRLLED EKRAEDYKGV KLLDRYYLS 60
 FINDVLHSIK LKNLNNYISL FRKKTREKE NKELENLEIN LRKEIAKAFK GNEGYSKLPK 120
 KDIIETILPE FLDDKDEIAL VNSFNGFTTA FTGFFDNREN MFSEEAKESTS IAFRCINENL 180
 TRYISNMDF EKVDALFDKH EVQEIKEKIL NSDYDVEDFF EGEFFNFVLT QEGIDVYNAI 240
 IGGFVTESE KIKGLNEYIN LYNQKTQKL PKFKPLYKQV LSDRESLSFY GEGYTSDEEV 300
 LEVFRNTLNK NSEIFSSIKK LEKLKFNDE YSSAGIFVKV GPAISTISKD IFGEWNVIRD 360
 KWNAEYDDIH LKKKAVVTEK YEDDRKSPK KIGSFSLEQL QEYADADLSV VEKLKEIIQ 420
 KVDEIYKVYG SSEKLFDAF VLEKSLKND AVVAIMKDLL DSVKSPENYI KAFFGEGKET 480
 NRDESFGDF VLAYDILLK DHIYDAIRNY VTQKPYSKDK FKLYFQNPQF MGGWDKDKET 540
 DYRATILRYG SKYYLAIMDK KYAKCLQKID KDDVNGNYEK INYKLLPGPN KMLPKVFFSK 600
 KWMAYNPSE DIQKIYKNGT FKKGDMFNLN DCHKLIDFFK DSISRYPKWS NAYDFNFSSET 660
 EKYKDIAGFY REVVEQGYKV SFESASKKEV DKLVEEGKLY MFQIYNKDFS DKSHGTPNLH 720
 TMYFKLLPDE NNHGQIRLSG GAELFMRAS LKKEELVVHP ANSPIANKP DNPKKTTLT 780
 YDVYKDKRFS EDQYELHPI AINKCPKNIF KINTEVRVLL KHDNPVYVIG IDRGERNLLY 840
 IVVVDGKGNV VEQYSLNEII NNFNGIRIKT DYHSLDKKE KERFEARQNW TSINIENKELK 900
 AGYISQVVKH ICELVEKYDA VIALEDLNSG FKNSRVKVEK QVYQKFEKML IDKLNVMVDK 960
 KSNPCATGGA LKGYQITNKF ESPKSMSTQN GFIFYPYPAWL TSKIDPSTGF VNLKTKYTS 1020
 IADSKKFISS FDRIMYPVEE DLFEFALDYK NFSRTDADYI KKWLYSYGN RIRIFRNPCK 1080
 NNVFDWEEVC LTSAYKELFN KYGINYQGD IRALLCEQSD KAFYSSFMAL MSLMLQMRNS 1140
 ITGRTDVFDF ISPVKNSDGI FYSRNYEAQ ENAILPKNAD ANGAYNIARK VLWAIQGFKK 1200
 AEDEKLDKVK IAINKEWLE YAQTSVKH 1228

SEQ ID NO: 50
 FEATURE Location/Qualifiers
 source 1..41
 mol_type = other RNA

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                                organism = synthetic construct
SEQUENCE: 50
taatttctac taagtgtaga tccattacag taggagcata c                               41

SEQ ID NO: 51      moltype = RNA length = 56
FEATURE           Location/Qualifiers
source            1..56
                  mol_type = other RNA
                  organism = synthetic construct

SEQUENCE: 51
ctagcaatga ctcagaaatg tgtccccagt tgacacccat tacagtagga gcatac           56

SEQ ID NO: 52      moltype = RNA length = 56
FEATURE           Location/Qualifiers
source            1..56
                  mol_type = other RNA
                  organism = synthetic construct

SEQUENCE: 52
atcgcaacat cttagaaatc cgtccttagt tgacggccat tacagtagga gcatac           56

SEQ ID NO: 53      moltype = RNA length = 56
FEATURE           Location/Qualifiers
source            1..56
                  mol_type = other RNA
                  organism = synthetic construct

SEQUENCE: 53
tctcaacgat agtcagacat gtgtccccag tgacacccat tacagtagga gcatac           56

SEQ ID NO: 54      moltype = RNA length = 56
FEATURE           Location/Qualifiers
source            1..56
                  mol_type = other RNA
                  organism = synthetic construct

SEQUENCE: 54
ctcaaagtgt caaaagaatg tccttgctaa tgggacccat tacagtagga gcatac           56

SEQ ID NO: 55      moltype = RNA length = 57
FEATURE           Location/Qualifiers
source            1..57
                  mol_type = other RNA
                  organism = synthetic construct

SEQUENCE: 55
tcccaaagtg gcaaaagaat ctccctgtta atgggagcca ttacagtagg agcatac           57

SEQ ID NO: 56      moltype = RNA length = 56
FEATURE           Location/Qualifiers
source            1..56
                  mol_type = other RNA
                  organism = synthetic construct

SEQUENCE: 56
gtctaactgc catagaatcg tgctgcaat tggcacccat tacagtagga gcatac           56

SEQ ID NO: 57      moltype = RNA length = 55
FEATURE           Location/Qualifiers
source            1..55
                  mol_type = other RNA
                  organism = synthetic construct

SEQUENCE: 57
tcgggggcacc aaaataatct ccttggtaat gggagccatt acagtaggag catac           55

SEQ ID NO: 58      moltype = RNA length = 56
FEATURE           Location/Qualifiers
source            1..56
                  mol_type = other RNA
                  organism = synthetic construct

SEQUENCE: 58
ccacaacaac caaaagaatg tcctgaaag tgggacccat tacagtagga gcatac           56

SEQ ID NO: 59      moltype = RNA length = 56
FEATURE           Location/Qualifiers
source            1..56
                  mol_type = other RNA
                  organism = synthetic construct

SEQUENCE: 59
gtaacagtgg ctaagtaatg tgtcttccaa tgacacccat tacagtagga gcatac           56

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SEQ ID NO: 60	moltype = RNA length = 44	
FEATURE	Location/Qualifiers	
source	1..44	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 60		
gagagaatgt gtgcaaagtc acacccatta cagtaggagc atac		44
SEQ ID NO: 61	moltype = RNA length = 10	
FEATURE	Location/Qualifiers	
source	1..10	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 61		
ccattacagt		10
SEQ ID NO: 62	moltype = RNA length = 12	
FEATURE	Location/Qualifiers	
source	1..12	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 62		
ccattacagt ag		12
SEQ ID NO: 63	moltype = RNA length = 14	
FEATURE	Location/Qualifiers	
source	1..14	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 63		
ccattacagt agga		14
SEQ ID NO: 64	moltype = RNA length = 15	
FEATURE	Location/Qualifiers	
source	1..15	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 64		
ccattacagt aggag		15
SEQ ID NO: 65	moltype = RNA length = 16	
FEATURE	Location/Qualifiers	
source	1..16	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 65		
ccattacagt aggagc		16
SEQ ID NO: 66	moltype = RNA length = 17	
FEATURE	Location/Qualifiers	
source	1..17	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 66		
ccattacagt aggagca		17
SEQ ID NO: 67	moltype = RNA length = 18	
FEATURE	Location/Qualifiers	
source	1..18	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 67		
ccattacagt aggagcat		18
SEQ ID NO: 68	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 68		
ccattacagt aggagcata		19
SEQ ID NO: 69	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	

-continued

SEQUENCE: 69	organism = synthetic construct	
ccattacagt aggagcatac g		21
SEQ ID NO: 70	moltype = RNA length = 22	
FEATURE	Location/Qualifiers	
source	1..22	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 70		
ccattacagt aggagcatac gg		22
SEQ ID NO: 71	moltype = RNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 71		
ccattacagt aggagcatac ggg		23
SEQ ID NO: 72	moltype = RNA length = 24	
FEATURE	Location/Qualifiers	
source	1..24	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 72		
ccattacagt aggagcatac ggga		24
SEQ ID NO: 73	moltype = RNA length = 26	
FEATURE	Location/Qualifiers	
source	1..26	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 73		
ccattacagt aggagcatac gggaga		26
SEQ ID NO: 74	moltype = RNA length = 27	
FEATURE	Location/Qualifiers	
source	1..27	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 74		
ccattacagt aggagcatac gggagac		27
SEQ ID NO: 75	moltype = RNA length = 28	
FEATURE	Location/Qualifiers	
source	1..28	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 75		
ccattacagt aggagcatac gggagaca		28
SEQ ID NO: 76	moltype = RNA length = 30	
FEATURE	Location/Qualifiers	
source	1..30	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 76		
ccattacagt aggagcatac gggagacaag		30
SEQ ID NO: 77	moltype = RNA length = 32	
FEATURE	Location/Qualifiers	
source	1..32	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 77		
ccattacagt aggagcatac gggagacaag ct		32
SEQ ID NO: 78	moltype = RNA length = 35	
FEATURE	Location/Qualifiers	
source	1..35	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 78		
ccattacagt aggagcatac gggagacaag ctttg		35

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SEQ ID NO: 79	moltype = RNA length = 40	
FEATURE	Location/Qualifiers	
source	1..40	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 79		
ccattacagt aggagcatac gggagacaag ctttggccac		40
SEQ ID NO: 80	moltype = RNA length = 45	
FEATURE	Location/Qualifiers	
source	1..45	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 80		
ccattacagt aggagcatac gggagacaag ctttggccac ctacg		45
SEQ ID NO: 81	moltype = RNA length = 50	
FEATURE	Location/Qualifiers	
source	1..50	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 81		
ccattacagt aggagcatac gggagacaag ctttggccac ctacggcaag		50
SEQ ID NO: 82	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 82		
caaaaaccca aaatatttta		20
SEQ ID NO: 83	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 83		
gctcctactc agactgttac		20
SEQ ID NO: 84	moltype = DNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 84		
tgtcaccaga gtaacagtc		19
SEQ ID NO: 85	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 85		
cctcgctgga ctggtatttg		20
SEQ ID NO: 86	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 86		
tgtctgaagc tggccccgcg		20
SEQ ID NO: 87	moltype = DNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 87		
ccttcgactc ttctccttt g		21
SEQ ID NO: 88	moltype = DNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other DNA	

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SEQUENCE: 88	organism = synthetic construct	
ctcctttgcc tcgctggact g		21
SEQ ID NO: 89	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 89		
accatcagag gacatttggga		20
SEQ ID NO: 90	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 90		
gattctccag caccctgggc		20
SEQ ID NO: 91	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 91		
cagccacgtc tacagcaggg		20
SEQ ID NO: 92	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 92		
acagccacgt ctacagcagg		20
SEQ ID NO: 93	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 93		
gaacactttt acagccacgt		20
SEQ ID NO: 94	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 94		
aaaaagacct ctgagggatc		20
SEQ ID NO: 95	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 95		
aacactttta cagccacgtc		20
SEQ ID NO: 96	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 96		
tagaaggagt gtacagagta		20
SEQ ID NO: 97	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 97		
gcatttcccc gttccatgaa		20

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SEQ ID NO: 98	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 98		
tcattctgtgg tgagcccgtg		20
SEQ ID NO: 99	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 99		
gtgtccagtt ctactctgta		20
SEQ ID NO: 100	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 100		
cagtacgatt tgggtgccag		20
SEQ ID NO: 101	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 101		
tacaaacttc tcattctgtgg		20
SEQ ID NO: 102	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 102		
cacagccaac gactctggcc		20
SEQ ID NO: 103	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 103		
acagccaacg actctggcca		20
SEQ ID NO: 104	moltype = DNA length = 22	
FEATURE	Location/Qualifiers	
source	1..22	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 104		
ctgacgacag ccgtgggtgct gt		22
SEQ ID NO: 105	moltype = DNA length = 22	
FEATURE	Location/Qualifiers	
source	1..22	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 105		
aaaaagacct ctgagggatc ct		22
SEQ ID NO: 106	moltype = DNA length = 22	
FEATURE	Location/Qualifiers	
source	1..22	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 106		
agaaaggctg ctgatgacac ct		22
SEQ ID NO: 107	moltype = DNA length = 22	
FEATURE	Location/Qualifiers	
source	1..22	
	mol_type = other DNA	

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                                organism = synthetic construct
SEQUENCE: 107
tagaagggat atacaaagtg ga                                     22

SEQ ID NO: 108            moltype = DNA  length = 22
FEATURE                  Location/Qualifiers
source                  1..22
                        mol_type = other DNA
                        organism = synthetic construct

SEQUENCE: 108
caccacggct gtcgtcacca at                                     22

SEQ ID NO: 109            moltype = DNA  length = 22
FEATURE                  Location/Qualifiers
source                  1..22
                        mol_type = other DNA
                        organism = synthetic construct

SEQUENCE: 109
aatccaagtg tcctctgatg gt                                     22

SEQ ID NO: 110            moltype = DNA  length = 22
FEATURE                  Location/Qualifiers
source                  1..22
                        mol_type = other DNA
                        organism = synthetic construct

SEQUENCE: 110
aatgtggccg tgcattgtgt ca                                     22

SEQ ID NO: 111            moltype = DNA  length = 22
FEATURE                  Location/Qualifiers
source                  1..22
                        mol_type = other DNA
                        organism = synthetic construct

SEQUENCE: 111
tagatgctgt ccgaggcagt cc                                     22

SEQ ID NO: 112            moltype = DNA  length = 22
FEATURE                  Location/Qualifiers
source                  1..22
                        mol_type = other DNA
                        organism = synthetic construct

SEQUENCE: 112
gcatgggctc acaactgagg ag                                     22

SEQ ID NO: 113            moltype = DNA  length = 22
FEATURE                  Location/Qualifiers
source                  1..22
                        mol_type = other DNA
                        organism = synthetic construct

SEQUENCE: 113
tatacaaagt ggaaatagac ac                                     22

SEQ ID NO: 114            moltype = DNA  length = 22
FEATURE                  Location/Qualifiers
source                  1..22
                        mol_type = other DNA
                        organism = synthetic construct

SEQUENCE: 114
ctggaaggca cttggcatct cc                                     22

SEQ ID NO: 115            moltype = DNA  length = 22
FEATURE                  Location/Qualifiers
source                  1..22
                        mol_type = other DNA
                        organism = synthetic construct

SEQUENCE: 115
gcatctcccc attccatgag ca                                     22

SEQ ID NO: 116            moltype = DNA  length = 22
FEATURE                  Location/Qualifiers
source                  1..22
                        mol_type = other DNA
                        organism = synthetic construct

SEQUENCE: 116
acagccaacg actccggccc cc                                     22

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SEQ ID NO: 117	moltype = DNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 117		
cctggcacct acgtggtgg		19
SEQ ID NO: 118	moltype = DNA length = 18	
FEATURE	Location/Qualifiers	
source	1..18	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 118		
catggccttc ttctctggc		18
SEQ ID NO: 119	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 119		
ttctctggctt cctgggtgaag		20
SEQ ID NO: 120	moltype = length =	
SEQUENCE: 120		
000		
SEQ ID NO: 121	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 121		
aagttgcccc atgtcgacta		20
SEQ ID NO: 122	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 122		
cccagagcat cccgtggaac		20
SEQ ID NO: 123	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 123		
cagatacgaa cctaaacttt		20
SEQ ID NO: 124	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 124		
gagttcttca gctggtacac		20
SEQ ID NO: 125	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 125		
tctgtgatat acacatcaga		20
SEQ ID NO: 126	moltype = length =	
SEQUENCE: 126		
000		
SEQ ID NO: 127	moltype = DNA length = 22	
FEATURE	Location/Qualifiers	
source	1..22	
	mol_type = other DNA	

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                                organism = synthetic construct
SEQUENCE: 127
cagcaggctt ctacaaagtg ga                                     22

SEQ ID NO: 128            moltype = DNA  length = 22
FEATURE                  Location/Qualifiers
source                  1..22
                        mol_type = other DNA
                        organism = synthetic construct

SEQUENCE: 128
taaaagggat atacaatatg ta                                     22

SEQ ID NO: 129            moltype = DNA  length = 22
FEATURE                  Location/Qualifiers
source                  1..22
                        mol_type = other DNA
                        organism = synthetic construct

SEQUENCE: 129
tagaagggat atagaaagta tc                                     22

SEQ ID NO: 130            moltype = DNA  length = 22
FEATURE                  Location/Qualifiers
source                  1..22
                        mol_type = other DNA
                        organism = synthetic construct

SEQUENCE: 130
tagaagggat atacaaagtg ga                                     22

SEQ ID NO: 131            moltype = DNA  length = 23
FEATURE                  Location/Qualifiers
source                  1..23
                        mol_type = other DNA
                        organism = synthetic construct

SEQUENCE: 131
tgtctgaggc tggccctacg gtg                                   23

SEQ ID NO: 132            moltype = DNA  length = 23
FEATURE                  Location/Qualifiers
source                  1..23
                        mol_type = other DNA
                        organism = synthetic construct

SEQUENCE: 132
accatcagag gacacttgga ttc                                   23

SEQ ID NO: 133            moltype = DNA  length = 23
FEATURE                  Location/Qualifiers
source                  1..23
                        mol_type = other DNA
                        organism = synthetic construct

SEQUENCE: 133
tgaacacatg cacggccaca ttg                                   23

SEQ ID NO: 134            moltype = DNA  length = 23
FEATURE                  Location/Qualifiers
source                  1..23
                        mol_type = other DNA
                        organism = synthetic construct

SEQUENCE: 134
cctctgggta agttgccaaa gaa                                   23

SEQ ID NO: 135            moltype = DNA  length = 23
FEATURE                  Location/Qualifiers
source                  1..23
                        mol_type = other DNA
                        organism = synthetic construct

SEQUENCE: 135
gcaacttacc cagaggcaaa tgg                                   23

SEQ ID NO: 136            moltype = DNA  length = 23
FEATURE                  Location/Qualifiers
source                  1..23
                        mol_type = other DNA
                        organism = synthetic construct

SEQUENCE: 136
acacettata ggaaaaccag tga                                   23

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-continued

SEQ ID NO: 137	moltype = DNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 137		
ctataagggtg tgaaagtctg gat		23
SEQ ID NO: 138	moltype = DNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 138		
cctataagggt gtgaaagtctt gga		23
SEQ ID NO: 139	moltype = DNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 139		
tagaagggtat atacaaagtgt gaa		23
SEQ ID NO: 140	moltype = DNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 140		
tatatccctt ctacaaattc ctc		23
SEQ ID NO: 141	moltype = DNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 141		
cactttgtat atcccttcta caa		23
SEQ ID NO: 142	moltype = DNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 142		
gtgtctattt ccactttgta tat		23
SEQ ID NO: 143	moltype = DNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 143		
tgtctgaggc tggccctacg gtg		23
SEQ ID NO: 144	moltype = DNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 144		
accatcagag gacacttgga ttc		23
SEQ ID NO: 145	moltype = DNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 145		
tgaacacatg cacggccaca ttg		23
SEQ ID NO: 146	moltype = DNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = other DNA	

-continued

SEQUENCE: 146	organism = synthetic construct	
cctctgggta agttgccaaa gaa		23
SEQ ID NO: 147	moltype = DNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 147		
gcaacttacc cagaggcaaa tgg		23
SEQ ID NO: 148	moltype = DNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 148		
acaccttata ggaaaaccag tga		23
SEQ ID NO: 149	moltype = DNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 149		
ctataaggtg tgaaagtctg gat		23
SEQ ID NO: 150	moltype = DNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 150		
cctataaggt gtgaaagtct gga		23
SEQ ID NO: 151	moltype = DNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 151		
tagaagggat atacaaagtg gaa		23
SEQ ID NO: 152	moltype = DNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 152		
tatatccctt ctacaaattc etc		23
SEQ ID NO: 153	moltype = DNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 153		
cactttgtat atcccttcta caa		23
SEQ ID NO: 154	moltype = DNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 154		
gtgtctatatt ccactttgta tat		23
SEQ ID NO: 155	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 155		
ccacctatga gagaagacag		20

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SEQ ID NO: 156	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 156		
ggctgtcgtc accaatccca		20
SEQ ID NO: 157	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 157		
gacgacagcc gtggtggaat		20
SEQ ID NO: 158	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 158		
ctgaacacat gcacggccac		20
SEQ ID NO: 159	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 159		
caccagggc accggtgaat		20
SEQ ID NO: 160	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 160		
aatggtgtag cggcgggggc		20
SEQ ID NO: 161	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 161		
ctttggcaac ttaccagag		20
SEQ ID NO: 162	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 162		
tgtctgaggc tggccctacg		20
SEQ ID NO: 163	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 163		
tttgtgtctg aggctggccc		20
SEQ ID NO: 164	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 164		
attggtgacg acagccgtgg		20
SEQ ID NO: 165	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	

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SEQUENCE: 165	organism = synthetic construct	
aggagaagtc cctcattcct		20
SEQ ID NO: 166	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 166		
ccaagtgcct tccagtaaga		20
SEQ ID NO: 167	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 167		
acacaaatac cagtccagca		20
SEQ ID NO: 168	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 168		
ccagtccagc aaggcagagg		20
SEQ ID NO: 169	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 169		
gaagtccact cattcttggc		20
SEQ ID NO: 170	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 170		
aaagttctag atgctgtccg		20
SEQ ID NO: 171	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 171		
cccagaggca aatggctccc		20
SEQ ID NO: 172	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 172		
ttctttggca acttaccag		20
SEQ ID NO: 173	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 173		
actgaggagg aatttgtaga		20
SEQ ID NO: 174	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 174		
cccattccat gagcatgcag		20

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SEQ ID NO: 175	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 175		
gcatgggctc acaactgagg		20
SEQ ID NO: 176	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 176		
aataggagta ggggctcagc		20
SEQ ID NO: 177	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 177		
gacgacagcc gtggtggaat		20
SEQ ID NO: 178	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 178		
ggctgtcgtc accaatccca		20
SEQ ID NO: 179	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 179		
gtcaccaatc ccaaggaatg		20
SEQ ID NO: 180	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 180		
tgtgtctgag gctggcccta		20
SEQ ID NO: 181	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 181		
agcctttctg aacacatgca		20
SEQ ID NO: 182	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 182		
cagaggacac ttggattcac		20
SEQ ID NO: 183	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 183		
cattgatggc aggactgcct		20
SEQ ID NO: 184	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	

-continued

SEQUENCE: 184	organism = synthetic construct	
cttctctaca cccagggcac		20
SEQ ID NO: 185	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 185		
aatggtgtag cggcgggggc		20
SEQ ID NO: 186	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 186		
cccctactcc tattccacca		20
SEQ ID NO: 187	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 187		
gcagggcggc aatggtgtag		20
SEQ ID NO: 188	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 188		
ggagtagggg ctcagcaggg		20
SEQ ID NO: 189	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 189		
gtattcacag ccaacgactc		20
SEQ ID NO: 190	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 190		
tcacagaaac actcacgta		20
SEQ ID NO: 191	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 191		
aaaggctgct gatgacacct		20
SEQ ID NO: 192	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 192		
cttggattca ccggtgcctt		20
SEQ ID NO: 193	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 193		
gccgtggtgg aataggagta		20

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SEQ ID NO: 194	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 194		
gcggcaatgg tgtagcggcg		20
SEQ ID NO: 195	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 195		
ggagaagtcc ctcattcctt		20
SEQ ID NO: 196	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 196		
ggcggcaatg gtgtagcggc		20
SEQ ID NO: 197	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 197		
tcaccaatcc caaggaatga		20
SEQ ID NO: 198	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 198		
gcaacttacc cagaggcaaa		20
SEQ ID NO: 199	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 199		
aagtgccttc cagtaagatt		20
SEQ ID NO: 200	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 200		
acctctgcat gctcatggaa		20
SEQ ID NO: 201	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 201		
tactcacctc tgcattgctca		20
SEQ ID NO: 202	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 202		
tgtagaaggg atatacaaag		20
SEQ ID NO: 203	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	

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SEQUENCE: 203	organism = synthetic construct	
aggagaagtc cctcattcct		20
SEQ ID NO: 204	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 204		
attggtgacg acagccgtgg		20
SEQ ID NO: 205	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 205		
gcggcggggg ccggagtcgt		20
SEQ ID NO: 206	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 206		
gggattggtg acgacagccg		20
SEQ ID NO: 207	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 207		
ggggctcagc agggcggcaa		20
SEQ ID NO: 208	moltype = DNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 208		
tgtctgaggc tggccctacg gtg		23
SEQ ID NO: 209	moltype = DNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 209		
accatcagag gacacttgga ttc		23
SEQ ID NO: 210	moltype = DNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 210		
tgaacacatg cagggccaca ttg		23
SEQ ID NO: 211	moltype = DNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 211		
cctctgggta agttgccaaa gaa		23
SEQ ID NO: 212	moltype = DNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 212		
gcaacttacc cagaggcaaa tgg		23

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SEQ ID NO: 213	moltype = DNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 213		
acaccttata ggaaaaccag tga		23
SEQ ID NO: 214	moltype = DNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 214		
ctataaggtg tgaaagtctg gat		23
SEQ ID NO: 215	moltype = DNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 215		
cctataaggt gtgaaagtct gga		23
SEQ ID NO: 216	moltype = DNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 216		
tagaagggat atacaaagtg gaa		23
SEQ ID NO: 217	moltype = DNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 217		
tatatccctt ctacaaattc etc		23
SEQ ID NO: 218	moltype = DNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 218		
cactttgtat atcccttcta caa		23
SEQ ID NO: 219	moltype = DNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 219		
gtgtctattt ccactttgta tat		23
SEQ ID NO: 220	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 220		
ctggaaggca cttggcatct		20
SEQ ID NO: 221	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 221		
taggaaaacc agtgagtctg		20
SEQ ID NO: 222	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	

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SEQUENCE: 222	organism = synthetic construct	
tcatcgtctg ctcctcctct		20
SEQ ID NO: 223	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 223		
ttggcaggat ggcttctcat		20
SEQ ID NO: 224	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 224		
accggtgccc tgggtgtaga		20
SEQ ID NO: 225	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 225		
agaaaggctg ctgatgacac		20
SEQ ID NO: 226	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 226		
tagatgctgt ccgaggcagt		20
SEQ ID NO: 227	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 227		
tctacacca gggcaccggt		20
SEQ ID NO: 228	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 228		
tttgcaact taccagagg		20
SEQ ID NO: 229	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 229		
cagtaagatt tgggtcttat		20
SEQ ID NO: 230	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 230		
catgagcatg cagaggtgag		20
SEQ ID NO: 231	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 231		
ctcctcagtt gtgagcccat		20

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SEQ ID NO: 232	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 232		
tacaaattcc tctcagttg		20
SEQ ID NO: 233	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 233		
acagccaacg actccggccc		20
SEQ ID NO: 234	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 234		
caccacggct gtcgtcacca		20
SEQ ID NO: 235	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 235		
cttgggattg gtagcacag		20
SEQ ID NO: 236	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 236		
tctcataggt ggtattcaca		20
SEQ ID NO: 237	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 237		
ctggactggt atttgtgtct		20
SEQ ID NO: 238	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 238		
gcaggatggc ttctcatcgt		20
SEQ ID NO: 239	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 239		
atggcaggac tgcctcggac		20
SEQ ID NO: 240	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 240		
gattcaccgg tgccctgggt		20
SEQ ID NO: 241	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	

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SEQUENCE: 241	organism = synthetic construct	
gcattctcccc attccatgag		20
SEQ ID NO: 242	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 242		
tgagcccatg cagctctcca		20
SEQ ID NO: 243	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 243		
ccgccctgct gagccctac		20
SEQ ID NO: 244	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 244		
gctgtgaata ccacctatga		20
SEQ ID NO: 245	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 245		
ggatttgtga cgacagccgt		20
SEQ ID NO: 246	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 246		
gtgacgacag ccgtggtgga		20
SEQ ID NO: 247	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 247		
gtgtctgagg ctggccctac		20
SEQ ID NO: 248	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 248		
gaccatcaga ggacacttgg		20
SEQ ID NO: 249	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 249		
gcctctgggt aagttgcaa		20
SEQ ID NO: 250	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 250		
ggcaacttac ccagaggcaa		20

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SEQ ID NO: 251	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 251		
ggtgtctatt tccactttgt		20
SEQ ID NO: 252	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 252		
gtatatccct tctacaaatt		20
SEQ ID NO: 253	moltype = DNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 253		
tgtctgaggc tggccctacg gtg		23
SEQ ID NO: 254	moltype = DNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 254		
accatcagag gacacttga ttc		23
SEQ ID NO: 255	moltype = DNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 255		
tgaacacatg cacggccaca ttg		23
SEQ ID NO: 256	moltype = DNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 256		
cctctgggta agttgccaaa gaa		23
SEQ ID NO: 257	moltype = DNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 257		
acaccttata ggaaaaccag tga		23
SEQ ID NO: 258	moltype = DNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 258		
ctataaggtg tgaaagtctg gat		23
SEQ ID NO: 259	moltype = DNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 259		
cctataaggt gtgaaagtct gga		23
SEQ ID NO: 260	moltype = DNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = other DNA	

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SEQUENCE: 260	organism = synthetic construct	
tagaaggat atacaaagtg gaa		23
SEQ ID NO: 261	moltype = DNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 261		
tatatccctt ctacaaattc ctc		23
SEQ ID NO: 262	moltype = DNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 262		
cactttgtat atcccttcta caa		23
SEQ ID NO: 263	moltype = DNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 263		
gtgtctatct ccactttgta tat		23
SEQ ID NO: 264	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 264		
ctggaaggca cttggcatct		20
SEQ ID NO: 265	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 265		
taggaaaacc agtgagtctg		20
SEQ ID NO: 266	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 266		
tcacgtctg ctccctctct		20
SEQ ID NO: 267	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 267		
ttggcaggat ggcttctcat		20
SEQ ID NO: 268	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 268		
accggtgccc tgggtgtaga		20
SEQ ID NO: 269	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 269		
agaaaggctg ctgatgacac		20

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SEQ ID NO: 270	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 270		
tagatgctgt ccgaggcagt		20
SEQ ID NO: 271	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 271		
tttgcaact tacccagagg		20
SEQ ID NO: 272	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 272		
cagtaagatt tgggtctctat		20
SEQ ID NO: 273	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 273		
catgagcatg cagaggtgag		20
SEQ ID NO: 274	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 274		
ctctcagtt gtgagcccat		20
SEQ ID NO: 275	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 275		
tacaaattcc tcctcagttg		20
SEQ ID NO: 276	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 276		
acagccaacg actccggccc		20
SEQ ID NO: 277	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 277		
caccacggct gtcgtcacca		20
SEQ ID NO: 278	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 278		
cttgggattg gtgacgacag		20
SEQ ID NO: 279	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	

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SEQUENCE: 279	organism = synthetic construct	
tctcataggt ggtattcaca		20
SEQ ID NO: 280	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 280		
ctggactggg atttgtgtct		20
SEQ ID NO: 281	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 281		
gcaggatggc ttctcatcgt		20
SEQ ID NO: 282	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 282		
atggcaggac tgcctcggac		20
SEQ ID NO: 283	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 283		
gattcaccgg tgcctcgggt		20
SEQ ID NO: 284	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 284		
gcattctccc attccatgag		20
SEQ ID NO: 285	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 285		
tgagcccatg cagctctcca		20
SEQ ID NO: 286	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 286		
cgcacctgct gagccctac		20
SEQ ID NO: 287	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 287		
gctgtgaata ccacctatga		20
SEQ ID NO: 288	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 288		
ggattggtga cgacagccgt		20

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SEQ ID NO: 289	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 289		
gtgacgacag ccgtggtgga		20
SEQ ID NO: 290	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 290		
gtgtctgagg ctggccctac		20
SEQ ID NO: 291	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 291		
gaccatcaga ggacacttgg		20
SEQ ID NO: 292	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 292		
gcctctgggt aagttgcaa		20
SEQ ID NO: 293	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 293		
ggcaacttac ccagaggcaa		20
SEQ ID NO: 294	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 294		
ggtgtctatt tccactttgt		20
SEQ ID NO: 295	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 295		
gtatatccct tctacaaatt		20
SEQ ID NO: 296	moltype = DNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 296		
tgtctgaggc tggccctacg gtg		23
SEQ ID NO: 297	moltype = DNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 297		
accatcagag gacacttgga ttc		23
SEQ ID NO: 298	moltype = DNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = other DNA	

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                                organism = synthetic construct
SEQUENCE: 298
tgaacacatg cacggccaca ttg                                     23

SEQ ID NO: 299      moltype = DNA length = 23
FEATURE            Location/Qualifiers
source              1..23
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 299
cctctgggta agttgccaaa gaa                                     23

SEQ ID NO: 300      moltype = DNA length = 23
FEATURE            Location/Qualifiers
source              1..23
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 300
acaccttata ggaaaaccag tga                                     23

SEQ ID NO: 301      moltype = DNA length = 23
FEATURE            Location/Qualifiers
source              1..23
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 301
ctataaggtg tgaaagtctg gat                                     23

SEQ ID NO: 302      moltype = DNA length = 23
FEATURE            Location/Qualifiers
source              1..23
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 302
cctataaggt gtgaaagtct gga                                     23

SEQ ID NO: 303      moltype = DNA length = 23
FEATURE            Location/Qualifiers
source              1..23
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 303
tagaagggat atacaaagtg gaa                                     23

SEQ ID NO: 304      moltype = DNA length = 23
FEATURE            Location/Qualifiers
source              1..23
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 304
tatatccctt ctacaaattc etc                                     23

SEQ ID NO: 305      moltype = DNA length = 23
FEATURE            Location/Qualifiers
source              1..23
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 305
cactttgtat atcccttcta caa                                     23

SEQ ID NO: 306      moltype = DNA length = 23
FEATURE            Location/Qualifiers
source              1..23
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 306
gtgtctattt ccactttgta tat                                     23

SEQ ID NO: 307      moltype = DNA length = 20
FEATURE            Location/Qualifiers
source              1..20
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 307
ctggaaggca cttggcatct                                         20

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SEQ ID NO: 308	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 308		
taggaaaacc agtgagtctg		20
SEQ ID NO: 309	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 309		
tcatcgtctg ctctctctct		20
SEQ ID NO: 310	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 310		
ttggcaggat ggcttctcat		20
SEQ ID NO: 311	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 311		
accggtgccc tgggtgtaga		20
SEQ ID NO: 312	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 312		
agaaaggctg ctgatgacac		20
SEQ ID NO: 313	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 313		
tagatgctgt ccgaggcagt		20
SEQ ID NO: 314	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 314		
tttgcaact taccagagg		20
SEQ ID NO: 315	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 315		
cagtaagatt tgggtgtctat		20
SEQ ID NO: 316	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 316		
catgagcatg cagaggtgag		20
SEQ ID NO: 317	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	

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SEQUENCE: 317	organism = synthetic construct	
ctcctcagtt gtagcccat		20
SEQ ID NO: 318	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 318		
tacaaattcc tcctcagttg		20
SEQ ID NO: 319	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 319		
acagccaacg actccggccc		20
SEQ ID NO: 320	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 320		
caccacggct gtcgtcacca		20
SEQ ID NO: 321	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 321		
cttgggattg gtgacgacag		20
SEQ ID NO: 322	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 322		
tctcataggt ggtattcaca		20
SEQ ID NO: 323	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 323		
ctggactggt atttgtgtct		20
SEQ ID NO: 324	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 324		
gcaggatggc ttctcatcgt		20
SEQ ID NO: 325	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 325		
atggcaggac tgccctggac		20
SEQ ID NO: 326	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 326		
gattcacggg tgccctgggt		20

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SEQ ID NO: 327	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 327		
gcattctccc attccatgag		20
SEQ ID NO: 328	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 328		
tgagcccatg cagctctcca		20
SEQ ID NO: 329	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 329		
cggccctgct gagccctac		20
SEQ ID NO: 330	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 330		
gctgtgaata ccacatgag		20
SEQ ID NO: 331	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 331		
ggattggtga cgacagccgt		20
SEQ ID NO: 332	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 332		
gtgacgacag ccgtggtgga		20
SEQ ID NO: 333	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 333		
gtgtctgagg ctggccctac		20
SEQ ID NO: 334	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 334		
gaccatcaga ggacacttgg		20
SEQ ID NO: 335	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 335		
gcctctgggt aagttgcca		20
SEQ ID NO: 336	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	

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SEQUENCE: 336	organism = synthetic construct	
ggcaacttac ccagaggcaa		20
SEQ ID NO: 337	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 337		
gggtgtctatt tccactttgt		20
SEQ ID NO: 338	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 338		
gtatatccct tctacaaatt		20
SEQ ID NO: 339	moltype = DNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 339		
tgtctgaggc tggccctacg gtg		23
SEQ ID NO: 340	moltype = DNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 340		
accatcagag gacacttgga ttc		23
SEQ ID NO: 341	moltype = DNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 341		
tgaacacatg cacggccaca ttg		23
SEQ ID NO: 342	moltype = DNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 342		
cctctgggta agttgccaaa gaa		23
SEQ ID NO: 343	moltype = DNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 343		
acaccttata ggaaaaccag tga		23
SEQ ID NO: 344	moltype = DNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 344		
ctataagggtg tgaaagtctg gat		23
SEQ ID NO: 345	moltype = DNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 345		
cctataagggt gtgaaagtct gga		23

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SEQ ID NO: 346	moltype = DNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 346		
tagaagggat atacaaagtg gaa		23
SEQ ID NO: 347	moltype = DNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 347		
tatataccctt ctacaaattc ctc		23
SEQ ID NO: 348	moltype = DNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 348		
cactttgtat atcccttcta caa		23
SEQ ID NO: 349	moltype = DNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 349		
gtgtctattt ccactttgta tat		23
SEQ ID NO: 350	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 350		
ctggaaggca cttggcatct		20
SEQ ID NO: 351	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 351		
taggaaaacc agtgagtctg		20
SEQ ID NO: 352	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 352		
tcacgtctg ctctcctct		20
SEQ ID NO: 353	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 353		
ttggcaggat ggcttctcat		20
SEQ ID NO: 354	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 354		
accggtgccc tgggtgtaga		20
SEQ ID NO: 355	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	

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SEQUENCE: 355	organism = synthetic construct	
agaaaggctg ctgatgacac		20
SEQ ID NO: 356	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 356		
tagatgctgt ccgaggcagt		20
SEQ ID NO: 357	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 357		
tttgcaact taccagagg		20
SEQ ID NO: 358	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 358		
cagtaagatt tgggtgtctat		20
SEQ ID NO: 359	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 359		
catgagcatg cagaggtgag		20
SEQ ID NO: 360	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 360		
ctctcagtt gtgagcccat		20
SEQ ID NO: 361	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 361		
tacaaattcc tcctcagttg		20
SEQ ID NO: 362	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 362		
acagccaacg actccggccc		20
SEQ ID NO: 363	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 363		
caccacggct gtcgtcacca		20
SEQ ID NO: 364	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 364		
cttgggattg gtgacgacag		20

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SEQ ID NO: 365	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 365		
tctcataggt ggtattcaca		20
SEQ ID NO: 366	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 366		
ctggactggt atttgtgtct		20
SEQ ID NO: 367	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 367		
gcaggatggc ttctcatcgt		20
SEQ ID NO: 368	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 368		
atggcaggac tgcctcggac		20
SEQ ID NO: 369	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 369		
gattcaccgg tgcctcgggt		20
SEQ ID NO: 370	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 370		
gcatctcccc attccatgag		20
SEQ ID NO: 371	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 371		
tgagcccatg cagctctcca		20
SEQ ID NO: 372	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 372		
ccgccctgct gagcccctac		20
SEQ ID NO: 373	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 373		
gctgtgaata ccacctatga		20
SEQ ID NO: 374	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	

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SEQUENCE: 374	organism = synthetic construct	
ggattggtga cgacagccgt		20
SEQ ID NO: 375	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 375		
gtgacgacag ccgtggtgga		20
SEQ ID NO: 376	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 376		
gtgtctgagg ctggccctac		20
SEQ ID NO: 377	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 377		
gaccatcaga ggacacttgg		20
SEQ ID NO: 378	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 378		
gcctctgggt aagttgcaa		20
SEQ ID NO: 379	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 379		
ggcaacttac ccagaggcaa		20
SEQ ID NO: 380	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 380		
ggtgtctatt tccactttgt		20
SEQ ID NO: 381	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 381		
gtatatccct tctacaaatt		20
SEQ ID NO: 382	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 382		
cacataggcc attcagaaac		20
SEQ ID NO: 383	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 383		
catatgggtc attcagaaac		20

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SEQ ID NO: 384	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 384		
cacacaggcc tttcagcaac		20
SEQ ID NO: 385	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 385		
cacagatgcc attcagatac		20
SEQ ID NO: 386	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 386		
cacataggtc attcacaat		20
SEQ ID NO: 387	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 387		
cagatagtcc tttcagaaac		20
SEQ ID NO: 388	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 388		
cacataggct attcaaaatc		20
SEQ ID NO: 389	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 389		
cacatggccc atacagaaac		20
SEQ ID NO: 390	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 390		
ctcaaaggcc attcaggaac		20
SEQ ID NO: 391	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 391		
cacataggcc attcagaaac		20
SEQ ID NO: 392	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 392		
ccctgggtcc ctcaccacc		20
SEQ ID NO: 393	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	

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SEQUENCE: 393	organism = synthetic construct	
catatgggtc attcagaaac		20
SEQ ID NO: 394	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 394		
cacacaggcc tttcagcaac		20
SEQ ID NO: 395	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 395		
cacagatgcc attcagatac		20
SEQ ID NO: 396	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 396		
cacataggtc attcacaac		20
SEQ ID NO: 397	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 397		
cagatagtcc tttcagaaac		20
SEQ ID NO: 398	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 398		
agcaccacca cgtaggtgcc		20
SEQ ID NO: 399	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 399		
ctggtgaaga tgagtggcga		20
SEQ ID NO: 400	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 400		
aagttgcccc atgtcgacta		20
SEQ ID NO: 401	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 401		
ccccatgtcg actacatcga		20
SEQ ID NO: 402	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 402		
cccagagcat cccgtggaac		20

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SEQ ID NO: 403	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 403		
ccggtggtca ctctgtatgc		20
SEQ ID NO: 404	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 404		
agcacgcgca ggctgcgcac		20
SEQ ID NO: 405	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 405		
gcggcaccct cataggtgag		20
SEQ ID NO: 406	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 406		
gggccaccaa tgcccaggac		20
SEQ ID NO: 407	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 407		
gtggcccca ctgtgatgac		20
SEQ ID NO: 408	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 408		
gtgctccag cgactgcagc		20
SEQ ID NO: 409	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 409		
accctgcac caggcattgc		20
SEQ ID NO: 410	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 410		
cctgaggacc agcgggtact		20
SEQ ID NO: 411	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 411		
gtggcagtgg acacgggtcc		20
SEQ ID NO: 412	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	

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SEQUENCE: 412	organism = synthetic construct	
tctacggcgt aggcccccag		20
SEQ ID NO: 413	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 413		
caagtgtcct ctgatgtgca		20
SEQ ID NO: 414	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 414		
gtcaaagttc tagatgctgt		20
SEQ ID NO: 415	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 415		
ctgtccgagg cagtctgcc		20
SEQ ID NO: 416	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 416		
tggccgtgca tgtgttcaga		20
SEQ ID NO: 417	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 417		
tgttcagaaa ggctgctgat		20
SEQ ID NO: 418	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 418		
acacctggga gccatttgcc		20
SEQ ID NO: 419	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 419		
caccggtgcc ctgggtgtag		20
SEQ ID NO: 420	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 420		
agaggacact tggattcacc		20
SEQ ID NO: 421	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 421		
tagaactttg accatcagag		20

-continued

SEQ ID NO: 422	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 422		
gcaggactgc ctcggacagc		20
SEQ ID NO: 423	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 423		
gatggcagga ctgcctcgga		20
SEQ ID NO: 424	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 424		
cacggccaca ttgatggcag		20
SEQ ID NO: 425	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 425		
agcagccttt ctgaacacat		20
SEQ ID NO: 426	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 426		
tgatgggtcaa agttctagat		20
SEQ ID NO: 427	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 427		
atggtcaaag ttctagatgc		20
SEQ ID NO: 428	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 428		
tccgaggcag tccctgccatc		20
SEQ ID NO: 429	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 429		
atgacacctg ggagccattt		20
SEQ ID NO: 430	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 430		
ggagccattt gcctctgggt		20
SEQ ID NO: 431	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	

-continued

SEQUENCE: 431	organism = synthetic construct	
tgggtaagtt gccaaagaac		20
SEQ ID NO: 432	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 432		
ggattcacgc gtgccttggg		20
SEQ ID NO: 433	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 433		
tgaccatcag aggacacttg		20
SEQ ID NO: 434	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 434		
gaactttgac catcagagga		20
SEQ ID NO: 435	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 435		
ggacagcatc tagaactttg		20
SEQ ID NO: 436	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 436		
cctcggacag catctagaac		20
SEQ ID NO: 437	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 437		
ccaggtgtca tcagcagcct		20
SEQ ID NO: 438	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 438		
accagaggc aaatggctcc		20
SEQ ID NO: 439	moltype = AA length = 166	
FEATURE	Location/Qualifiers	
source	1..166	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 439		
SEVEFSHEYW MRHALTLAKR ARDEREVPVG AVLVLNNRVI GEGWNRAIGL HDPTAHAEIM	60	
ALRQGGLVMQ NYRLIDATLY VTFEPCVMCA GAMIHSRIGR VVFGWRNSKR GAAGSLMNVL	120	
NYPGMNHRVE ITEGILADEC AALLCDFYRM PRQVFNAQKK AQSSIN	166	
SEQ ID NO: 440	moltype = AA length = 199	
FEATURE	Location/Qualifiers	
source	1..199	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 440		

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MEASPASGPR HLMDPHIFTS NFNNGIGRHK TYLCYEVERL DNGTSVKMDQ HRGFLHNQAK 60
NLLCGFYGRH AELRFLDLVP SLQLDPAQIY RVTWFISYSP CFSAGCAGEV RAFLQENTHV 120
RLRIFAARIF DYDPLYKEAL QMLRDAGAQV SIMTYDEPKH CWDTFVDHQG CPFQPWDGLD 180
EHSQALSGRL RAILQNQGN 199

```

```

SEQ ID NO: 441      moltype = AA length = 83
FEATURE            Location/Qualifiers
source             1..83
                   mol_type = protein
                   organism = synthetic construct

```

```

SEQUENCE: 441
TNLSDIIEKE TGKQLVIQES ILMLPEEVVEE VIGNKPESDI LVHTAYDEST DENVMLLTSD 60
APEYKPWALV IQDSNGENKI KML 83

```

```

SEQ ID NO: 442      moltype = AA length = 16
FEATURE            Location/Qualifiers
source             1..16
                   mol_type = protein
                   organism = synthetic construct

```

```

SEQUENCE: 442
SGSETPGTSE SATPES 16

```

```

SEQ ID NO: 443      moltype = AA length = 18
FEATURE            Location/Qualifiers
source             1..18
                   mol_type = protein
                   organism = synthetic construct

```

```

SEQUENCE: 443
KRTADGSEFE SPKKKRKV 18

```

```

SEQ ID NO: 444      moltype = AA length = 7
FEATURE            Location/Qualifiers
source             1..7
                   mol_type = protein
                   organism = synthetic construct

```

```

SEQUENCE: 444
PKKKRKV 7

```

```

SEQ ID NO: 445      moltype = AA length = 16
FEATURE            Location/Qualifiers
source             1..16
                   mol_type = protein
                   organism = synthetic construct

```

```

SEQUENCE: 445
KRPAATKKAG QAKKKK 16

```

```

SEQ ID NO: 446      moltype = DNA length = 241
FEATURE            Location/Qualifiers
source             1..241
                   mol_type = other DNA
                   organism = synthetic construct

```

```

SEQUENCE: 446
gagggcctat ttcccatgat tccttcatat ttgcatatac gatacaaggc tgtagagag 60
ataattggaa ttaatttgac tgtaaacaca aagatattag tacaaaatac gtgacgtaga 120
aagtaataat ttcttgggta gtttcagatt ttaaaattat gttttaaata ggactatcat 180
atgcttaccg taacttgaaa gtatttcgat ttcttggctt tataatatctt gtggaaagga 240
c 241

```

```

SEQ ID NO: 447      moltype = DNA length = 204
FEATURE            Location/Qualifiers
source             1..204
                   mol_type = other DNA
                   organism = synthetic construct

```

```

SEQUENCE: 447
gtgatgagggt tttggcagta catcaatggg cgtggatagc ggtttgactc acggggattt 60
ccaagtctcc accccattga cgtcaatggg agtttgtttt ggcacaaaaa tcaacgggac 120
tttccaaaat gtcgtaacaa ctccgccccca ttgacgcaaa tgggaggtag gcgtgtacgg 180
tgggagggtct atataagcag agct 204

```

```

SEQ ID NO: 448      moltype = DNA length = 208
FEATURE            Location/Qualifiers
source             1..208
                   mol_type = other DNA
                   organism = synthetic construct

```

```

SEQUENCE: 448
ctgtgcttc tagttgccag ccatctgttg ttggccctc ccccggtgct tccttgacct 60

```

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tggaaggtgc	cactcccact	gtcctttcct	aataaaatga	ggaaattgca	tcgcattgtc	120
tgagtaggtg	tcattctatt	ctgggggggtg	gggtgggggca	ggacagcaag	ggggaggatt	180
gggaagagaa	tagcaggcat	gctggggga				208

SEQ ID NO: 449 moltype = AA length = 291
 FEATURE Location/Qualifiers
 source 1..291
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 449
 MSKSWGKFIE EEEAEMASRR NLMIVDGTNL GFRFKHNSK KPFASSYVST IQSLAKSYSA 60
 RTTIVLGDKG KSVFRLEHLP EYKGNRDEKY AQRTEEEKAL DEQFFEYLKD AFELCKTTFP 120
 TPTIRGVEAD DMAAYIVKLI GHLYDHVWLI STDGDWDTLL TDKVSRFSFT TRREYHLRDM 180
 YEHHNVDDVE QFISLKAIMG DLGDNIRGVE GIGAKRGYNI IREFGNVLDI IDQLPLPGKQ 240
 KYIQNLNASE ELLFRNLILV DLPTYCVDAL AAVGQDVLDK FTKDILEIAE Q 291

SEQ ID NO: 450 moltype = DNA length = 794
 FEATURE Location/Qualifiers
 source 1..794
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 450
 cgttacataa cttacggtaa atggcccgcc tggctgaccg cccaacgacc ccgcgccatt 60
 gacgtcaata gtaacgccaa tagggacttt ccattgacgt caatgggtgg agtattttacg 120
 gtaaaactgcc cacttggcag tacatcaagt gtatcatatg ccaagtacgc cccctattga 180
 cgtcaatgac ggtaaatggc ccgcctggca ttgtgcccag tacatgacct tatgggactt 240
 tcctacttgg cagtacatct acgtattagt catcgctatt accatggctcg aggtgagccc 300
 cacgtttcgc ttcactctcc ccatctcccc cccctcccca ccccaattt tgtatttatt 360
 tattttttaa ttattttgtg cagcgatggg ggcggggggg gggggggggc gcgcgccagg 420
 cggggcgggg cggggcgagg ggcggggcgg ggcgagggcg agaggtgcgg cggcagccaa 480
 tcagagcggc gcgctccgaa agtttccttt tatggcgagg cggcgggcgc ggcggcccta 540
 taaaaagcga agcgcgcggc gggcgggagt cgctgcgacg ctgccttcgc cccgtgcccc 600
 gctccgcgcg cgctcgcgcg cgcgcgcccc ggctctgact gaccgcgtta ctcccacagg 660
 tgagcgggcg ggacggccct tctcctccgg gctgtaatta gctgagcaag aggtaagggt 720
 ttaagggatg gttgggtggg ggggtattaa tgtttaatta cctggagcac ctgcctgaaa 780
 tcactttttt tcag 794

SEQ ID NO: 451 moltype = RNA length = 30
 FEATURE Location/Qualifiers
 source 1..30
 mol_type = other RNA
 organism = synthetic construct

SEQUENCE: 451
 atgactcaga aatgtgtccc cagttgacac 30

SEQ ID NO: 452 moltype = RNA length = 23
 FEATURE Location/Qualifiers
 source 1..23
 mol_type = other RNA
 organism = synthetic construct

SEQUENCE: 452
 agaaatgtgt cccagttga cac 23

SEQ ID NO: 453 moltype = RNA length = 23
 FEATURE Location/Qualifiers
 source 1..23
 mol_type = other RNA
 organism = synthetic construct

SEQUENCE: 453
 agaaatccgt ccttagttga cgg 23

SEQ ID NO: 454 moltype = RNA length = 22
 FEATURE Location/Qualifiers
 source 1..22
 mol_type = other RNA
 organism = synthetic construct

SEQUENCE: 454
 agacatgtgt cccagtgac ac 22

SEQ ID NO: 455 moltype = RNA length = 23
 FEATURE Location/Qualifiers
 source 1..23
 mol_type = other RNA
 organism = synthetic construct

SEQUENCE: 455
 agaaatgttt cccagttga aac 23

SEQ ID NO: 456	moltype = RNA	length = 23	
FEATURE	Location/Qualifiers		
source	1..23		
	mol_type = other RNA		
	organism = synthetic construct		
SEQUENCE: 456			
agaaatgtgt tcccagttaa cac			23
SEQ ID NO: 457	moltype = RNA	length = 23	
FEATURE	Location/Qualifiers		
source	1..23		
	mol_type = other RNA		
	organism = synthetic construct		
SEQUENCE: 457			
agaaatttgt ccccagttag caa			23
SEQ ID NO: 458	moltype = AA	length = 1080	
FEATURE	Location/Qualifiers		
source	1..1080		
	mol_type = protein		
	organism = synthetic construct		
SEQUENCE: 458			
MSSDVVRPYN TKLLPDNRKH NMFLQTFKRL NSISLNHFDL LICLYAAITN KKAEEYKSEK			60
EAHVTADSLC AINWFRPMSK RYSKYATTTF NMLELPEKEYS GHEPDAYSKN YLMSNIDSDR			120
FVWVDCRKFA KDFAYQOMELG FHEFTVLAET LLANSILVLN ESTKANWAWG TVSALYGGGD			180
KEDSTLKS KI LLAFLVDALNN HELKTREIL NQVCESLKYQ SYQDMYVDFR SVVDENGNKK			240
SPRGSMPIVT KPETDDLLIS NQRKAMISNF TKNAAAKAAK KPIPYLDRLK EHMVSLCDEY			300
NVYAAAAAIT NSNADVTARN TRNLTFIGEQ NSRRKLSVL QTTTNEKAKD ILNKINDNLI			360
QEVRYTPAPK HLGRLANLF DTLKEKDINN IENEEBKQNV INDCIEQVYD DCRSLNRNPI			420
AALLKHISRY YEDFSAKNFL DGAKLNVLTE VVNROKAHPT IWSEKAYTWI SKFDKNRRQA			480
NSSLVGWVVP PEEVHKEKIA GQQSMWVTL TLLDDGKWKV HHIPFSDRSY YSEVYAYNP			540
LPYLDGGIPR QSKFGNKPT NLTAESQALL ANSKYKKANK SFLRAKENAT HNVRSPTNS			600
LCIRLLKDSA GNQMPDKIGN VLFQMGINHK ITVGKPNYKI EVGDRFLGFD QNQSENHTYA			660
VLQRVSESSH DTHHFNGWDV KVLEKGKVT S DVIVRDEVYD QLSYEGVPYD SSKFAEWRDK			720
RRRFVLENL S IQLEEGKTF L TEFDKLNKDS LYRWNMNYLK LRRKAIRAGG KEPAKIAKTE			780
IFELAVERFG PINLGLSLQI SLKMIA SFPG VVQSYFVSFG CVDDASKKHA D SMLFTFMCA			840
AE EKRTNKRE EKTNRAASFI LQKAYLHGCK MIVCEDDLPV ADGKTGKAQN ARMDWCARA			900
LAKKVNDCGV AMSICYRAIP AYMSHQDPF VHMQDKKTSV LRPRMEVYNK DSTRDYHVAG			960
LRRLNLSKSD AGTSVYVYRQA ALHFCEALGV SPELVKNKKT HAAELGKHMG SAMLMPWRGG			1020
RVYIASKKLT SDAKSVKYCG EDMWQYHADE IAAVNIAMYE VCCQTGAFGK KQKKSDELPG			1080
SEQ ID NO: 459	moltype = AA	length = 1080	
FEATURE	Location/Qualifiers		
source	1..1080		
	mol_type = protein		
	organism = synthetic construct		
SEQUENCE: 459			
MSSDVVRPYN TKLLPDNRKH NMFLQTFKRL NSISLNHFDL LICLYAAITN KKAEEYKSEK			60
EAHVTADSLC AINWFRPMSK RYSKYATTTF NMLELPEKEYS GHEPDAYSKN YLMSNIDSDR			120
FVWVDCRKFA KDFAYQOMELG FHEFTVLAET LLANSILVLN ESTKANWAWG TVSALYGGGD			180
KEDSTLKS KI LLAFLVDALNN HELKTREIL NQVCESLKYQ SYQDMYVDFR SVVDENGNKK			240
SPRGSMPIVT KPETDDLLIS NQRKAMISNF TKNAAAKAAK KPIPYLDRLK EHMVSLCDEY			300
NVYAAAAAIT NSNADVTARN TRNLTFIGEQ NSRRKLSVL QTTTNEKAKD ILNKINDNLI			360
QEVRYTPAPK HLGRLANLF DTLKEKDINN IENEEBKQNV INDCIEQVYD DCRSLNRNPI			420
AALLKHISRY YEDFSAKNFL DGAKLNVLTE VVNROKAHPT IWSEKAYTWI SKFDKNRRQA			480
NSSLVGWVVP PEEVHKEKIA GQQSMWVTL TLLDDGKWKV HHIPFSDRSY YSEVYAYNP			540
LPYLDGGIPR QSKFGNKPT NLTAESQALL ANSKYKKANK SFLRAKENAT HNVRSPTNS			600
LCIRLLKDSA GNQMPDKIGN VLFQMGINHK ITVGKPNYKI EVGDRFLGFD QNQSENHTYA			660
VLQRVSESSH DTHHFNGWDV KVLEKGKVT S DVIVRDEVYD QLSYEGVPYD SSKFAEWRDK			720
RRRFVLENL S IQLEEGKTF L TEFDKLNKDS LYRWNMNYLK LRRKAIRAGG KEPAKIAKTE			780
IFELAVERFG PINLGLSLQI SLKMIA SFPG VVQSYFVSFG CVDDASKKHA D SMLFTFMCA			840
AE EKRTNKRE EKTNRAASFI LQKAYLHGCK MIVCEDDLPV ADGKTGKAQN ARMDWCARA			900
LAKKVNDCGV AMSICYRAIP AYMSHQDPF VHMQDKKTSV LRPRMEVYNK DSTRDYHVAG			960
LRRLNLSKSD AGTSVYVYRQA ALHFCEALGV SPELVKNKKT HAAELGKHMG SAMLMPWRGG			1020
RVYIASKKLT SDAKSVKYCG EDMWQYHADE IAAVNIAMYE VCCQTGAFGK KQKKSDELPG			1080
SEQ ID NO: 460	moltype = AA	length = 1080	
FEATURE	Location/Qualifiers		
source	1..1080		
	mol_type = protein		
	organism = synthetic construct		

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KEDSTLKS	KSKI	LLAFVDALNN	HELKTKREIL	NQVCESLKYQ	SYQDMYVD	SVVDENG	NKK	240
SPRGSMP	IVT	KFETDD	LISD	NQRKAMIS	NF	TKNAAAKA	K	300
NPYAWAA	AIT	NSNADVT	ARN	TRNLTFI	GEQ	NSRRKRL	SVL	360
QEVRYT	PAPK	HLGRDL	ANLF	DTLKEK	DINN	IENEEKQ	NV	420
AALLKHIS	RY	YEDFSA	KNFL	DGAKLNV	LTE	VVNRQKA	HT	480
NSSLVGV	WVVP	PEEVHKE	KIA	GQQSMM	MVTL	TLLDDG	KWVK	540
LPYLDGG	IPR	QSKFGN	KPTT	NLTAES	QALL	ANSKYK	KANK	600
LCIRLLK	KDSA	GNQMF	DKIGN	VLFGM	QINHK	ITVGKPN	YKI	660
VLQRVSE	SSH	DTHHF	NGWDV	KVLEK	GKVTS	DVIVRDE	VYD	720
RRRFVLE	NLS	IQLEEG	KTF	TEFDK	LNKDS	LYRWNM	NYLK	780
IFELAVE	RFG	PINLGS	LSQI	SLKMI	ASF	FKG	VVQSY	840
AEKRTN	KRE	EKTNR	AASFI	LQKAY	LHGCK	MIVCED	DLPV	900
LAKKVND	GCV	AMSICY	RAIP	AYMSSH	QDPF	VHMQDK	KTSV	960
LRRMLNS	KSD	AGTSV	YRQA	ALHFC	EALGV	SPELVK	NKKT	1020
RVYIASK	KL	SDAKSV	KYCG	EDMWQ	YHADE	IAAVNI	AMYE	1080
				VCCQT	GAFGK	KQKKS	DELPG	

SEQ ID NO: 461 moltype = AA length = 166

FEATURE Location/Qualifiers

source 1..166
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 461

SEVEFS	SHEYW	MRHALT	LAKR	ARDERE	VPVG	AVLVLN	NRVI	GEGWN	RAIGL	HDPTA	HAEIM	60
ALRQGG	LVLMQ	NYRLID	ATLY	VTPEPC	VMCA	GAMIHS	RIGR	VVFGVR	NSKR	GAAGS	LMNVL	120
NYPGMN	HRVE	ITEGIL	ADEC	AALLCD	FYRM	PRQVF	NAQKK	AQSSIN				166

SEQ ID NO: 462 moltype = AA length = 17

FEATURE Location/Qualifiers

source 1..17
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 462

KRTADG	SESE	PKKKRK	V									17
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SEQ ID NO: 463 moltype = AA length = 1315

FEATURE Location/Qualifiers

source 1..1315
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 463

MKRTAD	GSEF	ESPKKK	RRKVS	EVEFS	SHEYWM	RHALT	LAKRA	RDERE	VPVGA	VLVLNN	RVIG	60
EGWNRA	IGLH	DPTAHA	EIMA	LRQGG	LVLMQN	YRLID	ATLYV	TFPEPC	VMCAG	AMIHS	RIGRV	120
VFGVRN	SKRG	AAGSLM	NVNLN	YPGMN	HRVEI	TEGILA	DECA	ALLCD	FYRMP	RQVFNA	QKKA	180
QSSINS	GSS	GGSKRT	ADGS	EFESP	KKRK	VSGSS	GSSM	SSDVVR	PYNT	KLLPD	NRKHN	240
MPLQTF	KRLN	SISLNH	FDLL	ICLYAA	ITNK	KABEYK	SEKE	AHVTA	DSLCA	INWFR	PMSKR	300
YSKYAT	TTFN	MLELF	KEYSG	HEPDAY	SKNY	LMSNID	SDRF	VWVDC	RKFAK	DFAYQ	MELGF	360
HEFTVL	AETL	LANSIL	VLNE	STKAN	WAWGT	VSALY	GGGDK	EDSTL	KSKIL	LAFVD	DALNNH	420
ELKTKR	EILN	QVCE	SLKYQS	YQDMY	VDPRS	VVDENG	NKKS	PRGSMP	IVTK	FETDD	LISDN	480
QRKAMI	SNFT	KNAAAK	AAKK	PIPYLD	RDLKE	HMVSL	CDEYN	VYAWAA	AITN	SNADV	TARNT	540
RNLTFI	GEQN	SRKRL	SVLQ	TTTNEK	AKDI	LNKIND	NLIQ	EVRYT	PAPKH	LGRDL	ANLFD	600
TLKEKD	INNI	ENEEK	QNV	NDCEI	QYVDD	CRSLNR	NP	ALLKH	ISRY	EDFSA	KNFLD	660
GAKLNV	LTE	VNRQKA	HT	IFSKAY	TWIS	KFDKNR	RQAN	SSLVG	WVVP	EEVHKE	KIAG	720
QQSMM	MVTLT	LDDGK	WVKH	HIPFSD	SRYY	SEVYAY	NP	PYLDGG	IPRQ	SKFGN	KPTTN	780
LTAES	QALLA	NSKYK	KANKS	FLRAKEN	ATH	NVRVSP	NTSL	CIRLLK	DSAG	NQMF	DKIGNV	840
LFQM	QINHKI	TVGKPN	YKIE	VGDRFL	GFDQ	NQSENH	TYAV	LQRVSE	SSHD	THHFNG	WDVK	900
VLEKGV	TS	VIVRDE	VYDQ	LSYEG	VPYDS	SKFAE	WRDKR	RRFVLE	NLSI	QLEEG	KTF	960
EPDKLN	KDSL	YRWNM	NYLKL	LRKAIR	AGGK	EPAKIA	KTEI	FELAVE	RFGP	INLGS	LSQIS	1020
LKMIAS	FKGV	VQSYF	SVSGC	VDDASK	KAHD	SMLFT	FMCAA	BEKRTN	KREE	KTNR	AASFIL	1080
QKAYLH	GCKM	IVCED	DLPVA	DGKTG	KAQNA	DRMDW	CARAL	AKKVND	GCV	MSICY	RAIPA	1140
YMSSHQ	DPFV	HMQDK	KTSVL	RPRFM	EVNKD	SIRDYH	VAGL	RRMLNS	KSDA	GTSVY	RQAA	1200
LHFC	EALGVS	PELVKN	KTH	AELGK	HMG	AMLMP	WRGGR	VYIASK	KLTS	DAKSV	KYCGE	1260
DMWQY	HAAEI	AAVNI	AMYE	CCQTG	AFGKK	QKKS	DELPGK	RPAAT	KKAGQ	AKKKK		1315

SEQ ID NO: 464 moltype = AA length = 1453

FEATURE Location/Qualifiers

source 1..1453
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 464

MKRTAD	GSEF	ESPKKK	RRKVM	EASPA	SGRPH	LMDPH	IFTSN	FNNGI	GRHKT	YLCYE	VERLD	60
NGTSVK	MDQH	RGFLHN	QAKN	LLCGFY	GRHA	ELRFLD	LVPS	LQLDPA	QIYR	VTWFIS	YSPC	120
FSAGCA	GEVR	AFLQEN	THVR	LRIFA	ARIFD	YDPLY	KEALQ	MLRDAG	AQVS	IMTYDE	FKHC	180
WDTFV	DHQC	PFQPD	WGLDE	HSQAL	SGR	LRL	AILQN	QGNSG	GSSG	GSKRTA	DGSEF	240
KRKVS	GSSG	GSMSSD	VVRP	YNTKL	LDPNR	KHNMF	LQTFK	RLNSIS	LNHF	DLICLY	AAI	300
TNKA	AEYKS	PELKV	NKTH	LCAIN	WFRPM	SKRYS	KYATT	TFNM	LELFKE	YSGHE	P	360
KNYLM	SNIDS	DRFVW	DCRK	FAKDF	AYQME	LGFHE	FTVLA	ETLLAN	SILV	LNES	T	420

SEQ ID NO: 465	moltype = AA length = 34	
FEATURE	Location/Qualifiers	
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	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 465		
SGGSSGGSKR TADGSEFESP KKKRKVSGGS SGGSS		34
SEQ ID NO: 466	moltype = AA length = 1080	
FEATURE	Location/Qualifiers	
source	1..1080	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 466		
MSSDVVRPYN TKLLPDNRKH NMFLQTFKRL NSISLNHF DL LICYLAITN KKAEEKSEK		60
EAHVTADSLC AINWFRPM SK RYSKYATTF NMLBLFKEYS GHEPDAYSKN YLMSNIDSDR		120
FVWVDCRKFA KDFAYQME LG FHEFTVLAET LLANSILVLN ESTKANWAWG TVSALYGGGD		180
KEDSTLSKSI LFAFVDALNN HELKTKREIL NQVCESLKYQ SQDMYVDVFR SVVDENGNNK		240
SPRGSMPIVT KLETDDLISD NQRKAMISNF TNKAAAKAAK KPYPYLDRLK EHMVSLCDEY		300
NVYAAAAAIT NSNADVTARN TRNLTFIGE Q NSRRKRLSVL QTTTNEKAKD ILNKINDNLI		360
QEVRYTPAPK HGRDLANLF DTLKEKDINN IENEEKQVN INDCEIQVVD DCRSLNRNPI		420
AALLKHISRY YLDEFSAKNFL DGAKNLVLT E VVNQRKAHPT INSEKAYTWI SKFDKNRQA		480
NSSLVGWVVP PEEVHKIEKIA GQQSMWVVT L TLLDDGKWKV HHIPFSDRSY YSEVYAYNP		540
LPYLDGGIPR QSKFGNKPT NLTAESQALL ANSKYKAKN SF LRAKENAT HNVRVSPNTS		600
LCIRLLKDSA GNQMPDKIGN VLFPGQINHK IITVGKNYKI EVGDRFLGFD QNQSENHTYA		660
VLQRVSESSH DTHHFNGWDV KVLKKGKVT DVIVRDEVYD QLSYEGVFPYD SSKFAEWRDK		720
RRRFVLENLS IQLLEGKTFL TEFDKLNKLS LYRNMMNYLK LRLKAIRAGG KEFAKIAMTE		780
IFELAVERFG PINLGSLSQI SKLMIASFKG VVQYSFVSFG CVDDASKAKH D SMLFTFMCA		840
AEERKTNKRE EKTNRASAF LQKAYLHGCK MIVCEDLPV ADGKTGKAQN ADRMDWCARA		900
LAKKVNDCGV AMSICYRAIP AYMSHQDPF VHMQDKKTSV LRPFRMEVKN DSIRDYHVAG		960
LRRLMNSKSD AGTSVYYRQA ALHFCEALGV SPELVKNNKT HAAELGKMGK SAMLMPWRGK		1020
VVYIAKSKLT SDAKSVKYCG EDMWQYHAE IIAENVIAMYE VCCOTGAPGK KOKKSDELPG		1080

SEQ ID NO: 467	moltype = AA length = 1080				
FEATURE	Location/Qualifiers				
source	1..1080				
	mol_type = protein				
	organism = synthetic construct				
SEQUENCE: 467					
MSSDVVRPYN	TKLLPDNRKH	NMFLQTFKRL	NSISLNHFDL	LICLYAAITN	KKAEYKSEK 60
EAHTADSLC	AINWFRPMSK	RYSKYATTF	NMLELFPKEYS	GHEPDAYSKN	YLMNSIDSDR 120
FVWVDCRKFA	KDFAYQMBELG	FHEFTVLAET	LLANSILVLN	ESTKANWAWG	TVSALYGGGD 180
KEDSTLSKSI	LLAFVYDALNN	HELTKTKREL	NQVCESLKYQ	SYQDKMYVDFR	SVVDENGNKK 240
SPRGSMPIVT	KETFDTLISD	NQRKAMISNF	TQNAAKAAAK	KHIPYDLRLK	EHMVSLCDEY 300
NVYAAAAIT	NSNADVTARN	TRNLTFIGEIQ	NSRRKRLSVL	QTTTNEKAKD	ILNKINDNLI 360
QEVRYTPAPK	HLGRDLANLF	DTLKEKDINN	IENEEEEKQNV	INDCIEQVVD	DCRSLNRNPI 420
AALLKHISRY	YEDFSAKNFL	DGAKLMVLTE	VVNRQKAHP	IWSEKAYTWI	SKPDKNRRQA 480
NSSLGVVVVP	PEEVHKEKIA	GQQSMNVVTL	TLDDDGKWKV	KHIPFSDSRV	YSEVYAYNPN 540
LPYLDGGIPR	QSKFGNKPTT	NLTAESQALL	ANSKYKKANK	SPLRAKENAT	HNVRVSPNTS 600
LCIRLLKDSA	GNQMFDPKIGN	VLFGMQINH	ITVGKPNKYI	QVGRDRFLGFD	QNQSENHTYA 660
VLQVRVSSSH	DTTHFNGWDV	KVLEKEGKVS	DVIVREBVDY	ELSVYEGVPYD	SKKPAEWRDK 720
RRRFVLENLS	IQLBEGKTFL	TEPDKLNKDS	LYRWNNMYLK	LLRKAIRAGG	KEFAKIAKTE 780
IFELAVERTK	PINLGSLSQI	SLKMIASFQG	VVQSYFSVSG	CVDDGSKAKN	DSMLFTFMCA 840
AEEKRTNFKG	EKTNRAASF	LQKAYLHGCK	MIVCEDDLVP	ADAGSTGKAG	ADRMDCWARA 900
LAKKVNDCGV	AMTICYRAIP	AYMSSHODPF	VHMQDKTSV	LRPRFMEVKN	DSIRDYHVAG 960

-continued

LRRMLNSKSD	AGTSVYYRQA	ALHFCEALGV	SPELVKNKKT	HAELGKHM	SAMLMWRGG	1020
RVYIASKKLT	SDAKSVKYCG	EDMWQYHADE	IAAVNIAMYE	VCCQTGAFGK	KQKKSDELPG	1080

1. A Cas12i polypeptide comprising an amino acid substitution at E336, V880, G883, D892, and/or M923 of SEQ ID NO: 458; optionally, wherein the Cas12i polypeptide has a sequence identity of at least about 80%, at least about 85%, at least about 90%, at least about 95%, or at least about 99% to SEQ ID NO: 458.

2. The Cas12i polypeptide of claim 1, wherein the Cas12i polypeptide comprises an amino acid substitution at one position selected from the group consisting of E336, V880, G883, D892, and M923 of SEQ ID NO: 458; optionally, wherein the amino acid substitution is a substitution with a positively charged amino acid residue (such as, Lysine (Lys/K), Arginine (Arg/R), Histidine (His/H)), and optionally a substitution with Arginine (Arg/R).

3. The Cas12i polypeptide of claim 2, wherein the Cas12i polypeptide comprises an amino acid substitution E336R relative to SEQ ID NO: 458; optionally, wherein the Cas12i polypeptide comprises the amino acid sequence of SEQ ID NO: 467, or an amino acid sequence having a sequence identity of at least about 80% (e.g., at least about 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, 99.1%, 99.2%, 99.3%, 99.4%, 99.5%, 99.6%, 99.7%, 99.8%, 99.9%, or 100%) to the amino acid sequence of SEQ ID NO: 467.

4. The Cas12i polypeptide of claim 2, wherein the Cas12i polypeptide comprises one amino acid substitution selected from the group consisting of V880R, G883R, D892R, and M923R relative to SEQ ID NO: 458.

5. The Cas12i polypeptide of claim 1, wherein the Cas12i polypeptide comprises two amino acid substitutions at any two positions of E336, V880, G883, D892, and M923 of SEQ ID NO: 458; optionally, wherein each of the two amino acid substitutions is independently a substitution with a positively charged amino acid residue (such as, Lysine (Lys/K), Arginine (Arg/R), Histidine (His/H)), and optionally each a substitution with Arginine (Arg/R).

6. The Cas12i polypeptide of claim 5, wherein the Cas12i polypeptide comprises amino acid substitutions E336R and one amino acid substitution selected from the group consisting of V880R, G883R, D892R, and M923R relative to SEQ ID NO: 458.

7. The Cas12i polypeptide of claim 6, wherein the Cas12i polypeptide comprises amino acid substitutions E336R and D892R relative to SEQ ID NO: 458; optionally, wherein the Cas12i polypeptide comprises the amino acid sequence of SEQ ID NO: 459, or an amino acid sequence having a sequence identity of at least about 80% (e.g., at least about 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, 99.1%, 99.2%, 99.3%, 99.4%, 99.5%, 99.6%, 99.7%, 99.8%, 99.9%, or 100%) to the amino acid sequence of SEQ ID NO: 459.

8. The Cas12i polypeptide of claim 1, wherein the Cas12i polypeptide further comprises an additional amino acid substitution at a position selected from the group consisting of K109, L112, D125, 127, F144, L147, A148, L151, L157, V195, Y226, F252, I258, M293, W305, A308, I309, S312, A314, D315, V316, A318, L324, I327, A348, L352, Y365, L372, L376, L379, L383, I405, L424, I427, A436, F439,

A443, V447, A457, H458, P459, T460, S463, S814, F859, A864, H867, Y977, S1031, A1053, and F1068 of SEQ ID NO: 458; optionally, wherein the additional amino acid substitution is a substitution with a positively charged amino acid residue (such as, Lysine (Lys/K), Arginine (Arg/R), Histidine (His/H)), and optionally a substitution with Arginine (Arg/R).

9. The Cas12i polypeptide of claim 1, wherein the Cas12i polypeptide has spacer sequence-specific (on-target) dsDNA cleavage activity; optionally, wherein the Cas12i polypeptide substantially retains the spacer sequence-specific (on-target) dsDNA cleavage activity of SEQ ID NO: 458 or SEQ ID NO: 1; and/or optionally, wherein the Cas12i polypeptide has an increased spacer sequence-specific (on-target) dsDNA cleavage activity compared to that of SEQ ID NO: 458 or SEQ ID NO: 1 when both are used in combination with a same guide nucleic acid, e.g., an increase by at least about 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 100%, 110%, 120%, 130%, 140%, 150%, 160%, 170%, 180%, 190%, 200%, 210%, 220%, 230%, 240%, 250%, 260%, 270%, 280%, 290%, 300%, or more.

10. The Cas12i polypeptide of claim 1, wherein the Cas12i polypeptide substantially lacks spacer sequence-independent (off-target) dsDNA cleavage activity; optionally, wherein the Cas12i polypeptide substantially lacks the spacer sequence-independent (off-target) dsDNA cleavage activity of SEQ ID NO: 458 or SEQ ID NO: 1; and/or optionally, wherein the Cas12i polypeptide has a decreased spacer sequence-independent (off-target) dsDNA cleavage activity compared to that of SEQ ID NO: 458 or SEQ ID NO: 1 when both are used in combination with a same guide nucleic acid, e.g., a decrease by at least about 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, or 100%.

11. The Cas12i polypeptide of claim 1, wherein the Cas12i polypeptide is further engineered to substantially lack spacer sequence-specific (on-target) dsDNA cleavage activity; optionally, wherein the Cas12i polypeptide substantially lacks the spacer sequence-specific (on-target) dsDNA cleavage activity of SEQ ID NO: 458 or SEQ ID NO: 1; and/or optionally, wherein the Cas12i polypeptide has a decreased spacer sequence-specific (on-target) dsDNA cleavage activity compared to that of SEQ ID NO: 458 or SEQ ID NO: 1 when both used in combination with a same guide nucleic acid, e.g., a decrease by at least about 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, or 100%.

12. The Cas12i polypeptide of claim 11, wherein the Cas12i polypeptide comprise a further amino acid substitution at a position selected from the group consisting of D650, D700, E875, and D1049 of SEQ ID NO: 458; optionally, wherein the further amino acid substitution is a substitution with a non-polar amino acid residue (such as, Glycine (Gly/G), Alanine (Ala/A), Valine (Val/V), Cysteine (Cys/C), Proline (Pro/P), Leucine (Leu/L), Isoleucine (Ile/I), Methionine (Met/M), Tryptophan (Trp/W), Phenylalanine (Phe/F)), and optionally a substitution with Alanine (Ala/A).

13. The Cas12i polypeptide of claim 12, wherein the Cas12i polypeptide comprises amino acid substitutions E336R and D1049A relative to SEQ ID NO: 458; optionally, wherein the Cas12i polypeptide comprises the amino acid sequence of SEQ ID NO: 466, or an amino acid sequence having a sequence identity of at least about 80% (e.g., at least about 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, 99.1%, 99.2%, 99.3%, 99.4%, 99.5%, 99.6%, 99.7%, 99.8%, 99.9%, or 100%) to the amino acid sequence of SEQ ID NO: 466.

14. The Cas12i polypeptide of claim 1, wherein the Cas12i polypeptide is further engineered to be a nickase.

15. The Cas12i polypeptide of claim 14, wherein the Cas12i polypeptide comprise a further amino acid substitution at a position selected from the group consisting of W896, S924, and S925 to of SEQ ID NO: 458; optionally, wherein the Cas12i polypeptide comprise a further amino acid substitution selected from the group consisting of W896R, W896P, W896K, S924R, S924F, S924D, S924E, S924H, S925R, and S925T relative to SEQ ID NO: 458.

16. The Cas12i polypeptide of claim 1, wherein the Cas12i polypeptide further comprises a functional domain fused to the Cas12i polypeptide; optionally, wherein the functional domain is selected from the group consisting of a nuclear localization signal (NLS), a nuclear export signal (NES), a base editing domain, for example, a deaminase or a catalytic domain thereof, a base excising domain, an uracil glycosylase inhibitor (UGI) or a catalytic domain thereof, an uracil glycosylase (UNG) or a catalytic domain thereof, a methylpurine glycosylase (MPG) or a catalytic domain thereof, a methylase or a catalytic domain thereof, a demethylase or a catalytic domain thereof, an transcription activating domain (e.g., VP64 or VPR), an transcription inhibiting domain (e.g., KRAB moiety or SID moiety), a reverse transcriptase or a catalytic domain thereof, an exonuclease (e.g., T5E (SEQ ID NO: 449)) or a catalytic domain thereof, a destabilized domain (e.g., destabilized domains (DD) of *E. coli* dihydrofolate reductase (ecDHFR)), a histone residue modification domain, a nuclease catalytic domain (e.g., FokI), a transcription modification factor, a light gating factor, a chemical inducible factor, a chromatin visualization factor, a targeting polypeptide for providing binding to a cell surface portion on a target cell or a target cell type, a reporter (e.g., fluorescent) polypeptide or a detection label (e.g., GST, HRP, CAT, GFP, HcRed, DsRed, CFP, YFP, BFP), a localization signal, a polypeptide targeting moiety, a DNA binding domain (e.g., MBP, Lex A DBD, Gal4 DBD), an epitope tag (e.g., His, myc, V5, FLAG, HA, VSV-G, Trx, etc), a transcription release factor, an HDAC, a moiety having ssRNA cleavage activity, a moiety having dsRNA cleavage activity, a moiety having ssDNA cleavage activity, a moiety having dsDNA cleavage activity, a DNA or RNA ligase, a functional domain exhibiting activity to modify a target DNA, selected from the group consisting of: methyltransferase activity, DNA repair activity, DNA damage activity, dismutase activity, alkylation activity, dealkylation activity, depurination activity, oxidation activity, deoxidation activity, pyrimidine dimer forming activity, integrase activity, transposase activity, recombinase activity, polymerase activity, ligase activity, helicase activity, photolyase activity, glycosylase activity, acetyl transferase activity, deacetylase activity, kinase activity, phosphatase activity, ubiquitin ligase activity, deubiquitination activity, adenylation activity, deadenylation activity, SUMOylation

activity, deSUMOylation activity, ribosylation activity, deribosylation activity, myristoylation activity, demyristoylation activity, glycosylation activity (e.g., from O-GlcNAc transferase), deglycosylation activity, and a catalytic domain thereof, and a functional fragment (e.g., a functional truncation) thereof, and any combination thereof;

optionally, wherein the NLS comprises or is SV40 NLS (SEQ ID NO: 444), bpSV40 NLS (BP NLS, bpNLS, SEQ ID NO: 443 or 462), or NP NLS (*Xenopus laevis* Nucleoplasmin NLS, nucleoplasmin NLS, SEQ ID NO: 445);

optionally, wherein the deaminase or catalytic domain thereof is an adenine deaminase (e.g., TadA, such as, TadA8e, TadA8.17, TadA8.20, TadA9) or a catalytic domain thereof, for example, TadA8e-V106W (SEQ ID NO: 439), TadA8e-W106V (SEQ ID NO: 461);

optionally, wherein the deaminase or catalytic domain thereof is a cytidine deaminase (e.g., APOBEC, such as, APOBEC3, for example, APOBEC3A, APOBEC3B, APOBEC3C; DddA) or a catalytic domain thereof, for example, hAPOBEC3-W104A (SEQ ID NO: 440); and/or

optionally, wherein the UGI is human UGI domain (such as, SEQ ID NO: 441).

17. The Cas12i polypeptide of claim 16, wherein the Cas12i polypeptide comprises amino acid substitutions E336R and D1049A relative to SEQ ID NO: 458, and a base editing domain, for example, a deaminase or a catalytic domain thereof.

18. The Cas12i polypeptide of claim 17, wherein the Cas12i polypeptide comprises the amino acid sequence of SEQ ID NO: 463, or an amino acid sequence having a sequence identity of at least about 80% (e.g., at least about 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, 99.1%, 99.2%, 99.3%, 99.4%, 99.5%, 99.6%, 99.7%, 99.8%, 99.9%, or 100%) to the amino acid sequence of SEQ ID NO: 463.

19. The Cas12i polypeptide of claim 17, wherein the Cas12i polypeptide comprises the amino acid sequence of SEQ ID NO: 464, or an amino acid sequence having a sequence identity of at least about 80% (e.g., at least about 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, 99.1%, 99.2%, 99.3%, 99.4%, 99.5%, 99.6%, 99.7%, 99.8%, 99.9%, or 100%) to the amino acid sequence of SEQ ID NO: 464.

20. A system comprising:

- (1) the Cas12i polypeptide of claim 1 or a polynucleotide encoding the Cas12i polypeptide; and
- (2) a guide nucleic acid or a polynucleotide encoding the guide nucleic acid, the guide nucleic acid comprising:
 - (i) a direct repeat (DR) sequence capable of forming a complex with the Cas12i polypeptide; and
 - (ii) a spacer sequence capable of hybridizing to a target sequence of a target DNA, thereby guiding the complex to the target DNA;

optionally, wherein the direct repeat sequence is 5' to the spacer sequence; and/or

optionally, wherein the guide nucleic acid is a guide RNA (gRNA).

21. The system of claim 20, wherein the direct repeat sequence has substantially the same secondary structure as the secondary structure of any one of SEQ ID NOs: 11 and 451-457;

optionally, wherein the direct repeat sequence:

(1) comprises the polynucleotide sequence of any one of SEQ ID NOs: 11 and 451-457; or

(2) comprises a polynucleotide sequence having a sequence identity of at least about 80% (e.g., at least about 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, 99.1%, 99.2%, 99.3%, 99.4%, 99.5%, 99.6%, 99.7%, 99.8%, 99.9%, or 100%) to any one of SEQ ID NOs: 11 and 451-457;

optionally, wherein the direct repeat sequence comprises the polynucleotide sequence of SEQ ID NO: 452.

22. The system of claim **20**, wherein the target sequence comprises about or at least about 16 contiguous nucleotides of the target DNA, e.g., about or at least about 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, or more contiguous nucleotides of the target DNA, or in a numerical range between any two of the preceding values, e.g., from about 16 to about 50, or from about 17 to about 22 contiguous nucleotides of the target DNA; optionally, wherein the target sequence comprises about 20 contiguous nucleotides of the target DNA.

23. The system of claim **20**, wherein the reversely complementary sequence of the target sequence is immediately 3' to a protospacer adjacent motif (PAM); optionally the PAM is 5'-TN, 5'-TTN, or 5'-GCC, wherein N is A, T, G, or C.

24. The system of claim **20**, wherein the spacer sequence is about or at least about 16 nucleotides in length, e.g., about or at least about 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, or more nucleotides in length, or in a length of a numerical range between any two of the preceding values, e.g., in a length of from about 16 to about 50 nucleotides, or from about 17 to about 22 nucleotides; optionally, wherein the spacer sequence is about 20 nucleotides in length.

25. The system of claim **20**, wherein the guide nucleic acid comprises a plurality (e.g., 2, 3, 4, 5 or more) of the spacer sequences capable of hybridizing to a plurality of the target sequences, respectively.

26. The system of claim **25**, wherein the guide nucleic acid comprises, from 5' to 3', the direct repeat sequence, the spacer sequence, the direct repeat sequence, the spacer sequence, and the direct repeat sequence.

27. The system of claim **20**, wherein the target DNA is a dsDNA, such as, a eukaryotic dsDNA, e.g., a gene in a eukaryotic cell.

28. A polynucleotide encoding the Cas12i polypeptide of claim **1**.

29. A vector comprising the polynucleotide of claim **28**; optionally wherein the vector is a plasmid vector, a recombinant AAV (rAAV) vector, or a recombinant lentivirus vector.

30. A ribonucleoprotein (RNP) comprising the Cas12i polypeptide of claim **1** and a guide nucleic acid.

31. A lipid nanoparticle (LNP) comprising the Cas12i polypeptide of claim **1**.

32. A method for modifying a target DNA, comprising contacting the target DNA with the system of claim **20**,

wherein the spacer sequence is capable of hybridizing to a target sequence of the target DNA, wherein the target DNA is modified by the complex.

33. The method of claim **32**, wherein the target DNA is in a cell;

optionally, wherein the cell is a eukaryotic cell (e.g., an animal cell, a vertebrate cell, a mammalian cell, a non-human mammalian cell, a non-human primate cell, a rodent (e.g., mouse or rat) cell, a human cell, a plant cell, or a yeast cell) or a prokaryotic cell (e.g., a bacteria cell);

optionally, wherein the cell is from a plant or an animal; optionally, wherein the plant is a dicotyledon; optionally selected from the group consisting of soybean, cabbage (e.g., Chinese cabbage), rapeseed, *brassica*, watermelon, melon, potato, tomato, tobacco, eggplant, pepper, cucumber, cotton, alfalfa, eggplant, grape;

optionally, wherein the plant is a monocotyledon; optionally selected from the group consisting of rice, corn, wheat, barley, oat, sorghum, millet, grasses, Poaceae, *Zizania*, *Avena*, *Coix*, *Hordeum*, *Oryza*, *Panicum* (e.g., *Panicum miliaceum*), *Secale*, *Setaria* (e.g., *Setaria italica*), *Sorghum*, *Triticum*, *Zea*, *Cymbopogon*, *Saccharum* (e.g., *Saccharum officinarum*), *Phyllostachys*, *Dendrocalamus*, *Bambusa*, *Yushania*; and/or

optionally, wherein the animal is selected from the group consisting of pig, ox, sheep, goat, mouse, rat, alpaca, monkey, rabbit, chicken, duck, goose, fish (e.g., zebra fish).

34. The method of claim **32**, wherein the modification comprises one or more of cleavage, base editing, repairing, and exogenous sequence insertion or integration of the target DNA.

35. A cell modified by the method of claim **32**.

36. A pharmaceutical composition comprising (1) the system of claim **20**; and (2) a pharmaceutically acceptable excipient.

37. A method for diagnosing, preventing, or treating a disease in a subject in need thereof, comprising administering to the subject the pharmaceutical composition of claim **36**, wherein the disease is associated with a target DNA, wherein the spacer sequence is capable of hybridizing to a target sequence of the target DNA, wherein the target DNA is modified by the complex, and wherein the modification of the target DNA diagnose, prevents, or treats the disease.

38. The method of claim **37**, wherein the disease is selected from the group consisting of Angelman syndrome (AS), Alzheimer's disease (AD), transthyretin amyloidosis (ATTR), transthyretin amyloid cardiomyopathy (ATTR-CM), cystic fibrosis (CF), hereditary angioedema, diabetes, progressive pseudohypertrophic muscular dystrophy, Duchenne muscular dystrophy (DMD), Becker muscular dystrophy (BMD), spinal muscular atrophy (SMA), alpha-1-antitrypsin deficiency, Pompe disease, myotonic dystrophy, Huntington's disease (HTT), fragile X syndrome, Friedreich ataxia, amyotrophic lateral sclerosis (ALS), frontotemporal dementia, hereditary chronic kidney disease, hyperlipidemia, Leber congenital amaurosis (LCA), sickle cell disease, thalassemia (e.g., β -thalassemia), Parkinson's disease (PD), myelodysplastic syndrome (MDS), retinitis pigmentosa (RP), age-related macular degeneration (AMD), Hepatitis B, nonalcoholic fatty liver disease (NAFLD), Acquired Immune Deficiency Syndrome, corneal dystrophy (CD),

hypercholesterolemia, familial hypercholesterolemia (FH), heart disease (e.g., hypertrophic cardiomyopathy (HCM)), and cancer.

39. A method of detecting a target DNA, comprising contacting the target DNA with the system of claim **20**, wherein the target DNA is modified by the complex, and wherein the modification detects the target DNA; optionally wherein the modification generates a detectable signal, e.g., a fluorescent signal.

* * * * *